# TextAugLLMEdge

November 8, 2026

[1]:
```
# TextAugLLMEdge

# Supplementary Results

# Partha Pratim Ray, 15/6/2025

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Path to your CSV file (change if needed)
csv_path = r"D:\technical writing\textaugmentation\result.csv"

# Load data
df = pd.read_csv(csv_path)
print("Columns:", df.columns.tolist())
print(df.head())
```

```
Columns: ['timestamp', 'prompt_id', 'prompt', 'augmentation_type', 'model',
'augmented_text', 'total_duration_ns', 'load_duration_ns', 'prompt_eval_count',
'prompt_eval_duration_ns', 'eval_count', 'eval_duration_ns',
'tokens_per_second', 'levenshtein_similarity', 'jaccard_similarity',
'length_ratio', 'bleu', 'cosine_similarity', 'wer', 'char_diversity',
'type_token_ratio', 'bigram_overlap']
                   timestamp  prompt_id  \
0  2025-06-15T09:50:22.829584          1
1  2025-06-15T09:50:33.601336          2
2  2025-06-15T09:50:41.541565          3
3  2025-06-15T09:50:49.370159          4
4  2025-06-15T09:50:57.445704          5

                                              prompt augmentation_type  \
0  Although agriculture remains the backbone of o…        paraphrase
1  The rapid advancement of technology has not on…        paraphrase
2  Despite receiving a prestigious scholarship to…        paraphrase
3  In the aftermath of the devastating earthquake…        paraphrase
4  The increasing adoption of renewable energy so…        paraphrase
```

```
        model                                      augmented_text  \
0  qwen2:0.5b  Despite maintaining a substantial portion of i…
1  qwen2:0.5b  The rapid technological progress over the last…
2  qwen2:0.5b  Despite being offered an impressive scholarshi…
3  qwen2:0.5b  In response to a severe natural disaster that …
4  qwen2:0.5b  Renewable energy sources like solar and wind a…

   total_duration_ns  load_duration_ns  prompt_eval_count  \
0        10743499160        2422096170                 62
1         7918451598          53620750                 58
2         7807772750          51113778                 54
3         8053528241          49385191                 62
4         8027890100          52459502                 61

   prompt_eval_duration_ns  …  tokens_per_second  levenshtein_similarity  \
0               1835488062  …           9.253171                0.008955
1               1415348964  …           9.304752                0.050445
2               1302542405  …           9.300377                0.049261
3               1530142101  …           9.269372                0.213873
4               1501011644  …           9.268947                0.006299

   jaccard_similarity  length_ratio      bleu  cosine_similarity       wer  \
0            0.338710      1.359155  0.216514           0.612094  1.054054
1            0.178082      1.442029  0.023705           0.413481  1.394737
2            0.164384      1.537500  0.018897           0.303488  1.500000
3            0.148649      1.261438  0.024377           0.390257  1.250000
4            0.219178      1.369403  0.084801           0.403823  1.119048

   char_diversity  type_token_ratio  bigram_overlap
0        0.107143          0.924528        0.173333
1        0.038462          0.896552        0.032967
2        0.242424          0.981818        0.011628
3        0.068966          0.945455        0.033333
4        0.193548          0.980769        0.082353

[5 rows x 22 columns]
```

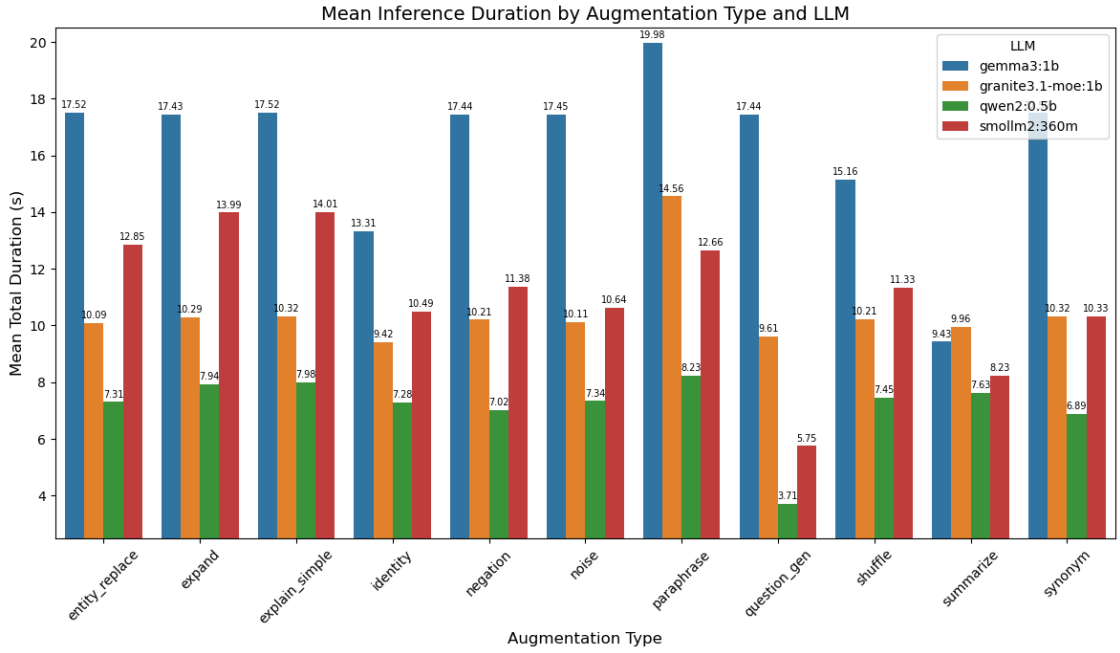
[2]:
```
# BAR PLOT
df['total_duration_s'] = df['total_duration_ns'] / 1e9
agg = df.groupby(['augmentation_type', 'model'])['total_duration_s'].mean().
  ↪reset_index()
augmentation_order = sorted(df['augmentation_type'].unique())

plt.figure(figsize=(12,7))
bar = sns.barplot(
    data=agg,
    x='augmentation_type',
```

```
    y='total_duration_s',
    hue='model',
    order=augmentation_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.2f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Mean Total Duration (s)", fontsize=12)
plt.title("Mean Inference Duration by Augmentation Type and LLM", fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.ylim(2.5,20.5)
plt.show()

print("Mean Total Duration (s) Table:\n")
print(agg.to_string(index=False))

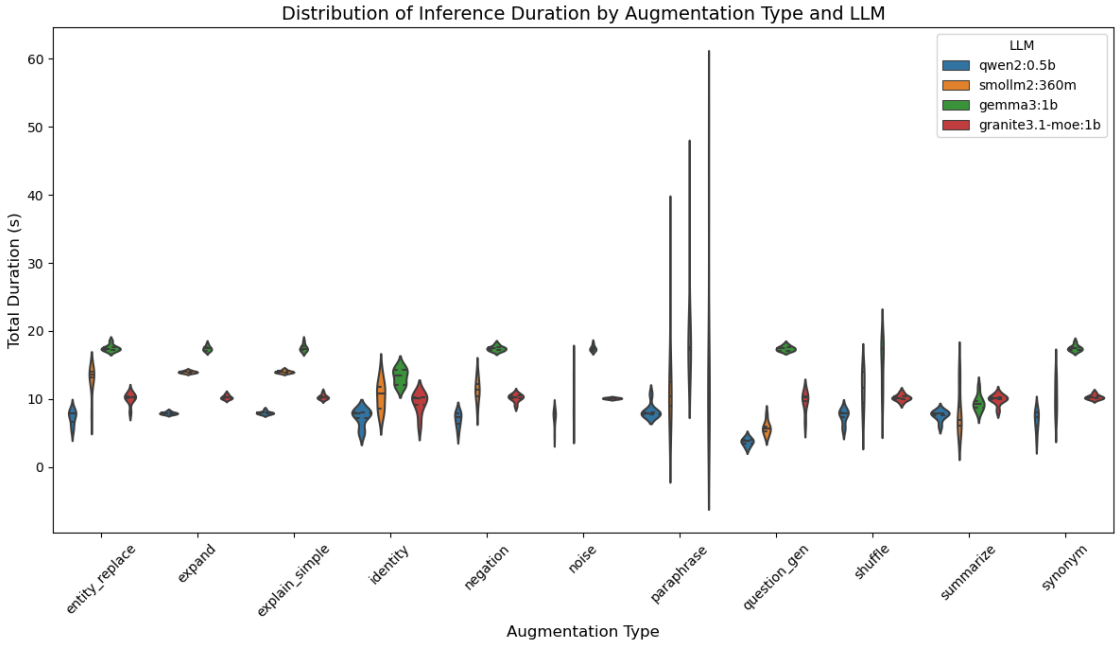
# VIOLIN PLOT
plt.figure(figsize=(12,7))
violin = sns.violinplot(
    data=df,
    x='augmentation_type',
    y='total_duration_s',
    hue='model',
    order=augmentation_order,
    palette='tab10',
    inner="quartile"  # Remove split=True!
)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Total Duration (s)", fontsize=12)
plt.title("Distribution of Inference Duration by Augmentation Type and LLM",␣
 ↪fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()
```




Mean Total Duration (s) Table:

| augmentation_type | model | total_duration_s |
|---|---|---|
| entity_replace | gemma3:1b | 17.519171 |
| entity_replace | granite3.1-moe:1b | 10.087887 |
| entity_replace | qwen2:0.5b | 7.306076 |
| entity_replace | smollm2:360m | 12.848805 |
| expand | gemma3:1b | 17.434952 |
| expand | granite3.1-moe:1b | 10.289341 |
| expand | qwen2:0.5b | 7.936603 |
| expand | smollm2:360m | 13.985258 |
| explain_simple | gemma3:1b | 17.521006 |
| explain_simple | granite3.1-moe:1b | 10.324334 |
| explain_simple | qwen2:0.5b | 7.984643 |
| explain_simple | smollm2:360m | 14.012370 |
| identity | gemma3:1b | 13.313809 |
| identity | granite3.1-moe:1b | 9.422094 |
| identity | qwen2:0.5b | 7.275286 |
| identity | smollm2:360m | 10.487829 |
| negation | gemma3:1b | 17.443342 |
| negation | granite3.1-moe:1b | 10.205199 |
| negation | qwen2:0.5b | 7.019680 |
| negation | smollm2:360m | 11.375481 |
| noise | gemma3:1b | 17.448303 |
| noise | granite3.1-moe:1b | 10.109262 |
| noise | qwen2:0.5b | 7.336289 |

```
          noise      smollm2:360m         10.640014
     paraphrase         gemma3:1b         19.976331
     paraphrase granite3.1-moe:1b         14.560397
     paraphrase        qwen2:0.5b          8.228610
     paraphrase      smollm2:360m         12.661435
  question_gen         gemma3:1b         17.435018
  question_gen granite3.1-moe:1b          9.610098
  question_gen        qwen2:0.5b          3.707023
  question_gen      smollm2:360m          5.754444
       shuffle         gemma3:1b         15.157816
       shuffle granite3.1-moe:1b         10.209583
       shuffle        qwen2:0.5b          7.451169
       shuffle      smollm2:360m         11.327010
     summarize         gemma3:1b          9.429700
     summarize granite3.1-moe:1b          9.956269
     summarize        qwen2:0.5b          7.627685
     summarize      smollm2:360m          8.226404
       synonym         gemma3:1b         17.496148
       synonym granite3.1-moe:1b         10.318432
       synonym        qwen2:0.5b          6.892995
       synonym      smollm2:360m         10.327454
```

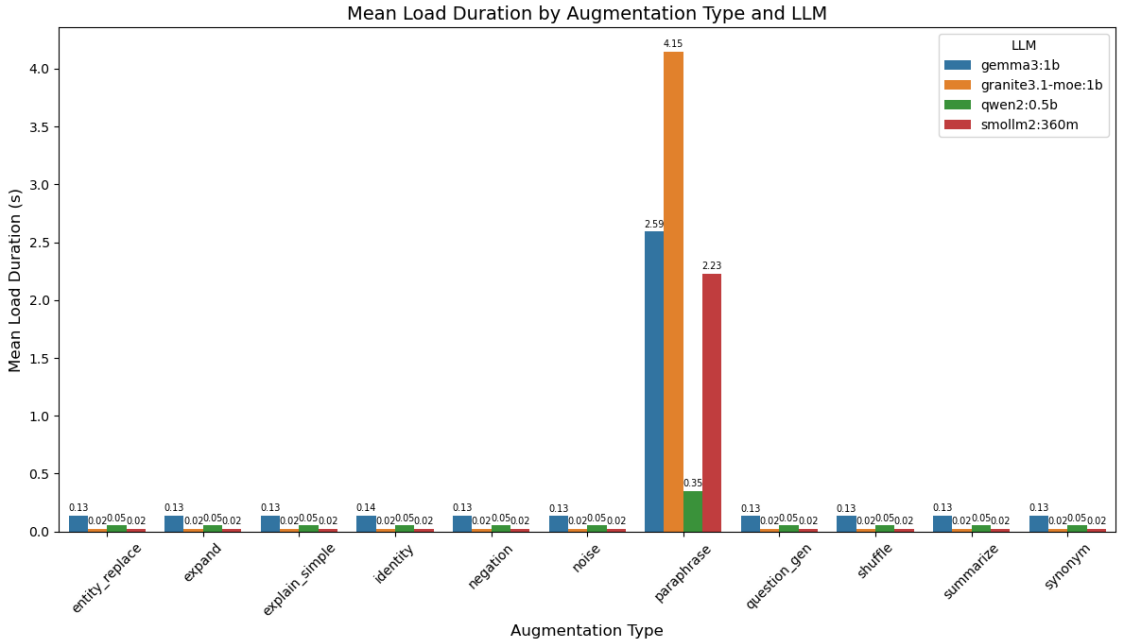



[3]:
```
# BAR PLOT
df['load_duration_s'] = df['load_duration_ns'] / 1e9
agg = df.groupby(['augmentation_type', 'model'])['load_duration_s'].mean().
 ↪reset_index()
```

```
plt.figure(figsize=(12,7))
bar = sns.barplot(
    data=agg,
    x='augmentation_type',
    y='load_duration_s',
    hue='model',
    order=augmentation_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.2f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Mean Load Duration (s)", fontsize=12)
plt.title("Mean Load Duration by Augmentation Type and LLM", fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()

print("Mean Load Duration (s) Table:\n")
print(agg.to_string(index=False))

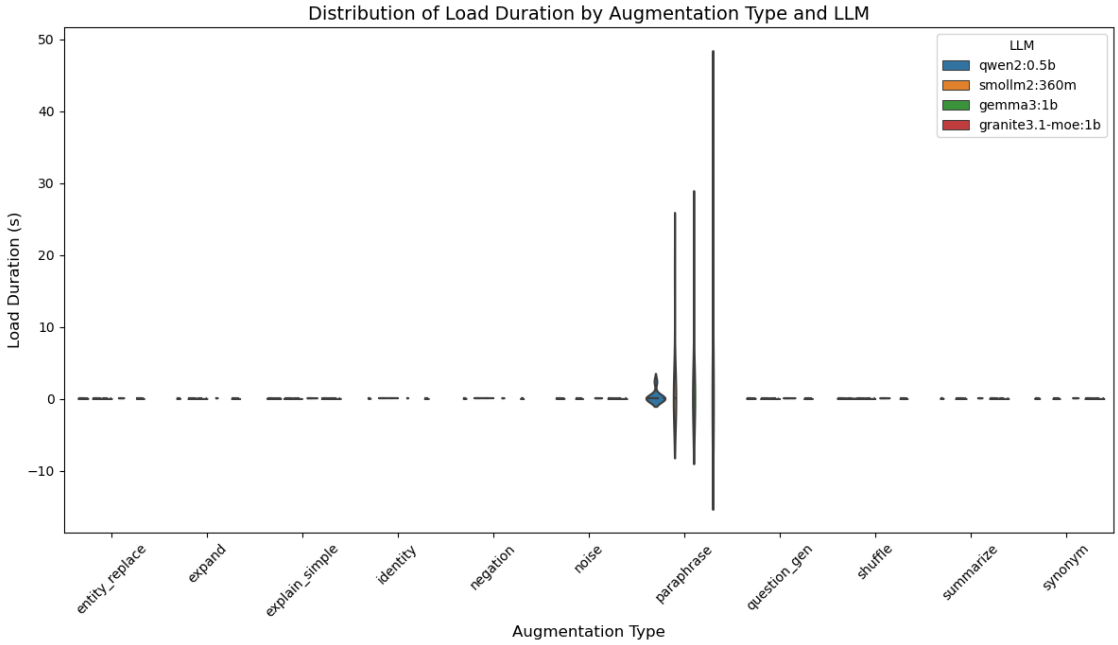
# VIOLIN PLOT
plt.figure(figsize=(12,7))
violin = sns.violinplot(
    data=df,
    x='augmentation_type',
    y='load_duration_s',
    hue='model',
    order=augmentation_order,
    palette='tab10',
    inner="quartile"
)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Load Duration (s)", fontsize=12)
plt.title("Distribution of Load Duration by Augmentation Type and LLM",␣
 ↪fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()
```




Mean Load Duration (s) Table:

| augmentation_type | model | load_duration_s |
|---|---|---|
| entity_replace | gemma3:1b | 0.134435 |
| entity_replace | granite3.1-moe:1b | 0.022135 |
| entity_replace | qwen2:0.5b | 0.051694 |
| entity_replace | smollm2:360m | 0.020649 |
| expand | gemma3:1b | 0.134711 |
| expand | granite3.1-moe:1b | 0.020546 |
| expand | qwen2:0.5b | 0.050972 |
| expand | smollm2:360m | 0.020424 |
| explain_simple | gemma3:1b | 0.134660 |
| explain_simple | granite3.1-moe:1b | 0.021282 |
| explain_simple | qwen2:0.5b | 0.051691 |
| explain_simple | smollm2:360m | 0.021615 |
| identity | gemma3:1b | 0.135201 |
| identity | granite3.1-moe:1b | 0.021608 |
| identity | qwen2:0.5b | 0.051934 |
| identity | smollm2:360m | 0.020424 |
| negation | gemma3:1b | 0.134433 |
| negation | granite3.1-moe:1b | 0.021027 |
| negation | qwen2:0.5b | 0.052432 |
| negation | smollm2:360m | 0.020509 |
| noise | gemma3:1b | 0.133162 |
| noise | granite3.1-moe:1b | 0.020497 |
| noise | qwen2:0.5b | 0.051306 |

```
      noise      smollm2:360m         0.022082
 paraphrase         gemma3:1b         2.591088
 paraphrase granite3.1-moe:1b         4.148439
 paraphrase        qwen2:0.5b         0.348289
 paraphrase      smollm2:360m         2.230602
question_gen         gemma3:1b         0.133863
question_gen granite3.1-moe:1b         0.021682
question_gen        qwen2:0.5b         0.052184
question_gen      smollm2:360m         0.020965
     shuffle         gemma3:1b         0.132988
     shuffle granite3.1-moe:1b         0.021605
     shuffle        qwen2:0.5b         0.051072
     shuffle      smollm2:360m         0.021318
   summarize         gemma3:1b         0.134930
   summarize granite3.1-moe:1b         0.020477
   summarize        qwen2:0.5b         0.052047
   summarize      smollm2:360m         0.021119
     synonym         gemma3:1b         0.134818
     synonym granite3.1-moe:1b         0.021068
     synonym        qwen2:0.5b         0.052437
     synonym      smollm2:360m         0.022486
```

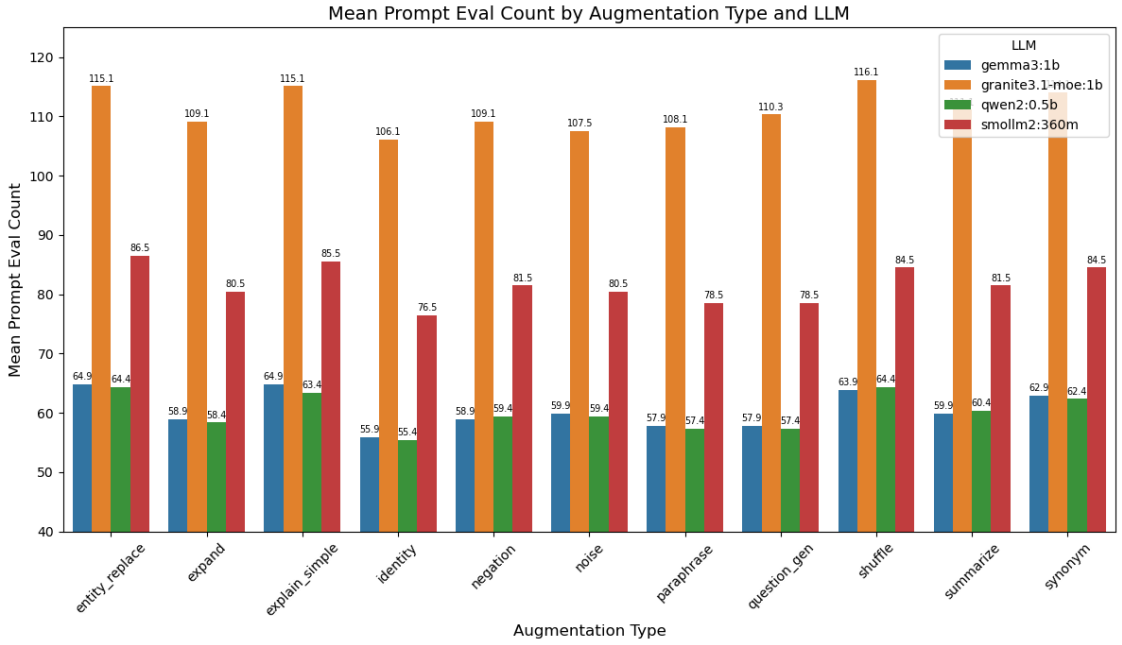



[4]:
```
# BAR PLOT
agg = df.groupby(['augmentation_type', 'model'])['prompt_eval_count'].mean().
 ↪reset_index()
```

```
plt.figure(figsize=(12,7))
bar = sns.barplot(
    data=agg,
    x='augmentation_type',
    y='prompt_eval_count',
    hue='model',
    order=augmentation_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.1f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Mean Prompt Eval Count", fontsize=12)
plt.title("Mean Prompt Eval Count by Augmentation Type and LLM", fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.ylim(40,125)
plt.show()

print("Mean Prompt Eval Count Table:\n")
print(agg.to_string(index=False))

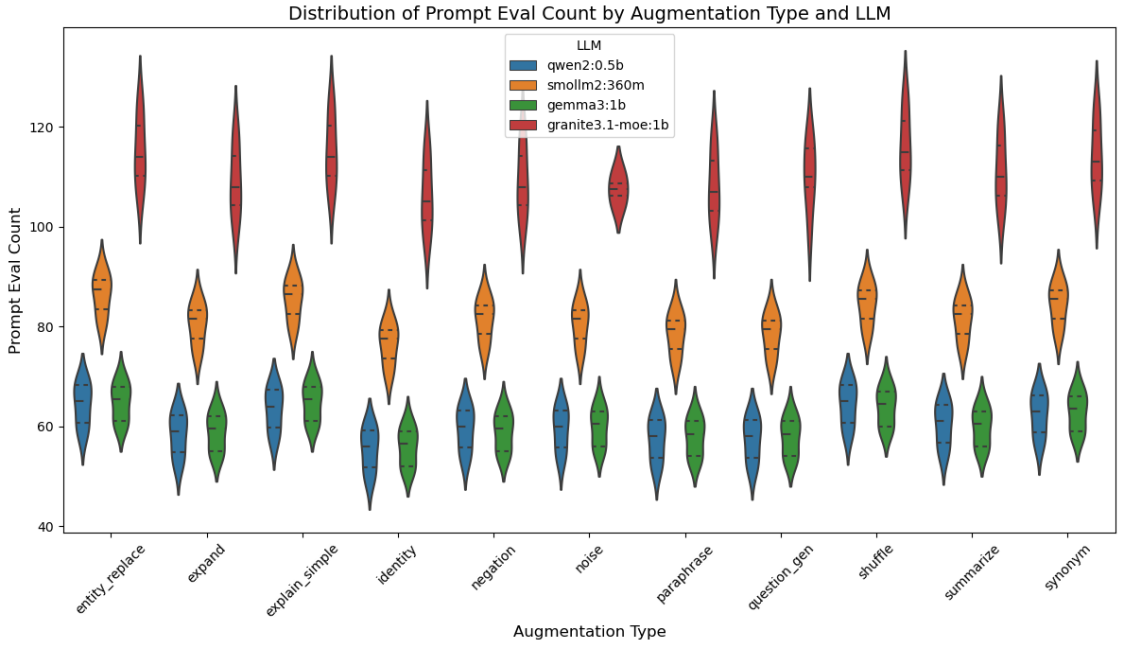
# VIOLIN PLOT
plt.figure(figsize=(12,7))
violin = sns.violinplot(
    data=df,
    x='augmentation_type',
    y='prompt_eval_count',
    hue='model',
    order=augmentation_order,
    palette='tab10',
    inner="quartile"
)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Prompt Eval Count", fontsize=12)
plt.title("Distribution of Prompt Eval Count by Augmentation Type and LLM",␣
 ↪fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()
```




Mean Prompt Eval Count Table:

| augmentation_type | model | prompt_eval_count |
|---|---|---|
| entity_replace | gemma3:1b | 64.875000 |
| entity_replace | granite3.1-moe:1b | 115.125000 |
| entity_replace | qwen2:0.5b | 64.375000 |
| entity_replace | smollm2:360m | 86.500000 |
| expand | gemma3:1b | 58.875000 |
| expand | granite3.1-moe:1b | 109.125000 |
| expand | qwen2:0.5b | 58.375000 |
| expand | smollm2:360m | 80.500000 |
| explain_simple | gemma3:1b | 64.875000 |
| explain_simple | granite3.1-moe:1b | 115.125000 |
| explain_simple | qwen2:0.5b | 63.375000 |
| explain_simple | smollm2:360m | 85.500000 |
| identity | gemma3:1b | 55.875000 |
| identity | granite3.1-moe:1b | 106.125000 |
| identity | qwen2:0.5b | 55.375000 |
| identity | smollm2:360m | 76.500000 |
| negation | gemma3:1b | 58.875000 |
| negation | granite3.1-moe:1b | 109.125000 |
| negation | qwen2:0.5b | 59.375000 |
| negation | smollm2:360m | 81.500000 |
| noise | gemma3:1b | 59.875000 |
| noise | granite3.1-moe:1b | 107.500000 |
| noise | qwen2:0.5b | 59.375000 |

```
      noise      smollm2:360m          80.500000
 paraphrase         gemma3:1b          57.875000
 paraphrase granite3.1-moe:1b         108.125000
 paraphrase        qwen2:0.5b          57.375000
 paraphrase      smollm2:360m          78.500000
question_gen         gemma3:1b          57.875000
question_gen granite3.1-moe:1b         110.333333
question_gen        qwen2:0.5b          57.375000
question_gen      smollm2:360m          78.500000
     shuffle         gemma3:1b          63.875000
     shuffle granite3.1-moe:1b         116.125000
     shuffle        qwen2:0.5b          64.375000
     shuffle      smollm2:360m          84.500000
   summarize         gemma3:1b          59.875000
   summarize granite3.1-moe:1b         111.125000
   summarize        qwen2:0.5b          60.375000
   summarize      smollm2:360m          81.500000
     synonym         gemma3:1b          62.875000
     synonym granite3.1-moe:1b         114.125000
     synonym        qwen2:0.5b          62.375000
     synonym      smollm2:360m          84.500000
```

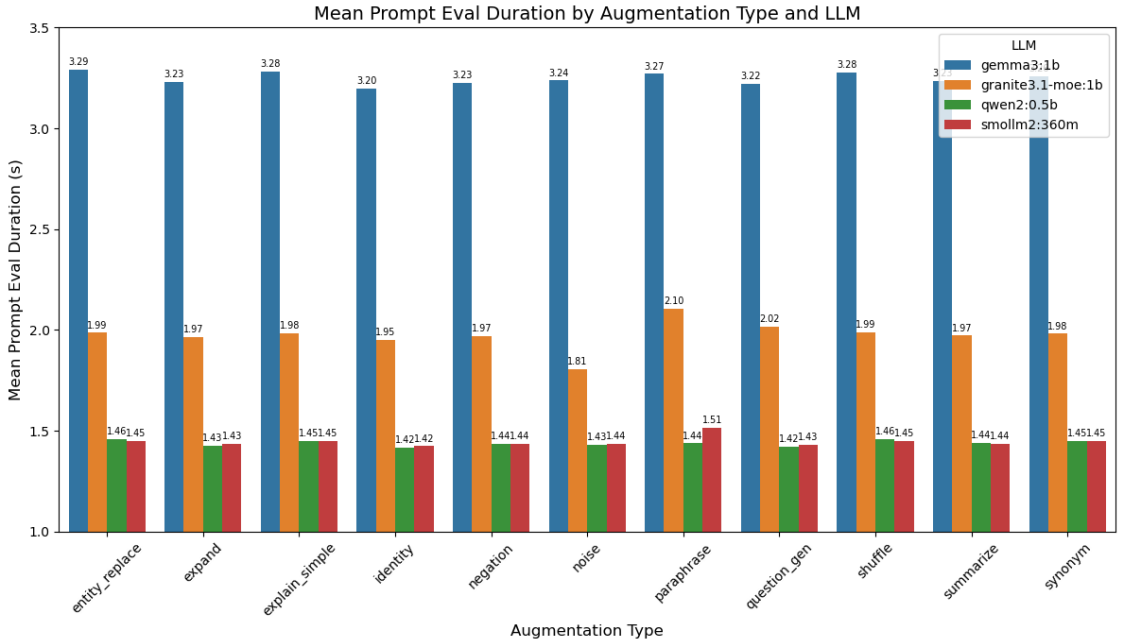



[5]:
```
# BAR PLOT
df['prompt_eval_duration_s'] = df['prompt_eval_duration_ns'] / 1e9
agg = df.groupby(['augmentation_type', 'model'])['prompt_eval_duration_s'].
 ↪mean().reset_index()
```

```
plt.figure(figsize=(12,7))
bar = sns.barplot(
    data=agg,
    x='augmentation_type',
    y='prompt_eval_duration_s',
    hue='model',
    order=augmentation_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.2f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Mean Prompt Eval Duration (s)", fontsize=12)
plt.title("Mean Prompt Eval Duration by Augmentation Type and LLM", fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.ylim(1,3.5)
plt.show()

print("Mean Prompt Eval Duration (s) Table:\n")
print(agg.to_string(index=False))

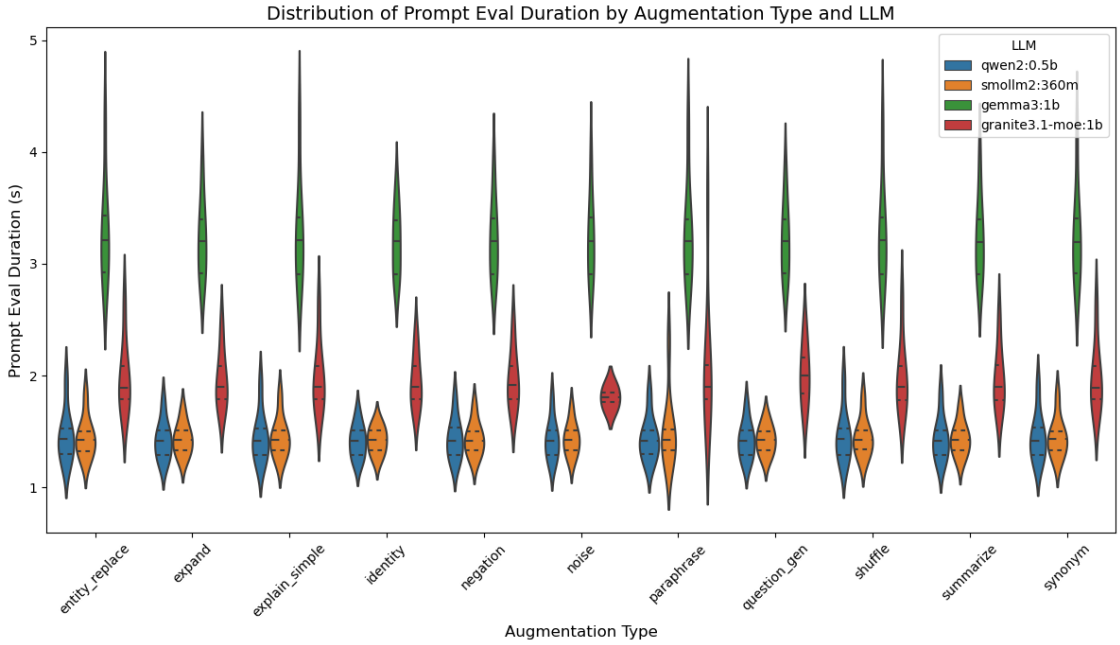
# VIOLIN PLOT
plt.figure(figsize=(12,7))
violin = sns.violinplot(
    data=df,
    x='augmentation_type',
    y='prompt_eval_duration_s',
    hue='model',
    order=augmentation_order,
    palette='tab10',
    inner="quartile"
)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Prompt Eval Duration (s)", fontsize=12)
plt.title("Distribution of Prompt Eval Duration by Augmentation Type and LLM",␣
 ↪fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()
```




Mean Prompt Eval Duration (s) Table:

| augmentation_type | model | prompt_eval_duration_s |
|---|---|---|
| entity_replace | gemma3:1b | 3.289920 |
| entity_replace | granite3.1-moe:1b | 1.986179 |
| entity_replace | qwen2:0.5b | 1.458739 |
| entity_replace | smollm2:360m | 1.448993 |
| expand | gemma3:1b | 3.230149 |
| expand | granite3.1-moe:1b | 1.965222 |
| expand | qwen2:0.5b | 1.425312 |
| expand | smollm2:360m | 1.434685 |
| explain_simple | gemma3:1b | 3.282262 |
| explain_simple | granite3.1-moe:1b | 1.984986 |
| explain_simple | qwen2:0.5b | 1.450086 |
| explain_simple | smollm2:360m | 1.450368 |
| identity | gemma3:1b | 3.198626 |
| identity | granite3.1-moe:1b | 1.952127 |
| identity | qwen2:0.5b | 1.415143 |
| identity | smollm2:360m | 1.423411 |
| negation | gemma3:1b | 3.226121 |
| negation | granite3.1-moe:1b | 1.969167 |
| negation | qwen2:0.5b | 1.436152 |
| negation | smollm2:360m | 1.435887 |
| noise | gemma3:1b | 3.237751 |
| noise | granite3.1-moe:1b | 1.806109 |
| noise | qwen2:0.5b | 1.429667 |

```
      noise      smollm2:360m              1.435769
 paraphrase         gemma3:1b              3.269960
 paraphrase granite3.1-moe:1b              2.104556
 paraphrase        qwen2:0.5b              1.437472
 paraphrase      smollm2:360m              1.514472
question_gen         gemma3:1b              3.219902
question_gen granite3.1-moe:1b              2.017061
question_gen        qwen2:0.5b              1.422472
question_gen      smollm2:360m              1.428817
    shuffle         gemma3:1b              3.277141
    shuffle granite3.1-moe:1b              1.988787
    shuffle        qwen2:0.5b              1.459124
    shuffle      smollm2:360m              1.448311
  summarize         gemma3:1b              3.234136
  summarize granite3.1-moe:1b              1.971836
  summarize        qwen2:0.5b              1.438362
  summarize      smollm2:360m              1.436426
    synonym         gemma3:1b              3.260029
    synonym granite3.1-moe:1b              1.981388
    synonym        qwen2:0.5b              1.450563
    synonym      smollm2:360m              1.449356
```

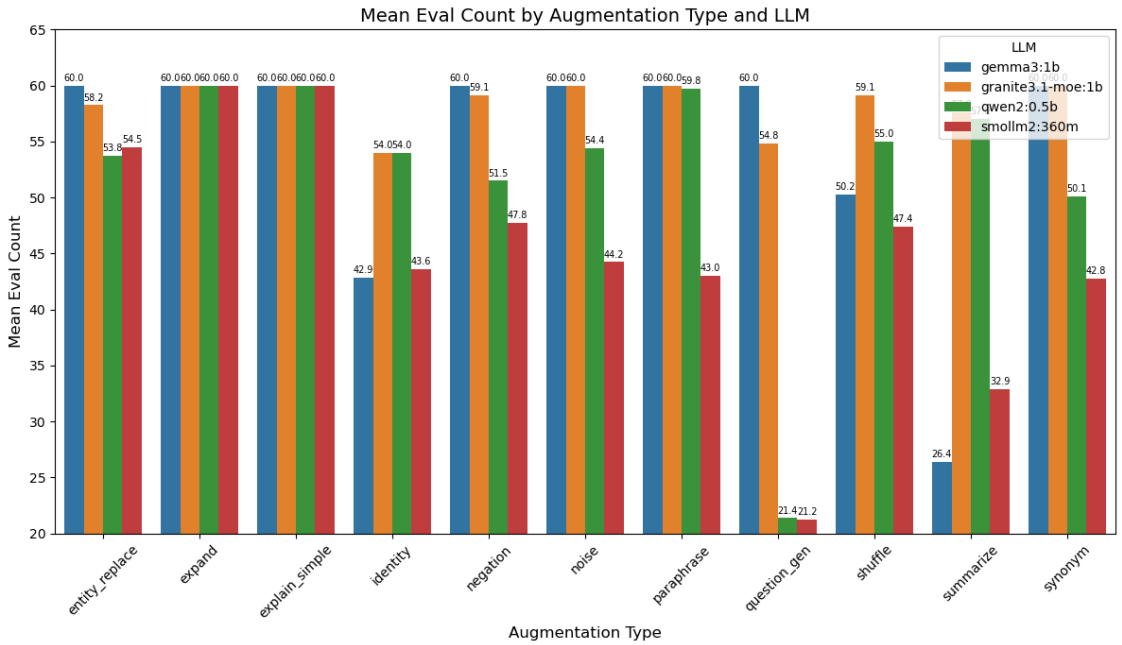



[6]:
```
# BAR PLOT
agg = df.groupby(['augmentation_type', 'model'])['eval_count'].mean().
 ↪reset_index()
```

```
plt.figure(figsize=(12,7))
bar = sns.barplot(
    data=agg,
    x='augmentation_type',
    y='eval_count',
    hue='model',
    order=augmentation_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.1f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Mean Eval Count", fontsize=12)
plt.title("Mean Eval Count by Augmentation Type and LLM", fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.ylim(20,65)
plt.show()

print("Mean Eval Count Table:\n")
print(agg.to_string(index=False))

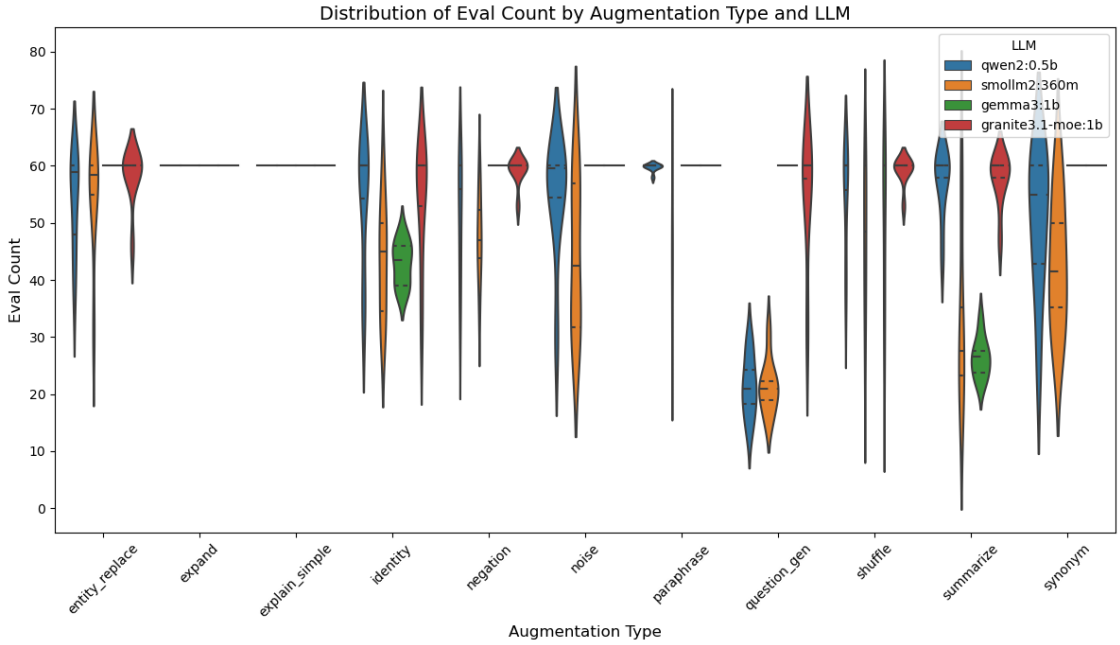
# VIOLIN PLOT
plt.figure(figsize=(12,7))
violin = sns.violinplot(
    data=df,
    x='augmentation_type',
    y='eval_count',
    hue='model',
    order=augmentation_order,
    palette='tab10',
    inner="quartile"
)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Eval Count", fontsize=12)
plt.title("Distribution of Eval Count by Augmentation Type and LLM",␣
 ↪fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()
```




Mean Eval Count Table:

| augmentation_type | model | eval_count |
|---|---|---|
| entity_replace | gemma3:1b | 60.000000 |
| entity_replace | granite3.1-moe:1b | 58.250000 |
| entity_replace | qwen2:0.5b | 53.750000 |
| entity_replace | smollm2:360m | 54.500000 |
| expand | gemma3:1b | 60.000000 |
| expand | granite3.1-moe:1b | 60.000000 |
| expand | qwen2:0.5b | 60.000000 |
| expand | smollm2:360m | 60.000000 |
| explain_simple | gemma3:1b | 60.000000 |
| explain_simple | granite3.1-moe:1b | 60.000000 |
| explain_simple | qwen2:0.5b | 60.000000 |
| explain_simple | smollm2:360m | 60.000000 |
| identity | gemma3:1b | 42.875000 |
| identity | granite3.1-moe:1b | 54.000000 |
| identity | qwen2:0.5b | 54.000000 |
| identity | smollm2:360m | 43.625000 |
| negation | gemma3:1b | 60.000000 |
| negation | granite3.1-moe:1b | 59.125000 |
| negation | qwen2:0.5b | 51.500000 |
| negation | smollm2:360m | 47.750000 |
| noise | gemma3:1b | 60.000000 |
| noise | granite3.1-moe:1b | 60.000000 |
| noise | qwen2:0.5b | 54.375000 |

```
          noise      smollm2:360m   44.250000
     paraphrase         gemma3:1b   60.000000
     paraphrase granite3.1-moe:1b   60.000000
     paraphrase        qwen2:0.5b   59.750000
     paraphrase      smollm2:360m   43.000000
  question_gen         gemma3:1b   60.000000
  question_gen granite3.1-moe:1b   54.833333
  question_gen        qwen2:0.5b   21.375000
  question_gen      smollm2:360m   21.250000
       shuffle         gemma3:1b   50.250000
       shuffle granite3.1-moe:1b   59.125000
       shuffle        qwen2:0.5b   55.000000
       shuffle      smollm2:360m   47.375000
     summarize         gemma3:1b   26.375000
     summarize granite3.1-moe:1b   57.625000
     summarize        qwen2:0.5b   57.000000
     summarize      smollm2:360m   32.875000
       synonym         gemma3:1b   60.000000
       synonym granite3.1-moe:1b   60.000000
       synonym        qwen2:0.5b   50.125000
       synonym      smollm2:360m   42.750000
```

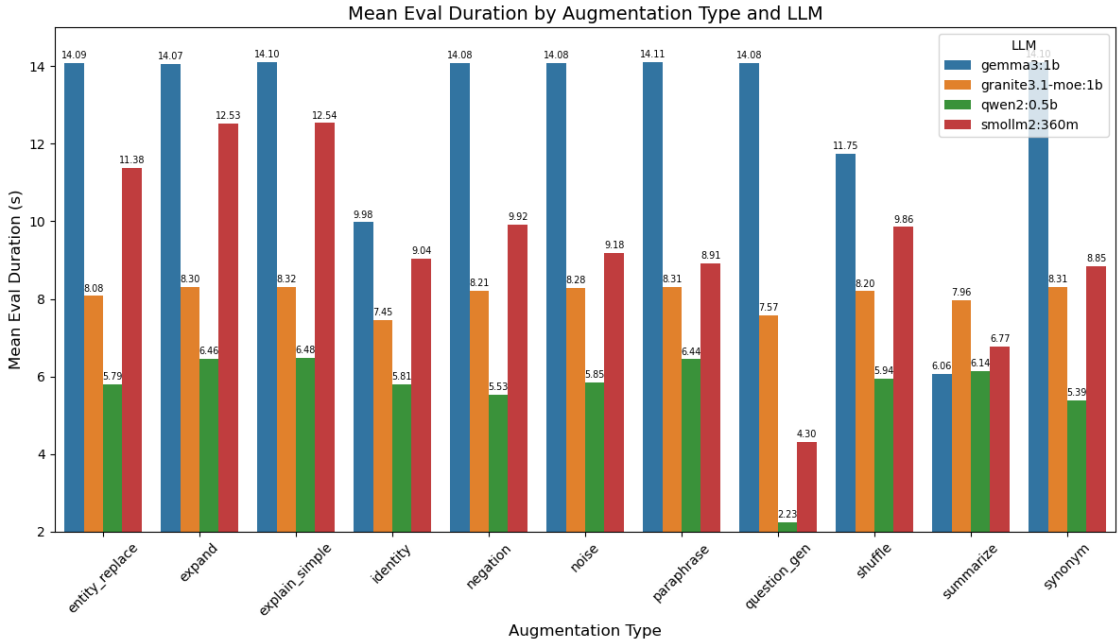



[7]:
```
# BAR PLOT
df['eval_duration_s'] = df['eval_duration_ns'] / 1e9
agg = df.groupby(['augmentation_type', 'model'])['eval_duration_s'].mean().
 ↪reset_index()
```

```
plt.figure(figsize=(12,7))
bar = sns.barplot(
    data=agg,
    x='augmentation_type',
    y='eval_duration_s',
    hue='model',
    order=augmentation_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.2f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Mean Eval Duration (s)", fontsize=12)
plt.title("Mean Eval Duration by Augmentation Type and LLM", fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.ylim(2,15)
plt.show()

print("Mean Eval Duration (s) Table:\n")
print(agg.to_string(index=False))

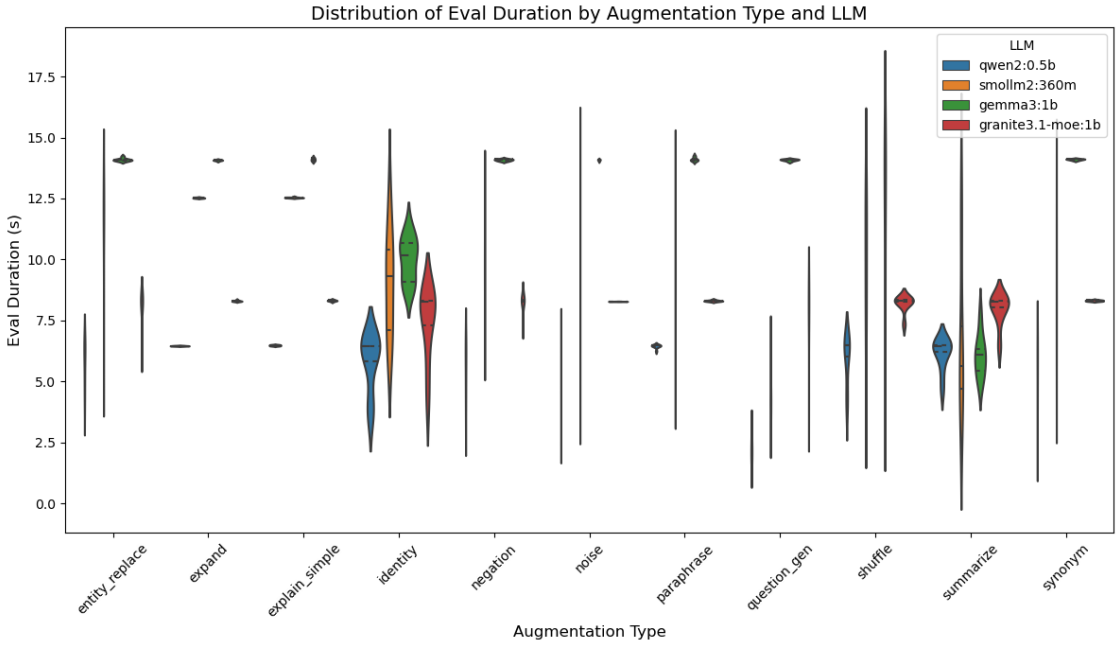
# VIOLIN PLOT
plt.figure(figsize=(12,7))
violin = sns.violinplot(
    data=df,
    x='augmentation_type',
    y='eval_duration_s',
    hue='model',
    order=augmentation_order,
    palette='tab10',
    inner="quartile"
)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Eval Duration (s)", fontsize=12)
plt.title("Distribution of Eval Duration by Augmentation Type and LLM",␣
 ↪fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()
```




Mean Eval Duration (s) Table:

| augmentation_type | model | eval_duration_s |
|---|---|---|
| entity_replace | gemma3:1b | 14.093728 |
| entity_replace | granite3.1-moe:1b | 8.077532 |
| entity_replace | qwen2:0.5b | 5.794070 |
| entity_replace | smollm2:360m | 11.377713 |
| expand | gemma3:1b | 14.069005 |
| expand | granite3.1-moe:1b | 8.301877 |
| expand | qwen2:0.5b | 6.459140 |
| expand | smollm2:360m | 12.528693 |
| explain_simple | gemma3:1b | 14.102958 |
| explain_simple | granite3.1-moe:1b | 8.316195 |
| explain_simple | qwen2:0.5b | 6.481418 |
| explain_simple | smollm2:360m | 12.538618 |
| identity | gemma3:1b | 9.978718 |
| identity | granite3.1-moe:1b | 7.446409 |
| identity | qwen2:0.5b | 5.806615 |
| identity | smollm2:360m | 9.042256 |
| negation | gemma3:1b | 14.081699 |
| negation | granite3.1-moe:1b | 8.212668 |
| negation | qwen2:0.5b | 5.529797 |
| negation | smollm2:360m | 9.917460 |
| noise | gemma3:1b | 14.076263 |
| noise | granite3.1-moe:1b | 8.280945 |
| noise | qwen2:0.5b | 5.853783 |

```
         noise      smollm2:360m          9.180232
    paraphrase         gemma3:1b         14.113944
    paraphrase granite3.1-moe:1b          8.305655
    paraphrase        qwen2:0.5b          6.441398
    paraphrase      smollm2:360m          8.914208
  question_gen         gemma3:1b         14.080092
  question_gen granite3.1-moe:1b          7.569508
  question_gen        qwen2:0.5b          2.231112
  question_gen      smollm2:360m          4.302042
       shuffle         gemma3:1b         11.746517
       shuffle granite3.1-moe:1b          8.197225
       shuffle        qwen2:0.5b          5.939719
       shuffle      smollm2:360m          9.855855
     summarize         gemma3:1b          6.059394
     summarize granite3.1-moe:1b          7.962053
     summarize        qwen2:0.5b          6.136115
     summarize      smollm2:360m          6.767067
       synonym         gemma3:1b         14.100174
       synonym granite3.1-moe:1b          8.314038
       synonym        qwen2:0.5b          5.388443
       synonym      smollm2:360m          8.853510
```

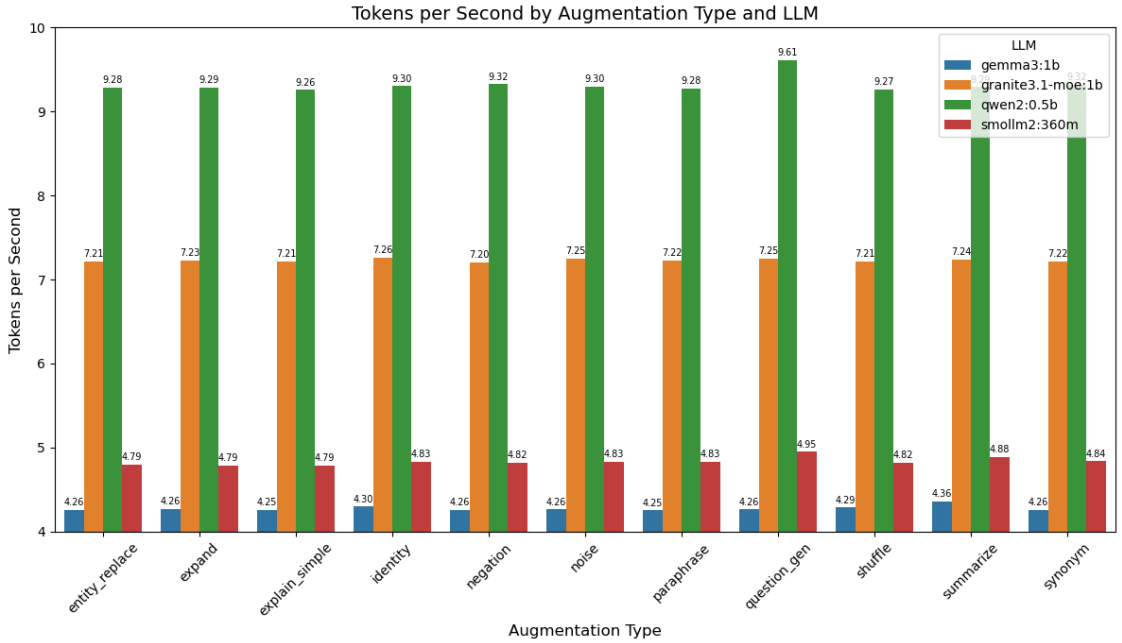



[8]:
```
# BAR PLOT
agg = df.groupby(['augmentation_type', 'model'])['tokens_per_second'].mean().
  ↪reset_index()
```

```
plt.figure(figsize=(12,7))
bar = sns.barplot(
    data=agg,
    x='augmentation_type',
    y='tokens_per_second',
    hue='model',
    order=augmentation_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.2f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Tokens per Second", fontsize=12)
plt.title("Tokens per Second by Augmentation Type and LLM", fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.ylim(4,10)
plt.show()

print("Tokens per Second Table:\n")
print(agg.to_string(index=False))

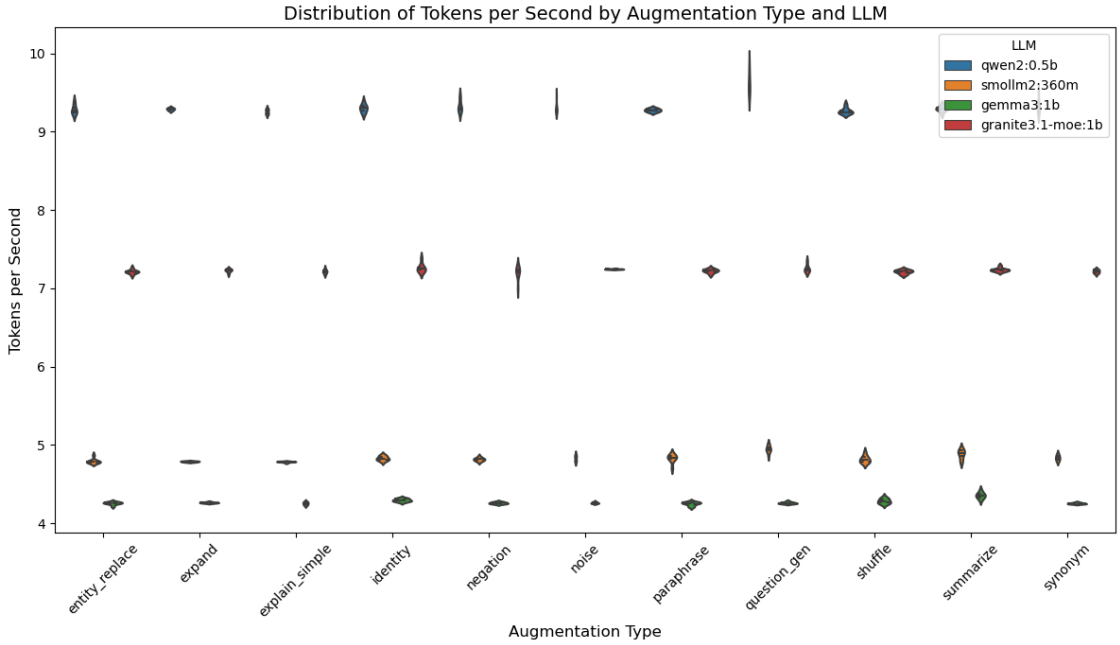
# VIOLIN PLOT
plt.figure(figsize=(12,7))
violin = sns.violinplot(
    data=df,
    x='augmentation_type',
    y='tokens_per_second',
    hue='model',
    order=augmentation_order,
    palette='tab10',
    inner="quartile"
)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Tokens per Second", fontsize=12)
plt.title("Distribution of Tokens per Second by Augmentation Type and LLM",␣
 ↪fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()
```




Tokens per Second Table:

| augmentation_type | model | tokens_per_second |
|---|---|---|
| entity_replace | gemma3:1b | 4.257286 |
| entity_replace | granite3.1-moe:1b | 7.212979 |
| entity_replace | qwen2:0.5b | 9.283693 |
| entity_replace | smollm2:360m | 4.794924 |
| expand | gemma3:1b | 4.264705 |
| expand | granite3.1-moe:1b | 7.227345 |
| expand | qwen2:0.5b | 9.289185 |
| expand | smollm2:360m | 4.789015 |
| explain_simple | gemma3:1b | 4.254498 |
| explain_simple | granite3.1-moe:1b | 7.214922 |
| explain_simple | qwen2:0.5b | 9.257323 |
| explain_simple | smollm2:360m | 4.785226 |
| identity | gemma3:1b | 4.297115 |
| identity | granite3.1-moe:1b | 7.261247 |
| identity | qwen2:0.5b | 9.301166 |
| identity | smollm2:360m | 4.830443 |
| negation | gemma3:1b | 4.260880 |
| negation | granite3.1-moe:1b | 7.201331 |
| negation | qwen2:0.5b | 9.324690 |
| negation | smollm2:360m | 4.817815 |
| noise | gemma3:1b | 4.262515 |
| noise | granite3.1-moe:1b | 7.245551 |
| noise | qwen2:0.5b | 9.299473 |

```
      noise      smollm2:360m          4.829795
 paraphrase         gemma3:1b          4.251231
 paraphrase granite3.1-moe:1b          7.224071
 paraphrase        qwen2:0.5b          9.276019
 paraphrase      smollm2:360m          4.830580
question_gen         gemma3:1b          4.261365
question_gen granite3.1-moe:1b          7.252370
question_gen        qwen2:0.5b          9.610416
question_gen      smollm2:360m          4.948760
    shuffle         gemma3:1b          4.285316
    shuffle granite3.1-moe:1b          7.212863
    shuffle        qwen2:0.5b          9.265605
    shuffle      smollm2:360m          4.817816
  summarize         gemma3:1b          4.356848
  summarize granite3.1-moe:1b          7.238769
  summarize        qwen2:0.5b          9.291481
  summarize      smollm2:360m          4.882819
    synonym         gemma3:1b          4.255283
    synonym granite3.1-moe:1b          7.216768
    synonym        qwen2:0.5b          9.320334
    synonym      smollm2:360m          4.836850
```

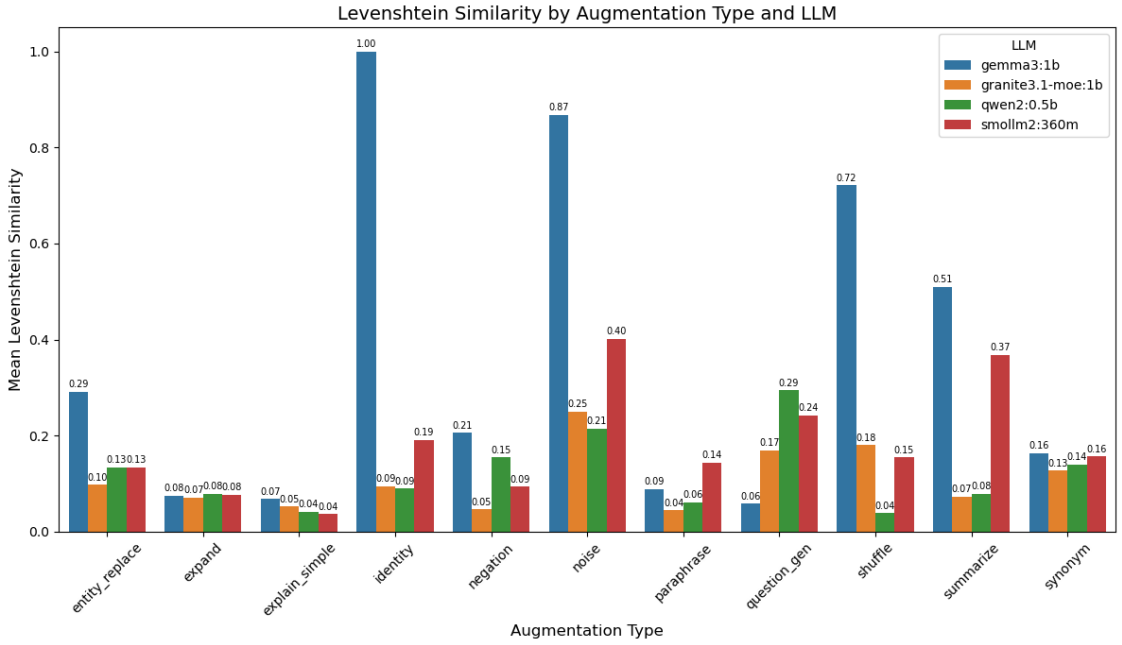



[9]:
```
agg = df.groupby(['augmentation_type', 'model'])['levenshtein_similarity'].
↪mean().reset_index()
augmentation_order = sorted(df['augmentation_type'].unique())
```

```
plt.figure(figsize=(12,7))
bar = sns.barplot(
    data=agg,
    x='augmentation_type',
    y='levenshtein_similarity',
    hue='model',
    order=augmentation_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.2f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Mean Levenshtein Similarity", fontsize=12)
plt.title("Levenshtein Similarity by Augmentation Type and LLM", fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()

print("Mean Levenshtein Similarity Table:\n")
print(agg.to_string(index=False))

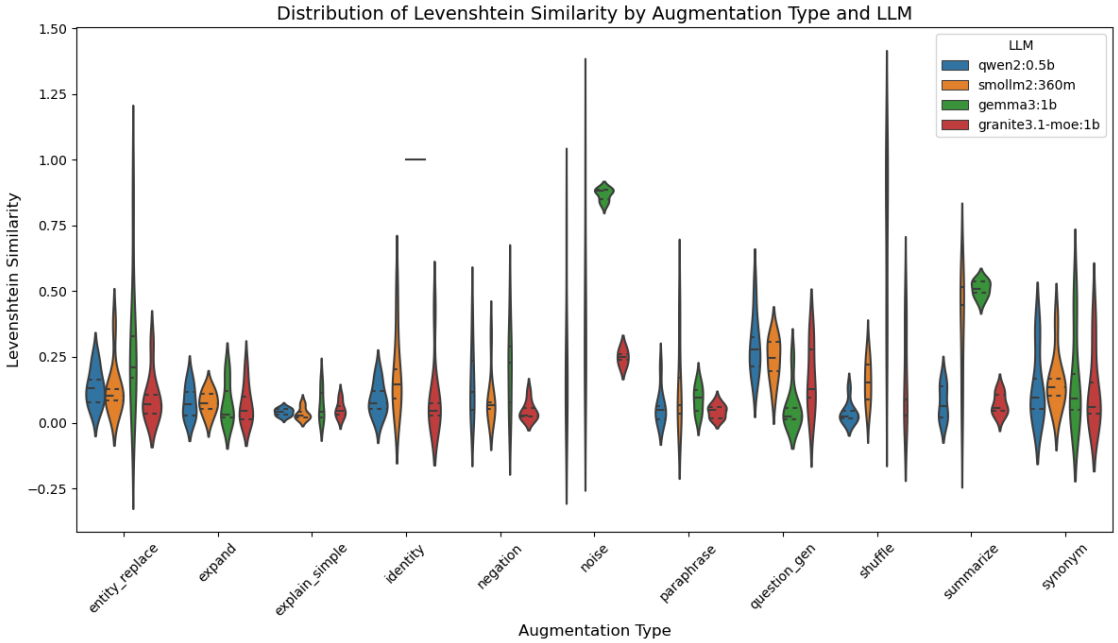
# VIOLIN PLOT
plt.figure(figsize=(12,7))
sns.violinplot(
    data=df,
    x='augmentation_type',
    y='levenshtein_similarity',
    hue='model',
    order=augmentation_order,
    palette='tab10',
    inner="quartile"
)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Levenshtein Similarity", fontsize=12)
plt.title("Distribution of Levenshtein Similarity by Augmentation Type and␣
 ↪LLM", fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()
```




Mean Levenshtein Similarity Table:

| augmentation_type | model | levenshtein_similarity |
|---|---|---|
| entity_replace | gemma3:1b | 0.290348 |
| entity_replace | granite3.1-moe:1b | 0.097073 |
| entity_replace | qwen2:0.5b | 0.132908 |
| entity_replace | smollm2:360m | 0.132863 |
| expand | gemma3:1b | 0.075134 |
| expand | granite3.1-moe:1b | 0.069932 |
| expand | qwen2:0.5b | 0.078037 |
| expand | smollm2:360m | 0.077145 |
| explain_simple | gemma3:1b | 0.067784 |
| explain_simple | granite3.1-moe:1b | 0.051873 |
| explain_simple | qwen2:0.5b | 0.040759 |
| explain_simple | smollm2:360m | 0.036005 |
| identity | gemma3:1b | 1.000000 |
| identity | granite3.1-moe:1b | 0.094511 |
| identity | qwen2:0.5b | 0.089047 |
| identity | smollm2:360m | 0.190436 |
| negation | gemma3:1b | 0.205183 |
| negation | granite3.1-moe:1b | 0.045812 |
| negation | qwen2:0.5b | 0.154084 |
| negation | smollm2:360m | 0.093264 |
| noise | gemma3:1b | 0.868107 |
| noise | granite3.1-moe:1b | 0.249687 |
| noise | qwen2:0.5b | 0.214436 |

| | | |
|---|---|---|
| noise | smollm2:360m | 0.400869 |
| paraphrase | gemma3:1b | 0.088593 |
| paraphrase | granite3.1-moe:1b | 0.044057 |
| paraphrase | qwen2:0.5b | 0.060148 |
| paraphrase | smollm2:360m | 0.144007 |
| question_gen | gemma3:1b | 0.057572 |
| question_gen | granite3.1-moe:1b | 0.168588 |
| question_gen | qwen2:0.5b | 0.293763 |
| question_gen | smollm2:360m | 0.241747 |
| shuffle | gemma3:1b | 0.721207 |
| shuffle | granite3.1-moe:1b | 0.179992 |
| shuffle | qwen2:0.5b | 0.038025 |
| shuffle | smollm2:360m | 0.154309 |
| summarize | gemma3:1b | 0.509782 |
| summarize | granite3.1-moe:1b | 0.071765 |
| summarize | qwen2:0.5b | 0.078576 |
| summarize | smollm2:360m | 0.367185 |
| synonym | gemma3:1b | 0.163364 |
| synonym | granite3.1-moe:1b | 0.126834 |
| synonym | qwen2:0.5b | 0.138962 |
| synonym | smollm2:360m | 0.156136 |



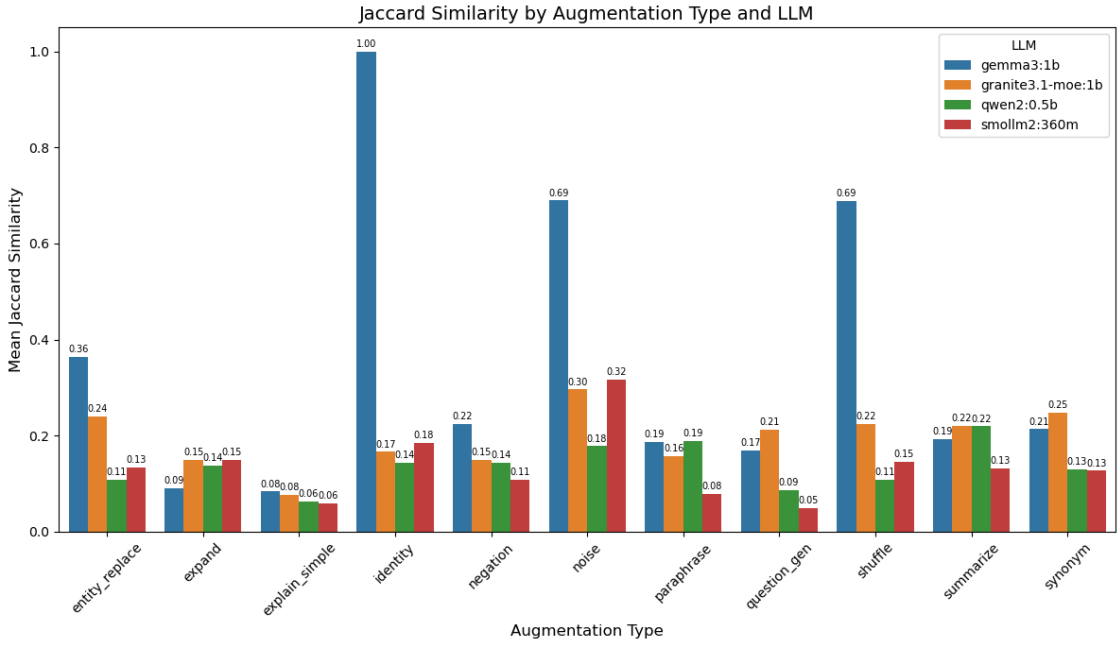

[10]:
```
agg = df.groupby(['augmentation_type', 'model'])['jaccard_similarity'].mean().
↪reset_index()

plt.figure(figsize=(12,7))
```

```
bar = sns.barplot(
    data=agg,
    x='augmentation_type',
    y='jaccard_similarity',
    hue='model',
    order=augmentation_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.2f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Mean Jaccard Similarity", fontsize=12)
plt.title("Jaccard Similarity by Augmentation Type and LLM", fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()

print("Mean Jaccard Similarity Table:\n")
print(agg.to_string(index=False))

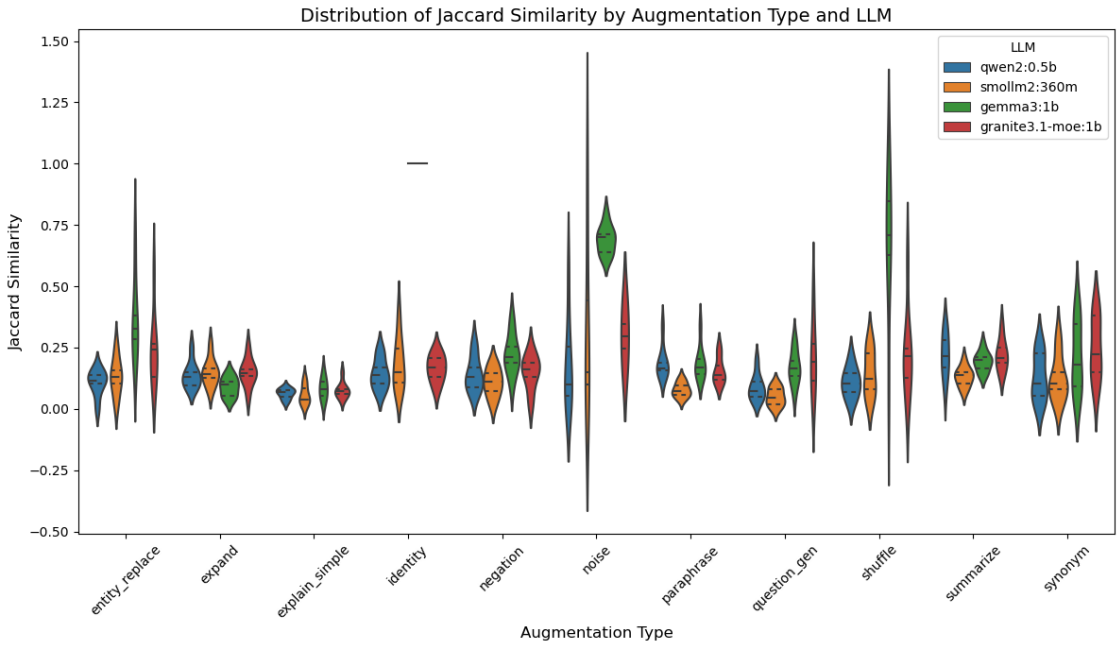

plt.figure(figsize=(12,7))
sns.violinplot(
    data=df,
    x='augmentation_type',
    y='jaccard_similarity',
    hue='model',
    order=augmentation_order,
    palette='tab10',
    inner="quartile"
)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Jaccard Similarity", fontsize=12)
plt.title("Distribution of Jaccard Similarity by Augmentation Type and LLM",␣
 ↪fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()
```




Mean Jaccard Similarity Table:

| augmentation_type | model | jaccard_similarity |
|---|---|---|
| entity_replace | gemma3:1b | 0.364388 |
| entity_replace | granite3.1-moe:1b | 0.239889 |
| entity_replace | qwen2:0.5b | 0.107304 |
| entity_replace | smollm2:360m | 0.132787 |
| expand | gemma3:1b | 0.090450 |
| expand | granite3.1-moe:1b | 0.148763 |
| expand | qwen2:0.5b | 0.136878 |
| expand | smollm2:360m | 0.149658 |
| explain_simple | gemma3:1b | 0.083227 |
| explain_simple | granite3.1-moe:1b | 0.076358 |
| explain_simple | qwen2:0.5b | 0.062661 |
| explain_simple | smollm2:360m | 0.058572 |
| identity | gemma3:1b | 1.000000 |
| identity | granite3.1-moe:1b | 0.166029 |
| identity | qwen2:0.5b | 0.143552 |
| identity | smollm2:360m | 0.184490 |
| negation | gemma3:1b | 0.223519 |
| negation | granite3.1-moe:1b | 0.149267 |
| negation | qwen2:0.5b | 0.143467 |
| negation | smollm2:360m | 0.108570 |
| noise | gemma3:1b | 0.689598 |
| noise | granite3.1-moe:1b | 0.295714 |
| noise | qwen2:0.5b | 0.177881 |

| | | |
|---|---|---|
| noise | smollm2:360m | 0.316545 |
| paraphrase | gemma3:1b | 0.186556 |
| paraphrase | granite3.1-moe:1b | 0.156475 |
| paraphrase | qwen2:0.5b | 0.188939 |
| paraphrase | smollm2:360m | 0.078176 |
| question_gen | gemma3:1b | 0.169422 |
| question_gen | granite3.1-moe:1b | 0.211565 |
| question_gen | qwen2:0.5b | 0.086832 |
| question_gen | smollm2:360m | 0.049405 |
| shuffle | gemma3:1b | 0.688624 |
| shuffle | granite3.1-moe:1b | 0.224040 |
| shuffle | qwen2:0.5b | 0.108495 |
| shuffle | smollm2:360m | 0.146016 |
| summarize | gemma3:1b | 0.191970 |
| summarize | granite3.1-moe:1b | 0.219977 |
| summarize | qwen2:0.5b | 0.219177 |
| summarize | smollm2:360m | 0.131202 |
| synonym | gemma3:1b | 0.213119 |
| synonym | granite3.1-moe:1b | 0.247364 |
| synonym | qwen2:0.5b | 0.129532 |
| synonym | smollm2:360m | 0.126717 |



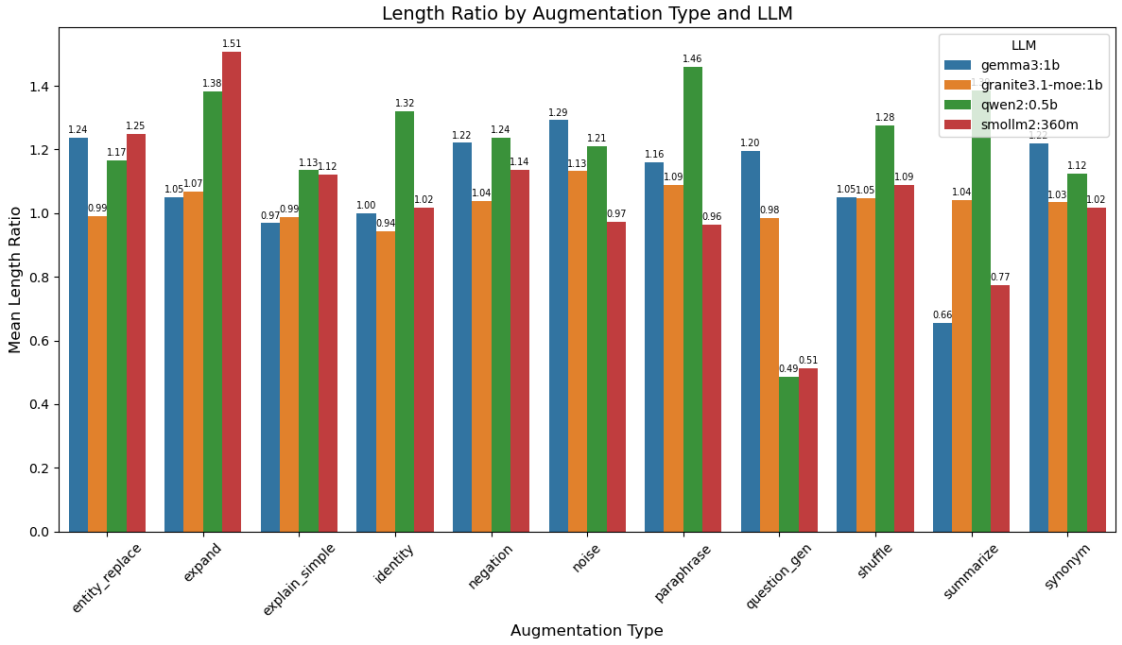

[11]:
```
agg = df.groupby(['augmentation_type', 'model'])['length_ratio'].mean().
↪reset_index()

plt.figure(figsize=(12,7))
```

```
bar = sns.barplot(
    data=agg,
    x='augmentation_type',
    y='length_ratio',
    hue='model',
    order=augmentation_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.2f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Mean Length Ratio", fontsize=12)
plt.title("Length Ratio by Augmentation Type and LLM", fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()

print("Mean Length Ratio Table:\n")
print(agg.to_string(index=False))

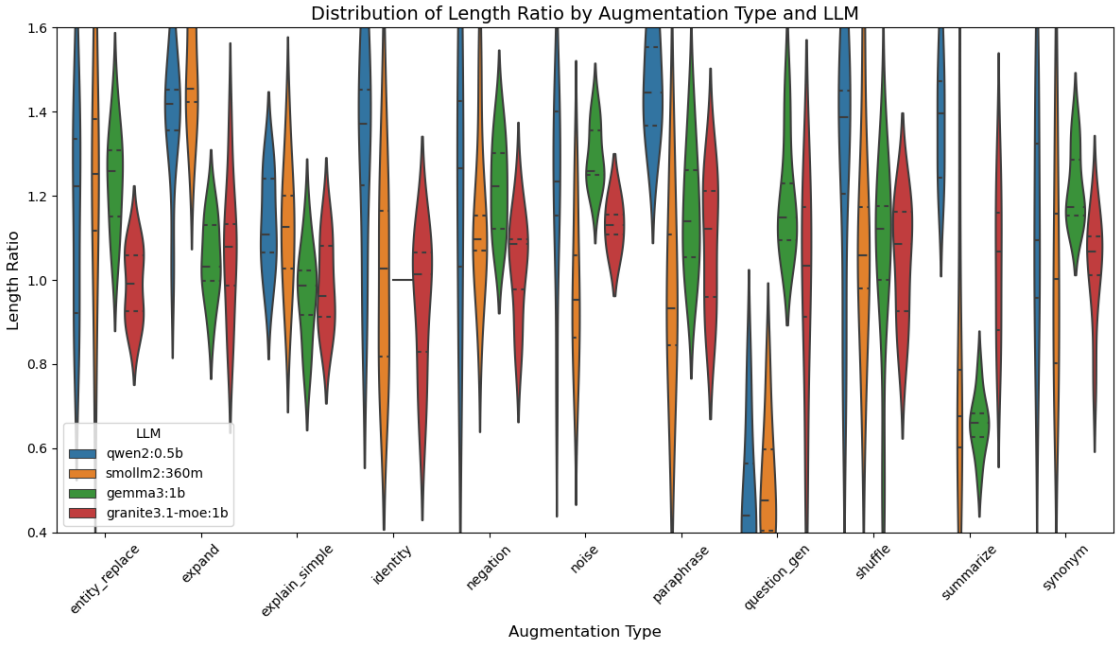
plt.figure(figsize=(12,7))
sns.violinplot(
    data=df,
    x='augmentation_type',
    y='length_ratio',
    hue='model',
    order=augmentation_order,
    palette='tab10',
    inner="quartile"
)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Length Ratio", fontsize=12)
plt.title("Distribution of Length Ratio by Augmentation Type and LLM",␣
 ↪fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.ylim(.4, 1.6)
plt.show()
```




Mean Length Ratio Table:

| augmentation_type | model | length_ratio |
|---|---|---|
| entity_replace | gemma3:1b | 1.236984 |
| entity_replace | granite3.1-moe:1b | 0.990239 |
| entity_replace | qwen2:0.5b | 1.166508 |
| entity_replace | smollm2:360m | 1.249348 |
| expand | gemma3:1b | 1.049079 |
| expand | granite3.1-moe:1b | 1.067523 |
| expand | qwen2:0.5b | 1.382795 |
| expand | smollm2:360m | 1.507696 |
| explain_simple | gemma3:1b | 0.968811 |
| explain_simple | granite3.1-moe:1b | 0.988082 |
| explain_simple | qwen2:0.5b | 1.134455 |
| explain_simple | smollm2:360m | 1.122041 |
| identity | gemma3:1b | 1.000000 |
| identity | granite3.1-moe:1b | 0.943784 |
| identity | qwen2:0.5b | 1.320280 |
| identity | smollm2:360m | 1.016836 |
| negation | gemma3:1b | 1.220938 |
| negation | granite3.1-moe:1b | 1.038055 |
| negation | qwen2:0.5b | 1.237072 |
| negation | smollm2:360m | 1.137183 |
| noise | gemma3:1b | 1.292165 |
| noise | granite3.1-moe:1b | 1.131486 |
| noise | qwen2:0.5b | 1.211395 |

```
      noise      smollm2:360m      0.973453
paraphrase         gemma3:1b      1.159645
paraphrase granite3.1-moe:1b      1.089286
paraphrase        qwen2:0.5b      1.458942
paraphrase      smollm2:360m      0.963935
question_gen       gemma3:1b      1.195989
question_gen granite3.1-moe:1b      0.984485
question_gen      qwen2:0.5b      0.487344
question_gen    smollm2:360m      0.513460
   shuffle         gemma3:1b      1.050232
   shuffle granite3.1-moe:1b      1.047250
   shuffle      qwen2:0.5b      1.275040
   shuffle    smollm2:360m      1.087972
 summarize         gemma3:1b      0.656102
 summarize granite3.1-moe:1b      1.041472
 summarize      qwen2:0.5b      1.385018
 summarize    smollm2:360m      0.773369
   synonym         gemma3:1b      1.219601
   synonym granite3.1-moe:1b      1.033883
   synonym      qwen2:0.5b      1.123736
   synonym    smollm2:360m      1.018208
```



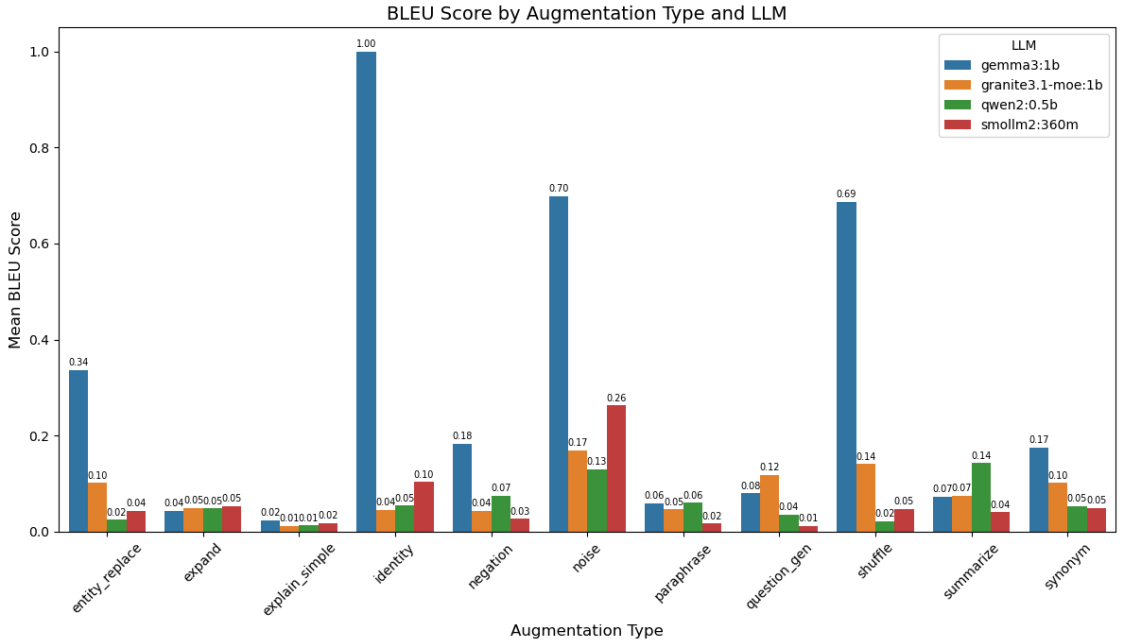

[12]:
```
agg = df.groupby(['augmentation_type', 'model'])['bleu'].mean().reset_index()

plt.figure(figsize=(12,7))
bar = sns.barplot(
```

```
    data=agg,
    x='augmentation_type',
    y='bleu',
    hue='model',
    order=augmentation_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.2f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Mean BLEU Score", fontsize=12)
plt.title("BLEU Score by Augmentation Type and LLM", fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()

print("Mean BLEU Score Table:\n")
print(agg.to_string(index=False))

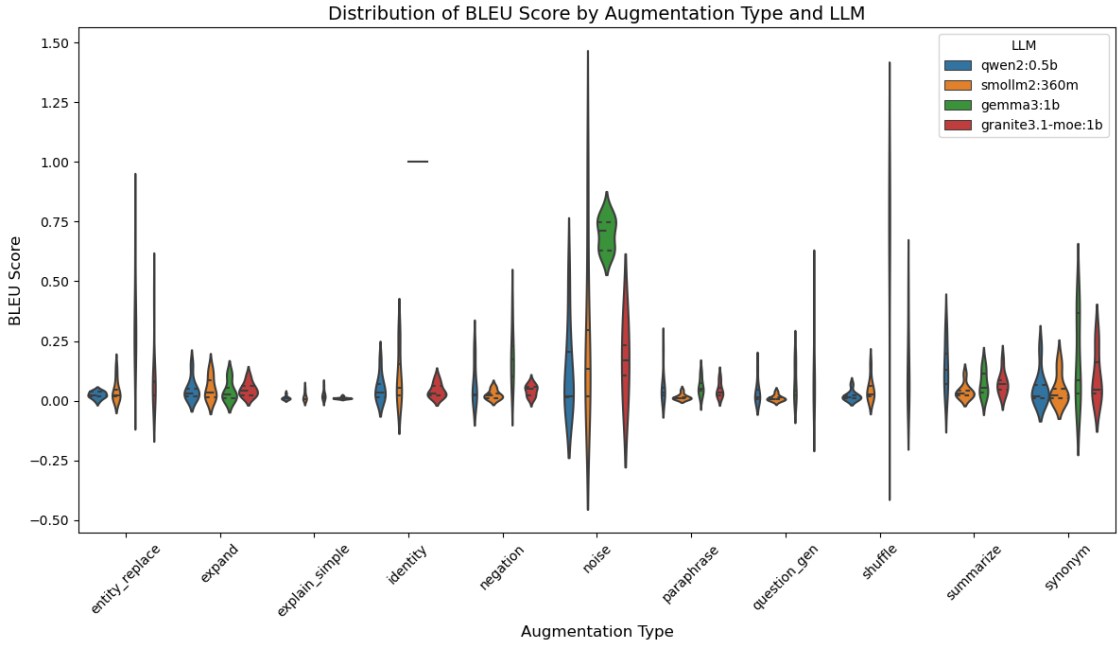
plt.figure(figsize=(12,7))
sns.violinplot(
    data=df,
    x='augmentation_type',
    y='bleu',
    hue='model',
    order=augmentation_order,
    palette='tab10',
    inner="quartile"
)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("BLEU Score", fontsize=12)
plt.title("Distribution of BLEU Score by Augmentation Type and LLM",␣
 ↪fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()
```




Mean BLEU Score Table:

| augmentation_type | model | bleu |
|---|---|---|
| entity_replace | gemma3:1b | 0.336007 |
| entity_replace | granite3.1-moe:1b | 0.100904 |
| entity_replace | qwen2:0.5b | 0.024408 |
| entity_replace | smollm2:360m | 0.042657 |
| expand | gemma3:1b | 0.042257 |
| expand | granite3.1-moe:1b | 0.048258 |
| expand | qwen2:0.5b | 0.047998 |
| expand | smollm2:360m | 0.053188 |
| explain_simple | gemma3:1b | 0.023838 |
| explain_simple | granite3.1-moe:1b | 0.011359 |
| explain_simple | qwen2:0.5b | 0.012483 |
| explain_simple | smollm2:360m | 0.017801 |
| identity | gemma3:1b | 1.000000 |
| identity | granite3.1-moe:1b | 0.044345 |
| identity | qwen2:0.5b | 0.053984 |
| identity | smollm2:360m | 0.102974 |
| negation | gemma3:1b | 0.182515 |
| negation | granite3.1-moe:1b | 0.043616 |
| negation | qwen2:0.5b | 0.074165 |
| negation | smollm2:360m | 0.026137 |
| noise | gemma3:1b | 0.698746 |
| noise | granite3.1-moe:1b | 0.168748 |
| noise | qwen2:0.5b | 0.130235 |

```
      noise      smollm2:360m 0.262345
 paraphrase         gemma3:1b 0.059311
 paraphrase granite3.1-moe:1b 0.046070
 paraphrase        qwen2:0.5b 0.059510
 paraphrase      smollm2:360m 0.016563
question_gen         gemma3:1b 0.079287
question_gen granite3.1-moe:1b 0.118030
question_gen        qwen2:0.5b 0.035539
question_gen      smollm2:360m 0.012183
    shuffle         gemma3:1b 0.687404
    shuffle granite3.1-moe:1b 0.140264
    shuffle        qwen2:0.5b 0.021556
    shuffle      smollm2:360m 0.047154
  summarize         gemma3:1b 0.072126
  summarize granite3.1-moe:1b 0.074009
  summarize        qwen2:0.5b 0.142341
  summarize      smollm2:360m 0.039963
    synonym         gemma3:1b 0.174645
    synonym granite3.1-moe:1b 0.101196
    synonym        qwen2:0.5b 0.052701
    synonym      smollm2:360m 0.049614
```



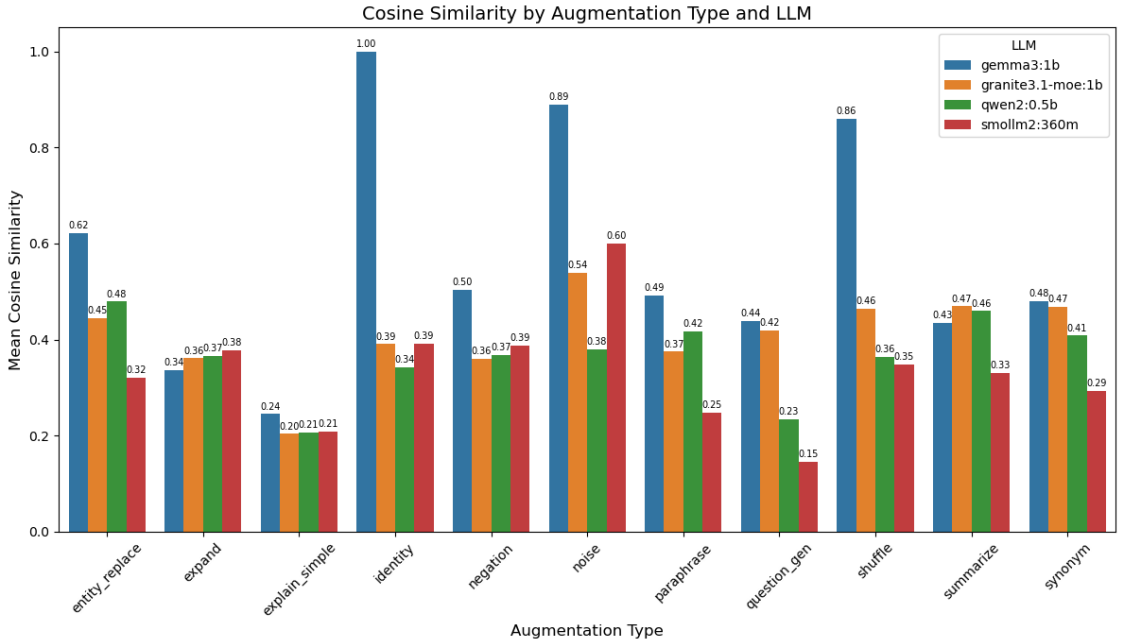

[13]:
```
agg = df.groupby(['augmentation_type', 'model'])['cosine_similarity'].mean().
↪reset_index()

plt.figure(figsize=(12,7))
```

```
bar = sns.barplot(
    data=agg,
    x='augmentation_type',
    y='cosine_similarity',
    hue='model',
    order=augmentation_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.2f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Mean Cosine Similarity", fontsize=12)
plt.title("Cosine Similarity by Augmentation Type and LLM", fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()

print("Mean Cosine Similarity Table:\n")
print(agg.to_string(index=False))

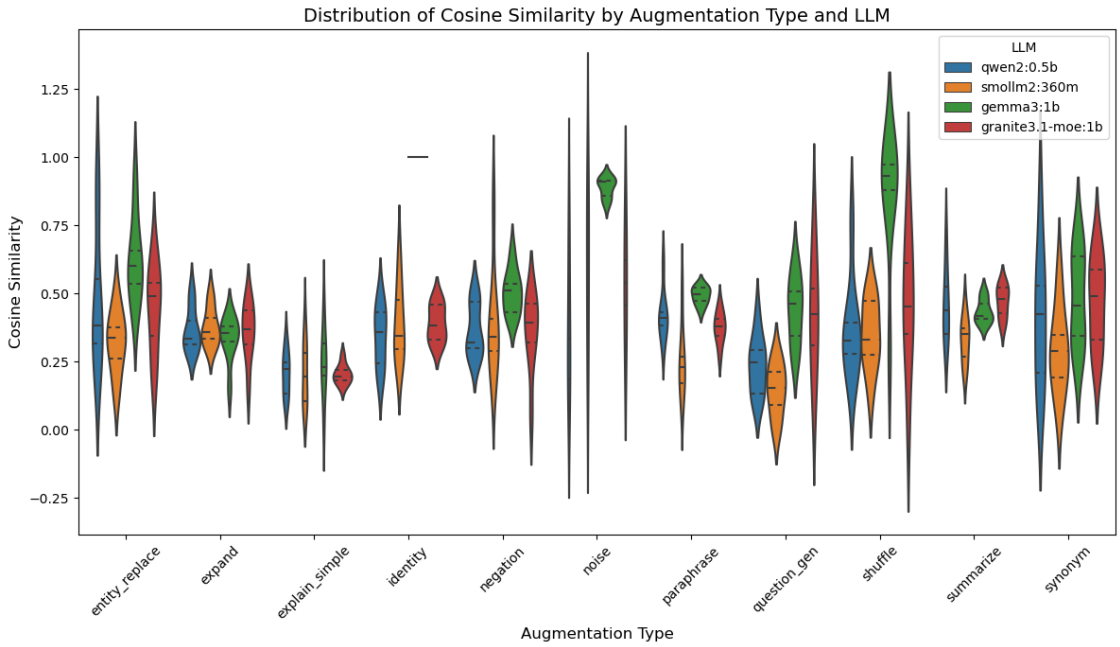
plt.figure(figsize=(12,7))
sns.violinplot(
    data=df,
    x='augmentation_type',
    y='cosine_similarity',
    hue='model',
    order=augmentation_order,
    palette='tab10',
    inner="quartile"
)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Cosine Similarity", fontsize=12)
plt.title("Distribution of Cosine Similarity by Augmentation Type and LLM",␣
 ↪fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()
```




Mean Cosine Similarity Table:

| augmentation_type | model | cosine_similarity |
|---|---|---|
| entity_replace | gemma3:1b | 0.622095 |
| entity_replace | granite3.1-moe:1b | 0.445180 |
| entity_replace | qwen2:0.5b | 0.479066 |
| entity_replace | smollm2:360m | 0.321294 |
| expand | gemma3:1b | 0.335939 |
| expand | granite3.1-moe:1b | 0.361077 |
| expand | qwen2:0.5b | 0.365863 |
| expand | smollm2:360m | 0.377856 |
| explain_simple | gemma3:1b | 0.244547 |
| explain_simple | granite3.1-moe:1b | 0.203341 |
| explain_simple | qwen2:0.5b | 0.206513 |
| explain_simple | smollm2:360m | 0.208020 |
| identity | gemma3:1b | 1.000000 |
| identity | granite3.1-moe:1b | 0.390596 |
| identity | qwen2:0.5b | 0.342094 |
| identity | smollm2:360m | 0.391382 |
| negation | gemma3:1b | 0.504206 |
| negation | granite3.1-moe:1b | 0.360369 |
| negation | qwen2:0.5b | 0.367078 |
| negation | smollm2:360m | 0.386901 |
| noise | gemma3:1b | 0.889123 |
| noise | granite3.1-moe:1b | 0.539366 |
| noise | qwen2:0.5b | 0.379859 |

```
          noise      smollm2:360m          0.600188
     paraphrase         gemma3:1b          0.492292
     paraphrase granite3.1-moe:1b          0.374676
     paraphrase        qwen2:0.5b          0.417718
     paraphrase      smollm2:360m          0.247645
  question_gen         gemma3:1b          0.439280
  question_gen granite3.1-moe:1b          0.419343
  question_gen        qwen2:0.5b          0.234261
  question_gen      smollm2:360m          0.145924
       shuffle         gemma3:1b          0.859389
       shuffle granite3.1-moe:1b          0.464123
       shuffle        qwen2:0.5b          0.364328
       shuffle      smollm2:360m          0.348602
     summarize         gemma3:1b          0.434552
     summarize granite3.1-moe:1b          0.469117
     summarize        qwen2:0.5b          0.459336
     summarize      smollm2:360m          0.330073
       synonym         gemma3:1b          0.480473
       synonym granite3.1-moe:1b          0.467419
       synonym        qwen2:0.5b          0.408312
       synonym      smollm2:360m          0.293056
```



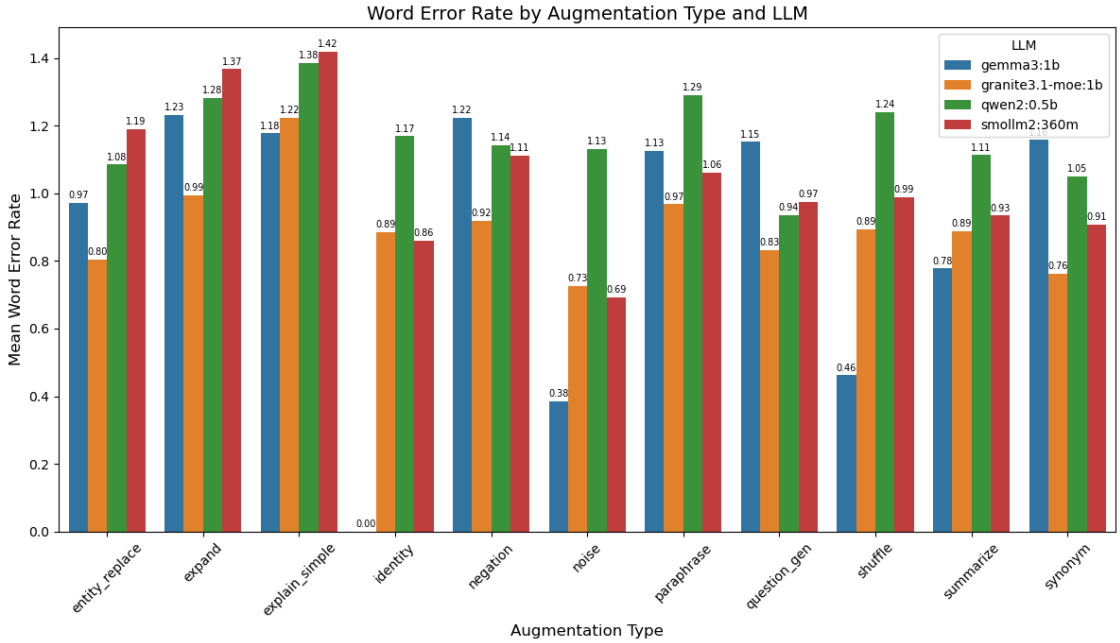

[14]:
```
agg = df.groupby(['augmentation_type', 'model'])['wer'].mean().reset_index()

plt.figure(figsize=(12,7))
bar = sns.barplot(
```

```
    data=agg,
    x='augmentation_type',
    y='wer',
    hue='model',
    order=augmentation_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.2f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Mean Word Error Rate", fontsize=12)
plt.title("Word Error Rate by Augmentation Type and LLM", fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()

print("Mean Word Error Rate Table:\n")
print(agg.to_string(index=False))

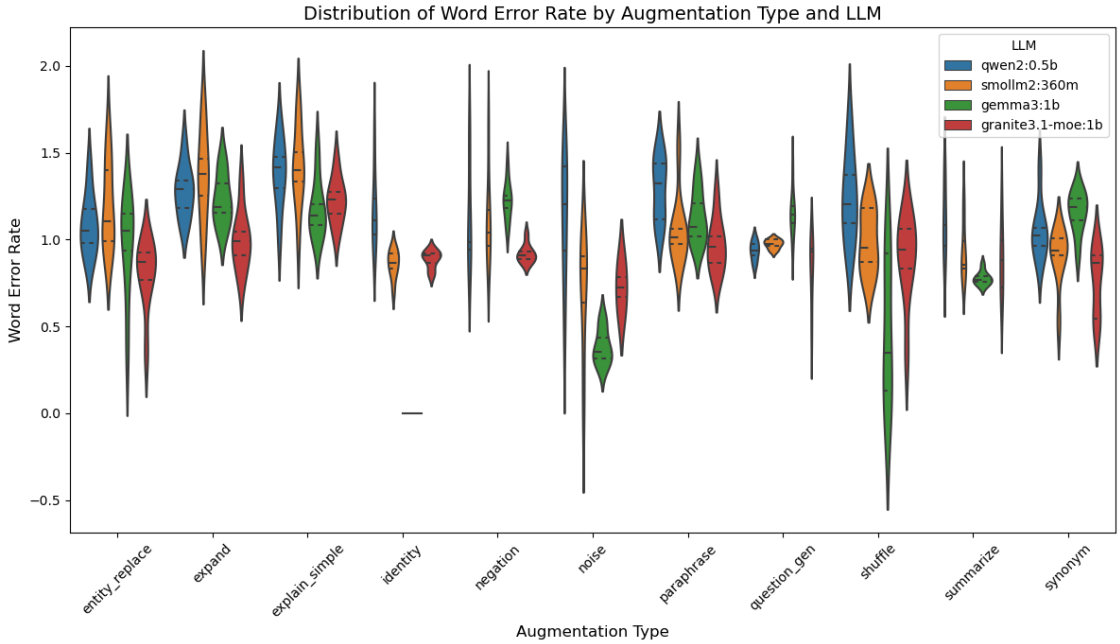
plt.figure(figsize=(12,7))
sns.violinplot(
    data=df,
    x='augmentation_type',
    y='wer',
    hue='model',
    order=augmentation_order,
    palette='tab10',
    inner="quartile"
)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Word Error Rate", fontsize=12)
plt.title("Distribution of Word Error Rate by Augmentation Type and LLM",␣
 ↪fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()
```




Mean Word Error Rate Table:

| augmentation_type | model | wer |
|---|---|---|
| entity_replace | gemma3:1b | 0.970939 |
| entity_replace | granite3.1-moe:1b | 0.804647 |
| entity_replace | qwen2:0.5b | 1.084771 |
| entity_replace | smollm2:360m | 1.190478 |
| expand | gemma3:1b | 1.230516 |
| expand | granite3.1-moe:1b | 0.993683 |
| expand | qwen2:0.5b | 1.280821 |
| expand | smollm2:360m | 1.367130 |
| explain_simple | gemma3:1b | 1.177038 |
| explain_simple | granite3.1-moe:1b | 1.223285 |
| explain_simple | qwen2:0.5b | 1.384482 |
| explain_simple | smollm2:360m | 1.419171 |
| identity | gemma3:1b | 0.000000 |
| identity | granite3.1-moe:1b | 0.885898 |
| identity | qwen2:0.5b | 1.168764 |
| identity | smollm2:360m | 0.859945 |
| negation | gemma3:1b | 1.222498 |
| negation | granite3.1-moe:1b | 0.919054 |
| negation | qwen2:0.5b | 1.141896 |
| negation | smollm2:360m | 1.111944 |
| noise | gemma3:1b | 0.384143 |
| noise | granite3.1-moe:1b | 0.725806 |
| noise | qwen2:0.5b | 1.131913 |

```
      noise      smollm2:360m 0.692547
 paraphrase         gemma3:1b 1.125937
 paraphrase granite3.1-moe:1b 0.967314
 paraphrase        qwen2:0.5b 1.290791
 paraphrase      smollm2:360m 1.060419
question_gen         gemma3:1b 1.151979
question_gen granite3.1-moe:1b 0.833109
question_gen        qwen2:0.5b 0.935138
question_gen      smollm2:360m 0.974449
    shuffle         gemma3:1b 0.462177
    shuffle granite3.1-moe:1b 0.893410
    shuffle        qwen2:0.5b 1.240501
    shuffle      smollm2:360m 0.989375
  summarize         gemma3:1b 0.777333
  summarize granite3.1-moe:1b 0.887743
  summarize        qwen2:0.5b 1.113069
  summarize      smollm2:360m 0.933830
    synonym         gemma3:1b 1.157594
    synonym granite3.1-moe:1b 0.763213
    synonym        qwen2:0.5b 1.049095
    synonym      smollm2:360m 0.906838
```



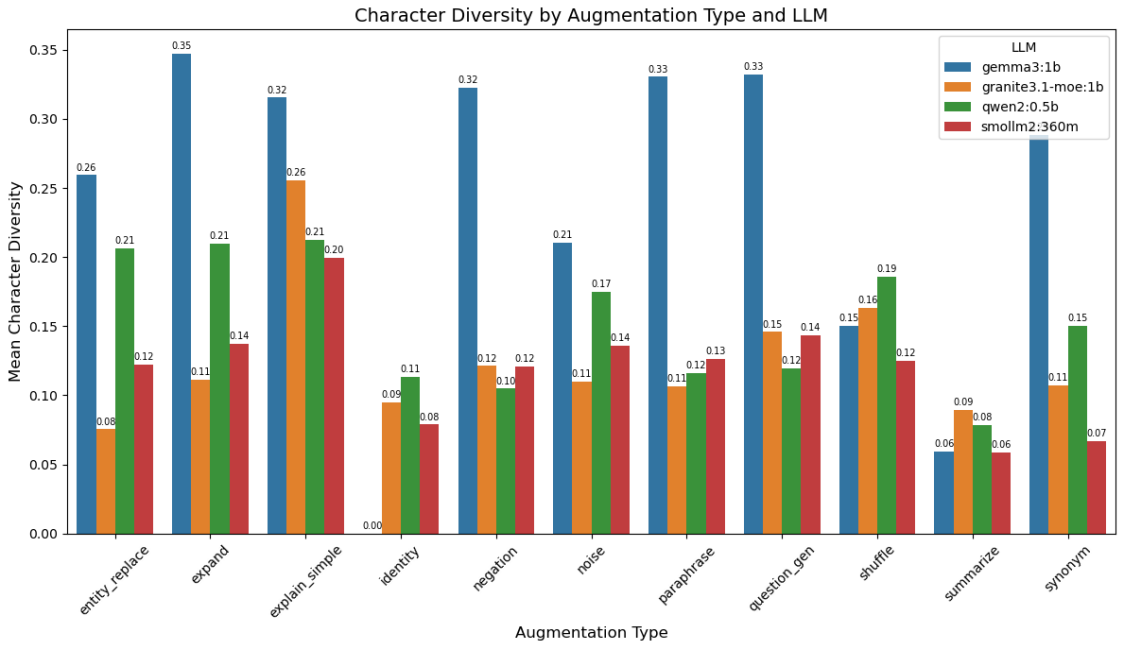

[15]:
```
agg = df.groupby(['augmentation_type', 'model'])['char_diversity'].mean().
  ↪reset_index()

plt.figure(figsize=(12,7))
```

```
bar = sns.barplot(
    data=agg,
    x='augmentation_type',
    y='char_diversity',
    hue='model',
    order=augmentation_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.2f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Mean Character Diversity", fontsize=12)
plt.title("Character Diversity by Augmentation Type and LLM", fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()

print("Mean Character Diversity Table:\n")
print(agg.to_string(index=False))

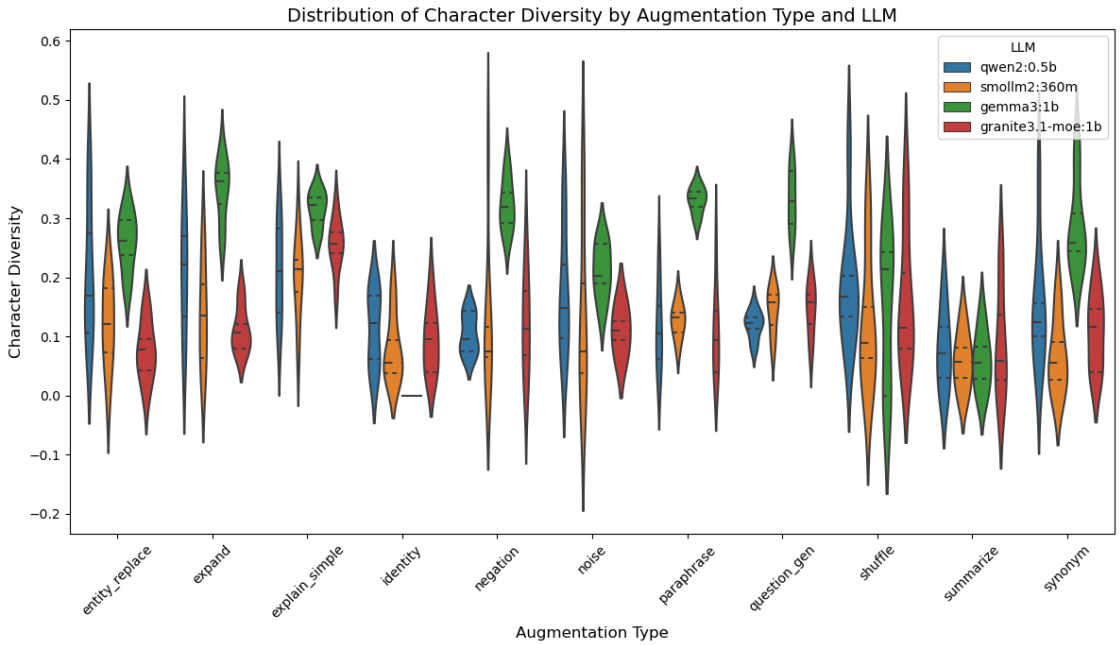
plt.figure(figsize=(12,7))
sns.violinplot(
    data=df,
    x='augmentation_type',
    y='char_diversity',
    hue='model',
    order=augmentation_order,
    palette='tab10',
    inner="quartile"
)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Character Diversity", fontsize=12)
plt.title("Distribution of Character Diversity by Augmentation Type and LLM",␣
 ↪fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()
```




Mean Character Diversity Table:

| augmentation_type | model | char_diversity |
|---|---|---|
| entity_replace | gemma3:1b | 0.259408 |
| entity_replace | granite3.1-moe:1b | 0.075425 |
| entity_replace | qwen2:0.5b | 0.206214 |
| entity_replace | smollm2:360m | 0.122106 |
| expand | gemma3:1b | 0.347327 |
| expand | granite3.1-moe:1b | 0.111536 |
| expand | qwen2:0.5b | 0.209976 |
| expand | smollm2:360m | 0.137446 |
| explain_simple | gemma3:1b | 0.315454 |
| explain_simple | granite3.1-moe:1b | 0.255851 |
| explain_simple | qwen2:0.5b | 0.212264 |
| explain_simple | smollm2:360m | 0.199737 |
| identity | gemma3:1b | 0.000000 |
| identity | granite3.1-moe:1b | 0.094899 |
| identity | qwen2:0.5b | 0.113213 |
| identity | smollm2:360m | 0.078906 |
| negation | gemma3:1b | 0.322866 |
| negation | granite3.1-moe:1b | 0.121212 |
| negation | qwen2:0.5b | 0.104828 |
| negation | smollm2:360m | 0.120919 |
| noise | gemma3:1b | 0.210512 |
| noise | granite3.1-moe:1b | 0.109890 |
| noise | qwen2:0.5b | 0.174991 |

```
      noise      smollm2:360m        0.135840
 paraphrase         gemma3:1b        0.330481
 paraphrase granite3.1-moe:1b        0.106477
 paraphrase        qwen2:0.5b        0.116298
 paraphrase      smollm2:360m        0.126579
question_gen         gemma3:1b        0.332157
question_gen granite3.1-moe:1b        0.145877
question_gen        qwen2:0.5b        0.119569
question_gen      smollm2:360m        0.143274
    shuffle         gemma3:1b        0.150104
    shuffle granite3.1-moe:1b        0.163039
    shuffle        qwen2:0.5b        0.186026
    shuffle      smollm2:360m        0.124883
  summarize         gemma3:1b        0.059246
  summarize granite3.1-moe:1b        0.089673
  summarize        qwen2:0.5b        0.078301
  summarize      smollm2:360m        0.058926
    synonym         gemma3:1b        0.288397
    synonym granite3.1-moe:1b        0.107062
    synonym        qwen2:0.5b        0.150192
    synonym      smollm2:360m        0.067023
```



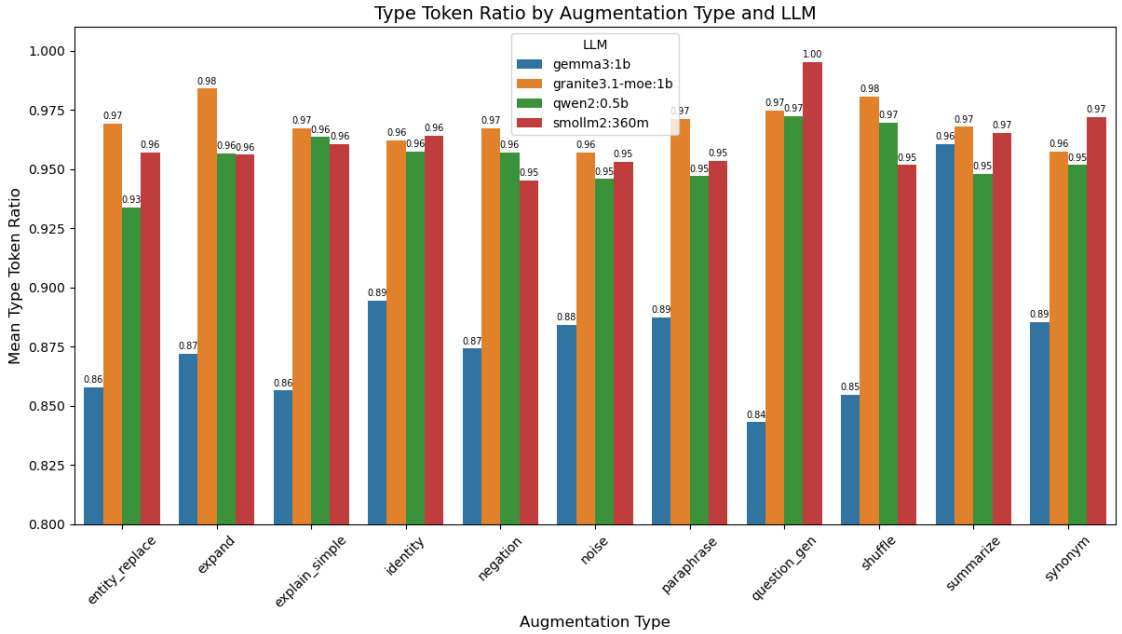

[16]:
```
agg = df.groupby(['augmentation_type', 'model'])['type_token_ratio'].mean().
 ↪reset_index()

plt.figure(figsize=(12,7))
```

```
bar = sns.barplot(
    data=agg,
    x='augmentation_type',
    y='type_token_ratio',
    hue='model',
    order=augmentation_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.2f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Mean Type Token Ratio", fontsize=12)
plt.title("Type Token Ratio by Augmentation Type and LLM", fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.ylim(.8,1.01)
plt.show()

print("Mean Type Token Ratio Table:\n")
print(agg.to_string(index=False))

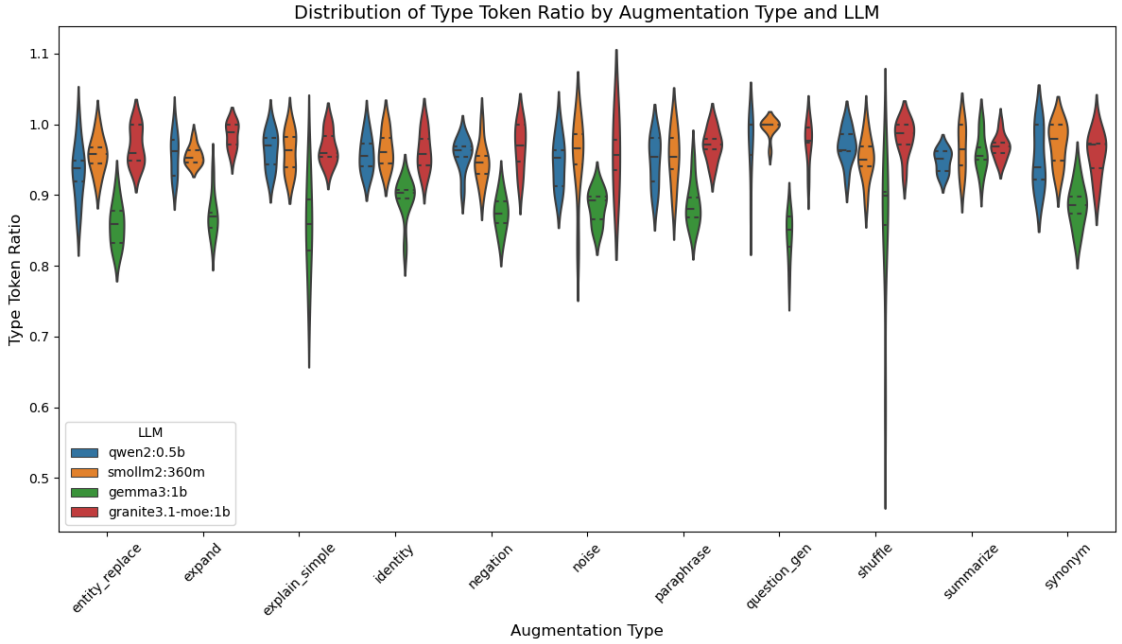
plt.figure(figsize=(12,7))
sns.violinplot(
    data=df,
    x='augmentation_type',
    y='type_token_ratio',
    hue='model',
    order=augmentation_order,
    palette='tab10',
    inner="quartile"
)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Type Token Ratio", fontsize=12)
plt.title("Distribution of Type Token Ratio by Augmentation Type and LLM",␣
 ↪fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()
```




Mean Type Token Ratio Table:

| augmentation_type | model | type_token_ratio |
|---|---|---|
| entity_replace | gemma3:1b | 0.857892 |
| entity_replace | granite3.1-moe:1b | 0.969279 |
| entity_replace | qwen2:0.5b | 0.933791 |
| entity_replace | smollm2:360m | 0.956961 |
| expand | gemma3:1b | 0.871875 |
| expand | granite3.1-moe:1b | 0.983983 |
| expand | qwen2:0.5b | 0.956655 |
| expand | smollm2:360m | 0.956069 |
| explain_simple | gemma3:1b | 0.856441 |
| explain_simple | granite3.1-moe:1b | 0.967255 |
| explain_simple | qwen2:0.5b | 0.963499 |
| explain_simple | smollm2:360m | 0.960689 |
| identity | gemma3:1b | 0.894546 |
| identity | granite3.1-moe:1b | 0.962174 |
| identity | qwen2:0.5b | 0.957552 |
| identity | smollm2:360m | 0.964159 |
| negation | gemma3:1b | 0.874110 |
| negation | granite3.1-moe:1b | 0.967172 |
| negation | qwen2:0.5b | 0.956909 |
| negation | smollm2:360m | 0.945347 |
| noise | gemma3:1b | 0.884083 |
| noise | granite3.1-moe:1b | 0.957143 |
| noise | qwen2:0.5b | 0.945767 |
| noise | smollm2:360m | 0.953060 |

```
   paraphrase          gemma3:1b          0.887194
   paraphrase granite3.1-moe:1b          0.971228
   paraphrase         qwen2:0.5b          0.947000
   paraphrase       smollm2:360m          0.953424
question_gen          gemma3:1b          0.843012
question_gen granite3.1-moe:1b          0.974845
question_gen         qwen2:0.5b          0.972431
question_gen       smollm2:360m          0.995192
     shuffle          gemma3:1b          0.854640
     shuffle granite3.1-moe:1b          0.980633
     shuffle         qwen2:0.5b          0.969498
     shuffle       smollm2:360m          0.951689
   summarize          gemma3:1b          0.960620
   summarize granite3.1-moe:1b          0.967855
   summarize         qwen2:0.5b          0.947847
   summarize       smollm2:360m          0.965432
     synonym          gemma3:1b          0.885274
     synonym granite3.1-moe:1b          0.957266
     synonym         qwen2:0.5b          0.951701
     synonym       smollm2:360m          0.971937
```



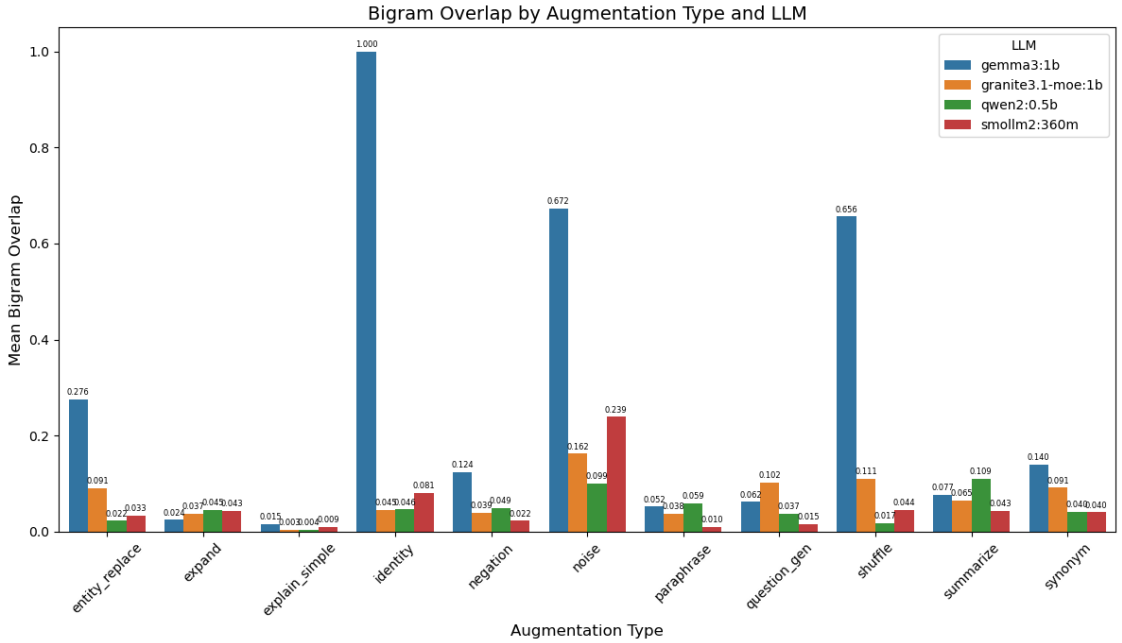

[17]:
```
agg = df.groupby(['augmentation_type', 'model'])['bigram_overlap'].mean().
↪reset_index()

plt.figure(figsize=(12,7))
bar = sns.barplot(
```

```
    data=agg,
    x='augmentation_type',
    y='bigram_overlap',
    hue='model',
    order=augmentation_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.3f', label_type='edge', padding=2,␣
 ↪fontsize=6)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Mean Bigram Overlap", fontsize=12)
plt.title("Bigram Overlap by Augmentation Type and LLM", fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()

print("Mean Bigram Overlap Table:\n")
print(agg.to_string(index=False))

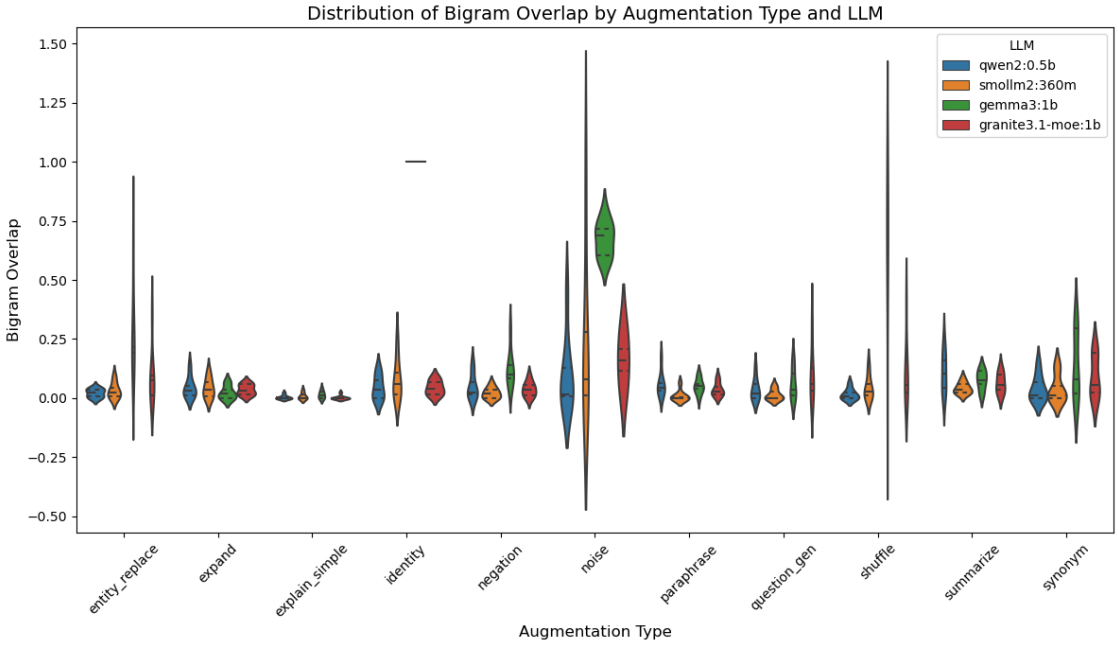
plt.figure(figsize=(12,7))
sns.violinplot(
    data=df,
    x='augmentation_type',
    y='bigram_overlap',
    hue='model',
    order=augmentation_order,
    palette='tab10',
    inner="quartile"
)
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Bigram Overlap", fontsize=12)
plt.title("Distribution of Bigram Overlap by Augmentation Type and LLM",␣
 ↪fontsize=14)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()
```




Mean Bigram Overlap Table:

| augmentation_type | model | bigram_overlap |
|---|---|---|
| entity_replace | gemma3:1b | 0.276008 |
| entity_replace | granite3.1-moe:1b | 0.090576 |
| entity_replace | qwen2:0.5b | 0.022345 |
| entity_replace | smollm2:360m | 0.032654 |
| expand | gemma3:1b | 0.024316 |
| expand | granite3.1-moe:1b | 0.036781 |
| expand | qwen2:0.5b | 0.045052 |
| expand | smollm2:360m | 0.043334 |
| explain_simple | gemma3:1b | 0.015429 |
| explain_simple | granite3.1-moe:1b | 0.003049 |
| explain_simple | qwen2:0.5b | 0.004320 |
| explain_simple | smollm2:360m | 0.008956 |
| identity | gemma3:1b | 1.000000 |
| identity | granite3.1-moe:1b | 0.044572 |
| identity | qwen2:0.5b | 0.046083 |
| identity | smollm2:360m | 0.081047 |
| negation | gemma3:1b | 0.124283 |
| negation | granite3.1-moe:1b | 0.039028 |
| negation | qwen2:0.5b | 0.049130 |
| negation | smollm2:360m | 0.022086 |
| noise | gemma3:1b | 0.672246 |
| noise | granite3.1-moe:1b | 0.161934 |
| noise | qwen2:0.5b | 0.099316 |

```
          noise      smollm2:360m        0.238900
     paraphrase         gemma3:1b        0.051615
     paraphrase granite3.1-moe:1b        0.037508
     paraphrase        qwen2:0.5b        0.058800
     paraphrase      smollm2:360m        0.010150
   question_gen         gemma3:1b        0.061532
   question_gen granite3.1-moe:1b        0.102173
   question_gen        qwen2:0.5b        0.036526
   question_gen      smollm2:360m        0.014815
        shuffle         gemma3:1b        0.655958
        shuffle granite3.1-moe:1b        0.110579
        shuffle        qwen2:0.5b        0.016694
        shuffle      smollm2:360m        0.044234
      summarize         gemma3:1b        0.076663
      summarize granite3.1-moe:1b        0.064816
      summarize        qwen2:0.5b        0.109437
      summarize      smollm2:360m        0.042743
        synonym         gemma3:1b        0.139534
        synonym granite3.1-moe:1b        0.091272
        synonym        qwen2:0.5b        0.040477
        synonym      smollm2:360m        0.040125
```




[18]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Load CSV if the DataFrame is not currently available
if 'df' not in globals():
    csv_path = r"D:\technical writing\textaugmentation\result.csv"
    df = pd.read_csv(csv_path)
    print(f"Dataset loaded successfully: {df.shape}")

# Ensure duration columns exist (nanoseconds to seconds)
duration_columns = {
    'total_duration_s': 'total_duration_ns',
    'load_duration_s': 'load_duration_ns',
    'prompt_eval_duration_s': 'prompt_eval_duration_ns'
}

for seconds_col, nanos_col in duration_columns.items():
    if seconds_col not in df.columns and nanos_col in df.columns:
        df[seconds_col] = pd.to_numeric(
            df[nanos_col], errors='coerce'
        ) / 1e9

# Define correlation metrics
metrics = [
    'levenshtein_similarity',
    'jaccard_similarity',
    'length_ratio',
    'bleu',
    'cosine_similarity',
    'wer',
    'char_diversity',
    'type_token_ratio',
    'bigram_overlap',
    'total_duration_s',
    'load_duration_s',
    'prompt_eval_duration_s',
    'tokens_per_second'
]

# Check availability
available_metrics = [m for m in metrics if m in df.columns]
missing_metrics = [m for m in metrics if m not in df.columns]

if missing_metrics:
    print("Missing metrics:", missing_metrics)

if len(available_metrics) < 2:
    raise ValueError("At least two metric columns are required.")

# Convert available metrics to numeric values
```

```
metric_data = df[available_metrics].apply(
    pd.to_numeric, errors='coerce'
)

# Pearson correlation matrix
corr = metric_data.corr(method='pearson')

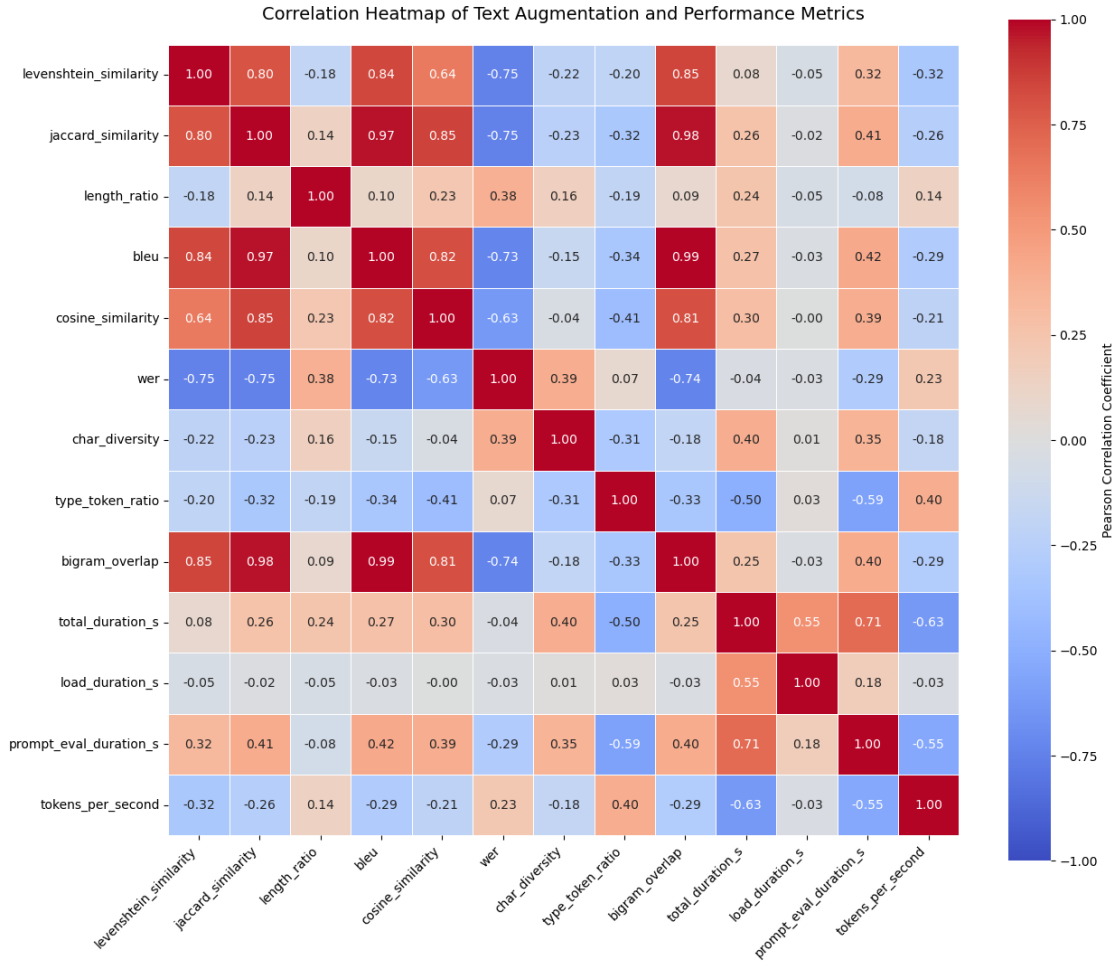
# Plot heatmap
plt.figure(figsize=(13, 11))

sns.heatmap(
    corr,
    annot=True,
    fmt=".2f",
    cmap="coolwarm",
    center=0,
    vmin=-1,
    vmax=1,
    square=True,
    linewidths=0.5,
    cbar_kws={'label': 'Pearson Correlation Coefficient'}
)

plt.title(
    "Correlation Heatmap of Text Augmentation and Performance Metrics",
    fontsize=14,
    pad=20
)

plt.xticks(rotation=45, ha='right')
plt.yticks(rotation=0)
plt.tight_layout()
plt.show()

print("\nCorrelation Matrix:")
display(corr.round(3))
```




Correlation Matrix:

| | levenshtein_similarity | jaccard_similarity | \ |
|---|---|---|---|
| levenshtein_similarity | 1.000 | 0.795 | |
| jaccard_similarity | 0.795 | 1.000 | |
| length_ratio | -0.185 | 0.144 | |
| bleu | 0.841 | 0.972 | |
| cosine_similarity | 0.641 | 0.852 | |
| wer | -0.746 | -0.749 | |
| char_diversity | -0.216 | -0.234 | |
| type_token_ratio | -0.196 | -0.325 | |
| bigram_overlap | 0.850 | 0.976 | |
| total_duration_s | 0.079 | 0.261 | |
| load_duration_s | -0.047 | -0.022 | |
| prompt_eval_duration_s | 0.321 | 0.405 | |
| tokens_per_second | -0.322 | -0.261 | |

| | length_ratio | bleu | cosine_similarity | wer | \ |
|---|---|---|---|---|---|
| levenshtein_similarity | -0.185 | 0.841 | 0.641 | -0.746 | |
| jaccard_similarity | 0.144 | 0.972 | 0.852 | -0.749 | |
| length_ratio | 1.000 | 0.105 | 0.230 | 0.376 | |
| bleu | 0.105 | 1.000 | 0.818 | -0.727 | |
| cosine_similarity | 0.230 | 0.818 | 1.000 | -0.633 | |
| wer | 0.376 | -0.727 | -0.633 | 1.000 | |
| char_diversity | 0.160 | -0.153 | -0.043 | 0.395 | |
| type_token_ratio | -0.191 | -0.343 | -0.407 | 0.075 | |
| bigram_overlap | 0.086 | 0.994 | 0.807 | -0.742 | |
| total_duration_s | 0.242 | 0.271 | 0.296 | -0.036 | |
| load_duration_s | -0.048 | -0.026 | -0.004 | -0.026 | |
| prompt_eval_duration_s | -0.085 | 0.417 | 0.392 | -0.291 | |
| tokens_per_second | 0.137 | -0.291 | -0.215 | 0.226 | |

| | char_diversity | type_token_ratio | bigram_overlap | \ |
|---|---|---|---|---|
| levenshtein_similarity | -0.216 | -0.196 | 0.850 | |
| jaccard_similarity | -0.234 | -0.325 | 0.976 | |
| length_ratio | 0.160 | -0.191 | 0.086 | |
| bleu | -0.153 | -0.343 | 0.994 | |
| cosine_similarity | -0.043 | -0.407 | 0.807 | |
| wer | 0.395 | 0.075 | -0.742 | |
| char_diversity | 1.000 | -0.313 | -0.181 | |
| type_token_ratio | -0.313 | 1.000 | -0.329 | |
| bigram_overlap | -0.181 | -0.329 | 1.000 | |
| total_duration_s | 0.398 | -0.503 | 0.250 | |
| load_duration_s | 0.015 | 0.028 | -0.028 | |
| prompt_eval_duration_s | 0.353 | -0.586 | 0.405 | |
| tokens_per_second | -0.178 | 0.396 | -0.287 | |

| | total_duration_s | load_duration_s | \ |
|---|---|---|---|
| levenshtein_similarity | 0.079 | -0.047 | |
| jaccard_similarity | 0.261 | -0.022 | |
| length_ratio | 0.242 | -0.048 | |
| bleu | 0.271 | -0.026 | |
| cosine_similarity | 0.296 | -0.004 | |
| wer | -0.036 | -0.026 | |
| char_diversity | 0.398 | 0.015 | |
| type_token_ratio | -0.503 | 0.028 | |
| bigram_overlap | 0.250 | -0.028 | |
| total_duration_s | 1.000 | 0.547 | |
| load_duration_s | 0.547 | 1.000 | |
| prompt_eval_duration_s | 0.710 | 0.176 | |
| tokens_per_second | -0.630 | -0.032 | |

| | prompt_eval_duration_s | tokens_per_second |
|---|---|---|
| levenshtein_similarity | 0.321 | -0.322 |
| jaccard_similarity | 0.405 | -0.261 |

```
length_ratio                -0.085          0.137
bleu                         0.417         -0.291
cosine_similarity            0.392         -0.215
wer                         -0.291          0.226
char_diversity               0.353         -0.178
type_token_ratio            -0.586          0.396
bigram_overlap               0.405         -0.287
total_duration_s             0.710         -0.630
load_duration_s              0.176         -0.032
prompt_eval_duration_s       1.000         -0.545
tokens_per_second           -0.545          1.000
```

[19]:
```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Define augmentation order
augmentation_order = sorted(
    df['augmentation_type'].dropna().unique()
)

# 2. Normalize tokens per second
tps_min = df['tokens_per_second'].min()
tps_max = df['tokens_per_second'].max()

if tps_max > tps_min:
    df['tokens_per_second_norm'] = (
        (df['tokens_per_second'] - tps_min) /
        (tps_max - tps_min)
    )
else:
    df['tokens_per_second_norm'] = 0.0

# 3. Compute composite score
w1, w2, w3 = 0.4, 0.3, 0.3

df['composite_score'] = (
    w1 * df['levenshtein_similarity'] +
    w2 * (1 - df['bigram_overlap']) +
    w3 * df['tokens_per_second_norm']
)

# 4. Aggregate by augmentation type and model
agg = (
    df.groupby(['augmentation_type', 'model'])['composite_score']
    .mean()
```

```
    .reset_index()
)

# 5. Bar plot
plt.figure(figsize=(12, 7))

bar = sns.barplot(
    data=agg,
    x='augmentation_type',
    y='composite_score',
    hue='model',
    order=augmentation_order,
    palette='tab10'
)

for container in bar.containers:
    bar.bar_label(
        container,
        fmt='%.2f',
        label_type='edge',
        padding=2,
        fontsize=7
    )

plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Mean Composite Score", fontsize=12)
plt.title(
    "Composite Score (Fidelity, Novelty, Speed) by Augmentation and LLM",
    fontsize=14
)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()

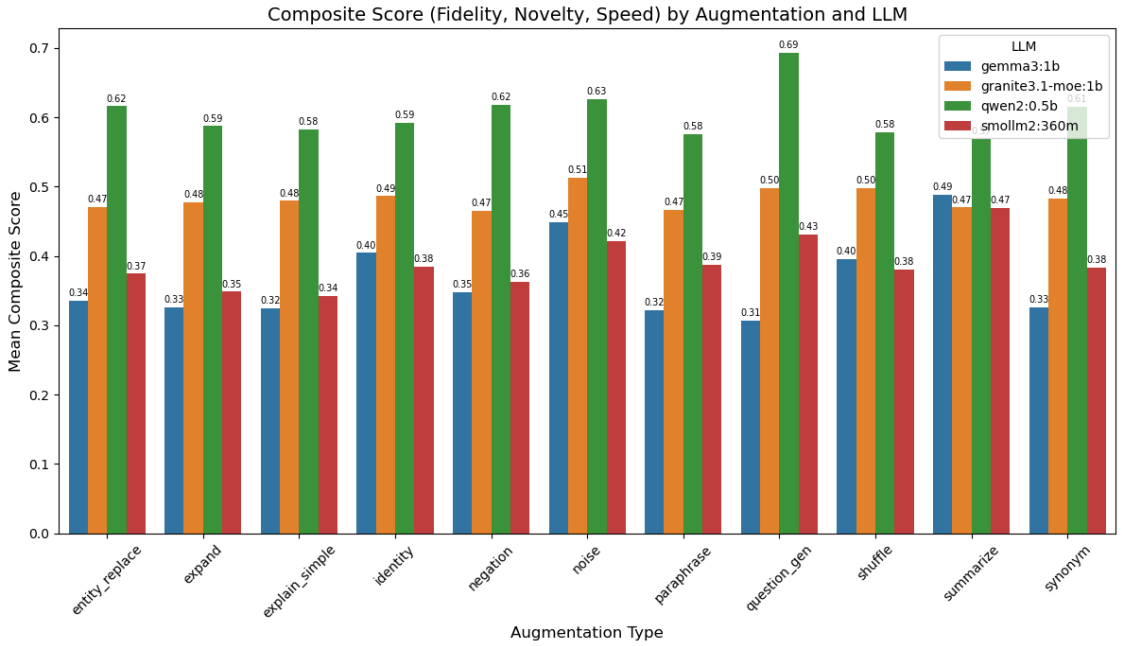
print("Mean Composite Score Table:\n")
print(agg.to_string(index=False))

# 6. Violin plot
plt.figure(figsize=(12, 7))

sns.violinplot(
    data=df,
    x='augmentation_type',
    y='composite_score',
    hue='model',
    order=augmentation_order,
```

```
    palette='tab10',
    inner="quartile"
)

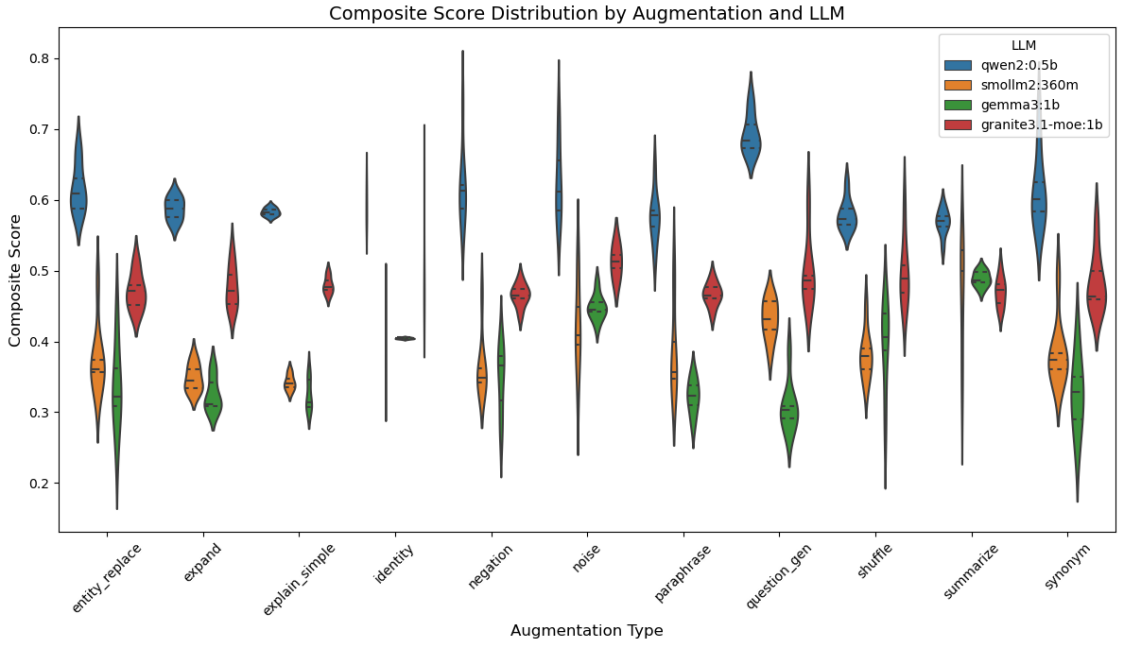
plt.xlabel("Augmentation Type", fontsize=12)
plt.ylabel("Composite Score", fontsize=12)
plt.title(
    "Composite Score Distribution by Augmentation and LLM",
    fontsize=14
)
plt.xticks(rotation=45)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()
```




Mean Composite Score Table:

| augmentation_type | model | composite_score |
|---|---|---|
| entity_replace | gemma3:1b | 0.335912 |
| entity_replace | granite3.1-moe:1b | 0.471249 |
| entity_replace | qwen2:0.5b | 0.616057 |
| entity_replace | smollm2:360m | 0.374485 |
| expand | gemma3:1b | 0.325728 |
| expand | granite3.1-moe:1b | 0.477295 |
| expand | qwen2:0.5b | 0.587588 |
| expand | smollm2:360m | 0.348680 |

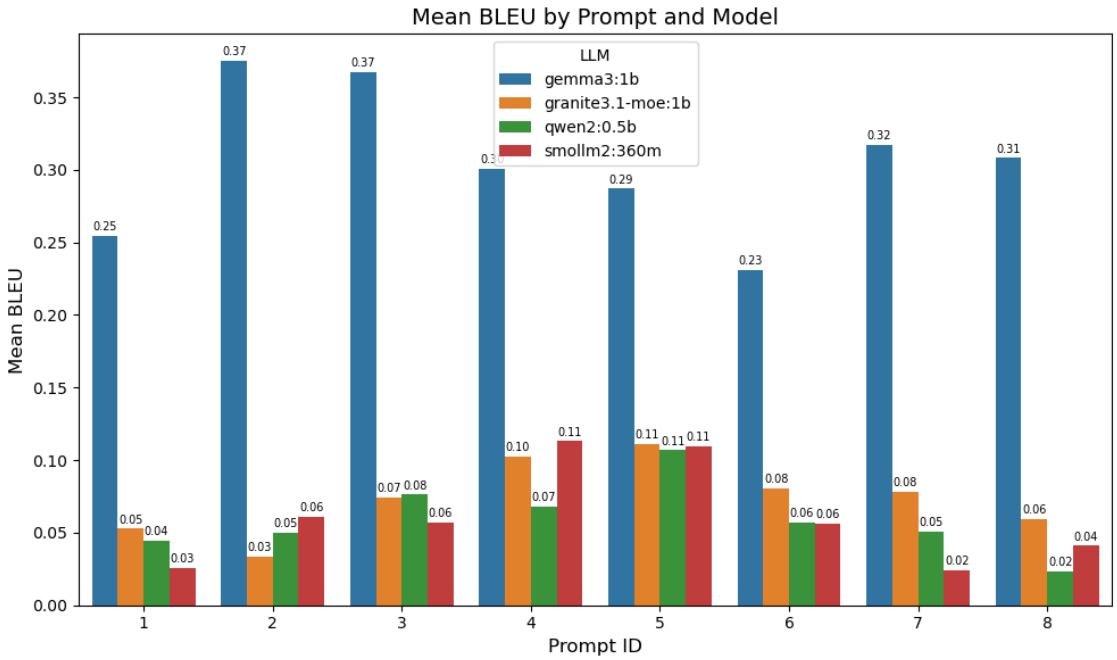
| | | |
|---|---|---|
| explain_simple | gemma3:1b | 0.324912 |
| explain_simple | granite3.1-moe:1b | 0.479531 |
| explain_simple | qwen2:0.5b | 0.583204 |
| explain_simple | smollm2:360m | 0.342336 |
| identity | gemma3:1b | 0.404691 |
| identity | granite3.1-moe:1b | 0.486590 |
| identity | qwen2:0.5b | 0.592319 |
| identity | smollm2:360m | 0.384884 |
| negation | gemma3:1b | 0.347554 |
| negation | granite3.1-moe:1b | 0.465591 |
| negation | qwen2:0.5b | 0.618670 |
| negation | smollm2:360m | 0.363032 |
| noise | gemma3:1b | 0.448422 |
| noise | granite3.1-moe:1b | 0.512618 |
| noise | qwen2:0.5b | 0.626415 |
| noise | smollm2:360m | 0.421667 |
| paraphrase | gemma3:1b | 0.322206 |
| paraphrase | granite3.1-moe:1b | 0.466552 |
| paraphrase | qwen2:0.5b | 0.575609 |
| paraphrase | smollm2:360m | 0.387589 |
| question_gen | gemma3:1b | 0.307361 |
| question_gen | granite3.1-moe:1b | 0.498469 |
| question_gen | qwen2:0.5b | 0.693501 |
| question_gen | smollm2:360m | 0.431563 |
| shuffle | gemma3:1b | 0.395759 |
| shuffle | granite3.1-moe:1b | 0.498410 |
| shuffle | qwen2:0.5b | 0.578838 |
| shuffle | smollm2:360m | 0.380806 |
| summarize | gemma3:1b | 0.488778 |
| summarize | granite3.1-moe:1b | 0.470224 |
| summarize | qwen2:0.5b | 0.568610 |
| summarize | smollm2:360m | 0.469857 |
| synonym | gemma3:1b | 0.325954 |
| synonym | granite3.1-moe:1b | 0.483146 |
| synonym | qwen2:0.5b | 0.614986 |
| synonym | smollm2:360m | 0.383781 |




[20]:
```
metric_col = 'bleu'
agg = df.groupby(['prompt_id', 'model'])[metric_col].mean().reset_index()
prompt_order = sorted(df['prompt_id'].unique())

plt.figure(figsize=(10,6))
bar = sns.barplot(
    data=agg,
    x='prompt_id',
    y=metric_col,
    hue='model',
    order=prompt_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.2f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Prompt ID", fontsize=12)
plt.ylabel("Mean BLEU", fontsize=12)
plt.title("Mean BLEU by Prompt and Model", fontsize=14)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()

print("Mean BLEU Table by Prompt and Model:\n")
print(agg.to_string(index=False))
```




Mean BLEU Table by Prompt and Model:

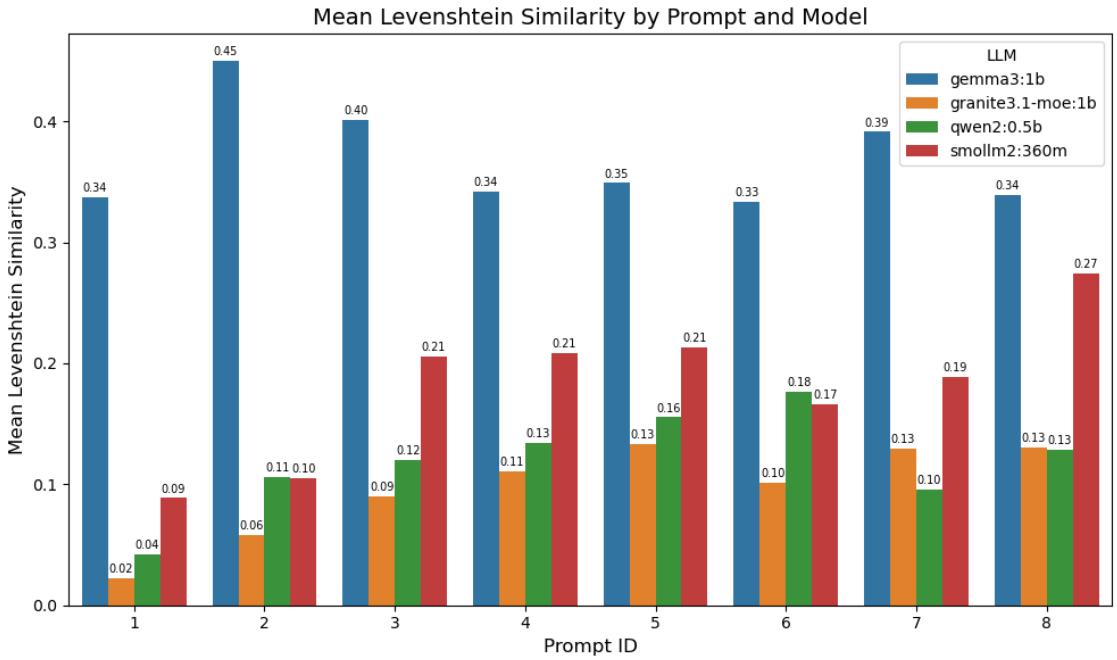
| prompt_id | model | bleu |
|---|---|---|
| 1 | gemma3:1b | 0.254414 |
| 1 | granite3.1-moe:1b | 0.052784 |
| 1 | qwen2:0.5b | 0.044227 |
| 1 | smollm2:360m | 0.025847 |
| 2 | gemma3:1b | 0.374889 |
| 2 | granite3.1-moe:1b | 0.033505 |
| 2 | qwen2:0.5b | 0.049927 |
| 2 | smollm2:360m | 0.060754 |
| 3 | gemma3:1b | 0.367525 |
| 3 | granite3.1-moe:1b | 0.074351 |
| 3 | qwen2:0.5b | 0.076296 |
| 3 | smollm2:360m | 0.056945 |
| 4 | gemma3:1b | 0.300483 |
| 4 | granite3.1-moe:1b | 0.102675 |
| 4 | qwen2:0.5b | 0.067952 |
| 4 | smollm2:360m | 0.113136 |
| 5 | gemma3:1b | 0.287023 |
| 5 | granite3.1-moe:1b | 0.110838 |
| 5 | qwen2:0.5b | 0.106843 |
| 5 | smollm2:360m | 0.109319 |
| 6 | gemma3:1b | 0.231133 |
| 6 | granite3.1-moe:1b | 0.080830 |
| 6 | qwen2:0.5b | 0.056787 |

```
6      smollm2:360m 0.056255
7         gemma3:1b 0.317204
7 granite3.1-moe:1b 0.078031
7        qwen2:0.5b 0.050871
7      smollm2:360m 0.024435
8         gemma3:1b 0.308155
8 granite3.1-moe:1b 0.059570
8        qwen2:0.5b 0.023400
8      smollm2:360m 0.041003
```

[21]:
```
metric_col = 'levenshtein_similarity'
agg = df.groupby(['prompt_id', 'model'])[metric_col].mean().reset_index()
prompt_order = sorted(df['prompt_id'].unique())

plt.figure(figsize=(10,6))
bar = sns.barplot(
    data=agg,
    x='prompt_id',
    y=metric_col,
    hue='model',
    order=prompt_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.2f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Prompt ID", fontsize=12)
plt.ylabel("Mean Levenshtein Similarity", fontsize=12)
plt.title("Mean Levenshtein Similarity by Prompt and Model", fontsize=14)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()

print("Mean Levenshtein Similarity Table by Prompt and Model:\n")
print(agg.to_string(index=False))
```




Mean Levenshtein Similarity Table by Prompt and Model:

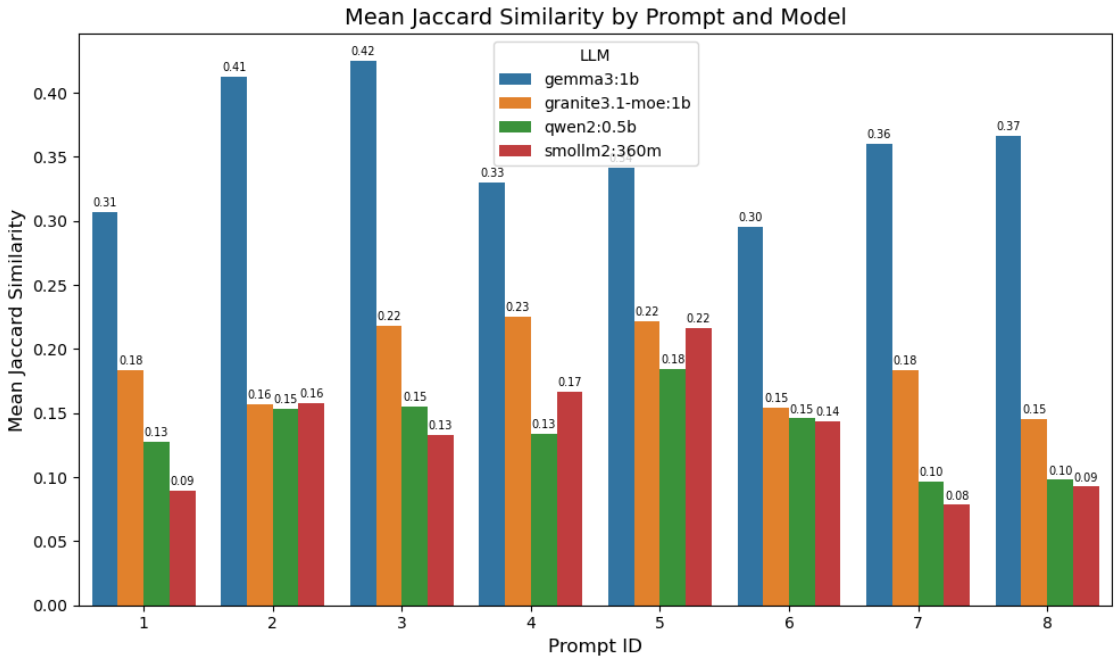
| prompt_id | model | levenshtein_similarity |
|---|---|---|
| 1 | gemma3:1b | 0.336949 |
| 1 | granite3.1-moe:1b | 0.022555 |
| 1 | qwen2:0.5b | 0.042446 |
| 1 | smollm2:360m | 0.088772 |
| 2 | gemma3:1b | 0.449745 |
| 2 | granite3.1-moe:1b | 0.058244 |
| 2 | qwen2:0.5b | 0.106182 |
| 2 | smollm2:360m | 0.104981 |
| 3 | gemma3:1b | 0.401415 |
| 3 | granite3.1-moe:1b | 0.090178 |
| 3 | qwen2:0.5b | 0.119815 |
| 3 | smollm2:360m | 0.206051 |
| 4 | gemma3:1b | 0.342165 |
| 4 | granite3.1-moe:1b | 0.110445 |
| 4 | qwen2:0.5b | 0.134457 |
| 4 | smollm2:360m | 0.208253 |
| 5 | gemma3:1b | 0.349078 |
| 5 | granite3.1-moe:1b | 0.133652 |
| 5 | qwen2:0.5b | 0.155324 |
| 5 | smollm2:360m | 0.213184 |
| 6 | gemma3:1b | 0.333702 |
| 6 | granite3.1-moe:1b | 0.101035 |
| 6 | qwen2:0.5b | 0.176270 |

```
6      smollm2:360m                0.165891
7         gemma3:1b                0.391250
7 granite3.1-moe:1b                0.129848
7        qwen2:0.5b                0.095614
7      smollm2:360m                0.188834
8         gemma3:1b                0.339023
8 granite3.1-moe:1b                0.130611
8        qwen2:0.5b                0.128982
8      smollm2:360m                0.274192
```

[22]:
```
metric_col = 'jaccard_similarity'
agg = df.groupby(['prompt_id', 'model'])[metric_col].mean().reset_index()
prompt_order = sorted(df['prompt_id'].unique())

plt.figure(figsize=(10,6))
bar = sns.barplot(
    data=agg,
    x='prompt_id',
    y=metric_col,
    hue='model',
    order=prompt_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.2f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Prompt ID", fontsize=12)
plt.ylabel("Mean Jaccard Similarity", fontsize=12)
plt.title("Mean Jaccard Similarity by Prompt and Model", fontsize=14)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()

print("Mean Jaccard Similarity Table by Prompt and Model:\n")
print(agg.to_string(index=False))
```




Mean Jaccard Similarity Table by Prompt and Model:

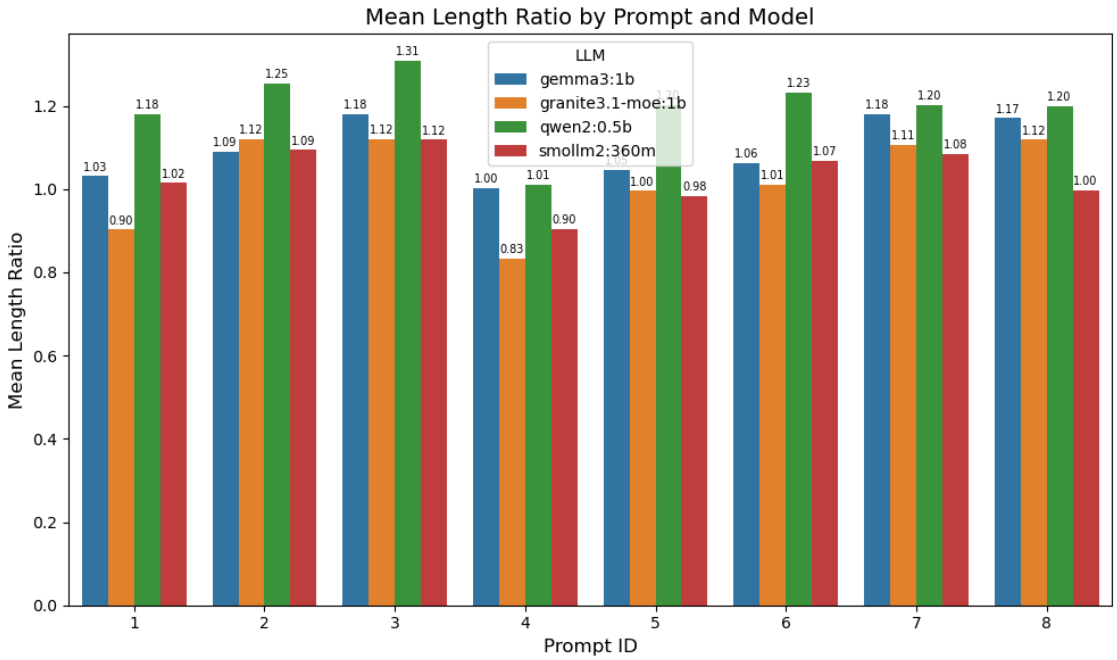
| prompt_id | model | jaccard_similarity |
|---|---|---|
| 1 | gemma3:1b | 0.307219 |
| 1 | granite3.1-moe:1b | 0.183757 |
| 1 | qwen2:0.5b | 0.127423 |
| 1 | smollm2:360m | 0.089247 |
| 2 | gemma3:1b | 0.412321 |
| 2 | granite3.1-moe:1b | 0.157192 |
| 2 | qwen2:0.5b | 0.153279 |
| 2 | smollm2:360m | 0.157645 |
| 3 | gemma3:1b | 0.424638 |
| 3 | granite3.1-moe:1b | 0.218134 |
| 3 | qwen2:0.5b | 0.154935 |
| 3 | smollm2:360m | 0.133321 |
| 4 | gemma3:1b | 0.329740 |
| 4 | granite3.1-moe:1b | 0.225126 |
| 4 | qwen2:0.5b | 0.133691 |
| 4 | smollm2:360m | 0.166607 |
| 5 | gemma3:1b | 0.341227 |
| 5 | granite3.1-moe:1b | 0.221673 |
| 5 | qwen2:0.5b | 0.184770 |
| 5 | smollm2:360m | 0.216383 |
| 6 | gemma3:1b | 0.295691 |
| 6 | granite3.1-moe:1b | 0.153952 |
| 6 | qwen2:0.5b | 0.145783 |

```
6      smollm2:360m            0.143711
7         gemma3:1b            0.359963
7 granite3.1-moe:1b            0.183441
7        qwen2:0.5b            0.096504
7      smollm2:360m            0.078461
8         gemma3:1b            0.366199
8 granite3.1-moe:1b            0.145337
8        qwen2:0.5b            0.097956
8      smollm2:360m            0.092543
```

[23]:
```
metric_col = 'length_ratio'
agg = df.groupby(['prompt_id', 'model'])[metric_col].mean().reset_index()
prompt_order = sorted(df['prompt_id'].unique())

plt.figure(figsize=(10,6))
bar = sns.barplot(
    data=agg,
    x='prompt_id',
    y=metric_col,
    hue='model',
    order=prompt_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.2f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Prompt ID", fontsize=12)
plt.ylabel("Mean Length Ratio", fontsize=12)
plt.title("Mean Length Ratio by Prompt and Model", fontsize=14)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()

print("Mean Length Ratio Table by Prompt and Model:\n")
print(agg.to_string(index=False))
```




Mean Length Ratio Table by Prompt and Model:

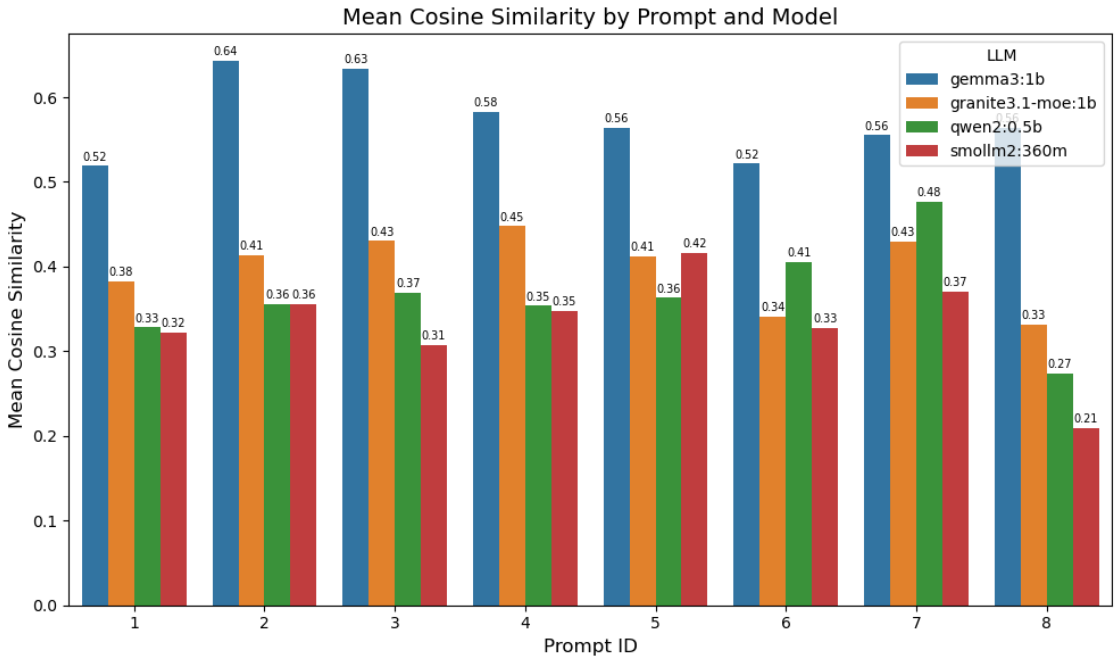
| prompt_id | model | length_ratio |
|---|---|---|
| 1 | gemma3:1b | 1.031050 |
| 1 | granite3.1-moe:1b | 0.902465 |
| 1 | qwen2:0.5b | 1.180218 |
| 1 | smollm2:360m | 1.015045 |
| 2 | gemma3:1b | 1.090909 |
| 2 | granite3.1-moe:1b | 1.121014 |
| 2 | qwen2:0.5b | 1.253623 |
| 2 | smollm2:360m | 1.094203 |
| 3 | gemma3:1b | 1.181061 |
| 3 | granite3.1-moe:1b | 1.120000 |
| 3 | qwen2:0.5b | 1.307576 |
| 3 | smollm2:360m | 1.118561 |
| 4 | gemma3:1b | 1.002674 |
| 4 | granite3.1-moe:1b | 0.832026 |
| 4 | qwen2:0.5b | 1.011586 |
| 4 | smollm2:360m | 0.903446 |
| 5 | gemma3:1b | 1.045455 |
| 5 | granite3.1-moe:1b | 0.995522 |
| 5 | qwen2:0.5b | 1.201832 |
| 5 | smollm2:360m | 0.983039 |
| 6 | gemma3:1b | 1.062328 |
| 6 | granite3.1-moe:1b | 1.011742 |
| 6 | qwen2:0.5b | 1.231061 |

```
6      smollm2:360m      1.068526
7         gemma3:1b      1.179189
7 granite3.1-moe:1b      1.107025
7        qwen2:0.5b      1.202855
7      smollm2:360m      1.084899
8         gemma3:1b      1.170641
8 granite3.1-moe:1b      1.118852
8        qwen2:0.5b      1.198584
8      smollm2:360m      0.996647
```

[24]:
```
metric_col = 'cosine_similarity'
agg = df.groupby(['prompt_id', 'model'])[metric_col].mean().reset_index()
prompt_order = sorted(df['prompt_id'].unique())

plt.figure(figsize=(10,6))
bar = sns.barplot(
    data=agg,
    x='prompt_id',
    y=metric_col,
    hue='model',
    order=prompt_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.2f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Prompt ID", fontsize=12)
plt.ylabel("Mean Cosine Similarity", fontsize=12)
plt.title("Mean Cosine Similarity by Prompt and Model", fontsize=14)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()

print("Mean Cosine Similarity Table by Prompt and Model:\n")
print(agg.to_string(index=False))
```




Mean Cosine Similarity Table by Prompt and Model:

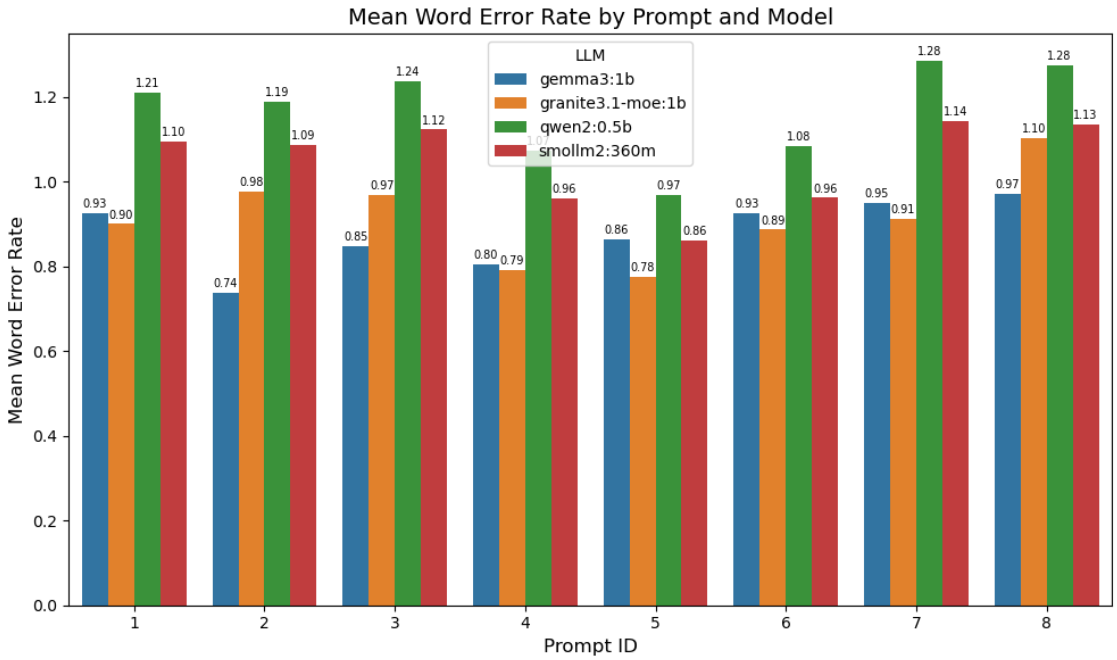
| prompt_id | model | cosine_similarity |
|---|---|---|
| 1 | gemma3:1b | 0.518970 |
| 1 | granite3.1-moe:1b | 0.382670 |
| 1 | qwen2:0.5b | 0.328296 |
| 1 | smollm2:360m | 0.322072 |
| 2 | gemma3:1b | 0.642691 |
| 2 | granite3.1-moe:1b | 0.413292 |
| 2 | qwen2:0.5b | 0.355949 |
| 2 | smollm2:360m | 0.355266 |
| 3 | gemma3:1b | 0.633511 |
| 3 | granite3.1-moe:1b | 0.430231 |
| 3 | qwen2:0.5b | 0.369297 |
| 3 | smollm2:360m | 0.307824 |
| 4 | gemma3:1b | 0.582984 |
| 4 | granite3.1-moe:1b | 0.447602 |
| 4 | qwen2:0.5b | 0.353659 |
| 4 | smollm2:360m | 0.347627 |
| 5 | gemma3:1b | 0.564296 |
| 5 | granite3.1-moe:1b | 0.412355 |
| 5 | qwen2:0.5b | 0.363213 |
| 5 | smollm2:360m | 0.415689 |
| 6 | gemma3:1b | 0.521634 |
| 6 | granite3.1-moe:1b | 0.341044 |
| 6 | qwen2:0.5b | 0.405859 |

```
6      smollm2:360m           0.327513
7         gemma3:1b           0.555280
7 granite3.1-moe:1b           0.429507
7        qwen2:0.5b           0.477016
7      smollm2:360m           0.370274
8         gemma3:1b           0.563831
8 granite3.1-moe:1b           0.331496
8        qwen2:0.5b           0.273568
8      smollm2:360m           0.208965
```

[25]:
```
metric_col = 'wer'
agg = df.groupby(['prompt_id', 'model'])[metric_col].mean().reset_index()
prompt_order = sorted(df['prompt_id'].unique())

plt.figure(figsize=(10,6))
bar = sns.barplot(
    data=agg,
    x='prompt_id',
    y=metric_col,
    hue='model',
    order=prompt_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.2f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Prompt ID", fontsize=12)
plt.ylabel("Mean Word Error Rate", fontsize=12)
plt.title("Mean Word Error Rate by Prompt and Model", fontsize=14)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()

print("Mean Word Error Rate Table by Prompt and Model:\n")
print(agg.to_string(index=False))
```




Mean Word Error Rate Table by Prompt and Model:

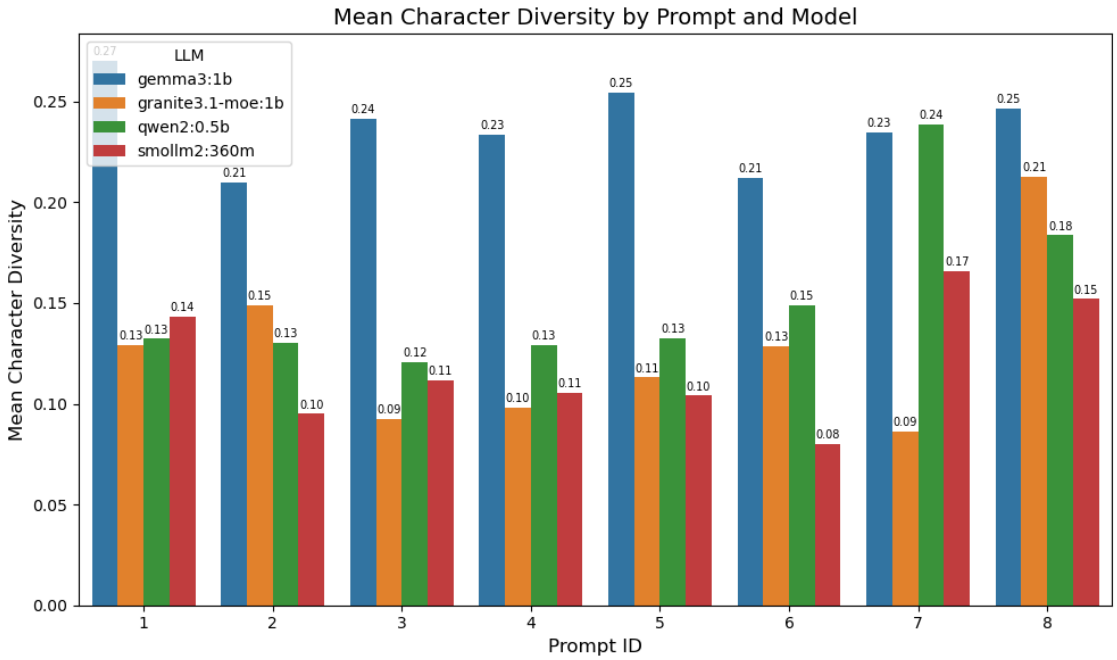
| prompt_id | model | wer |
|---|---|---|
| 1 | gemma3:1b | 0.926290 |
| 1 | granite3.1-moe:1b | 0.900000 |
| 1 | qwen2:0.5b | 1.211302 |
| 1 | smollm2:360m | 1.095823 |
| 2 | gemma3:1b | 0.736842 |
| 2 | granite3.1-moe:1b | 0.976316 |
| 2 | qwen2:0.5b | 1.188995 |
| 2 | smollm2:360m | 1.086124 |
| 3 | gemma3:1b | 0.847594 |
| 3 | granite3.1-moe:1b | 0.967647 |
| 3 | qwen2:0.5b | 1.237968 |
| 3 | smollm2:360m | 1.122995 |
| 4 | gemma3:1b | 0.804545 |
| 4 | granite3.1-moe:1b | 0.792500 |
| 4 | qwen2:0.5b | 1.072727 |
| 4 | smollm2:360m | 0.961364 |
| 5 | gemma3:1b | 0.863636 |
| 5 | granite3.1-moe:1b | 0.776190 |
| 5 | qwen2:0.5b | 0.969697 |
| 5 | smollm2:360m | 0.861472 |
| 6 | gemma3:1b | 0.925837 |
| 6 | granite3.1-moe:1b | 0.886842 |
| 6 | qwen2:0.5b | 1.083732 |

```
6      smollm2:360m 0.961722
7         gemma3:1b 0.950147
7 granite3.1-moe:1b 0.912903
7        qwen2:0.5b 1.284457
7      smollm2:360m 1.143695
8         gemma3:1b 0.970674
8 granite3.1-moe:1b 1.103226
8        qwen2:0.5b 1.275660
8      smollm2:360m 1.134897
```

[26]:
```
metric_col = 'char_diversity'
agg = df.groupby(['prompt_id', 'model'])[metric_col].mean().reset_index()
prompt_order = sorted(df['prompt_id'].unique())

plt.figure(figsize=(10,6))
bar = sns.barplot(
    data=agg,
    x='prompt_id',
    y=metric_col,
    hue='model',
    order=prompt_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.2f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Prompt ID", fontsize=12)
plt.ylabel("Mean Character Diversity", fontsize=12)
plt.title("Mean Character Diversity by Prompt and Model", fontsize=14)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()

print("Mean Character Diversity Table by Prompt and Model:\n")
print(agg.to_string(index=False))
```




Mean Character Diversity Table by Prompt and Model:

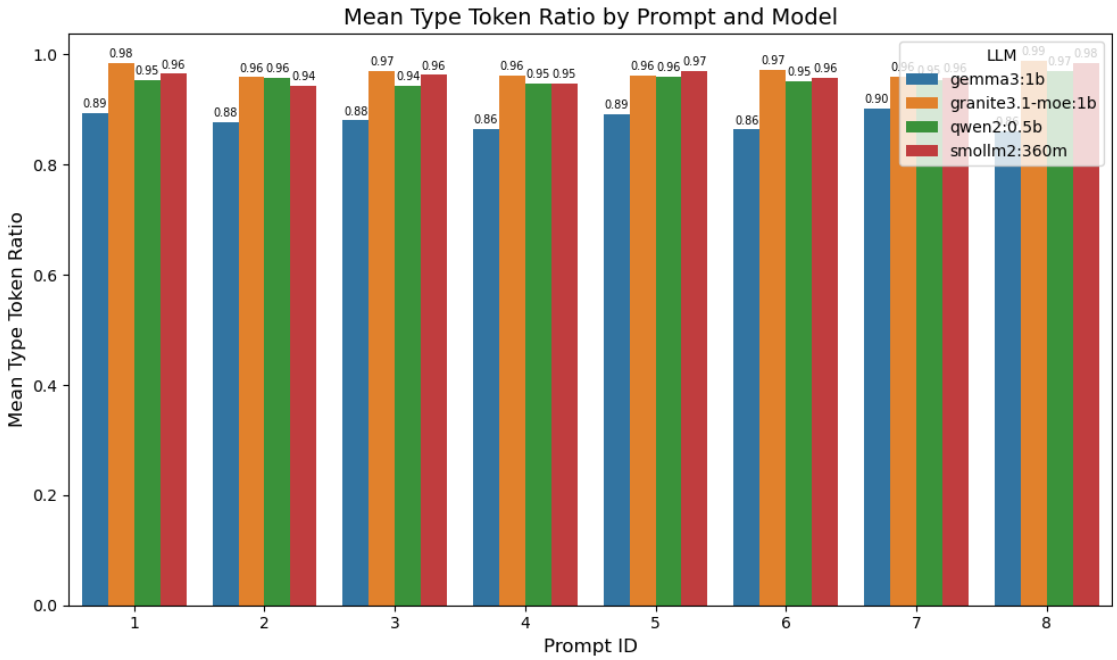
| prompt_id | model | char_diversity |
|---|---|---|
| 1 | gemma3:1b | 0.270020 |
| 1 | granite3.1-moe:1b | 0.129324 |
| 1 | qwen2:0.5b | 0.132288 |
| 1 | smollm2:360m | 0.143191 |
| 2 | gemma3:1b | 0.209836 |
| 2 | granite3.1-moe:1b | 0.148893 |
| 2 | qwen2:0.5b | 0.130082 |
| 2 | smollm2:360m | 0.095010 |
| 3 | gemma3:1b | 0.241285 |
| 3 | granite3.1-moe:1b | 0.092397 |
| 3 | qwen2:0.5b | 0.120813 |
| 3 | smollm2:360m | 0.111742 |
| 4 | gemma3:1b | 0.233601 |
| 4 | granite3.1-moe:1b | 0.098295 |
| 4 | qwen2:0.5b | 0.128973 |
| 4 | smollm2:360m | 0.105373 |
| 5 | gemma3:1b | 0.254620 |
| 5 | granite3.1-moe:1b | 0.113102 |
| 5 | qwen2:0.5b | 0.132592 |
| 5 | smollm2:360m | 0.103992 |
| 6 | gemma3:1b | 0.212050 |
| 6 | granite3.1-moe:1b | 0.128421 |
| 6 | qwen2:0.5b | 0.149011 |

```
6      smollm2:360m        0.079962
7         gemma3:1b        0.234534
7 granite3.1-moe:1b        0.086330
7        qwen2:0.5b        0.238583
7      smollm2:360m        0.165616
8         gemma3:1b        0.246564
8 granite3.1-moe:1b        0.212882
8        qwen2:0.5b        0.183565
8      smollm2:360m        0.151941
```

[27]:
```
metric_col = 'type_token_ratio'
agg = df.groupby(['prompt_id', 'model'])[metric_col].mean().reset_index()
prompt_order = sorted(df['prompt_id'].unique())

plt.figure(figsize=(10,6))
bar = sns.barplot(
    data=agg,
    x='prompt_id',
    y=metric_col,
    hue='model',
    order=prompt_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.2f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Prompt ID", fontsize=12)
plt.ylabel("Mean Type Token Ratio", fontsize=12)
plt.title("Mean Type Token Ratio by Prompt and Model", fontsize=14)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()

print("Mean Type Token Ratio Table by Prompt and Model:\n")
print(agg.to_string(index=False))
```




Mean Type Token Ratio Table by Prompt and Model:

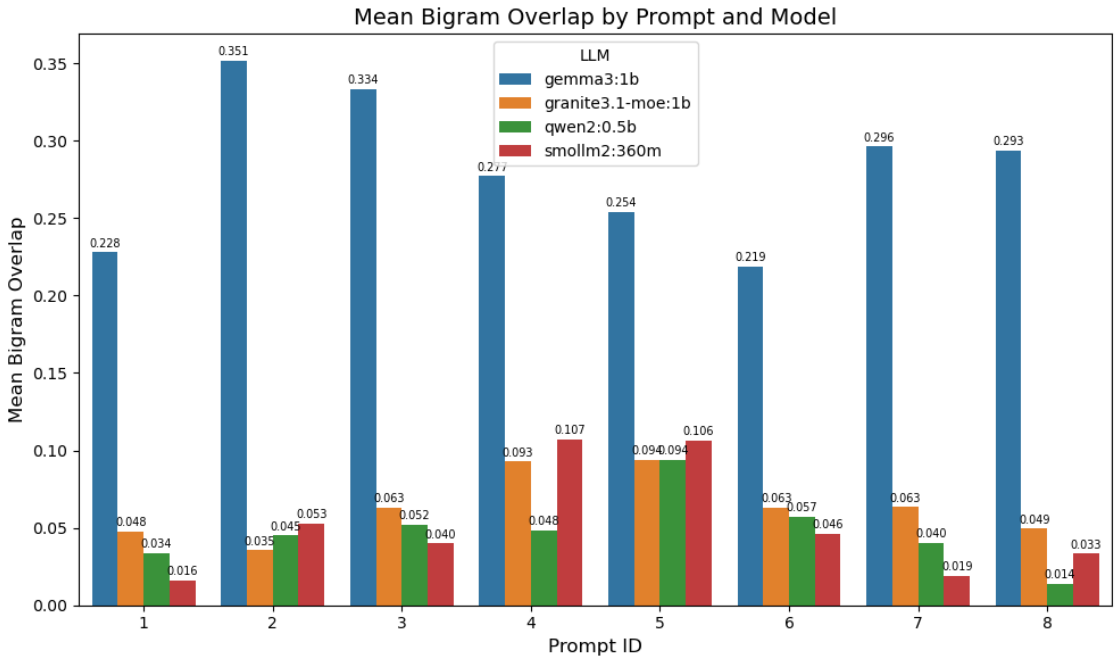
| prompt_id | model | type_token_ratio |
|---|---|---|
| 1 | gemma3:1b | 0.893512 |
| 1 | granite3.1-moe:1b | 0.984078 |
| 1 | qwen2:0.5b | 0.954470 |
| 1 | smollm2:360m | 0.964942 |
| 2 | gemma3:1b | 0.877512 |
| 2 | granite3.1-moe:1b | 0.958874 |
| 2 | qwen2:0.5b | 0.957332 |
| 2 | smollm2:360m | 0.943816 |
| 3 | gemma3:1b | 0.880468 |
| 3 | granite3.1-moe:1b | 0.969680 |
| 3 | qwen2:0.5b | 0.942743 |
| 3 | smollm2:360m | 0.963048 |
| 4 | gemma3:1b | 0.864202 |
| 4 | granite3.1-moe:1b | 0.962176 |
| 4 | qwen2:0.5b | 0.948129 |
| 4 | smollm2:360m | 0.948189 |
| 5 | gemma3:1b | 0.890984 |
| 5 | granite3.1-moe:1b | 0.962562 |
| 5 | qwen2:0.5b | 0.959796 |
| 5 | smollm2:360m | 0.970213 |
| 6 | gemma3:1b | 0.863806 |
| 6 | granite3.1-moe:1b | 0.972466 |
| 6 | qwen2:0.5b | 0.952460 |

```
6      smollm2:360m          0.958267
7         gemma3:1b          0.901537
7 granite3.1-moe:1b          0.959901
7        qwen2:0.5b          0.953417
7      smollm2:360m          0.957546
8         gemma3:1b          0.860478
8 granite3.1-moe:1b          0.988074
8        qwen2:0.5b          0.969944
8      smollm2:360m          0.984129
```

[28]:
```
metric_col = 'bigram_overlap'
agg = df.groupby(['prompt_id', 'model'])[metric_col].mean().reset_index()
prompt_order = sorted(df['prompt_id'].unique())

plt.figure(figsize=(10,6))
bar = sns.barplot(
    data=agg,
    x='prompt_id',
    y=metric_col,
    hue='model',
    order=prompt_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.3f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Prompt ID", fontsize=12)
plt.ylabel("Mean Bigram Overlap", fontsize=12)
plt.title("Mean Bigram Overlap by Prompt and Model", fontsize=14)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()

print("Mean Bigram Overlap Table by Prompt and Model:\n")
print(agg.to_string(index=False))
```




Mean Bigram Overlap Table by Prompt and Model:

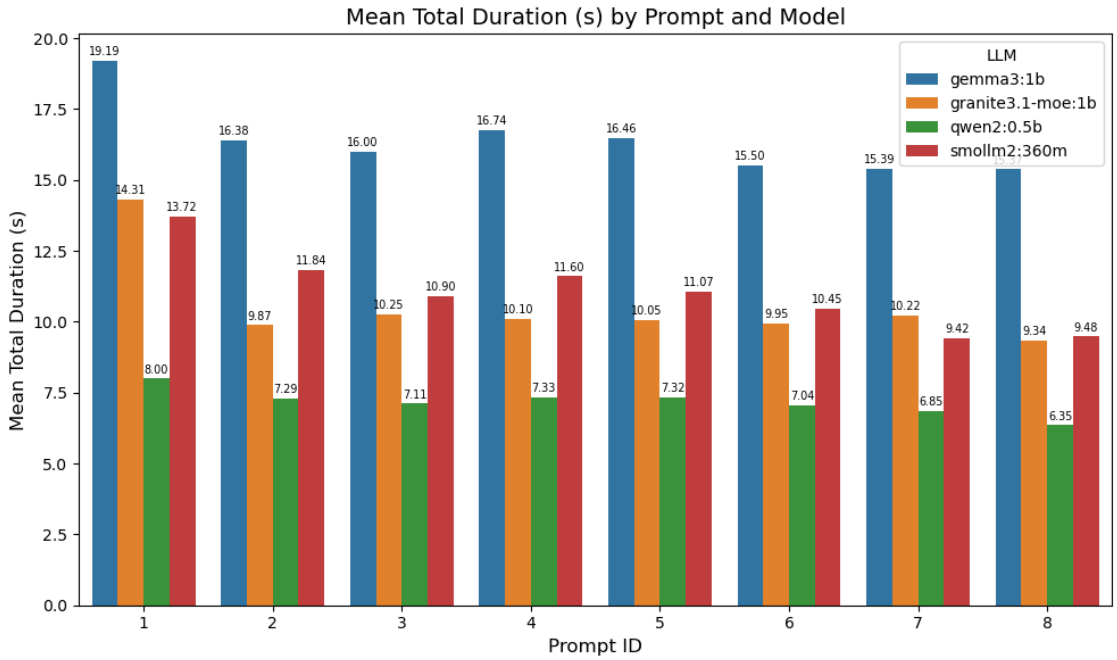
| prompt_id | model | bigram_overlap |
|---|---|---|
| 1 | gemma3:1b | 0.227948 |
| 1 | granite3.1-moe:1b | 0.047773 |
| 1 | qwen2:0.5b | 0.033781 |
| 1 | smollm2:360m | 0.016246 |
| 2 | gemma3:1b | 0.351405 |
| 2 | granite3.1-moe:1b | 0.035467 |
| 2 | qwen2:0.5b | 0.045038 |
| 2 | smollm2:360m | 0.052751 |
| 3 | gemma3:1b | 0.333533 |
| 3 | granite3.1-moe:1b | 0.062834 |
| 3 | qwen2:0.5b | 0.052062 |
| 3 | smollm2:360m | 0.039926 |
| 4 | gemma3:1b | 0.277089 |
| 4 | granite3.1-moe:1b | 0.092721 |
| 4 | qwen2:0.5b | 0.048092 |
| 4 | smollm2:360m | 0.107319 |
| 5 | gemma3:1b | 0.254168 |
| 5 | granite3.1-moe:1b | 0.093665 |
| 5 | qwen2:0.5b | 0.094014 |
| 5 | smollm2:360m | 0.106278 |
| 6 | gemma3:1b | 0.218936 |
| 6 | granite3.1-moe:1b | 0.063029 |
| 6 | qwen2:0.5b | 0.057376 |

```
6      smollm2:360m        0.046410
7         gemma3:1b        0.296233
7 granite3.1-moe:1b        0.063329
7        qwen2:0.5b        0.040025
7      smollm2:360m        0.018833
8         gemma3:1b        0.293477
8 granite3.1-moe:1b        0.049417
8        qwen2:0.5b        0.013744
8      smollm2:360m        0.033360
```

[29]:
```
metric_col = 'total_duration_s'
agg = df.groupby(['prompt_id', 'model'])[metric_col].mean().reset_index()
prompt_order = sorted(df['prompt_id'].unique())

plt.figure(figsize=(10,6))
bar = sns.barplot(
    data=agg,
    x='prompt_id',
    y=metric_col,
    hue='model',
    order=prompt_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.2f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Prompt ID", fontsize=12)
plt.ylabel("Mean Total Duration (s)", fontsize=12)
plt.title("Mean Total Duration (s) by Prompt and Model", fontsize=14)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()

print("Mean Total Duration (s) Table by Prompt and Model:\n")
print(agg.to_string(index=False))
```




Mean Total Duration (s) Table by Prompt and Model:

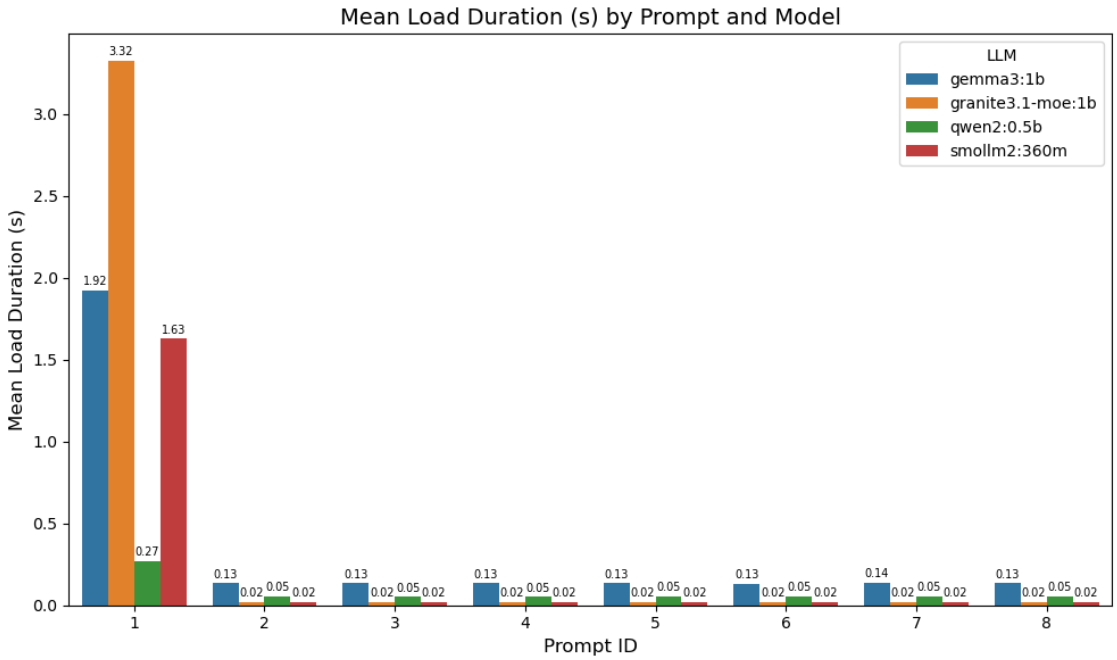
| prompt_id | model | total_duration_s |
|---|---|---|
| 1 | gemma3:1b | 19.188144 |
| 1 | granite3.1-moe:1b | 14.308235 |
| 1 | qwen2:0.5b | 7.998199 |
| 1 | smollm2:360m | 13.717496 |
| 2 | gemma3:1b | 16.380431 |
| 2 | granite3.1-moe:1b | 9.874609 |
| 2 | qwen2:0.5b | 7.288003 |
| 2 | smollm2:360m | 11.835116 |
| 3 | gemma3:1b | 16.000983 |
| 3 | granite3.1-moe:1b | 10.246667 |
| 3 | qwen2:0.5b | 7.114733 |
| 3 | smollm2:360m | 10.895771 |
| 4 | gemma3:1b | 16.743796 |
| 4 | granite3.1-moe:1b | 10.095100 |
| 4 | qwen2:0.5b | 7.329973 |
| 4 | smollm2:360m | 11.602065 |
| 5 | gemma3:1b | 16.463278 |
| 5 | granite3.1-moe:1b | 10.051996 |
| 5 | qwen2:0.5b | 7.319335 |
| 5 | smollm2:360m | 11.066141 |
| 6 | gemma3:1b | 15.499161 |
| 6 | granite3.1-moe:1b | 9.947697 |
| 6 | qwen2:0.5b | 7.036435 |

```
6      smollm2:360m         10.453246
7         gemma3:1b         15.390212
7 granite3.1-moe:1b         10.223190
7        qwen2:0.5b          6.846961
7      smollm2:360m          9.418833
8         gemma3:1b         15.370792
8 granite3.1-moe:1b          9.339246
8        qwen2:0.5b          6.350768
8      smollm2:360m          9.481517
```

[30]:
```
df['load_duration_s'] = df['load_duration_ns'] / 1e9
metric_col = 'load_duration_s'
agg = df.groupby(['prompt_id', 'model'])[metric_col].mean().reset_index()
prompt_order = sorted(df['prompt_id'].unique())

plt.figure(figsize=(10,6))
bar = sns.barplot(
    data=agg,
    x='prompt_id',
    y=metric_col,
    hue='model',
    order=prompt_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.2f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Prompt ID", fontsize=12)
plt.ylabel("Mean Load Duration (s)", fontsize=12)
plt.title("Mean Load Duration (s) by Prompt and Model", fontsize=14)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()

print("Mean Load Duration (s) Table by Prompt and Model:\n")
print(agg.to_string(index=False))
```




Mean Load Duration (s) Table by Prompt and Model:

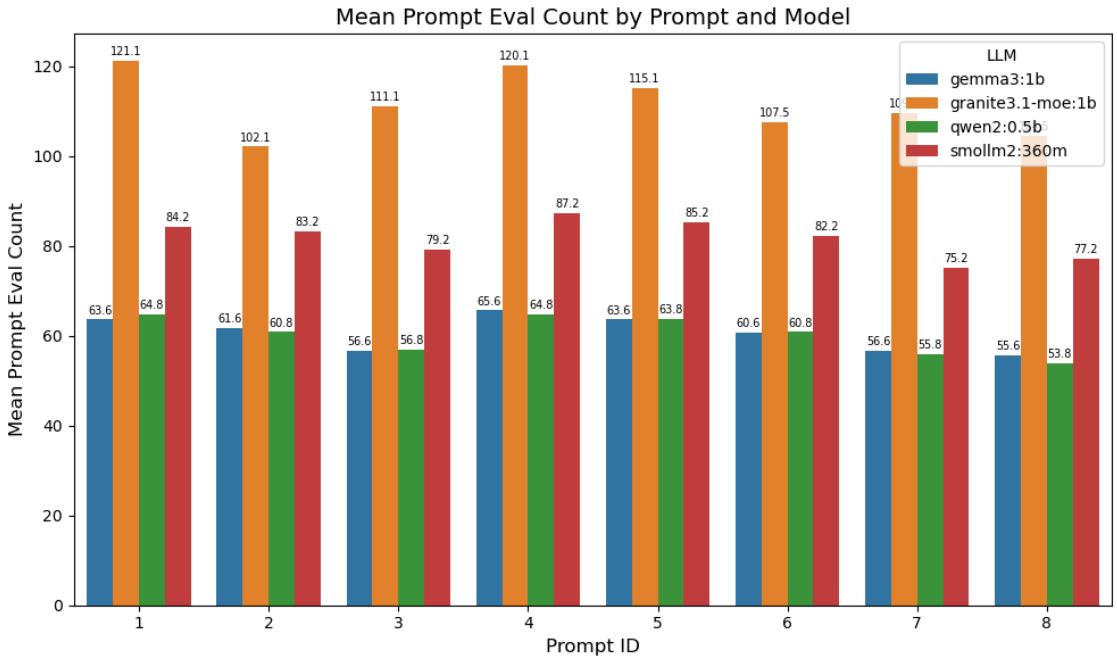
| prompt_id | model | load_duration_s |
|---|---|---|
| 1 | gemma3:1b | 1.920152 |
| 1 | granite3.1-moe:1b | 3.322415 |
| 1 | qwen2:0.5b | 0.267021 |
| 1 | smollm2:360m | 1.627052 |
| 2 | gemma3:1b | 0.134766 |
| 2 | granite3.1-moe:1b | 0.021508 |
| 2 | qwen2:0.5b | 0.051730 |
| 2 | smollm2:360m | 0.021827 |
| 3 | gemma3:1b | 0.134548 |
| 3 | granite3.1-moe:1b | 0.021360 |
| 3 | qwen2:0.5b | 0.051673 |
| 3 | smollm2:360m | 0.021094 |
| 4 | gemma3:1b | 0.134743 |
| 4 | granite3.1-moe:1b | 0.020843 |
| 4 | qwen2:0.5b | 0.051226 |
| 4 | smollm2:360m | 0.021520 |
| 5 | gemma3:1b | 0.134060 |
| 5 | granite3.1-moe:1b | 0.021788 |
| 5 | qwen2:0.5b | 0.052652 |
| 5 | smollm2:360m | 0.021343 |
| 6 | gemma3:1b | 0.133452 |
| 6 | granite3.1-moe:1b | 0.021591 |
| 6 | qwen2:0.5b | 0.051767 |

```
6      smollm2:360m         0.021137
7         gemma3:1b         0.135561
7 granite3.1-moe:1b         0.021237
7        qwen2:0.5b         0.052055
7      smollm2:360m         0.020865
8         gemma3:1b         0.134019
8 granite3.1-moe:1b         0.020916
8        qwen2:0.5b         0.051736
8      smollm2:360m         0.021303
```

[31]:
```
metric_col = 'prompt_eval_count'
agg = df.groupby(['prompt_id', 'model'])[metric_col].mean().reset_index()
prompt_order = sorted(df['prompt_id'].unique())

plt.figure(figsize=(10,6))
bar = sns.barplot(
    data=agg,
    x='prompt_id',
    y=metric_col,
    hue='model',
    order=prompt_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.1f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Prompt ID", fontsize=12)
plt.ylabel("Mean Prompt Eval Count", fontsize=12)
plt.title("Mean Prompt Eval Count by Prompt and Model", fontsize=14)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()

print("Mean Prompt Eval Count Table by Prompt and Model:\n")
print(agg.to_string(index=False))
```




Mean Prompt Eval Count Table by Prompt and Model:

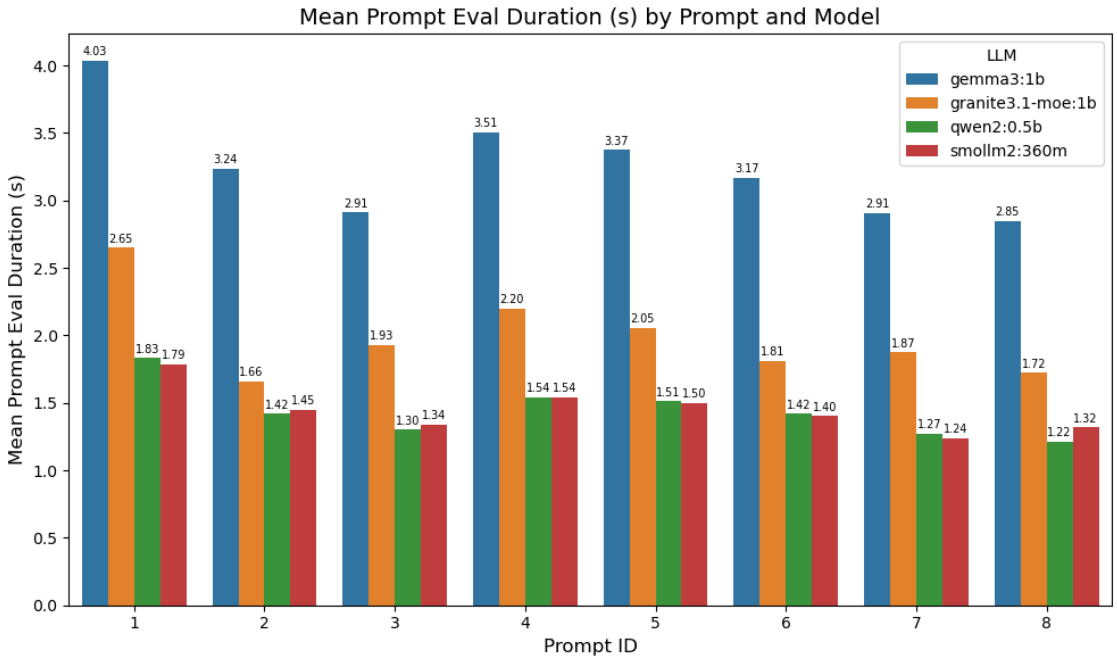
| prompt_id | model | prompt_eval_count |
|---|---|---|
| 1 | gemma3:1b | 63.636364 |
| 1 | granite3.1-moe:1b | 121.100000 |
| 1 | qwen2:0.5b | 64.818182 |
| 1 | smollm2:360m | 84.181818 |
| 2 | gemma3:1b | 61.636364 |
| 2 | granite3.1-moe:1b | 102.100000 |
| 2 | qwen2:0.5b | 60.818182 |
| 2 | smollm2:360m | 83.181818 |
| 3 | gemma3:1b | 56.636364 |
| 3 | granite3.1-moe:1b | 111.100000 |
| 3 | qwen2:0.5b | 56.818182 |
| 3 | smollm2:360m | 79.181818 |
| 4 | gemma3:1b | 65.636364 |
| 4 | granite3.1-moe:1b | 120.100000 |
| 4 | qwen2:0.5b | 64.818182 |
| 4 | smollm2:360m | 87.181818 |
| 5 | gemma3:1b | 63.636364 |
| 5 | granite3.1-moe:1b | 115.100000 |
| 5 | qwen2:0.5b | 63.818182 |
| 5 | smollm2:360m | 85.181818 |
| 6 | gemma3:1b | 60.636364 |
| 6 | granite3.1-moe:1b | 107.500000 |
| 6 | qwen2:0.5b | 60.818182 |

```
6      smollm2:360m          82.181818
7         gemma3:1b          56.636364
7 granite3.1-moe:1b         109.500000
7        qwen2:0.5b          55.818182
7      smollm2:360m          75.181818
8         gemma3:1b          55.636364
8 granite3.1-moe:1b         104.500000
8        qwen2:0.5b          53.818182
8      smollm2:360m          77.181818
```

[32]:
```
df['prompt_eval_duration_s'] = df['prompt_eval_duration_ns'] / 1e9
metric_col = 'prompt_eval_duration_s'
agg = df.groupby(['prompt_id', 'model'])[metric_col].mean().reset_index()
prompt_order = sorted(df['prompt_id'].unique())

plt.figure(figsize=(10,6))
bar = sns.barplot(
    data=agg,
    x='prompt_id',
    y=metric_col,
    hue='model',
    order=prompt_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.2f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Prompt ID", fontsize=12)
plt.ylabel("Mean Prompt Eval Duration (s)", fontsize=12)
plt.title("Mean Prompt Eval Duration (s) by Prompt and Model", fontsize=14)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()

print("Mean Prompt Eval Duration (s) Table by Prompt and Model:\n")
print(agg.to_string(index=False))
```




Mean Prompt Eval Duration (s) Table by Prompt and Model:

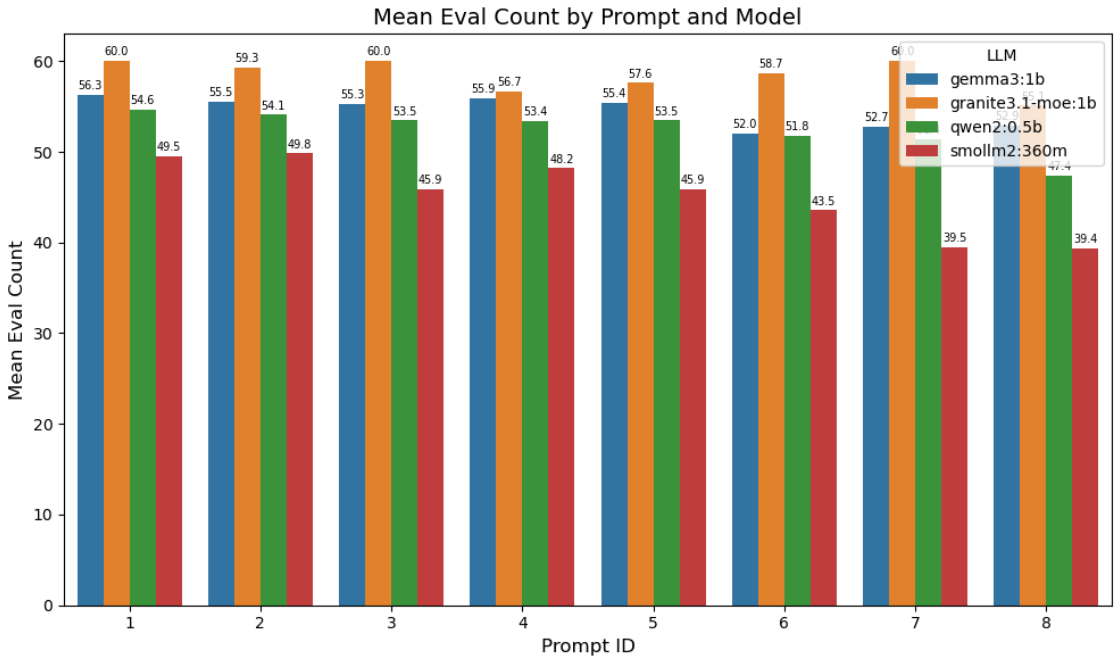
| prompt_id | model | prompt_eval_duration_s |
|---|---|---|
| 1 | gemma3:1b | 4.032869 |
| 1 | granite3.1-moe:1b | 2.649561 |
| 1 | qwen2:0.5b | 1.831723 |
| 1 | smollm2:360m | 1.786276 |
| 2 | gemma3:1b | 3.236039 |
| 2 | granite3.1-moe:1b | 1.662311 |
| 2 | qwen2:0.5b | 1.418149 |
| 2 | smollm2:360m | 1.450954 |
| 3 | gemma3:1b | 2.910115 |
| 3 | granite3.1-moe:1b | 1.926919 |
| 3 | qwen2:0.5b | 1.301053 |
| 3 | smollm2:360m | 1.338266 |
| 4 | gemma3:1b | 3.505857 |
| 4 | granite3.1-moe:1b | 2.198182 |
| 4 | qwen2:0.5b | 1.537797 |
| 4 | smollm2:360m | 1.537981 |
| 5 | gemma3:1b | 3.373812 |
| 5 | granite3.1-moe:1b | 2.052317 |
| 5 | qwen2:0.5b | 1.512005 |
| 5 | smollm2:360m | 1.497412 |
| 6 | gemma3:1b | 3.169544 |
| 6 | granite3.1-moe:1b | 1.810106 |
| 6 | qwen2:0.5b | 1.418276 |

```
6      smollm2:360m               1.402899
7         gemma3:1b               2.909144
7 granite3.1-moe:1b               1.873770
7        qwen2:0.5b               1.273654
7      smollm2:360m               1.237074
8         gemma3:1b               2.845163
8 granite3.1-moe:1b               1.721691
8        qwen2:0.5b               1.215047
8      smollm2:360m               1.317496
```

[33]:
```
metric_col = 'eval_count'
agg = df.groupby(['prompt_id', 'model'])[metric_col].mean().reset_index()
prompt_order = sorted(df['prompt_id'].unique())

plt.figure(figsize=(10,6))
bar = sns.barplot(
    data=agg,
    x='prompt_id',
    y=metric_col,
    hue='model',
    order=prompt_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.1f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Prompt ID", fontsize=12)
plt.ylabel("Mean Eval Count", fontsize=12)
plt.title("Mean Eval Count by Prompt and Model", fontsize=14)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()

print("Mean Eval Count Table by Prompt and Model:\n")
print(agg.to_string(index=False))
```




Mean Eval Count Table by Prompt and Model:

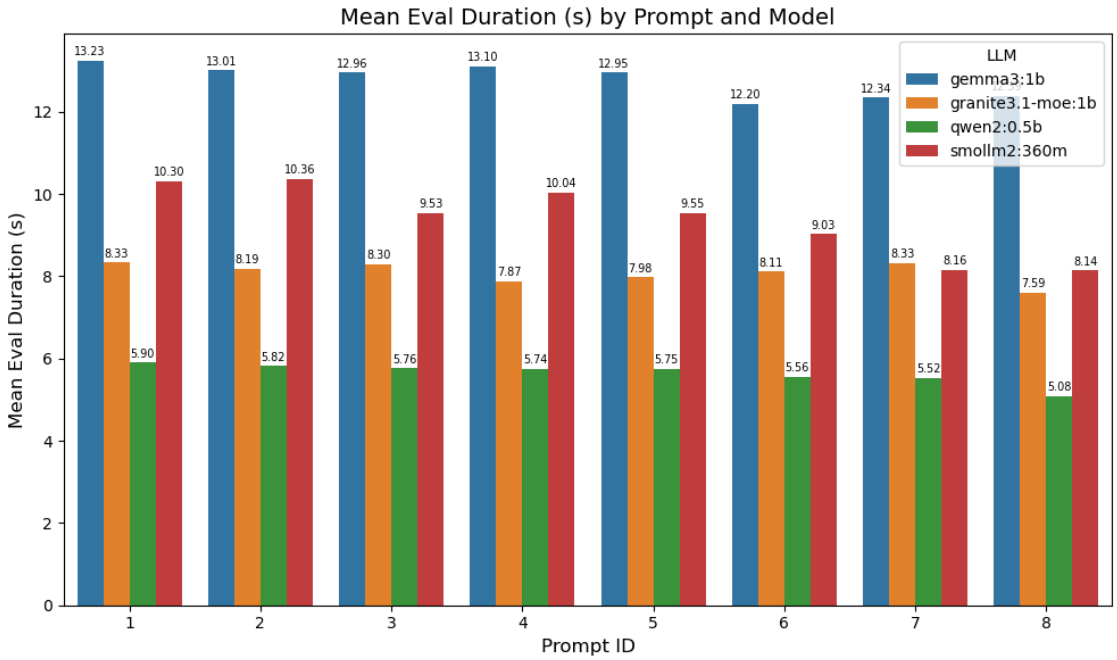
| prompt_id | model | eval_count |
|---|---|---|
| 1 | gemma3:1b | 56.272727 |
| 1 | granite3.1-moe:1b | 60.000000 |
| 1 | qwen2:0.5b | 54.636364 |
| 1 | smollm2:360m | 49.545455 |
| 2 | gemma3:1b | 55.545455 |
| 2 | granite3.1-moe:1b | 59.300000 |
| 2 | qwen2:0.5b | 54.090909 |
| 2 | smollm2:360m | 49.818182 |
| 3 | gemma3:1b | 55.272727 |
| 3 | granite3.1-moe:1b | 60.000000 |
| 3 | qwen2:0.5b | 53.454545 |
| 3 | smollm2:360m | 45.909091 |
| 4 | gemma3:1b | 55.909091 |
| 4 | granite3.1-moe:1b | 56.700000 |
| 4 | qwen2:0.5b | 53.363636 |
| 4 | smollm2:360m | 48.181818 |
| 5 | gemma3:1b | 55.363636 |
| 5 | granite3.1-moe:1b | 57.600000 |
| 5 | qwen2:0.5b | 53.454545 |
| 5 | smollm2:360m | 45.909091 |
| 6 | gemma3:1b | 52.000000 |
| 6 | granite3.1-moe:1b | 58.700000 |
| 6 | qwen2:0.5b | 51.818182 |

```
6      smollm2:360m   43.545455
7         gemma3:1b   52.727273
7 granite3.1-moe:1b   60.000000
7        qwen2:0.5b   51.363636
7      smollm2:360m   39.454545
8         gemma3:1b   52.909091
8 granite3.1-moe:1b   55.100000
8        qwen2:0.5b   47.363636
8      smollm2:360m   39.363636
```

[34]:
```
df['eval_duration_s'] = df['eval_duration_ns'] / 1e9
metric_col = 'eval_duration_s'
agg = df.groupby(['prompt_id', 'model'])[metric_col].mean().reset_index()
prompt_order = sorted(df['prompt_id'].unique())

plt.figure(figsize=(10,6))
bar = sns.barplot(
    data=agg,
    x='prompt_id',
    y=metric_col,
    hue='model',
    order=prompt_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.2f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Prompt ID", fontsize=12)
plt.ylabel("Mean Eval Duration (s)", fontsize=12)
plt.title("Mean Eval Duration (s) by Prompt and Model", fontsize=14)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()

print("Mean Eval Duration (s) Table by Prompt and Model:\n")
print(agg.to_string(index=False))
```




Mean Eval Duration (s) Table by Prompt and Model:

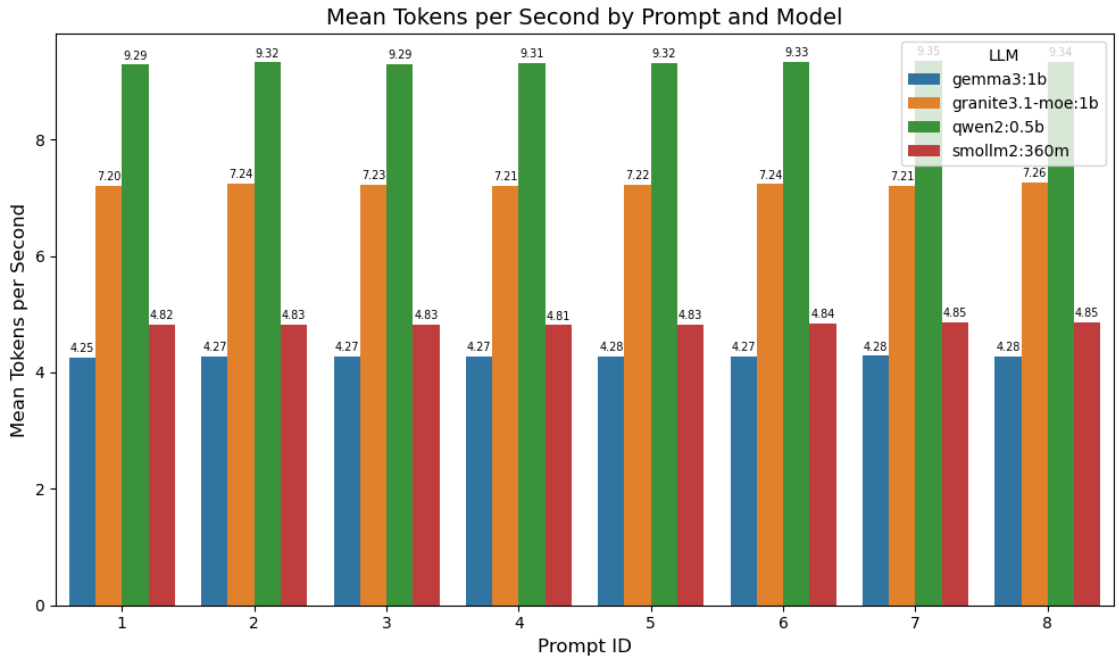
| prompt_id | model | eval_duration_s |
|---|---|---|
| 1 | gemma3:1b | 13.233804 |
| 1 | granite3.1-moe:1b | 8.333593 |
| 1 | qwen2:0.5b | 5.898187 |
| 1 | smollm2:360m | 10.302506 |
| 2 | gemma3:1b | 13.008460 |
| 2 | granite3.1-moe:1b | 8.188722 |
| 2 | qwen2:0.5b | 5.816410 |
| 2 | smollm2:360m | 10.360541 |
| 3 | gemma3:1b | 12.955185 |
| 3 | granite3.1-moe:1b | 8.296848 |
| 3 | qwen2:0.5b | 5.760703 |
| 3 | smollm2:360m | 9.534744 |
| 4 | gemma3:1b | 13.102083 |
| 4 | granite3.1-moe:1b | 7.874251 |
| 4 | qwen2:0.5b | 5.739488 |
| 4 | smollm2:360m | 10.040729 |
| 5 | gemma3:1b | 12.954279 |
| 5 | granite3.1-moe:1b | 7.976062 |
| 5 | qwen2:0.5b | 5.753469 |
| 5 | smollm2:360m | 9.545345 |
| 6 | gemma3:1b | 12.195011 |
| 6 | granite3.1-moe:1b | 8.114146 |
| 6 | qwen2:0.5b | 5.564658 |

```
6      smollm2:360m         9.027609
7         gemma3:1b        12.344349
7 granite3.1-moe:1b         8.326613
7        qwen2:0.5b         5.520027
7      smollm2:360m         8.158739
8         gemma3:1b        12.390459
8 granite3.1-moe:1b         7.594581
8        qwen2:0.5b         5.082774
8      smollm2:360m         8.140808
```

[35]:
```
metric_col = 'tokens_per_second'
agg = df.groupby(['prompt_id', 'model'])[metric_col].mean().reset_index()
prompt_order = sorted(df['prompt_id'].unique())

plt.figure(figsize=(10,6))
bar = sns.barplot(
    data=agg,
    x='prompt_id',
    y=metric_col,
    hue='model',
    order=prompt_order,
    palette='tab10'
)
for container in bar.containers:
    bar.bar_label(container, fmt='%.2f', label_type='edge', padding=2,␣
 ↪fontsize=7)
plt.xlabel("Prompt ID", fontsize=12)
plt.ylabel("Mean Tokens per Second", fontsize=12)
plt.title("Mean Tokens per Second by Prompt and Model", fontsize=14)
plt.legend(title="LLM")
plt.tight_layout()
plt.show()

print("Mean Tokens per Second Table by Prompt and Model:\n")
print(agg.to_string(index=False))
```




Mean Tokens per Second Table by Prompt and Model:

| prompt_id | model | tokens_per_second |
|---|---|---|
| 1 | gemma3:1b | 4.254656 |
| 1 | granite3.1-moe:1b | 7.199806 |
| 1 | qwen2:0.5b | 9.286185 |
| 1 | smollm2:360m | 4.818912 |
| 2 | gemma3:1b | 4.274260 |
| 2 | granite3.1-moe:1b | 7.241827 |
| 2 | qwen2:0.5b | 9.324803 |
| 2 | smollm2:360m | 4.825393 |
| 3 | gemma3:1b | 4.272150 |
| 3 | granite3.1-moe:1b | 7.231682 |
| 3 | qwen2:0.5b | 9.291864 |
| 3 | smollm2:360m | 4.827106 |
| 4 | gemma3:1b | 4.271214 |
| 4 | granite3.1-moe:1b | 7.207941 |
| 4 | qwen2:0.5b | 9.314725 |
| 4 | smollm2:360m | 4.813005 |
| 5 | gemma3:1b | 4.280101 |
| 5 | granite3.1-moe:1b | 7.223503 |
| 5 | qwen2:0.5b | 9.322892 |
| 5 | smollm2:360m | 4.829543 |
| 6 | gemma3:1b | 4.274892 |
| 6 | granite3.1-moe:1b | 7.235559 |
| 6 | qwen2:0.5b | 9.331980 |

```
6      smollm2:360m           4.842847
7         gemma3:1b           4.283511
7 granite3.1-moe:1b           7.206529
7        qwen2:0.5b           9.347458
7      smollm2:360m           4.854049
8         gemma3:1b           4.276154
8 granite3.1-moe:1b           7.261920
8        qwen2:0.5b           9.339646
8      smollm2:360m           4.853902
```

[36]:
```
from scipy.stats import f_oneway
import pandas as pd

metric_col = 'bleu'
anova_results = []
for aug in sorted(df['augmentation_type'].unique()):
    subset = df[df['augmentation_type'] == aug]
    groups = [subset[subset['model'] == m][metric_col].dropna().values
              for m in subset['model'].unique()]
    if sum([len(g) > 1 for g in groups]) >= 2:
        stat, pval = f_oneway(*groups)
        anova_results.append({'augmentation_type': aug, 'F-statistic': stat,␣
 ↪'p-value': pval})
    else:
        anova_results.append({'augmentation_type': aug, 'F-statistic':␣
 ↪float('nan'), 'p-value': float('nan')})

anova_df = pd.DataFrame(anova_results)
anova_df['Significant'] = anova_df['p-value'] < 0.05
display(anova_df)

print("\nSignificance Interpretation (BLEU):")
for idx, row in anova_df.iterrows():
    if pd.notna(row['p-value']):
        if row['Significant']:
            print(f"Augmentation '{row['augmentation_type']}': Significant␣
 ↪difference in mean BLEU across LLMs (p={row['p-value']:.4g}).")
        else:
            print(f"Augmentation '{row['augmentation_type']}': No significant␣
 ↪difference in mean BLEU across LLMs (p={row['p-value']:.4g}).")
    else:
        print(f"Augmentation '{row['augmentation_type']}': Insufficient data␣
 ↪for ANOVA.")
```

```
   augmentation_type  F-statistic       p-value  Significant
0     entity_replace    12.031344  3.074649e-05         True
1             expand     0.088237  9.659122e-01        False
2     explain_simple     1.439620  2.522929e-01        False
```

```
3            identity   418.946728  2.337463e-23         True
4            negation     7.310758  9.090059e-04         True
5               noise     9.195276  3.905748e-04         True
6         paraphrase     1.888940  1.543782e-01        False
7       question_gen     2.234300  1.080536e-01        False
8             shuffle    24.313615  5.957550e-08         True
9           summarize     3.417430  3.085063e-02         True
10            synonym     2.062183  1.279123e-01        False


Significance Interpretation (BLEU):
Augmentation 'entity_replace': Significant difference in mean BLEU across LLMs
(p=3.075e-05).
Augmentation 'expand': No significant difference in mean BLEU across LLMs
(p=0.9659).
Augmentation 'explain_simple': No significant difference in mean BLEU across
LLMs (p=0.2523).
Augmentation 'identity': Significant difference in mean BLEU across LLMs
(p=2.337e-23).
Augmentation 'negation': Significant difference in mean BLEU across LLMs
(p=0.000909).
Augmentation 'noise': Significant difference in mean BLEU across LLMs
(p=0.0003906).
Augmentation 'paraphrase': No significant difference in mean BLEU across LLMs
(p=0.1544).
Augmentation 'question_gen': No significant difference in mean BLEU across LLMs
(p=0.1081).
Augmentation 'shuffle': Significant difference in mean BLEU across LLMs
(p=5.958e-08).
Augmentation 'summarize': Significant difference in mean BLEU across LLMs
(p=0.03085).
Augmentation 'synonym': No significant difference in mean BLEU across LLMs
(p=0.1279).
```

[37]:
```
metric_col = 'levenshtein_similarity'
anova_results = []
for aug in sorted(df['augmentation_type'].unique()):
    subset = df[df['augmentation_type'] == aug]
    groups = [subset[subset['model'] == m][metric_col].dropna().values
              for m in subset['model'].unique()]
    if sum([len(g) > 1 for g in groups]) >= 2:
        stat, pval = f_oneway(*groups)
        anova_results.append({'augmentation_type': aug, 'F-statistic': stat,␣
 ↪'p-value': pval})
    else:
        anova_results.append({'augmentation_type': aug, 'F-statistic':␣
 ↪float('nan'), 'p-value': float('nan')})
```

```
anova_df = pd.DataFrame(anova_results)
anova_df['Significant'] = anova_df['p-value'] < 0.05
display(anova_df)

print("\nSignificance Interpretation (Levenshtein Similarity):")
for idx, row in anova_df.iterrows():
    if pd.notna(row['p-value']):
        if row['Significant']:
            print(f"Augmentation '{row['augmentation_type']}': Significant␣
↪difference in mean Levenshtein similarity across LLMs (p={row['p-value']:.
↪4g}).")
        else:
            print(f"Augmentation '{row['augmentation_type']}': No significant␣
↪difference in mean Levenshtein similarity across LLMs (p={row['p-value']:.
↪4g}).")
    else:
        print(f"Augmentation '{row['augmentation_type']}': Insufficient data␣
↪for ANOVA.")
```

| | augmentation_type | F-statistic | p-value | Significant |
|---|---|---|---|---|
| 0 | entity_replace | 2.625844 | 6.997952e-02 | False |
| 1 | expand | 0.024014 | 9.948452e-01 | False |
| 2 | explain_simple | 1.190140 | 3.314228e-01 | False |
| 3 | identity | 132.481680 | 1.200935e-16 | True |
| 4 | negation | 2.854926 | 5.501458e-02 | False |
| 5 | noise | 14.193399 | 2.299913e-05 | True |
| 6 | paraphrase | 1.649674 | 2.004514e-01 | False |
| 7 | question_gen | 8.240009 | 5.109847e-04 | True |
| 8 | shuffle | 20.934390 | 2.570053e-07 | True |
| 9 | summarize | 31.819413 | 3.665627e-09 | True |
| 10 | synonym | 0.103508 | 9.573280e-01 | False |

```
Significance Interpretation (Levenshtein Similarity):
Augmentation 'entity_replace': No significant difference in mean Levenshtein
similarity across LLMs (p=0.06998).
Augmentation 'expand': No significant difference in mean Levenshtein similarity
across LLMs (p=0.9948).
Augmentation 'explain_simple': No significant difference in mean Levenshtein
similarity across LLMs (p=0.3314).
Augmentation 'identity': Significant difference in mean Levenshtein similarity
across LLMs (p=1.201e-16).
Augmentation 'negation': No significant difference in mean Levenshtein
similarity across LLMs (p=0.05501).
Augmentation 'noise': Significant difference in mean Levenshtein similarity
across LLMs (p=2.3e-05).
Augmentation 'paraphrase': No significant difference in mean Levenshtein
similarity across LLMs (p=0.2005).
```

```
Augmentation 'question_gen': Significant difference in mean Levenshtein
similarity across LLMs (p=0.000511).
Augmentation 'shuffle': Significant difference in mean Levenshtein similarity
across LLMs (p=2.57e-07).
Augmentation 'summarize': Significant difference in mean Levenshtein similarity
across LLMs (p=3.666e-09).
Augmentation 'synonym': No significant difference in mean Levenshtein similarity
across LLMs (p=0.9573).
```

[38]:
```python
metric_col = 'jaccard_similarity'
anova_results = []
for aug in sorted(df['augmentation_type'].unique()):
    subset = df[df['augmentation_type'] == aug]
    groups = [subset[subset['model'] == m][metric_col].dropna().values
              for m in subset['model'].unique()]
    if sum([len(g) > 1 for g in groups]) >= 2:
        stat, pval = f_oneway(*groups)
        anova_results.append({'augmentation_type': aug, 'F-statistic': stat,␣
 ↪'p-value': pval})
    else:
        anova_results.append({'augmentation_type': aug, 'F-statistic':␣
 ↪float('nan'), 'p-value': float('nan')})

anova_df = pd.DataFrame(anova_results)
anova_df['Significant'] = anova_df['p-value'] < 0.05
display(anova_df)

print("\nSignificance Interpretation (Jaccard Similarity):")
for idx, row in anova_df.iterrows():
    if pd.notna(row['p-value']):
        if row['Significant']:
            print(f"Augmentation '{row['augmentation_type']}': Significant␣
 ↪difference in mean Jaccard similarity across LLMs (p={row['p-value']:.4g}).")
        else:
            print(f"Augmentation '{row['augmentation_type']}': No significant␣
 ↪difference in mean Jaccard similarity across LLMs (p={row['p-value']:.4g}).")
    else:
        print(f"Augmentation '{row['augmentation_type']}': Insufficient data␣
 ↪for ANOVA.")
```

| | augmentation_type | F-statistic | p-value | Significant |
|---|---|---|---|---|
| 0 | entity_replace | 7.304946 | 9.132850e-04 | True |
| 1 | expand | 2.343734 | 9.446474e-02 | False |
| 2 | explain_simple | 0.829942 | 4.886380e-01 | False |
| 3 | identity | 316.158954 | 1.086267e-21 | True |
| 4 | negation | 3.746211 | 2.218612e-02 | True |
| 5 | noise | 7.561195 | 1.180207e-03 | True |
| 6 | paraphrase | 6.759171 | 1.429236e-03 | True |

```
7       question_gen     5.763377  3.678599e-03         True
8            shuffle    17.924767  1.088418e-06         True
9          summarize     3.590329  2.591959e-02         True
10           synonym     2.171286  1.136909e-01        False


Significance Interpretation (Jaccard Similarity):
Augmentation 'entity_replace': Significant difference in mean Jaccard similarity
across LLMs (p=0.0009133).
Augmentation 'expand': No significant difference in mean Jaccard similarity
across LLMs (p=0.09446).
Augmentation 'explain_simple': No significant difference in mean Jaccard
similarity across LLMs (p=0.4886).
Augmentation 'identity': Significant difference in mean Jaccard similarity
across LLMs (p=1.086e-21).
Augmentation 'negation': Significant difference in mean Jaccard similarity
across LLMs (p=0.02219).
Augmentation 'noise': Significant difference in mean Jaccard similarity across
LLMs (p=0.00118).
Augmentation 'paraphrase': Significant difference in mean Jaccard similarity
across LLMs (p=0.001429).
Augmentation 'question_gen': Significant difference in mean Jaccard similarity
across LLMs (p=0.003679).
Augmentation 'shuffle': Significant difference in mean Jaccard similarity across
LLMs (p=1.088e-06).
Augmentation 'summarize': Significant difference in mean Jaccard similarity
across LLMs (p=0.02592).
Augmentation 'synonym': No significant difference in mean Jaccard similarity
across LLMs (p=0.1137).
```

[39]:
```python
metric_col = 'length_ratio'
anova_results = []
for aug in sorted(df['augmentation_type'].unique()):
    subset = df[df['augmentation_type'] == aug]
    groups = [subset[subset['model'] == m][metric_col].dropna().values
              for m in subset['model'].unique()]
    if sum([len(g) > 1 for g in groups]) >= 2:
        stat, pval = f_oneway(*groups)
        anova_results.append({'augmentation_type': aug, 'F-statistic': stat,␣
 ↪'p-value': pval})
    else:
        anova_results.append({'augmentation_type': aug, 'F-statistic':␣
 ↪float('nan'), 'p-value': float('nan')})

anova_df = pd.DataFrame(anova_results)
anova_df['Significant'] = anova_df['p-value'] < 0.05
display(anova_df)
```

```
print("\nSignificance Interpretation (Length Ratio):")
for idx, row in anova_df.iterrows():
    if pd.notna(row['p-value']):
        if row['Significant']:
            print(f"Augmentation '{row['augmentation_type']}': Significant␣
 ↪difference in mean Length Ratio across LLMs (p={row['p-value']:.4g}).")
        else:
            print(f"Augmentation '{row['augmentation_type']}': No significant␣
 ↪difference in mean Length Ratio across LLMs (p={row['p-value']:.4g}).")
    else:
        print(f"Augmentation '{row['augmentation_type']}': Insufficient data␣
 ↪for ANOVA.")
```

| | augmentation_type | F-statistic | p-value | Significant |
|---|---|---|---|---|
| 0 | entity_replace | 2.566639 | 7.450275e-02 | False |
| 1 | expand | 19.571886 | 4.850757e-07 | True |
| 2 | explain_simple | 3.906114 | 1.894207e-02 | True |
| 3 | identity | 5.717529 | 3.494327e-03 | True |
| 4 | negation | 1.480470 | 2.412523e-01 | False |
| 5 | noise | 4.947161 | 8.949647e-03 | True |
| 6 | paraphrase | 11.201796 | 5.277643e-05 | True |
| 7 | question_gen | 30.531595 | 1.125467e-08 | True |
| 8 | shuffle | 1.587574 | 2.145492e-01 | False |
| 9 | summarize | 20.727578 | 2.825033e-07 | True |
| 10 | synonym | 1.586254 | 2.148594e-01 | False |

```
Significance Interpretation (Length Ratio):
Augmentation 'entity_replace': No significant difference in mean Length Ratio
across LLMs (p=0.0745).
Augmentation 'expand': Significant difference in mean Length Ratio across LLMs
(p=4.851e-07).
Augmentation 'explain_simple': Significant difference in mean Length Ratio
across LLMs (p=0.01894).
Augmentation 'identity': Significant difference in mean Length Ratio across LLMs
(p=0.003494).
Augmentation 'negation': No significant difference in mean Length Ratio across
LLMs (p=0.2413).
Augmentation 'noise': Significant difference in mean Length Ratio across LLMs
(p=0.00895).
Augmentation 'paraphrase': Significant difference in mean Length Ratio across
LLMs (p=5.278e-05).
Augmentation 'question_gen': Significant difference in mean Length Ratio across
LLMs (p=1.125e-08).
Augmentation 'shuffle': No significant difference in mean Length Ratio across
LLMs (p=0.2145).
Augmentation 'summarize': Significant difference in mean Length Ratio across
LLMs (p=2.825e-07).
```

Augmentation 'synonym': No significant difference in mean Length Ratio across LLMs (p=0.2149).

[40]:
```
metric_col = 'cosine_similarity'
anova_results = []
for aug in sorted(df['augmentation_type'].unique()):
    subset = df[df['augmentation_type'] == aug]
    groups = [subset[subset['model'] == m][metric_col].dropna().values
              for m in subset['model'].unique()]
    if sum([len(g) > 1 for g in groups]) >= 2:
        stat, pval = f_oneway(*groups)
        anova_results.append({'augmentation_type': aug, 'F-statistic': stat,␣
↪'p-value': pval})
    else:
        anova_results.append({'augmentation_type': aug, 'F-statistic':␣
↪float('nan'), 'p-value': float('nan')})

anova_df = pd.DataFrame(anova_results)
anova_df['Significant'] = anova_df['p-value'] < 0.05
display(anova_df)

print("\nSignificance Interpretation (Cosine Similarity):")
for idx, row in anova_df.iterrows():
    if pd.notna(row['p-value']):
        if row['Significant']:
            print(f"Augmentation '{row['augmentation_type']}': Significant␣
↪difference in mean Cosine Similarity across LLMs (p={row['p-value']:.4g}).")
        else:
            print(f"Augmentation '{row['augmentation_type']}': No significant␣
↪difference in mean Cosine Similarity across LLMs (p={row['p-value']:.4g}).")
    else:
        print(f"Augmentation '{row['augmentation_type']}': Insufficient data␣
↪for ANOVA.")
```

| | augmentation_type | F-statistic | p-value | Significant |
|---|---|---|---|---|
| 0 | entity_replace | 3.811782 | 2.078983e-02 | True |
| 1 | expand | 0.350688 | 7.889392e-01 | False |
| 2 | explain_simple | 0.315831 | 8.137917e-01 | False |
| 3 | identity | 88.400634 | 2.170128e-14 | True |
| 4 | negation | 1.984077 | 1.392127e-01 | False |
| 5 | noise | 6.981303 | 1.795858e-03 | True |
| 6 | paraphrase | 11.692212 | 3.825179e-05 | True |
| 7 | question_gen | 8.245998 | 5.087193e-04 | True |
| 8 | shuffle | 10.959296 | 6.205005e-05 | True |
| 9 | summarize | 4.278674 | 1.318119e-02 | True |
| 10 | synonym | 1.635780 | 2.035221e-01 | False |

```
Significance Interpretation (Cosine Similarity):
Augmentation 'entity_replace': Significant difference in mean Cosine Similarity
across LLMs (p=0.02079).
Augmentation 'expand': No significant difference in mean Cosine Similarity
across LLMs (p=0.7889).
Augmentation 'explain_simple': No significant difference in mean Cosine
Similarity across LLMs (p=0.8138).
Augmentation 'identity': Significant difference in mean Cosine Similarity across
LLMs (p=2.17e-14).
Augmentation 'negation': No significant difference in mean Cosine Similarity
across LLMs (p=0.1392).
Augmentation 'noise': Significant difference in mean Cosine Similarity across
LLMs (p=0.001796).
Augmentation 'paraphrase': Significant difference in mean Cosine Similarity
across LLMs (p=3.825e-05).
Augmentation 'question_gen': Significant difference in mean Cosine Similarity
across LLMs (p=0.0005087).
Augmentation 'shuffle': Significant difference in mean Cosine Similarity across
LLMs (p=6.205e-05).
Augmentation 'summarize': Significant difference in mean Cosine Similarity
across LLMs (p=0.01318).
Augmentation 'synonym': No significant difference in mean Cosine Similarity
across LLMs (p=0.2035).
```

[41]:
```
metric_col = 'wer'
anova_results = []
for aug in sorted(df['augmentation_type'].unique()):
    subset = df[df['augmentation_type'] == aug]
    groups = [subset[subset['model'] == m][metric_col].dropna().values
              for m in subset['model'].unique()]
    if sum([len(g) > 1 for g in groups]) >= 2:
        stat, pval = f_oneway(*groups)
        anova_results.append({'augmentation_type': aug, 'F-statistic': stat,␣
↪'p-value': pval})
    else:
        anova_results.append({'augmentation_type': aug, 'F-statistic':␣
↪float('nan'), 'p-value': float('nan')})

anova_df = pd.DataFrame(anova_results)
anova_df['Significant'] = anova_df['p-value'] < 0.05
display(anova_df)

print("\nSignificance Interpretation (Word Error Rate):")
for idx, row in anova_df.iterrows():
    if pd.notna(row['p-value']):
        if row['Significant']:
```

```
            print(f"Augmentation '{row['augmentation_type']}': Significant␣
↪difference in mean WER across LLMs (p={row['p-value']:.4g}).")
        else:
            print(f"Augmentation '{row['augmentation_type']}': No significant␣
↪difference in mean WER across LLMs (p={row['p-value']:.4g}).")
    else:
        print(f"Augmentation '{row['augmentation_type']}': Insufficient data␣
↪for ANOVA.")
```

| | augmentation_type | F-statistic | p-value | Significant |
|---|---|---|---|---|
| 0 | entity_replace | 4.048903 | 1.646887e-02 | True |
| 1 | expand | 5.770317 | 3.335208e-03 | True |
| 2 | explain_simple | 3.311658 | 3.434822e-02 | True |
| 3 | identity | 141.124096 | 5.255988e-17 | True |
| 4 | negation | 3.193739 | 3.874627e-02 | True |
| 5 | noise | 9.000695 | 4.430880e-04 | True |
| 6 | paraphrase | 4.997235 | 6.697015e-03 | True |
| 7 | question_gen | 9.699730 | 1.803215e-04 | True |
| 8 | shuffle | 10.068562 | 1.142896e-04 | True |
| 9 | summarize | 5.380230 | 4.722528e-03 | True |
| 10 | synonym | 8.927697 | 2.600684e-04 | True |

```
Significance Interpretation (Word Error Rate):
Augmentation 'entity_replace': Significant difference in mean WER across LLMs
(p=0.01647).
Augmentation 'expand': Significant difference in mean WER across LLMs
(p=0.003335).
Augmentation 'explain_simple': Significant difference in mean WER across LLMs
(p=0.03435).
Augmentation 'identity': Significant difference in mean WER across LLMs
(p=5.256e-17).
Augmentation 'negation': Significant difference in mean WER across LLMs
(p=0.03875).
Augmentation 'noise': Significant difference in mean WER across LLMs
(p=0.0004431).
Augmentation 'paraphrase': Significant difference in mean WER across LLMs
(p=0.006697).
Augmentation 'question_gen': Significant difference in mean WER across LLMs
(p=0.0001803).
Augmentation 'shuffle': Significant difference in mean WER across LLMs
(p=0.0001143).
Augmentation 'summarize': Significant difference in mean WER across LLMs
(p=0.004723).
Augmentation 'synonym': Significant difference in mean WER across LLMs
(p=0.0002601).
```

[42]:
```
metric_col = 'char_diversity'
anova_results = []
for aug in sorted(df['augmentation_type'].unique()):
    subset = df[df['augmentation_type'] == aug]
    groups = [subset[subset['model'] == m][metric_col].dropna().values
              for m in subset['model'].unique()]
    if sum([len(g) > 1 for g in groups]) >= 2:
        stat, pval = f_oneway(*groups)
        anova_results.append({'augmentation_type': aug, 'F-statistic': stat,␣
 ↪'p-value': pval})
    else:
        anova_results.append({'augmentation_type': aug, 'F-statistic':␣
 ↪float('nan'), 'p-value': float('nan')})

anova_df = pd.DataFrame(anova_results)
anova_df['Significant'] = anova_df['p-value'] < 0.05
display(anova_df)

print("\nSignificance Interpretation (Character Diversity):")
for idx, row in anova_df.iterrows():
    if pd.notna(row['p-value']):
        if row['Significant']:
            print(f"Augmentation '{row['augmentation_type']}': Significant␣
 ↪difference in mean Character Diversity across LLMs (p={row['p-value']:.4g}).
 ↪")
        else:
            print(f"Augmentation '{row['augmentation_type']}': No significant␣
 ↪difference in mean Character Diversity across LLMs (p={row['p-value']:.4g}).
 ↪")
    else:
        print(f"Augmentation '{row['augmentation_type']}': Insufficient data␣
 ↪for ANOVA.")
```

| | augmentation_type | F-statistic | p-value | Significant |
|---|---|---|---|---|
| 0 | entity_replace | 9.366203 | 1.885479e-04 | True |
| 1 | expand | 15.506217 | 3.904275e-06 | True |
| 2 | explain_simple | 6.145506 | 2.404549e-03 | True |
| 3 | identity | 7.637759 | 6.995811e-04 | True |
| 4 | negation | 13.222636 | 1.464390e-05 | True |
| 5 | noise | 0.882166 | 4.655683e-01 | False |
| 6 | paraphrase | 29.943815 | 6.997905e-09 | True |
| 7 | question_gen | 46.338710 | 1.425922e-10 | True |
| 8 | shuffle | 0.392716 | 7.591815e-01 | False |
| 9 | summarize | 0.401780 | 7.528100e-01 | False |
| 10 | synonym | 11.899835 | 3.345085e-05 | True |

```
Significance Interpretation (Character Diversity):
```

```
Augmentation 'entity_replace': Significant difference in mean Character
Diversity across LLMs (p=0.0001885).
Augmentation 'expand': Significant difference in mean Character Diversity across
LLMs (p=3.904e-06).
Augmentation 'explain_simple': Significant difference in mean Character
Diversity across LLMs (p=0.002405).
Augmentation 'identity': Significant difference in mean Character Diversity
across LLMs (p=0.0006996).
Augmentation 'negation': Significant difference in mean Character Diversity
across LLMs (p=1.464e-05).
Augmentation 'noise': No significant difference in mean Character Diversity
across LLMs (p=0.4656).
Augmentation 'paraphrase': Significant difference in mean Character Diversity
across LLMs (p=6.998e-09).
Augmentation 'question_gen': Significant difference in mean Character Diversity
across LLMs (p=1.426e-10).
Augmentation 'shuffle': No significant difference in mean Character Diversity
across LLMs (p=0.7592).
Augmentation 'summarize': No significant difference in mean Character Diversity
across LLMs (p=0.7528).
Augmentation 'synonym': Significant difference in mean Character Diversity
across LLMs (p=3.345e-05).
```

[43]:
```
metric_col = 'type_token_ratio'
anova_results = []
for aug in sorted(df['augmentation_type'].unique()):
    subset = df[df['augmentation_type'] == aug]
    groups = [subset[subset['model'] == m][metric_col].dropna().values
              for m in subset['model'].unique()]
    if sum([len(g) > 1 for g in groups]) >= 2:
        stat, pval = f_oneway(*groups)
        anova_results.append({'augmentation_type': aug, 'F-statistic': stat,␣
 ↪'p-value': pval})
    else:
        anova_results.append({'augmentation_type': aug, 'F-statistic':␣
 ↪float('nan'), 'p-value': float('nan')})

anova_df = pd.DataFrame(anova_results)
anova_df['Significant'] = anova_df['p-value'] < 0.05
display(anova_df)

print("\nSignificance Interpretation (Type Token Ratio):")
for idx, row in anova_df.iterrows():
    if pd.notna(row['p-value']):
        if row['Significant']:
            print(f"Augmentation '{row['augmentation_type']}': Significant␣
 ↪difference in mean Type Token Ratio across LLMs (p={row['p-value']:.4g}).")
```

```
        else:
            print(f"Augmentation '{row['augmentation_type']}': No significant␣
↪difference in mean Type Token Ratio across LLMs (p={row['p-value']:.4g}).")
    else:
        print(f"Augmentation '{row['augmentation_type']}': Insufficient data␣
↪for ANOVA.")
```

| | augmentation_type | F-statistic | p-value | Significant |
|---|---|---|---|---|
| 0 | entity_replace | 20.346446 | 3.368696e-07 | True |
| 1 | expand | 33.260904 | 2.273976e-09 | True |
| 2 | explain_simple | 14.251822 | 7.948207e-06 | True |
| 3 | identity | 12.132673 | 2.882260e-05 | True |
| 4 | negation | 18.305278 | 8.993081e-07 | True |
| 5 | noise | 4.669504 | 1.134595e-02 | True |
| 6 | paraphrase | 9.965862 | 1.228372e-04 | True |
| 7 | question_gen | 36.764313 | 1.681870e-09 | True |
| 8 | shuffle | 7.630062 | 7.038690e-04 | True |
| 9 | summarize | 1.061596 | 3.811208e-01 | False |
| 10 | synonym | 10.262494 | 9.983533e-05 | True |

```
Significance Interpretation (Type Token Ratio):
Augmentation 'entity_replace': Significant difference in mean Type Token Ratio
across LLMs (p=3.369e-07).
Augmentation 'expand': Significant difference in mean Type Token Ratio across
LLMs (p=2.274e-09).
Augmentation 'explain_simple': Significant difference in mean Type Token Ratio
across LLMs (p=7.948e-06).
Augmentation 'identity': Significant difference in mean Type Token Ratio across
LLMs (p=2.882e-05).
Augmentation 'negation': Significant difference in mean Type Token Ratio across
LLMs (p=8.993e-07).
Augmentation 'noise': Significant difference in mean Type Token Ratio across
LLMs (p=0.01135).
Augmentation 'paraphrase': Significant difference in mean Type Token Ratio
across LLMs (p=0.0001228).
Augmentation 'question_gen': Significant difference in mean Type Token Ratio
across LLMs (p=1.682e-09).
Augmentation 'shuffle': Significant difference in mean Type Token Ratio across
LLMs (p=0.0007039).
Augmentation 'summarize': No significant difference in mean Type Token Ratio
across LLMs (p=0.3811).
Augmentation 'synonym': Significant difference in mean Type Token Ratio across
LLMs (p=9.984e-05).
```

[44]:
```
metric_col = 'bigram_overlap'
anova_results = []
for aug in sorted(df['augmentation_type'].unique()):
```

```
    subset = df[df['augmentation_type'] == aug]
    groups = [subset[subset['model'] == m][metric_col].dropna().values
              for m in subset['model'].unique()]
    if sum([len(g) > 1 for g in groups]) >= 2:
        stat, pval = f_oneway(*groups)
        anova_results.append({'augmentation_type': aug, 'F-statistic': stat,␣
↪'p-value': pval})
    else:
        anova_results.append({'augmentation_type': aug, 'F-statistic':␣
↪float('nan'), 'p-value': float('nan')})

anova_df = pd.DataFrame(anova_results)
anova_df['Significant'] = anova_df['p-value'] < 0.05
display(anova_df)

print("\nSignificance Interpretation (Bigram Overlap):")
for idx, row in anova_df.iterrows():
    if pd.notna(row['p-value']):
        if row['Significant']:
            print(f"Augmentation '{row['augmentation_type']}': Significant␣
↪difference in mean Bigram Overlap across LLMs (p={row['p-value']:.4g}).")
        else:
            print(f"Augmentation '{row['augmentation_type']}': No significant␣
↪difference in mean Bigram Overlap across LLMs (p={row['p-value']:.4g}).")
    else:
        print(f"Augmentation '{row['augmentation_type']}': Insufficient data␣
↪for ANOVA.")
```

| | augmentation_type | F-statistic | p-value | Significant |
|---|---|---|---|---|
| 0 | entity_replace | 8.484103 | 3.627449e-04 | True |
| 1 | expand | 0.573622 | 6.370908e-01 | False |
| 2 | explain_simple | 1.664026 | 1.973288e-01 | False |
| 3 | identity | 645.730192 | 6.116771e-26 | True |
| 4 | negation | 6.273466 | 2.154121e-03 | True |
| 5 | noise | 9.714903 | 2.807608e-04 | True |
| 6 | paraphrase | 2.946594 | 5.000476e-02 | False |
| 7 | question_gen | 2.092720 | 1.256118e-01 | False |
| 8 | shuffle | 23.042056 | 1.014598e-07 | True |
| 9 | summarize | 2.147089 | 1.166971e-01 | False |
| 10 | synonym | 2.112927 | 1.210820e-01 | False |

```
Significance Interpretation (Bigram Overlap):
Augmentation 'entity_replace': Significant difference in mean Bigram Overlap
across LLMs (p=0.0003627).
Augmentation 'expand': No significant difference in mean Bigram Overlap across
LLMs (p=0.6371).
Augmentation 'explain_simple': No significant difference in mean Bigram Overlap
```

```
across LLMs (p=0.1973).
Augmentation 'identity': Significant difference in mean Bigram Overlap across
LLMs (p=6.117e-26).
Augmentation 'negation': Significant difference in mean Bigram Overlap across
LLMs (p=0.002154).
Augmentation 'noise': Significant difference in mean Bigram Overlap across LLMs
(p=0.0002808).
Augmentation 'paraphrase': No significant difference in mean Bigram Overlap
across LLMs (p=0.05).
Augmentation 'question_gen': No significant difference in mean Bigram Overlap
across LLMs (p=0.1256).
Augmentation 'shuffle': Significant difference in mean Bigram Overlap across
LLMs (p=1.015e-07).
Augmentation 'summarize': No significant difference in mean Bigram Overlap
across LLMs (p=0.1167).
Augmentation 'synonym': No significant difference in mean Bigram Overlap across
LLMs (p=0.1211).
```

[45]:
```
metric_col = 'total_duration_s'
anova_results = []
for aug in sorted(df['augmentation_type'].unique()):
    subset = df[df['augmentation_type'] == aug]
    groups = [subset[subset['model'] == m][metric_col].dropna().values
              for m in subset['model'].unique()]
    if sum([len(g) > 1 for g in groups]) >= 2:
        stat, pval = f_oneway(*groups)
        anova_results.append({'augmentation_type': aug, 'F-statistic': stat,␣
 ↪'p-value': pval})
    else:
        anova_results.append({'augmentation_type': aug, 'F-statistic':␣
 ↪float('nan'), 'p-value': float('nan')})

anova_df = pd.DataFrame(anova_results)
anova_df['Significant'] = anova_df['p-value'] < 0.05
display(anova_df)

print("\nSignificance Interpretation (Total Duration (s)):")
for idx, row in anova_df.iterrows():
    if pd.notna(row['p-value']):
        if row['Significant']:
            print(f"Augmentation '{row['augmentation_type']}': Significant␣
 ↪difference in mean total duration (s) across LLMs (p={row['p-value']:.4g}).")
        else:
            print(f"Augmentation '{row['augmentation_type']}': No significant␣
 ↪difference in mean total duration (s) across LLMs (p={row['p-value']:.4g}).")
    else:
```

```
        print(f"Augmentation '{row['augmentation_type']}': Insufficient data␣
↪for ANOVA.")
```

```
   augmentation_type  F-statistic       p-value  Significant
0     entity_replace    88.881725  2.026440e-14         True
1             expand  2069.061530  5.865327e-33         True
2     explain_simple  1272.918072  5.061652e-30         True
3           identity    19.903110  4.145660e-07         True
4           negation   136.157393  8.400300e-17         True
5              noise    44.356838  1.690792e-09         True
6         paraphrase     2.900621  5.245446e-02        False
7       question_gen   349.253167  4.069736e-21         True
8            shuffle    14.522880  6.795688e-06         True
9          summarize     2.706015  6.430778e-02        False
10           synonym    70.628896  3.564170e-13         True


Significance Interpretation (Total Duration (s)):
Augmentation 'entity_replace': Significant difference in mean total duration (s)
across LLMs (p=2.026e-14).
Augmentation 'expand': Significant difference in mean total duration (s) across
LLMs (p=5.865e-33).
Augmentation 'explain_simple': Significant difference in mean total duration (s)
across LLMs (p=5.062e-30).
Augmentation 'identity': Significant difference in mean total duration (s)
across LLMs (p=4.146e-07).
Augmentation 'negation': Significant difference in mean total duration (s)
across LLMs (p=8.4e-17).
Augmentation 'noise': Significant difference in mean total duration (s) across
LLMs (p=1.691e-09).
Augmentation 'paraphrase': No significant difference in mean total duration (s)
across LLMs (p=0.05245).
Augmentation 'question_gen': Significant difference in mean total duration (s)
across LLMs (p=4.07e-21).
Augmentation 'shuffle': Significant difference in mean total duration (s) across
LLMs (p=6.796e-06).
Augmentation 'summarize': No significant difference in mean total duration (s)
across LLMs (p=0.06431).
Augmentation 'synonym': Significant difference in mean total duration (s) across
LLMs (p=3.564e-13).
```

[46]:
```
metric_col = 'prompt_eval_count'
anova_results = []
for aug in sorted(df['augmentation_type'].unique()):
    subset = df[df['augmentation_type'] == aug]
    groups = [subset[subset['model'] == m][metric_col].dropna().values
              for m in subset['model'].unique()]
    if sum([len(g) > 1 for g in groups]) >= 2:
```

```
        stat, pval = f_oneway(*groups)
        anova_results.append({'augmentation_type': aug, 'F-statistic': stat,␣
↪'p-value': pval})
    else:
        anova_results.append({'augmentation_type': aug, 'F-statistic':␣
↪float('nan'), 'p-value': float('nan')})

anova_df = pd.DataFrame(anova_results)
anova_df['Significant'] = anova_df['p-value'] < 0.05
display(anova_df)

print("\nSignificance Interpretation (Prompt Eval Count):")
for idx, row in anova_df.iterrows():
    if pd.notna(row['p-value']):
        if row['Significant']:
            print(f"Augmentation '{row['augmentation_type']}': Significant␣
↪difference in mean Prompt Eval Count across LLMs (p={row['p-value']:.4g}).")
        else:
            print(f"Augmentation '{row['augmentation_type']}': No significant␣
↪difference in mean Prompt Eval Count across LLMs (p={row['p-value']:.4g}).")
    else:
        print(f"Augmentation '{row['augmentation_type']}': Insufficient data␣
↪for ANOVA.")
```

| | augmentation_type | F-statistic | p-value | Significant |
|---|---|---|---|---|
| 0 | entity_replace | 184.551363 | 1.519865e-18 | True |
| 1 | expand | 184.551363 | 1.519865e-18 | True |
| 2 | explain_simple | 187.784906 | 1.206164e-18 | True |
| 3 | identity | 183.820126 | 1.602289e-18 | True |
| 4 | negation | 181.532495 | 1.892597e-18 | True |
| 5 | noise | 109.941443 | 2.146687e-13 | True |
| 6 | paraphrase | 183.820126 | 1.602289e-18 | True |
| 7 | question_gen | 180.483303 | 1.606188e-17 | True |
| 8 | shuffle | 194.251992 | 7.681570e-19 | True |
| 9 | summarize | 187.677568 | 1.215382e-18 | True |
| 10 | synonym | 191.588679 | 9.233930e-19 | True |

```
Significance Interpretation (Prompt Eval Count):
Augmentation 'entity_replace': Significant difference in mean Prompt Eval Count
across LLMs (p=1.52e-18).
Augmentation 'expand': Significant difference in mean Prompt Eval Count across
LLMs (p=1.52e-18).
Augmentation 'explain_simple': Significant difference in mean Prompt Eval Count
across LLMs (p=1.206e-18).
Augmentation 'identity': Significant difference in mean Prompt Eval Count across
LLMs (p=1.602e-18).
Augmentation 'negation': Significant difference in mean Prompt Eval Count across
```

```
LLMs (p=1.893e-18).
Augmentation 'noise': Significant difference in mean Prompt Eval Count across
LLMs (p=2.147e-13).
Augmentation 'paraphrase': Significant difference in mean Prompt Eval Count
across LLMs (p=1.602e-18).
Augmentation 'question_gen': Significant difference in mean Prompt Eval Count
across LLMs (p=1.606e-17).
Augmentation 'shuffle': Significant difference in mean Prompt Eval Count across
LLMs (p=7.682e-19).
Augmentation 'summarize': Significant difference in mean Prompt Eval Count
across LLMs (p=1.215e-18).
Augmentation 'synonym': Significant difference in mean Prompt Eval Count across
LLMs (p=9.234e-19).
```

[47]:
```
metric_col = 'prompt_eval_duration_ns'
anova_results = []
for aug in sorted(df['augmentation_type'].unique()):
    subset = df[df['augmentation_type'] == aug]
    groups = [subset[subset['model'] == m][metric_col].dropna().values
              for m in subset['model'].unique()]
    if sum([len(g) > 1 for g in groups]) >= 2:
        stat, pval = f_oneway(*groups)
        anova_results.append({'augmentation_type': aug, 'F-statistic': stat,␣
↪'p-value': pval})
    else:
        anova_results.append({'augmentation_type': aug, 'F-statistic':␣
↪float('nan'), 'p-value': float('nan')})

anova_df = pd.DataFrame(anova_results)
anova_df['Significant'] = anova_df['p-value'] < 0.05
display(anova_df)

print("\nSignificance Interpretation (Prompt Eval Duration [ns]):")
for idx, row in anova_df.iterrows():
    if pd.notna(row['p-value']):
        if row['Significant']:
            print(f"Augmentation '{row['augmentation_type']}': Significant␣
↪difference in mean Prompt Eval Duration (ns) across LLMs (p={row['p-value']:.
↪4g}).")
        else:
            print(f"Augmentation '{row['augmentation_type']}': No significant␣
↪difference in mean Prompt Eval Duration (ns) across LLMs (p={row['p-value']:.
↪4g}).")
    else:
        print(f"Augmentation '{row['augmentation_type']}': Insufficient data␣
↪for ANOVA.")
```

```
  augmentation_type  F-statistic       p-value  Significant
```

```
0     entity_replace    58.697083  3.386839e-12         True
1             expand    93.546078  1.060680e-14         True
2     explain_simple    58.958322  3.210801e-12         True
3           identity   117.871546  5.481998e-16         True
4           negation    90.102476  1.705638e-14         True
5              noise    92.650919  1.234457e-12         True
6         paraphrase    30.775348  5.234439e-09         True
7       question_gen    99.581542  2.236007e-14         True
8            shuffle    59.311762  2.988070e-12         True
9          summarize    81.265498  6.247786e-14         True
10           synonym    64.419395  1.099528e-12         True


Significance Interpretation (Prompt Eval Duration [ns]):
Augmentation 'entity_replace': Significant difference in mean Prompt Eval
Duration (ns) across LLMs (p=3.387e-12).
Augmentation 'expand': Significant difference in mean Prompt Eval Duration (ns)
across LLMs (p=1.061e-14).
Augmentation 'explain_simple': Significant difference in mean Prompt Eval
Duration (ns) across LLMs (p=3.211e-12).
Augmentation 'identity': Significant difference in mean Prompt Eval Duration
(ns) across LLMs (p=5.482e-16).
Augmentation 'negation': Significant difference in mean Prompt Eval Duration
(ns) across LLMs (p=1.706e-14).
Augmentation 'noise': Significant difference in mean Prompt Eval Duration (ns)
across LLMs (p=1.234e-12).
Augmentation 'paraphrase': Significant difference in mean Prompt Eval Duration
(ns) across LLMs (p=5.234e-09).
Augmentation 'question_gen': Significant difference in mean Prompt Eval Duration
(ns) across LLMs (p=2.236e-14).
Augmentation 'shuffle': Significant difference in mean Prompt Eval Duration (ns)
across LLMs (p=2.988e-12).
Augmentation 'summarize': Significant difference in mean Prompt Eval Duration
(ns) across LLMs (p=6.248e-14).
Augmentation 'synonym': Significant difference in mean Prompt Eval Duration (ns)
across LLMs (p=1.1e-12).
```

[48]:
```
metric_col = 'eval_count'
anova_results = []
for aug in sorted(df['augmentation_type'].unique()):
    subset = df[df['augmentation_type'] == aug]
    groups = [subset[subset['model'] == m][metric_col].dropna().values
              for m in subset['model'].unique()]
    if sum([len(g) > 1 for g in groups]) >= 2:
        stat, pval = f_oneway(*groups)
        anova_results.append({'augmentation_type': aug, 'F-statistic': stat,␣
 ↪'p-value': pval})
    else:
```

```
        anova_results.append({'augmentation_type': aug, 'F-statistic':␣
 ↪float('nan'), 'p-value': float('nan')})

anova_df = pd.DataFrame(anova_results)
anova_df['Significant'] = anova_df['p-value'] < 0.05
display(anova_df)

print("\nSignificance Interpretation (Eval Count):")
for idx, row in anova_df.iterrows():
    if pd.notna(row['p-value']):
        if row['Significant']:
            print(f"Augmentation '{row['augmentation_type']}': Significant␣
 ↪difference in mean Eval Count across LLMs (p={row['p-value']:.4g}).")
        else:
            print(f"Augmentation '{row['augmentation_type']}': No significant␣
 ↪difference in mean Eval Count across LLMs (p={row['p-value']:.4g}).")
    else:
        print(f"Augmentation '{row['augmentation_type']}': Insufficient data␣
 ↪for ANOVA.")
```

```
C:\Users\parth\anaconda3\Lib\site-packages\scipy\stats\_stats_py.py:4167:
ConstantInputWarning: Each of the input arrays is constant;the F statistic is
not defined or infinite
  warnings.warn(stats.ConstantInputWarning(msg))
```

| | augmentation_type | F-statistic | p-value | Significant |
|---|---|---|---|---|
| 0 | entity_replace | 1.449674 | 2.495296e-01 | False |
| 1 | expand | NaN | NaN | False |
| 2 | explain_simple | NaN | NaN | False |
| 3 | identity | 3.540955 | 2.723647e-02 | True |
| 4 | negation | 6.485087 | 1.799039e-03 | True |
| 5 | noise | 4.094457 | 1.883038e-02 | True |
| 6 | paraphrase | 21.558619 | 1.939062e-07 | True |
| 7 | question_gen | 88.272611 | 9.341158e-14 | True |
| 8 | shuffle | 1.874725 | 1.567859e-01 | False |
| 9 | summarize | 27.612183 | 1.633497e-08 | True |
| 10 | synonym | 7.764514 | 6.328400e-04 | True |

```
Significance Interpretation (Eval Count):
Augmentation 'entity_replace': No significant difference in mean Eval Count
across LLMs (p=0.2495).
Augmentation 'expand': Insufficient data for ANOVA.
Augmentation 'explain_simple': Insufficient data for ANOVA.
Augmentation 'identity': Significant difference in mean Eval Count across LLMs
(p=0.02724).
Augmentation 'negation': Significant difference in mean Eval Count across LLMs
(p=0.001799).
```

```
Augmentation 'noise': Significant difference in mean Eval Count across LLMs
(p=0.01883).
Augmentation 'paraphrase': Significant difference in mean Eval Count across LLMs
(p=1.939e-07).
Augmentation 'question_gen': Significant difference in mean Eval Count across
LLMs (p=9.341e-14).
Augmentation 'shuffle': No significant difference in mean Eval Count across LLMs
(p=0.1568).
Augmentation 'summarize': Significant difference in mean Eval Count across LLMs
(p=1.633e-08).
Augmentation 'synonym': Significant difference in mean Eval Count across LLMs
(p=0.0006328).
```

[49]:
```
metric_col = 'eval_duration_ns'
anova_results = []
for aug in sorted(df['augmentation_type'].unique()):
    subset = df[df['augmentation_type'] == aug]
    groups = [subset[subset['model'] == m][metric_col].dropna().values
              for m in subset['model'].unique()]
    if sum([len(g) > 1 for g in groups]) >= 2:
        stat, pval = f_oneway(*groups)
        anova_results.append({'augmentation_type': aug, 'F-statistic': stat,␣
 ↪'p-value': pval})
    else:
        anova_results.append({'augmentation_type': aug, 'F-statistic':␣
 ↪float('nan'), 'p-value': float('nan')})

anova_df = pd.DataFrame(anova_results)
anova_df['Significant'] = anova_df['p-value'] < 0.05
display(anova_df)

print("\nSignificance Interpretation (Eval Duration [ns]):")
for idx, row in anova_df.iterrows():
    if pd.notna(row['p-value']):
        if row['Significant']:
            print(f"Augmentation '{row['augmentation_type']}': Significant␣
 ↪difference in mean Eval Duration (ns) across LLMs (p={row['p-value']:.4g}).")
        else:
            print(f"Augmentation '{row['augmentation_type']}': No significant␣
 ↪difference in mean Eval Duration (ns) across LLMs (p={row['p-value']:.4g}).")
    else:
        print(f"Augmentation '{row['augmentation_type']}': Insufficient data␣
 ↪for ANOVA.")
```

| | augmentation_type | F-statistic | p-value | Significant |
|---|---|---|---|---|
| 0 | entity_replace | 72.447491 | 2.605049e-13 | True |
| 1 | expand | 239890.205769 | 7.888191e-62 | True |
| 2 | explain_simple | 73165.378112 | 1.307125e-54 | True |

```
3           identity      11.974073  3.189430e-05         True
4           negation      99.381020  4.911820e-15         True
5              noise      31.464285  3.869458e-08         True
6         paraphrase      72.540123  2.564261e-13         True
7       question_gen     254.463979  2.211799e-19         True
8            shuffle       9.738025  1.443391e-04         True
9          summarize       2.031106  1.322907e-01        False
10           synonym      52.952672  1.156670e-11         True


Significance Interpretation (Eval Duration [ns]):
Augmentation 'entity_replace': Significant difference in mean Eval Duration (ns)
across LLMs (p=2.605e-13).
Augmentation 'expand': Significant difference in mean Eval Duration (ns) across
LLMs (p=7.888e-62).
Augmentation 'explain_simple': Significant difference in mean Eval Duration (ns)
across LLMs (p=1.307e-54).
Augmentation 'identity': Significant difference in mean Eval Duration (ns)
across LLMs (p=3.189e-05).
Augmentation 'negation': Significant difference in mean Eval Duration (ns)
across LLMs (p=4.912e-15).
Augmentation 'noise': Significant difference in mean Eval Duration (ns) across
LLMs (p=3.869e-08).
Augmentation 'paraphrase': Significant difference in mean Eval Duration (ns)
across LLMs (p=2.564e-13).
Augmentation 'question_gen': Significant difference in mean Eval Duration (ns)
across LLMs (p=2.212e-19).
Augmentation 'shuffle': Significant difference in mean Eval Duration (ns) across
LLMs (p=0.0001443).
Augmentation 'summarize': No significant difference in mean Eval Duration (ns)
across LLMs (p=0.1323).
Augmentation 'synonym': Significant difference in mean Eval Duration (ns) across
LLMs (p=1.157e-11).
```

[50]:
```
metric_col = 'tokens_per_second'
anova_results = []
for aug in sorted(df['augmentation_type'].unique()):
    subset = df[df['augmentation_type'] == aug]
    groups = [subset[subset['model'] == m][metric_col].dropna().values
              for m in subset['model'].unique()]
    if sum([len(g) > 1 for g in groups]) >= 2:
        stat, pval = f_oneway(*groups)
        anova_results.append({'augmentation_type': aug, 'F-statistic': stat,␣
↪'p-value': pval})
    else:
        anova_results.append({'augmentation_type': aug, 'F-statistic':␣
↪float('nan'), 'p-value': float('nan')})
```

```
anova_df = pd.DataFrame(anova_results)
anova_df['Significant'] = anova_df['p-value'] < 0.05
display(anova_df)

print("\nSignificance Interpretation (Tokens per Second):")
for idx, row in anova_df.iterrows():
    if pd.notna(row['p-value']):
        if row['Significant']:
            print(f"Augmentation '{row['augmentation_type']}': Significant␣
↪difference in mean Tokens per Second across LLMs (p={row['p-value']:.4g}).")
        else:
            print(f"Augmentation '{row['augmentation_type']}': No significant␣
↪difference in mean Tokens per Second across LLMs (p={row['p-value']:.4g}).")
    else:
        print(f"Augmentation '{row['augmentation_type']}': Insufficient data␣
↪for ANOVA.")
```

| | augmentation_type | F-statistic | p-value | Significant |
|---|---|---|---|---|
| 0 | entity_replace | 30649.552539 | 2.544171e-49 | True |
| 1 | expand | 198606.280785 | 1.109686e-60 | True |
| 2 | explain_simple | 84203.953624 | 1.828294e-55 | True |
| 3 | identity | 24097.032617 | 7.370114e-48 | True |
| 4 | negation | 12476.274974 | 7.370716e-44 | True |
| 5 | noise | 21347.177609 | 3.029594e-38 | True |
| 6 | paraphrase | 37493.323921 | 1.515228e-50 | True |
| 7 | question_gen | 7603.696943 | 2.258063e-38 | True |
| 8 | shuffle | 31266.724459 | 1.924724e-49 | True |
| 9 | summarize | 23577.888763 | 9.996357e-48 | True |
| 10 | synonym | 19457.175842 | 1.469829e-46 | True |

```
Significance Interpretation (Tokens per Second):
Augmentation 'entity_replace': Significant difference in mean Tokens per Second
across LLMs (p=2.544e-49).
Augmentation 'expand': Significant difference in mean Tokens per Second across
LLMs (p=1.11e-60).
Augmentation 'explain_simple': Significant difference in mean Tokens per Second
across LLMs (p=1.828e-55).
Augmentation 'identity': Significant difference in mean Tokens per Second across
LLMs (p=7.37e-48).
Augmentation 'negation': Significant difference in mean Tokens per Second across
LLMs (p=7.371e-44).
Augmentation 'noise': Significant difference in mean Tokens per Second across
LLMs (p=3.03e-38).
Augmentation 'paraphrase': Significant difference in mean Tokens per Second
across LLMs (p=1.515e-50).
Augmentation 'question_gen': Significant difference in mean Tokens per Second
across LLMs (p=2.258e-38).
```

```
Augmentation 'shuffle': Significant difference in mean Tokens per Second across
LLMs (p=1.925e-49).
Augmentation 'summarize': Significant difference in mean Tokens per Second
across LLMs (p=9.996e-48).
Augmentation 'synonym': Significant difference in mean Tokens per Second across
LLMs (p=1.47e-46).
```

[51]:
```python
# This code runs the Kruskal-Wallis test for each metric, grouped by␣
 ↪augmentation_type, and does pairwise Mann-Whitney U tests if significant.
# It prints results for each metric and group.

# Interpretation:
# If p < 0.05 in Kruskal-Wallis, at least one model is different for that␣
 ↪metric/augmentation.
# Mann-Whitney U will tell you which LLM pairs differ significantly.

from scipy.stats import kruskal, mannwhitneyu
from itertools import combinations
import numpy as np

metrics = [
    'levenshtein_similarity', 'jaccard_similarity', 'length_ratio', 'bleu',
    'cosine_similarity', 'wer', 'char_diversity', 'type_token_ratio',␣
 ↪'bigram_overlap',
    'total_duration_ns', 'load_duration_ns', 'prompt_eval_count',␣
 ↪'prompt_eval_duration_ns',
    'eval_count', 'eval_duration_ns', 'tokens_per_second'
]

group_var = 'augmentation_type'  # Or 'prompt_id'

for metric_col in metrics:
    print(f"\n\n==== Kruskal-Wallis and Mann-Whitney U for {metric_col} (by␣
 ↪{group_var}) ====")
    for group_val in sorted(df[group_var].unique()):
        sub = df[df[group_var] == group_val]
        models = sub['model'].unique()
        data = [sub[sub['model'] == m][metric_col].dropna().values for m in␣
 ↪models]
        if len(models) > 1:
            # Flatten all values for all groups, ignore nan
            all_values = np.concatenate([d for d in data if len(d) > 0])
            if len(all_values) == 0:
                print(f"  {group_var.capitalize()} '{group_val}': No data␣
 ↪available.")
                continue
            # Check if all values are identical
```

```
            if np.all(all_values == all_values[0]):
                print(f"  {group_var.capitalize()} '{group_val}': All values␣
↪identical; cannot perform Kruskal-Wallis test.")
                continue
            stat, p = kruskal(*data)
            print(f"{group_var.capitalize()} '{group_val}': Kruskal-Wallis␣
↪H={stat:.3f}, p={p:.4g}")
            if p < 0.05:
                print("  Significant: Pairwise comparisons:")
                for m1, m2 in combinations(models, 2):
                    d1 = sub[sub['model'] == m1][metric_col].dropna()
                    d2 = sub[sub['model'] == m2][metric_col].dropna()
                    # Mann-Whitney also needs at least one unique value
                    if len(d1) > 0 and len(d2) > 0 and (not np.all(d1 == d1.
↪iloc[0]) or not np.all(d2 == d2.iloc[0])):
                        try:
                            u, p_mw = mannwhitneyu(d1, d2,␣
↪alternative='two-sided')
                            print(f"    {m1} vs {m2}: U={u:.1f}, p={p_mw:.4g}")
                        except ValueError as e:
                            print(f"    {m1} vs {m2}: Mann-Whitney not possible␣
↪({e})")
            else:
                print("  No significant difference among LLMs.")
        else:
            print(f"{group_var.capitalize()} '{group_val}': Only one model␣
↪present.")
```

```
==== Kruskal-Wallis and Mann-Whitney U for levenshtein_similarity (by
augmentation_type) ====
Augmentation_type 'entity_replace': Kruskal-Wallis H=6.463, p=0.09113
  No significant difference among LLMs.
Augmentation_type 'expand': Kruskal-Wallis H=0.509, p=0.917
  No significant difference among LLMs.
Augmentation_type 'explain_simple': Kruskal-Wallis H=1.656, p=0.6467
  No significant difference among LLMs.
Augmentation_type 'identity': Kruskal-Wallis H=21.072, p=0.0001017
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=17.0, p=0.1304
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0004099
    qwen2:0.5b vs granite3.1-moe:1b: U=40.0, p=0.4418
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0004099
    smollm2:360m vs granite3.1-moe:1b: U=54.0, p=0.02067
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0004099
Augmentation_type 'negation': Kruskal-Wallis H=5.327, p=0.1494
```

```
  No significant difference among LLMs.
Augmentation_type 'noise': Kruskal-Wallis H=14.105, p=0.002766
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=19.0, p=0.1949
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=6.0, p=0.7111
    smollm2:360m vs gemma3:1b: U=8.0, p=0.01041
    smollm2:360m vs granite3.1-moe:1b: U=9.0, p=0.8889
    gemma3:1b vs granite3.1-moe:1b: U=16.0, p=0.04444
Augmentation_type 'paraphrase': Kruskal-Wallis H=3.991, p=0.2624
  No significant difference among LLMs.
Augmentation_type 'question_gen': Kruskal-Wallis H=13.592, p=0.003516
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=37.0, p=0.6454
    qwen2:0.5b vs gemma3:1b: U=61.0, p=0.001088
    qwen2:0.5b vs granite3.1-moe:1b: U=36.0, p=0.1419
    smollm2:360m vs gemma3:1b: U=60.0, p=0.001865
    smollm2:360m vs granite3.1-moe:1b: U=32.0, p=0.345
    gemma3:1b vs granite3.1-moe:1b: U=8.0, p=0.04262
Augmentation_type 'shuffle': Kruskal-Wallis H=20.262, p=0.0001498
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=5.0, p=0.002953
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.000931
    qwen2:0.5b vs granite3.1-moe:1b: U=12.0, p=0.03792
    smollm2:360m vs gemma3:1b: U=1.0, p=0.001348
    smollm2:360m vs granite3.1-moe:1b: U=36.0, p=0.7209
    gemma3:1b vs granite3.1-moe:1b: U=58.0, p=0.007362
Augmentation_type 'summarize': Kruskal-Wallis H=18.702, p=0.0003151
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=9.0, p=0.01476
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=30.0, p=0.8785
    smollm2:360m vs gemma3:1b: U=18.0, p=0.1605
    smollm2:360m vs granite3.1-moe:1b: U=55.0, p=0.01476
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554
Augmentation_type 'synonym': Kruskal-Wallis H=2.645, p=0.4497
  No significant difference among LLMs.


==== Kruskal-Wallis and Mann-Whitney U for jaccard_similarity (by
augmentation_type) ====
Augmentation_type 'entity_replace': Kruskal-Wallis H=15.628, p=0.001352
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=24.0, p=0.4306
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.000931
    qwen2:0.5b vs granite3.1-moe:1b: U=14.0, p=0.06588
    smollm2:360m vs gemma3:1b: U=3.0, p=0.001088
    smollm2:360m vs granite3.1-moe:1b: U=17.0, p=0.1304
```

```
    gemma3:1b vs granite3.1-moe:1b: U=49.0, p=0.08298
Augmentation_type 'expand': Kruskal-Wallis H=6.976, p=0.07267
  No significant difference among LLMs.
Augmentation_type 'explain_simple': Kruskal-Wallis H=2.332, p=0.5064
  No significant difference among LLMs.
Augmentation_type 'identity': Kruskal-Wallis H=18.281, p=0.0003848
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=25.0, p=0.5054
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0004099
    qwen2:0.5b vs granite3.1-moe:1b: U=23.0, p=0.3823
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0004099
    smollm2:360m vs granite3.1-moe:1b: U=33.0, p=0.9591
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0004099
Augmentation_type 'negation': Kruskal-Wallis H=8.592, p=0.03524
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=41.0, p=0.3823
    qwen2:0.5b vs gemma3:1b: U=14.0, p=0.06496
    qwen2:0.5b vs granite3.1-moe:1b: U=26.5, p=0.5992
    smollm2:360m vs gemma3:1b: U=7.0, p=0.006993
    smollm2:360m vs granite3.1-moe:1b: U=20.0, p=0.2345
    gemma3:1b vs granite3.1-moe:1b: U=50.0, p=0.06496
Augmentation_type 'noise': Kruskal-Wallis H=12.904, p=0.004849
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=22.0, p=0.3282
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.000931
    qwen2:0.5b vs granite3.1-moe:1b: U=4.0, p=0.4
    smollm2:360m vs gemma3:1b: U=11.0, p=0.0312
    smollm2:360m vs granite3.1-moe:1b: U=5.0, p=0.5333
    gemma3:1b vs granite3.1-moe:1b: U=16.0, p=0.04949
Augmentation_type 'paraphrase': Kruskal-Wallis H=17.713, p=0.000504
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=64.0, p=0.000931
    qwen2:0.5b vs gemma3:1b: U=35.0, p=0.7927
    qwen2:0.5b vs granite3.1-moe:1b: U=46.0, p=0.1556
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=4.5, p=0.004427
    gemma3:1b vs granite3.1-moe:1b: U=44.0, p=0.2268
Augmentation_type 'question_gen': Kruskal-Wallis H=13.291, p=0.004047
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=46.0, p=0.1559
    qwen2:0.5b vs gemma3:1b: U=9.0, p=0.01796
    qwen2:0.5b vs granite3.1-moe:1b: U=12.0, p=0.1372
    smollm2:360m vs gemma3:1b: U=3.0, p=0.002742
    smollm2:360m vs granite3.1-moe:1b: U=5.0, p=0.01265
    gemma3:1b vs granite3.1-moe:1b: U=22.0, p=0.8463
Augmentation_type 'shuffle': Kruskal-Wallis H=12.812, p=0.005061
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=26.0, p=0.5737
```

```
    qwen2:0.5b vs gemma3:1b: U=5.0, p=0.005351
    qwen2:0.5b vs granite3.1-moe:1b: U=17.0, p=0.1304
    smollm2:360m vs gemma3:1b: U=6.0, p=0.007362
    smollm2:360m vs granite3.1-moe:1b: U=24.0, p=0.4418
    gemma3:1b vs granite3.1-moe:1b: U=56.0, p=0.01352
Augmentation_type 'summarize': Kruskal-Wallis H=10.275, p=0.01637
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=54.0, p=0.02385
    qwen2:0.5b vs gemma3:1b: U=38.0, p=0.5632
    qwen2:0.5b vs granite3.1-moe:1b: U=32.0, p=1
    smollm2:360m vs gemma3:1b: U=9.5, p=0.02058
    smollm2:360m vs granite3.1-moe:1b: U=5.0, p=0.005284
    gemma3:1b vs granite3.1-moe:1b: U=23.5, p=0.3991
Augmentation_type 'synonym': Kruskal-Wallis H=5.595, p=0.133
  No significant difference among LLMs.


==== Kruskal-Wallis and Mann-Whitney U for length_ratio (by augmentation_type)
====
Augmentation_type 'entity_replace': Kruskal-Wallis H=7.583, p=0.05546
  No significant difference among LLMs.
Augmentation_type 'expand': Kruskal-Wallis H=20.349, p=0.0001437
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=19.0, p=0.1949
    qwen2:0.5b vs gemma3:1b: U=60.0, p=0.001865
    qwen2:0.5b vs granite3.1-moe:1b: U=58.0, p=0.004662
    smollm2:360m vs gemma3:1b: U=64.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=63.0, p=0.0003108
    gemma3:1b vs granite3.1-moe:1b: U=30.0, p=0.8785
Augmentation_type 'explain_simple': Kruskal-Wallis H=8.957, p=0.02986
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=34.0, p=0.8785
    qwen2:0.5b vs gemma3:1b: U=54.0, p=0.02067
    qwen2:0.5b vs granite3.1-moe:1b: U=52.0, p=0.03792
    smollm2:360m vs gemma3:1b: U=51.0, p=0.04988
    smollm2:360m vs granite3.1-moe:1b: U=50.0, p=0.06496
    gemma3:1b vs granite3.1-moe:1b: U=30.0, p=0.8785
Augmentation_type 'identity': Kruskal-Wallis H=6.772, p=0.07954
  No significant difference among LLMs.
Augmentation_type 'negation': Kruskal-Wallis H=5.442, p=0.1421
  No significant difference among LLMs.
Augmentation_type 'noise': Kruskal-Wallis H=9.853, p=0.01986
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=51.5, p=0.04584
    qwen2:0.5b vs gemma3:1b: U=28.0, p=0.7209
    qwen2:0.5b vs granite3.1-moe:1b: U=11.0, p=0.5333
    smollm2:360m vs gemma3:1b: U=5.0, p=0.002953
    smollm2:360m vs granite3.1-moe:1b: U=4.0, p=0.4
```

```
    gemma3:1b vs granite3.1-moe:1b: U=16.0, p=0.04444
Augmentation_type 'paraphrase': Kruskal-Wallis H=17.026, p=0.0006982
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=62.0, p=0.0006216
    qwen2:0.5b vs gemma3:1b: U=60.0, p=0.001865
    qwen2:0.5b vs granite3.1-moe:1b: U=63.0, p=0.0003108
    smollm2:360m vs gemma3:1b: U=16.0, p=0.1049
    smollm2:360m vs granite3.1-moe:1b: U=20.0, p=0.2345
    gemma3:1b vs granite3.1-moe:1b: U=39.0, p=0.5054
Augmentation_type 'question_gen': Kruskal-Wallis H=20.380, p=0.0001416
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=28.0, p=0.7209
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=2.0, p=0.002664
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=3.0, p=0.004662
    gemma3:1b vs granite3.1-moe:1b: U=33.0, p=0.2824
Augmentation_type 'shuffle': Kruskal-Wallis H=4.444, p=0.2173
  No significant difference among LLMs.
Augmentation_type 'summarize': Kruskal-Wallis H=20.957, p=0.0001074
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=60.0, p=0.001865
    qwen2:0.5b vs gemma3:1b: U=64.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=59.0, p=0.002953
    smollm2:360m vs gemma3:1b: U=36.0, p=0.7209
    smollm2:360m vs granite3.1-moe:1b: U=12.0, p=0.03792
    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.0001554
Augmentation_type 'synonym': Kruskal-Wallis H=6.611, p=0.08539
  No significant difference among LLMs.


==== Kruskal-Wallis and Mann-Whitney U for bleu (by augmentation_type) ====
Augmentation_type 'entity_replace': Kruskal-Wallis H=16.699, p=0.000815
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=28.0, p=0.7209
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=17.0, p=0.1304
    smollm2:360m vs gemma3:1b: U=1.0, p=0.0003108
    smollm2:360m vs granite3.1-moe:1b: U=24.0, p=0.4418
    gemma3:1b vs granite3.1-moe:1b: U=58.0, p=0.004662
Augmentation_type 'expand': Kruskal-Wallis H=1.363, p=0.7142
  No significant difference among LLMs.
Augmentation_type 'explain_simple': Kruskal-Wallis H=4.384, p=0.2228
  No significant difference among LLMs.
Augmentation_type 'identity': Kruskal-Wallis H=18.515, p=0.0003443
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=22.0, p=0.3282
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0004099
```

```
    qwen2:0.5b vs granite3.1-moe:1b: U=26.0, p=0.5737
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0004099
    smollm2:360m vs granite3.1-moe:1b: U=39.0, p=0.5054
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0004099
Augmentation_type 'negation': Kruskal-Wallis H=12.764, p=0.005175
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=41.0, p=0.3823
    qwen2:0.5b vs gemma3:1b: U=13.0, p=0.04988
    qwen2:0.5b vs granite3.1-moe:1b: U=34.0, p=0.8785
    smollm2:360m vs gemma3:1b: U=1.0, p=0.0003108
    smollm2:360m vs granite3.1-moe:1b: U=20.0, p=0.2345
    gemma3:1b vs granite3.1-moe:1b: U=59.0, p=0.002953
Augmentation_type 'noise': Kruskal-Wallis H=13.708, p=0.003331
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=23.0, p=0.3823
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.000931
    qwen2:0.5b vs granite3.1-moe:1b: U=4.0, p=0.4
    smollm2:360m vs gemma3:1b: U=8.0, p=0.01352
    smollm2:360m vs granite3.1-moe:1b: U=7.0, p=0.8889
    gemma3:1b vs granite3.1-moe:1b: U=16.0, p=0.04949
Augmentation_type 'paraphrase': Kruskal-Wallis H=11.199, p=0.0107
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=54.0, p=0.02067
    qwen2:0.5b vs gemma3:1b: U=23.0, p=0.3823
    qwen2:0.5b vs granite3.1-moe:1b: U=30.0, p=0.8785
    smollm2:360m vs gemma3:1b: U=5.0, p=0.002953
    smollm2:360m vs granite3.1-moe:1b: U=8.0, p=0.01041
    gemma3:1b vs granite3.1-moe:1b: U=43.0, p=0.2786
Augmentation_type 'question_gen': Kruskal-Wallis H=9.025, p=0.02896
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=44.0, p=0.2345
    qwen2:0.5b vs gemma3:1b: U=18.0, p=0.1605
    qwen2:0.5b vs granite3.1-moe:1b: U=12.0, p=0.1419
    smollm2:360m vs gemma3:1b: U=8.0, p=0.01041
    smollm2:360m vs granite3.1-moe:1b: U=7.0, p=0.0293
    gemma3:1b vs granite3.1-moe:1b: U=21.0, p=0.7546
Augmentation_type 'shuffle': Kruskal-Wallis H=12.465, p=0.005948
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=20.0, p=0.2345
    qwen2:0.5b vs gemma3:1b: U=8.0, p=0.01352
    qwen2:0.5b vs granite3.1-moe:1b: U=13.0, p=0.04988
    smollm2:360m vs gemma3:1b: U=8.0, p=0.01352
    smollm2:360m vs granite3.1-moe:1b: U=21.0, p=0.2786
    gemma3:1b vs granite3.1-moe:1b: U=56.0, p=0.01352
Augmentation_type 'summarize': Kruskal-Wallis H=6.739, p=0.08071
  No significant difference among LLMs.
Augmentation_type 'synonym': Kruskal-Wallis H=5.062, p=0.1673
  No significant difference among LLMs.
```

```
==== Kruskal-Wallis and Mann-Whitney U for cosine_similarity (by
augmentation_type) ====
Augmentation_type 'entity_replace': Kruskal-Wallis H=10.707, p=0.01342
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=43.0, p=0.2786
    qwen2:0.5b vs gemma3:1b: U=15.0, p=0.08298
    qwen2:0.5b vs granite3.1-moe:1b: U=30.0, p=0.8785
    smollm2:360m vs gemma3:1b: U=2.0, p=0.0006216
    smollm2:360m vs granite3.1-moe:1b: U=16.0, p=0.1049
    gemma3:1b vs granite3.1-moe:1b: U=50.0, p=0.06496
Augmentation_type 'expand': Kruskal-Wallis H=0.474, p=0.9245
  No significant difference among LLMs.
Augmentation_type 'explain_simple': Kruskal-Wallis H=1.310, p=0.7268
  No significant difference among LLMs.
Augmentation_type 'identity': Kruskal-Wallis H=18.302, p=0.0003811
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=25.0, p=0.5054
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0004099
    qwen2:0.5b vs granite3.1-moe:1b: U=24.0, p=0.4418
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0004099
    smollm2:360m vs granite3.1-moe:1b: U=27.0, p=0.6454
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0004099
Augmentation_type 'negation': Kruskal-Wallis H=8.043, p=0.04514
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=35.0, p=0.7984
    qwen2:0.5b vs gemma3:1b: U=9.0, p=0.01476
    qwen2:0.5b vs granite3.1-moe:1b: U=30.0, p=0.8785
    smollm2:360m vs gemma3:1b: U=11.0, p=0.02813
    smollm2:360m vs granite3.1-moe:1b: U=25.0, p=0.5054
    gemma3:1b vs granite3.1-moe:1b: U=52.0, p=0.03792
Augmentation_type 'noise': Kruskal-Wallis H=14.062, p=0.002822
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=17.0, p=0.1304
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=4.0, p=0.4
    smollm2:360m vs gemma3:1b: U=10.0, p=0.02067
    smollm2:360m vs granite3.1-moe:1b: U=9.0, p=0.8889
    gemma3:1b vs granite3.1-moe:1b: U=16.0, p=0.04444
Augmentation_type 'paraphrase': Kruskal-Wallis H=18.014, p=0.0004369
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=56.0, p=0.01041
    qwen2:0.5b vs gemma3:1b: U=8.0, p=0.01041
    qwen2:0.5b vs granite3.1-moe:1b: U=43.0, p=0.2786
    smollm2:360m vs gemma3:1b: U=5.0, p=0.002953
    smollm2:360m vs granite3.1-moe:1b: U=9.0, p=0.01476
    gemma3:1b vs granite3.1-moe:1b: U=63.0, p=0.0003108
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Augmentation_type 'question_gen': Kruskal-Wallis H=14.934, p=0.001874
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=45.0, p=0.1949
    qwen2:0.5b vs gemma3:1b: U=7.0, p=0.006993
    qwen2:0.5b vs granite3.1-moe:1b: U=10.0, p=0.08125
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=6.0, p=0.01998
    gemma3:1b vs granite3.1-moe:1b: U=24.0, p=1
Augmentation_type 'shuffle': Kruskal-Wallis H=12.857, p=0.004955
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=31.0, p=0.9591
    qwen2:0.5b vs gemma3:1b: U=6.0, p=0.007362
    qwen2:0.5b vs granite3.1-moe:1b: U=21.0, p=0.2786
    smollm2:360m vs gemma3:1b: U=5.0, p=0.005351
    smollm2:360m vs granite3.1-moe:1b: U=19.0, p=0.1949
    gemma3:1b vs granite3.1-moe:1b: U=57.0, p=0.01003
Augmentation_type 'summarize': Kruskal-Wallis H=10.281, p=0.01632
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=48.0, p=0.1049
    qwen2:0.5b vs gemma3:1b: U=34.0, p=0.8785
    qwen2:0.5b vs granite3.1-moe:1b: U=26.0, p=0.5737
    smollm2:360m vs gemma3:1b: U=6.0, p=0.004662
    smollm2:360m vs granite3.1-moe:1b: U=4.0, p=0.001865
    gemma3:1b vs granite3.1-moe:1b: U=21.0, p=0.2786
Augmentation_type 'synonym': Kruskal-Wallis H=5.770, p=0.1234
  No significant difference among LLMs.


==== Kruskal-Wallis and Mann-Whitney U for wer (by augmentation_type) ====
Augmentation_type 'entity_replace': Kruskal-Wallis H=11.207, p=0.01066
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=25.0, p=0.4942
    qwen2:0.5b vs gemma3:1b: U=35.0, p=0.7923
    qwen2:0.5b vs granite3.1-moe:1b: U=58.0, p=0.004662
    smollm2:360m vs gemma3:1b: U=41.0, p=0.3706
    smollm2:360m vs granite3.1-moe:1b: U=61.0, p=0.001088
    gemma3:1b vs granite3.1-moe:1b: U=50.0, p=0.06588
Augmentation_type 'expand': Kruskal-Wallis H=12.001, p=0.00738
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=24.5, p=0.4619
    qwen2:0.5b vs gemma3:1b: U=40.0, p=0.4306
    qwen2:0.5b vs granite3.1-moe:1b: U=59.5, p=0.004545
    smollm2:360m vs gemma3:1b: U=41.5, p=0.3439
    smollm2:360m vs granite3.1-moe:1b: U=57.0, p=0.006993
    gemma3:1b vs granite3.1-moe:1b: U=56.0, p=0.01352
Augmentation_type 'explain_simple': Kruskal-Wallis H=7.563, p=0.05596
  No significant difference among LLMs.
Augmentation_type 'identity': Kruskal-Wallis H=26.338, p=8.104e-06
```

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  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=63.0, p=0.001348
    qwen2:0.5b vs gemma3:1b: U=64.0, p=0.0004099
    qwen2:0.5b vs granite3.1-moe:1b: U=64.0, p=0.0001554
    smollm2:360m vs gemma3:1b: U=64.0, p=0.0004054
    smollm2:360m vs granite3.1-moe:1b: U=29.5, p=0.8334
    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.0004099
Augmentation_type 'negation': Kruskal-Wallis H=12.225, p=0.006649
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=29.5, p=0.8335
    qwen2:0.5b vs gemma3:1b: U=20.0, p=0.2345
    qwen2:0.5b vs granite3.1-moe:1b: U=50.5, p=0.05852
    smollm2:360m vs gemma3:1b: U=17.0, p=0.1304
    smollm2:360m vs granite3.1-moe:1b: U=52.5, p=0.03556
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554
Augmentation_type 'noise': Kruskal-Wallis H=13.909, p=0.003032
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=53.0, p=0.0312
    qwen2:0.5b vs gemma3:1b: U=62.0, p=0.001933
    qwen2:0.5b vs granite3.1-moe:1b: U=13.0, p=0.2667
    smollm2:360m vs gemma3:1b: U=52.0, p=0.04027
    smollm2:360m vs granite3.1-moe:1b: U=10.0, p=0.6944
    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.04949
Augmentation_type 'paraphrase': Kruskal-Wallis H=12.062, p=0.007174
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=54.5, p=0.02077
    qwen2:0.5b vs gemma3:1b: U=51.0, p=0.04988
    qwen2:0.5b vs granite3.1-moe:1b: U=59.0, p=0.005351
    smollm2:360m vs gemma3:1b: U=20.5, p=0.2476
    smollm2:360m vs granite3.1-moe:1b: U=41.0, p=0.3706
    gemma3:1b vs granite3.1-moe:1b: U=51.0, p=0.05134
Augmentation_type 'question_gen': Kruskal-Wallis H=16.093, p=0.001085
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=17.5, p=0.1353
    qwen2:0.5b vs gemma3:1b: U=3.0, p=0.002742
    qwen2:0.5b vs granite3.1-moe:1b: U=30.0, p=0.4772
    smollm2:360m vs gemma3:1b: U=6.0, p=0.007232
    smollm2:360m vs granite3.1-moe:1b: U=41.0, p=0.03239
    gemma3:1b vs granite3.1-moe:1b: U=45.5, p=0.006646
Augmentation_type 'shuffle': Kruskal-Wallis H=14.024, p=0.002872
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=50.0, p=0.06496
    qwen2:0.5b vs gemma3:1b: U=61.0, p=0.002742
    qwen2:0.5b vs granite3.1-moe:1b: U=56.0, p=0.01041
    smollm2:360m vs gemma3:1b: U=52.0, p=0.04042
    smollm2:360m vs granite3.1-moe:1b: U=38.5, p=0.5283
    gemma3:1b vs granite3.1-moe:1b: U=13.0, p=0.05186
Augmentation_type 'summarize': Kruskal-Wallis H=12.826, p=0.005029
```

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  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=48.5, p=0.09265
    qwen2:0.5b vs gemma3:1b: U=63.0, p=0.001348
    qwen2:0.5b vs granite3.1-moe:1b: U=50.0, p=0.06496
    smollm2:360m vs gemma3:1b: U=59.0, p=0.005351
    smollm2:360m vs granite3.1-moe:1b: U=35.5, p=0.7525
    gemma3:1b vs granite3.1-moe:1b: U=24.0, p=0.4306
Augmentation_type 'synonym': Kruskal-Wallis H=17.847, p=0.000473
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=47.5, p=0.1146
    qwen2:0.5b vs gemma3:1b: U=15.0, p=0.0829
    qwen2:0.5b vs granite3.1-moe:1b: U=59.0, p=0.005351
    smollm2:360m vs gemma3:1b: U=4.5, p=0.004545
    smollm2:360m vs granite3.1-moe:1b: U=49.5, p=0.07399
    gemma3:1b vs granite3.1-moe:1b: U=62.5, p=0.001616


==== Kruskal-Wallis and Mann-Whitney U for char_diversity (by augmentation_type)
====
Augmentation_type 'entity_replace': Kruskal-Wallis H=16.813, p=0.0007721
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=44.0, p=0.2258
    qwen2:0.5b vs gemma3:1b: U=18.0, p=0.1559
    qwen2:0.5b vs granite3.1-moe:1b: U=58.0, p=0.007319
    smollm2:360m vs gemma3:1b: U=3.0, p=0.001088
    smollm2:360m vs granite3.1-moe:1b: U=42.5, p=0.2929
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.000931
Augmentation_type 'expand': Kruskal-Wallis H=17.293, p=0.000615
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=46.0, p=0.1559
    qwen2:0.5b vs gemma3:1b: U=7.0, p=0.01003
    qwen2:0.5b vs granite3.1-moe:1b: U=47.0, p=0.1256
    smollm2:360m vs gemma3:1b: U=0.5, p=0.001122
    smollm2:360m vs granite3.1-moe:1b: U=35.0, p=0.7923
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0009069
Augmentation_type 'explain_simple': Kruskal-Wallis H=13.851, p=0.003116
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=34.5, p=0.8335
    qwen2:0.5b vs gemma3:1b: U=8.0, p=0.01345
    qwen2:0.5b vs granite3.1-moe:1b: U=22.5, p=0.3442
    smollm2:360m vs gemma3:1b: U=2.5, p=0.002271
    smollm2:360m vs granite3.1-moe:1b: U=13.0, p=0.04988
    gemma3:1b vs granite3.1-moe:1b: U=57.0, p=0.009808
Augmentation_type 'identity': Kruskal-Wallis H=18.517, p=0.0003441
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=41.0, p=0.3696
    qwen2:0.5b vs gemma3:1b: U=64.0, p=0.0004054
    qwen2:0.5b vs granite3.1-moe:1b: U=35.0, p=0.7921
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    smollm2:360m vs gemma3:1b: U=64.0, p=0.0004054
    smollm2:360m vs granite3.1-moe:1b: U=23.0, p=0.3696
    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.0004054
Augmentation_type 'negation': Kruskal-Wallis H=14.712, p=0.00208
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=39.0, p=0.4847
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.000891
    qwen2:0.5b vs granite3.1-moe:1b: U=27.0, p=0.6335
    smollm2:360m vs gemma3:1b: U=8.0, p=0.01324
    smollm2:360m vs granite3.1-moe:1b: U=25.0, p=0.4929
    gemma3:1b vs granite3.1-moe:1b: U=63.0, p=0.001348
Augmentation_type 'noise': Kruskal-Wallis H=4.849, p=0.1832
  No significant difference among LLMs.
Augmentation_type 'paraphrase': Kruskal-Wallis H=18.096, p=0.0004202
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=24.0, p=0.4278
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0009229
    qwen2:0.5b vs granite3.1-moe:1b: U=30.0, p=0.8743
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0009148
    smollm2:360m vs granite3.1-moe:1b: U=40.5, p=0.398
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.000891
Augmentation_type 'question_gen': Kruskal-Wallis H=18.394, p=0.0003648
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=18.0, p=0.1553
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.000931
    qwen2:0.5b vs granite3.1-moe:1b: U=15.0, p=0.272
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=23.0, p=0.9484
    gemma3:1b vs granite3.1-moe:1b: U=48.0, p=0.000666
Augmentation_type 'shuffle': Kruskal-Wallis H=1.506, p=0.6808
  No significant difference among LLMs.
Augmentation_type 'summarize': Kruskal-Wallis H=0.333, p=0.9536
  No significant difference among LLMs.
Augmentation_type 'synonym': Kruskal-Wallis H=18.256, p=0.0003895
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=51.0, p=0.05117
    qwen2:0.5b vs gemma3:1b: U=6.0, p=0.007319
    qwen2:0.5b vs granite3.1-moe:1b: U=39.0, p=0.492
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0009229
    smollm2:360m vs granite3.1-moe:1b: U=20.0, p=0.2227
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.000891


==== Kruskal-Wallis and Mann-Whitney U for type_token_ratio (by
augmentation_type) ====
Augmentation_type 'entity_replace': Kruskal-Wallis H=19.190, p=0.0002498
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=18.0, p=0.1556
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    qwen2:0.5b vs gemma3:1b: U=60.0, p=0.001865
    qwen2:0.5b vs granite3.1-moe:1b: U=12.5, p=0.04441
    smollm2:360m vs gemma3:1b: U=64.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=24.5, p=0.4589
    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.0009069
Augmentation_type 'expand': Kruskal-Wallis H=20.286, p=0.0001481
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=32.5, p=1
    qwen2:0.5b vs gemma3:1b: U=61.0, p=0.002624
    qwen2:0.5b vs granite3.1-moe:1b: U=16.0, p=0.09796
    smollm2:360m vs gemma3:1b: U=64.0, p=0.0008832
    smollm2:360m vs granite3.1-moe:1b: U=7.0, p=0.009431
    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.000822
Augmentation_type 'explain_simple': Kruskal-Wallis H=14.494, p=0.002304
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=30.5, p=0.9161
    qwen2:0.5b vs gemma3:1b: U=61.0, p=0.001088
    qwen2:0.5b vs granite3.1-moe:1b: U=30.5, p=0.9161
    smollm2:360m vs gemma3:1b: U=60.0, p=0.003798
    smollm2:360m vs granite3.1-moe:1b: U=27.0, p=0.6348
    gemma3:1b vs granite3.1-moe:1b: U=2.0, p=0.001933
Augmentation_type 'identity': Kruskal-Wallis H=17.688, p=0.0005101
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=26.0, p=0.5624
    qwen2:0.5b vs gemma3:1b: U=64.0, p=0.0009229
    qwen2:0.5b vs granite3.1-moe:1b: U=30.0, p=0.8745
    smollm2:360m vs gemma3:1b: U=64.0, p=0.0009148
    smollm2:360m vs granite3.1-moe:1b: U=33.0, p=0.9578
    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.0009148
Augmentation_type 'negation': Kruskal-Wallis H=18.646, p=0.0003235
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=47.0, p=0.1304
    qwen2:0.5b vs gemma3:1b: U=63.0, p=0.001348
    qwen2:0.5b vs granite3.1-moe:1b: U=24.0, p=0.4295
    smollm2:360m vs gemma3:1b: U=63.0, p=0.001348
    smollm2:360m vs granite3.1-moe:1b: U=19.0, p=0.1857
    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.0008989
Augmentation_type 'noise': Kruskal-Wallis H=11.094, p=0.01123
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=24.0, p=0.4295
    qwen2:0.5b vs gemma3:1b: U=60.0, p=0.00385
    qwen2:0.5b vs granite3.1-moe:1b: U=5.5, p=0.6004
    smollm2:360m vs gemma3:1b: U=56.0, p=0.01345
    smollm2:360m vs granite3.1-moe:1b: U=8.0, p=1
    gemma3:1b vs granite3.1-moe:1b: U=1.0, p=0.08868
Augmentation_type 'paraphrase': Kruskal-Wallis H=13.738, p=0.003285
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=29.5, p=0.8333
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    qwen2:0.5b vs gemma3:1b: U=58.0, p=0.007276
    qwen2:0.5b vs granite3.1-moe:1b: U=21.0, p=0.2694
    smollm2:360m vs gemma3:1b: U=58.0, p=0.007276
    smollm2:360m vs granite3.1-moe:1b: U=25.0, p=0.4916
    gemma3:1b vs granite3.1-moe:1b: U=2.0, p=0.001903
Augmentation_type 'question_gen': Kruskal-Wallis H=20.201, p=0.0001542
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=22.5, p=0.2143
    qwen2:0.5b vs gemma3:1b: U=63.5, p=0.0009342
    qwen2:0.5b vs granite3.1-moe:1b: U=27.0, p=0.7304
    smollm2:360m vs gemma3:1b: U=64.0, p=0.0005479
    smollm2:360m vs granite3.1-moe:1b: U=36.0, p=0.08359
    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.002362
Augmentation_type 'shuffle': Kruskal-Wallis H=18.362, p=0.0003704
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=43.0, p=0.268
    qwen2:0.5b vs gemma3:1b: U=63.0, p=0.001326
    qwen2:0.5b vs granite3.1-moe:1b: U=20.0, p=0.2146
    smollm2:360m vs gemma3:1b: U=60.0, p=0.003824
    smollm2:360m vs granite3.1-moe:1b: U=13.5, p=0.05464
    gemma3:1b vs granite3.1-moe:1b: U=1.0, p=0.001241
Augmentation_type 'summarize': Kruskal-Wallis H=3.078, p=0.3798
  No significant difference among LLMs.
Augmentation_type 'synonym': Kruskal-Wallis H=15.644, p=0.001342
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=22.5, p=0.3314
    qwen2:0.5b vs gemma3:1b: U=59.0, p=0.005183
    qwen2:0.5b vs granite3.1-moe:1b: U=30.5, p=0.9155
    smollm2:360m vs gemma3:1b: U=63.0, p=0.001305
    smollm2:360m vs granite3.1-moe:1b: U=43.5, p=0.2441
    gemma3:1b vs granite3.1-moe:1b: U=3.0, p=0.002722


==== Kruskal-Wallis and Mann-Whitney U for bigram_overlap (by augmentation_type)
====
Augmentation_type 'entity_replace': Kruskal-Wallis H=16.003, p=0.001132
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=28.0, p=0.7112
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.000931
    qwen2:0.5b vs granite3.1-moe:1b: U=18.0, p=0.1529
    smollm2:360m vs gemma3:1b: U=1.0, p=0.001348
    smollm2:360m vs granite3.1-moe:1b: U=23.0, p=0.3681
    gemma3:1b vs granite3.1-moe:1b: U=55.0, p=0.01796
Augmentation_type 'expand': Kruskal-Wallis H=1.631, p=0.6524
  No significant difference among LLMs.
Augmentation_type 'explain_simple': Kruskal-Wallis H=5.506, p=0.1383
  No significant difference among LLMs.
Augmentation_type 'identity': Kruskal-Wallis H=18.329, p=0.0003762
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  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=22.5, p=0.341
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0003918
    qwen2:0.5b vs granite3.1-moe:1b: U=27.0, p=0.6355
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0004099
    smollm2:360m vs granite3.1-moe:1b: U=36.0, p=0.7209
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0004099
Augmentation_type 'negation': Kruskal-Wallis H=12.423, p=0.006065
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=41.5, p=0.341
    qwen2:0.5b vs gemma3:1b: U=12.0, p=0.03792
    qwen2:0.5b vs granite3.1-moe:1b: U=33.5, p=0.9163
    smollm2:360m vs gemma3:1b: U=1.0, p=0.001315
    smollm2:360m vs granite3.1-moe:1b: U=21.5, p=0.29
    gemma3:1b vs granite3.1-moe:1b: U=57.0, p=0.006993
Augmentation_type 'noise': Kruskal-Wallis H=13.667, p=0.003396
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=24.0, p=0.4275
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0009229
    qwen2:0.5b vs granite3.1-moe:1b: U=4.0, p=0.3593
    smollm2:360m vs gemma3:1b: U=8.0, p=0.01345
    smollm2:360m vs granite3.1-moe:1b: U=6.0, p=0.6944
    gemma3:1b vs granite3.1-moe:1b: U=16.0, p=0.04949
Augmentation_type 'paraphrase': Kruskal-Wallis H=10.377, p=0.01562
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=57.0, p=0.008244
    qwen2:0.5b vs gemma3:1b: U=32.0, p=1
    qwen2:0.5b vs granite3.1-moe:1b: U=40.0, p=0.4418
    smollm2:360m vs gemma3:1b: U=10.0, p=0.01833
    smollm2:360m vs granite3.1-moe:1b: U=8.5, p=0.01306
    gemma3:1b vs granite3.1-moe:1b: U=44.0, p=0.2268
Augmentation_type 'question_gen': Kruskal-Wallis H=5.139, p=0.1619
  No significant difference among LLMs.
Augmentation_type 'shuffle': Kruskal-Wallis H=13.706, p=0.003334
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=20.0, p=0.2149
    qwen2:0.5b vs gemma3:1b: U=6.0, p=0.006521
    qwen2:0.5b vs granite3.1-moe:1b: U=13.0, p=0.0486
    smollm2:360m vs gemma3:1b: U=7.0, p=0.009808
    smollm2:360m vs granite3.1-moe:1b: U=22.0, p=0.317
    gemma3:1b vs granite3.1-moe:1b: U=56.5, p=0.0116
Augmentation_type 'summarize': Kruskal-Wallis H=3.779, p=0.2863
  No significant difference among LLMs.
Augmentation_type 'synonym': Kruskal-Wallis H=4.229, p=0.2377
  No significant difference among LLMs.


==== Kruskal-Wallis and Mann-Whitney U for total_duration_ns (by
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augmentation_type) ====
Augmentation_type 'entity_replace': Kruskal-Wallis H=25.997, p=9.55e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=5.0, p=0.002953
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=1.0, p=0.0003108
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=56.0, p=0.01041
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554
Augmentation_type 'expand': Kruskal-Wallis H=29.091, p=2.143e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=0.0, p=0.0001554
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=0.0, p=0.0001554
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=64.0, p=0.0001554
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554
Augmentation_type 'explain_simple': Kruskal-Wallis H=29.091, p=2.143e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=0.0, p=0.0001554
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=0.0, p=0.0001554
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=64.0, p=0.0001554
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554
Augmentation_type 'identity': Kruskal-Wallis H=21.702, p=7.525e-05
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=6.0, p=0.004662
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=7.0, p=0.006993
    smollm2:360m vs gemma3:1b: U=7.0, p=0.006993
    smollm2:360m vs granite3.1-moe:1b: U=43.0, p=0.2786
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554
Augmentation_type 'negation': Kruskal-Wallis H=27.003, p=5.879e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=0.0, p=0.0001554
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=0.0, p=0.0001554
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=49.0, p=0.08298
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554
Augmentation_type 'noise': Kruskal-Wallis H=18.876, p=0.00029
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=13.0, p=0.04988
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=0.0, p=0.04444
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=8.0, p=1
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    gemma3:1b vs granite3.1-moe:1b: U=16.0, p=0.04444
Augmentation_type 'paraphrase': Kruskal-Wallis H=16.690, p=0.0008183
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=18.0, p=0.1605
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=7.0, p=0.006993
    smollm2:360m vs gemma3:1b: U=7.0, p=0.006993
    smollm2:360m vs granite3.1-moe:1b: U=31.0, p=0.9591
    gemma3:1b vs granite3.1-moe:1b: U=56.0, p=0.01041
Augmentation_type 'question_gen': Kruskal-Wallis H=26.972, p=5.969e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=0.0, p=0.0001554
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=0.0, p=0.000666
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=1.0, p=0.001332
    gemma3:1b vs granite3.1-moe:1b: U=48.0, p=0.000666
Augmentation_type 'shuffle': Kruskal-Wallis H=19.486, p=0.0002169
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=6.0, p=0.004662
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=0.0, p=0.0001554
    smollm2:360m vs gemma3:1b: U=15.0, p=0.08298
    smollm2:360m vs granite3.1-moe:1b: U=40.0, p=0.4418
    gemma3:1b vs granite3.1-moe:1b: U=56.0, p=0.01041
Augmentation_type 'summarize': Kruskal-Wallis H=13.213, p=0.004198
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=42.0, p=0.3282
    qwen2:0.5b vs gemma3:1b: U=2.0, p=0.0006216
    qwen2:0.5b vs granite3.1-moe:1b: U=1.0, p=0.0003108
    smollm2:360m vs gemma3:1b: U=16.0, p=0.1049
    smollm2:360m vs granite3.1-moe:1b: U=16.0, p=0.1049
    gemma3:1b vs granite3.1-moe:1b: U=15.0, p=0.08298
Augmentation_type 'synonym': Kruskal-Wallis H=24.207, p=2.261e-05
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=8.0, p=0.01041
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=0.0, p=0.0001554
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=31.0, p=0.9591
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554


==== Kruskal-Wallis and Mann-Whitney U for load_duration_ns (by
augmentation_type) ====
Augmentation_type 'entity_replace': Kruskal-Wallis H=27.318, p=5.049e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=64.0, p=0.0001554
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    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=64.0, p=0.0001554
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=12.0, p=0.03792
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554
Augmentation_type 'expand': Kruskal-Wallis H=26.227, p=8.547e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=64.0, p=0.0001554
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=64.0, p=0.0001554
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=28.0, p=0.7209
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554
Augmentation_type 'explain_simple': Kruskal-Wallis H=26.364, p=8.003e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=64.0, p=0.0001554
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=64.0, p=0.0001554
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=40.0, p=0.4418
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554
Augmentation_type 'identity': Kruskal-Wallis H=28.253, p=3.214e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=64.0, p=0.0001554
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=64.0, p=0.0001554
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=5.0, p=0.002953
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554
Augmentation_type 'negation': Kruskal-Wallis H=26.207, p=8.63e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=64.0, p=0.0001554
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=64.0, p=0.0001554
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=29.0, p=0.7984
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554
Augmentation_type 'noise': Kruskal-Wallis H=22.538, p=5.039e-05
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=64.0, p=0.0001554
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=16.0, p=0.04444
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=14.0, p=0.1778
    gemma3:1b vs granite3.1-moe:1b: U=16.0, p=0.04444
Augmentation_type 'paraphrase': Kruskal-Wallis H=15.372, p=0.001525
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=56.0, p=0.01041
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    qwen2:0.5b vs gemma3:1b: U=7.0, p=0.006993
    qwen2:0.5b vs granite3.1-moe:1b: U=56.0, p=0.01041
    smollm2:360m vs gemma3:1b: U=7.0, p=0.006993
    smollm2:360m vs granite3.1-moe:1b: U=40.0, p=0.4418
    gemma3:1b vs granite3.1-moe:1b: U=56.0, p=0.01041
Augmentation_type 'question_gen': Kruskal-Wallis H=25.015, p=1.533e-05
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=64.0, p=0.0001554
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=48.0, p=0.000666
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=21.0, p=0.7546
    gemma3:1b vs granite3.1-moe:1b: U=48.0, p=0.000666
Augmentation_type 'shuffle': Kruskal-Wallis H=26.466, p=7.618e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=64.0, p=0.0001554
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=64.0, p=0.0001554
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=42.0, p=0.3282
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554
Augmentation_type 'summarize': Kruskal-Wallis H=27.003, p=5.879e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=64.0, p=0.0001554
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=64.0, p=0.0001554
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=49.0, p=0.08298
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554
Augmentation_type 'synonym': Kruskal-Wallis H=26.412, p=7.819e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=64.0, p=0.0001554
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=64.0, p=0.0001554
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=41.0, p=0.3823
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554


==== Kruskal-Wallis and Mann-Whitney U for prompt_eval_count (by
augmentation_type) ====
Augmentation_type 'entity_replace': Kruskal-Wallis H=26.267, p=8.385e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=0.0, p=0.0009229
    qwen2:0.5b vs gemma3:1b: U=29.5, p=0.8319
    qwen2:0.5b vs granite3.1-moe:1b: U=0.0, p=0.0009229
    smollm2:360m vs gemma3:1b: U=64.0, p=0.0009229
    smollm2:360m vs granite3.1-moe:1b: U=0.0, p=0.0001554
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    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.0009229
Augmentation_type 'expand': Kruskal-Wallis H=26.267, p=8.385e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=0.0, p=0.0009229
    qwen2:0.5b vs gemma3:1b: U=29.5, p=0.8319
    qwen2:0.5b vs granite3.1-moe:1b: U=0.0, p=0.0009229
    smollm2:360m vs gemma3:1b: U=64.0, p=0.0009229
    smollm2:360m vs granite3.1-moe:1b: U=0.0, p=0.0001554
    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.0009229
Augmentation_type 'explain_simple': Kruskal-Wallis H=26.450, p=7.677e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=0.0, p=0.0009229
    qwen2:0.5b vs gemma3:1b: U=23.5, p=0.3963
    qwen2:0.5b vs granite3.1-moe:1b: U=0.0, p=0.0009229
    smollm2:360m vs gemma3:1b: U=64.0, p=0.0009229
    smollm2:360m vs granite3.1-moe:1b: U=0.0, p=0.0001554
    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.0009229
Augmentation_type 'identity': Kruskal-Wallis H=26.267, p=8.385e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=0.0, p=0.0009229
    qwen2:0.5b vs gemma3:1b: U=29.5, p=0.8319
    qwen2:0.5b vs granite3.1-moe:1b: U=0.0, p=0.0009229
    smollm2:360m vs gemma3:1b: U=64.0, p=0.0009229
    smollm2:360m vs granite3.1-moe:1b: U=0.0, p=0.0001554
    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.0009229
Augmentation_type 'negation': Kruskal-Wallis H=26.290, p=8.293e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=0.0, p=0.0009229
    qwen2:0.5b vs gemma3:1b: U=36.0, p=0.7105
    qwen2:0.5b vs granite3.1-moe:1b: U=0.0, p=0.0009229
    smollm2:360m vs gemma3:1b: U=64.0, p=0.0009229
    smollm2:360m vs granite3.1-moe:1b: U=0.0, p=0.0001554
    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.0009229
Augmentation_type 'noise': Kruskal-Wallis H=18.577, p=0.0003343
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=0.0, p=0.0009229
    qwen2:0.5b vs gemma3:1b: U=29.5, p=0.8319
    qwen2:0.5b vs granite3.1-moe:1b: U=0.0, p=0.0488
    smollm2:360m vs gemma3:1b: U=64.0, p=0.0009229
    smollm2:360m vs granite3.1-moe:1b: U=0.0, p=0.04444
    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.0488
Augmentation_type 'paraphrase': Kruskal-Wallis H=26.267, p=8.385e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=0.0, p=0.0009229
    qwen2:0.5b vs gemma3:1b: U=29.5, p=0.8319
    qwen2:0.5b vs granite3.1-moe:1b: U=0.0, p=0.0009229
    smollm2:360m vs gemma3:1b: U=64.0, p=0.0009229
    smollm2:360m vs granite3.1-moe:1b: U=0.0, p=0.0001554
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    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.0009229
Augmentation_type 'question_gen': Kruskal-Wallis H=23.945, p=2.565e-05
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=0.0, p=0.0009229
    qwen2:0.5b vs gemma3:1b: U=29.5, p=0.8319
    qwen2:0.5b vs granite3.1-moe:1b: U=0.0, p=0.002335
    smollm2:360m vs gemma3:1b: U=64.0, p=0.0009229
    smollm2:360m vs granite3.1-moe:1b: U=0.0, p=0.002388
    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.002335
Augmentation_type 'shuffle': Kruskal-Wallis H=26.290, p=8.293e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=0.0, p=0.0009229
    qwen2:0.5b vs gemma3:1b: U=36.0, p=0.7105
    qwen2:0.5b vs granite3.1-moe:1b: U=0.0, p=0.0009229
    smollm2:360m vs gemma3:1b: U=64.0, p=0.0009229
    smollm2:360m vs granite3.1-moe:1b: U=0.0, p=0.0001554
    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.0009229
Augmentation_type 'summarize': Kruskal-Wallis H=26.290, p=8.293e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=0.0, p=0.0009229
    qwen2:0.5b vs gemma3:1b: U=36.0, p=0.7105
    qwen2:0.5b vs granite3.1-moe:1b: U=0.0, p=0.0009229
    smollm2:360m vs gemma3:1b: U=64.0, p=0.0009229
    smollm2:360m vs granite3.1-moe:1b: U=0.0, p=0.0001554
    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.0009229
Augmentation_type 'synonym': Kruskal-Wallis H=26.267, p=8.385e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=0.0, p=0.0009229
    qwen2:0.5b vs gemma3:1b: U=29.5, p=0.8319
    qwen2:0.5b vs granite3.1-moe:1b: U=0.0, p=0.0009229
    smollm2:360m vs gemma3:1b: U=64.0, p=0.0009229
    smollm2:360m vs granite3.1-moe:1b: U=0.0, p=0.0001554
    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.0009229


==== Kruskal-Wallis and Mann-Whitney U for prompt_eval_duration_ns (by
augmentation_type) ====
Augmentation_type 'entity_replace': Kruskal-Wallis H=24.136, p=2.339e-05
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=31.0, p=0.9591
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=5.0, p=0.002953
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=3.0, p=0.001088
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554
Augmentation_type 'expand': Kruskal-Wallis H=25.389, p=1.28e-05
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=30.0, p=0.8785
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    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=2.0, p=0.0006216
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=1.0, p=0.0003108
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554
Augmentation_type 'explain_simple': Kruskal-Wallis H=24.136, p=2.339e-05
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=31.0, p=0.9591
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=5.0, p=0.002953
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=3.0, p=0.001088
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554
Augmentation_type 'identity': Kruskal-Wallis H=25.918, p=9.924e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=30.0, p=0.8785
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=1.0, p=0.0003108
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=0.0, p=0.0001554
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554
Augmentation_type 'negation': Kruskal-Wallis H=25.128, p=1.452e-05
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=31.0, p=0.9591
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=2.0, p=0.0006216
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=2.0, p=0.0006216
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554
Augmentation_type 'noise': Kruskal-Wallis H=18.173, p=0.0004051
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=30.0, p=0.8785
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=1.0, p=0.08889
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=0.0, p=0.04444
    gemma3:1b vs granite3.1-moe:1b: U=16.0, p=0.04444
Augmentation_type 'paraphrase': Kruskal-Wallis H=22.438, p=5.289e-05
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=28.0, p=0.7209
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=3.0, p=0.001088
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=7.0, p=0.006993
    gemma3:1b vs granite3.1-moe:1b: U=57.0, p=0.006993
Augmentation_type 'question_gen': Kruskal-Wallis H=23.572, p=3.069e-05
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=30.0, p=0.8785
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    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=1.0, p=0.001332
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=0.0, p=0.000666
    gemma3:1b vs granite3.1-moe:1b: U=48.0, p=0.000666
Augmentation_type 'shuffle': Kruskal-Wallis H=24.378, p=2.083e-05
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=31.0, p=0.9591
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=5.0, p=0.002953
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=2.0, p=0.0006216
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554
Augmentation_type 'summarize': Kruskal-Wallis H=25.125, p=1.454e-05
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=31.0, p=0.9591
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=3.0, p=0.001088
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=1.0, p=0.0003108
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554
Augmentation_type 'synonym': Kruskal-Wallis H=24.622, p=1.852e-05
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=31.0, p=0.9591
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=4.0, p=0.001865
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=2.0, p=0.0006216
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554


==== Kruskal-Wallis and Mann-Whitney U for eval_count (by augmentation_type)
====
Augmentation_type 'entity_replace': Kruskal-Wallis H=7.167, p=0.06675
  No significant difference among LLMs.
  Augmentation_type 'expand': All values identical; cannot perform
Kruskal-Wallis test.
  Augmentation_type 'explain_simple': All values identical; cannot perform
Kruskal-Wallis test.
Augmentation_type 'identity': Kruskal-Wallis H=7.916, p=0.04778
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=50.5, p=0.04809
    qwen2:0.5b vs gemma3:1b: U=48.0, p=0.09413
    qwen2:0.5b vs granite3.1-moe:1b: U=35.0, p=0.7496
    smollm2:360m vs gemma3:1b: U=34.5, p=0.8332
    smollm2:360m vs granite3.1-moe:1b: U=14.5, p=0.06657
    gemma3:1b vs granite3.1-moe:1b: U=11.5, p=0.03261
Augmentation_type 'negation': Kruskal-Wallis H=17.854, p=0.0004715
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  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=42.5, p=0.2897
    qwen2:0.5b vs gemma3:1b: U=16.0, p=0.03247
    qwen2:0.5b vs granite3.1-moe:1b: U=18.0, p=0.08474
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0004099
    smollm2:360m vs granite3.1-moe:1b: U=2.0, p=0.00122
    gemma3:1b vs granite3.1-moe:1b: U=36.0, p=0.3816
Augmentation_type 'noise': Kruskal-Wallis H=10.862, p=0.0125
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=45.0, p=0.177
    qwen2:0.5b vs gemma3:1b: U=16.0, p=0.03247
    qwen2:0.5b vs granite3.1-moe:1b: U=4.0, p=0.3032
    smollm2:360m vs gemma3:1b: U=8.0, p=0.004569
    smollm2:360m vs granite3.1-moe:1b: U=2.0, p=0.1384
Augmentation_type 'paraphrase': Kruskal-Wallis H=23.092, p=3.865e-05
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=59.5, p=0.002435
    qwen2:0.5b vs gemma3:1b: U=28.0, p=0.3816
    qwen2:0.5b vs granite3.1-moe:1b: U=28.0, p=0.3816
    smollm2:360m vs gemma3:1b: U=4.0, p=0.001446
    smollm2:360m vs granite3.1-moe:1b: U=4.0, p=0.001446
Augmentation_type 'question_gen': Kruskal-Wallis H=23.453, p=3.249e-05
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=31.5, p=1
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0004099
    qwen2:0.5b vs granite3.1-moe:1b: U=0.0, p=0.002157
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0004008
    smollm2:360m vs granite3.1-moe:1b: U=0.0, p=0.002107
    gemma3:1b vs granite3.1-moe:1b: U=32.0, p=0.1121
Augmentation_type 'shuffle': Kruskal-Wallis H=4.933, p=0.1768
  No significant difference among LLMs.
Augmentation_type 'summarize': Kruskal-Wallis H=19.974, p=0.0001719
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=57.0, p=0.007185
    qwen2:0.5b vs gemma3:1b: U=64.0, p=0.0006754
    qwen2:0.5b vs granite3.1-moe:1b: U=34.0, p=0.8481
    smollm2:360m vs gemma3:1b: U=35.0, p=0.7918
    smollm2:360m vs granite3.1-moe:1b: U=6.5, p=0.006978
    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.0007781
Augmentation_type 'synonym': Kruskal-Wallis H=17.606, p=0.0005304
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=44.0, p=0.2202
    qwen2:0.5b vs gemma3:1b: U=16.0, p=0.03247
    qwen2:0.5b vs granite3.1-moe:1b: U=16.0, p=0.03247
    smollm2:360m vs gemma3:1b: U=4.0, p=0.00146
    smollm2:360m vs granite3.1-moe:1b: U=4.0, p=0.00146
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==== Kruskal-Wallis and Mann-Whitney U for eval_duration_ns (by
augmentation_type) ====
Augmentation_type 'entity_replace': Kruskal-Wallis H=25.912, p=9.951e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=4.0, p=0.001865
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=4.0, p=0.001865
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=57.0, p=0.006993
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554
Augmentation_type 'expand': Kruskal-Wallis H=29.091, p=2.143e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=0.0, p=0.0001554
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=0.0, p=0.0001554
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=64.0, p=0.0001554
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554
Augmentation_type 'explain_simple': Kruskal-Wallis H=29.091, p=2.143e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=0.0, p=0.0001554
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=0.0, p=0.0001554
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=64.0, p=0.0001554
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554
Augmentation_type 'identity': Kruskal-Wallis H=18.659, p=0.0003216
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=6.0, p=0.004662
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=12.0, p=0.03792
    smollm2:360m vs gemma3:1b: U=22.0, p=0.3282
    smollm2:360m vs granite3.1-moe:1b: U=46.0, p=0.1605
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554
Augmentation_type 'negation': Kruskal-Wallis H=27.818, p=3.966e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=0.0, p=0.0001554
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=0.0, p=0.0001554
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=56.0, p=0.01041
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554
Augmentation_type 'noise': Kruskal-Wallis H=19.731, p=0.000193
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=9.0, p=0.01476
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=0.0, p=0.04444
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
```

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    smollm2:360m vs granite3.1-moe:1b: U=8.0, p=1
    gemma3:1b vs granite3.1-moe:1b: U=16.0, p=0.04444
Augmentation_type 'paraphrase': Kruskal-Wallis H=22.366, p=5.472e-05
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=16.0, p=0.1049
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=0.0, p=0.0001554
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=41.0, p=0.3823
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554
Augmentation_type 'question_gen': Kruskal-Wallis H=26.640, p=7.003e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=0.0, p=0.0001554
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=0.0, p=0.000666
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=3.0, p=0.004662
    gemma3:1b vs granite3.1-moe:1b: U=48.0, p=0.000666
Augmentation_type 'shuffle': Kruskal-Wallis H=16.656, p=0.0008316
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=6.0, p=0.004662
    qwen2:0.5b vs gemma3:1b: U=6.0, p=0.004662
    qwen2:0.5b vs granite3.1-moe:1b: U=0.0, p=0.0001554
    smollm2:360m vs gemma3:1b: U=17.0, p=0.1304
    smollm2:360m vs granite3.1-moe:1b: U=42.0, p=0.3282
    gemma3:1b vs granite3.1-moe:1b: U=56.0, p=0.01041
Augmentation_type 'summarize': Kruskal-Wallis H=10.261, p=0.01647
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=42.0, p=0.3282
    qwen2:0.5b vs gemma3:1b: U=39.0, p=0.5054
    qwen2:0.5b vs granite3.1-moe:1b: U=5.0, p=0.002953
    smollm2:360m vs gemma3:1b: U=24.0, p=0.4418
    smollm2:360m vs granite3.1-moe:1b: U=16.0, p=0.1049
    gemma3:1b vs granite3.1-moe:1b: U=3.0, p=0.001088
Augmentation_type 'synonym': Kruskal-Wallis H=24.182, p=2.289e-05
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=8.0, p=0.01041
    qwen2:0.5b vs gemma3:1b: U=0.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=0.0, p=0.0001554
    smollm2:360m vs gemma3:1b: U=0.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=32.0, p=1
    gemma3:1b vs granite3.1-moe:1b: U=64.0, p=0.0001554


==== Kruskal-Wallis and Mann-Whitney U for tokens_per_second (by
augmentation_type) ====
Augmentation_type 'entity_replace': Kruskal-Wallis H=29.091, p=2.143e-06
  Significant: Pairwise comparisons:
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    qwen2:0.5b vs smollm2:360m: U=64.0, p=0.0001554
    qwen2:0.5b vs gemma3:1b: U=64.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=64.0, p=0.0001554
    smollm2:360m vs gemma3:1b: U=64.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=0.0, p=0.0001554
    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.0001554
Augmentation_type 'expand': Kruskal-Wallis H=29.091, p=2.143e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=64.0, p=0.0001554
    qwen2:0.5b vs gemma3:1b: U=64.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=64.0, p=0.0001554
    smollm2:360m vs gemma3:1b: U=64.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=0.0, p=0.0001554
    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.0001554
Augmentation_type 'explain_simple': Kruskal-Wallis H=29.091, p=2.143e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=64.0, p=0.0001554
    qwen2:0.5b vs gemma3:1b: U=64.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=64.0, p=0.0001554
    smollm2:360m vs gemma3:1b: U=64.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=0.0, p=0.0001554
    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.0001554
Augmentation_type 'identity': Kruskal-Wallis H=29.091, p=2.143e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=64.0, p=0.0001554
    qwen2:0.5b vs gemma3:1b: U=64.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=64.0, p=0.0001554
    smollm2:360m vs gemma3:1b: U=64.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=0.0, p=0.0001554
    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.0001554
Augmentation_type 'negation': Kruskal-Wallis H=29.091, p=2.143e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=64.0, p=0.0001554
    qwen2:0.5b vs gemma3:1b: U=64.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=64.0, p=0.0001554
    smollm2:360m vs gemma3:1b: U=64.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=0.0, p=0.0001554
    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.0001554
Augmentation_type 'noise': Kruskal-Wallis H=22.838, p=4.366e-05
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=64.0, p=0.0001554
    qwen2:0.5b vs gemma3:1b: U=64.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=16.0, p=0.04444
    smollm2:360m vs gemma3:1b: U=64.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=0.0, p=0.04444
    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.04444
Augmentation_type 'paraphrase': Kruskal-Wallis H=29.091, p=2.143e-06
  Significant: Pairwise comparisons:
```

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    qwen2:0.5b vs smollm2:360m: U=64.0, p=0.0001554
    qwen2:0.5b vs gemma3:1b: U=64.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=64.0, p=0.0001554
    smollm2:360m vs gemma3:1b: U=64.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=0.0, p=0.0001554
    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.0001554
Augmentation_type 'question_gen': Kruskal-Wallis H=27.148, p=5.48e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=64.0, p=0.0001554
    qwen2:0.5b vs gemma3:1b: U=64.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=48.0, p=0.000666
    smollm2:360m vs gemma3:1b: U=64.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=0.0, p=0.000666
    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.000666
Augmentation_type 'shuffle': Kruskal-Wallis H=29.091, p=2.143e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=64.0, p=0.0001554
    qwen2:0.5b vs gemma3:1b: U=64.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=64.0, p=0.0001554
    smollm2:360m vs gemma3:1b: U=64.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=0.0, p=0.0001554
    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.0001554
Augmentation_type 'summarize': Kruskal-Wallis H=29.091, p=2.143e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=64.0, p=0.0001554
    qwen2:0.5b vs gemma3:1b: U=64.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=64.0, p=0.0001554
    smollm2:360m vs gemma3:1b: U=64.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=0.0, p=0.0001554
    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.0001554
Augmentation_type 'synonym': Kruskal-Wallis H=29.091, p=2.143e-06
  Significant: Pairwise comparisons:
    qwen2:0.5b vs smollm2:360m: U=64.0, p=0.0001554
    qwen2:0.5b vs gemma3:1b: U=64.0, p=0.0001554
    qwen2:0.5b vs granite3.1-moe:1b: U=64.0, p=0.0001554
    smollm2:360m vs gemma3:1b: U=64.0, p=0.0001554
    smollm2:360m vs granite3.1-moe:1b: U=0.0, p=0.0001554
    gemma3:1b vs granite3.1-moe:1b: U=0.0, p=0.0001554
```

[52]:
```
# Friedman's Test (by augmentation_type)

from scipy.stats import friedmanchisquare

metrics = [
    'levenshtein_similarity', 'jaccard_similarity', 'length_ratio', 'bleu',
    'cosine_similarity', 'wer', 'char_diversity', 'type_token_ratio',␣
 ↪'bigram_overlap',
```

```
    'total_duration_ns', 'load_duration_ns', 'prompt_eval_count',␣
 ↪'prompt_eval_duration_ns',
    'eval_count', 'eval_duration_ns', 'tokens_per_second'
]

group_var = 'augmentation_type'   # Or use 'prompt_id' if you want

for metric_col in metrics:
    print(f"\n=== Friedman's Test for {metric_col} (by {group_var}) ===")
    for group_val in sorted(df[group_var].unique()):
        sub = df[df[group_var] == group_val]
        pivot = sub.pivot(index='prompt_id', columns='model', values=metric_col)
        # Only run if all models have data for all prompt_id
        if pivot.notnull().all(axis=1).any() and pivot.shape[1] > 1:
            # Only keep rows (prompts) where all models present
            data = [pivot[m].dropna().values for m in pivot.columns]
            if all(len(x) == len(data[0]) for x in data):
                stat, p = friedmanchisquare(*data)
                print(f"{group_var.capitalize()} '{group_val}': Friedman's␣
 ↪chi2={stat:.3f}, p={p:.4g}")
            else:
                print(f"{group_var.capitalize()} '{group_val}': Not all models␣
 ↪have data for all prompts.")
        else:
            print(f"{group_var.capitalize()} '{group_val}': Not enough data for␣
 ↪Friedman's test.")
#
```

```
=== Friedman's Test for levenshtein_similarity (by augmentation_type) ===
Augmentation_type 'entity_replace': Friedman's chi2=6.450, p=0.09166
Augmentation_type 'expand': Friedman's chi2=1.350, p=0.7173
Augmentation_type 'explain_simple': Friedman's chi2=2.550, p=0.4663
Augmentation_type 'identity': Friedman's chi2=19.950, p=0.0001738
Augmentation_type 'negation': Friedman's chi2=4.200, p=0.2407
Augmentation_type 'noise': Not all models have data for all prompts.
Augmentation_type 'paraphrase': Friedman's chi2=1.050, p=0.7892
Augmentation_type 'question_gen': Not all models have data for all prompts.
Augmentation_type 'shuffle': Friedman's chi2=20.250, p=0.0001506
Augmentation_type 'summarize': Friedman's chi2=17.250, p=0.0006278
Augmentation_type 'synonym': Friedman's chi2=1.950, p=0.5828

=== Friedman's Test for jaccard_similarity (by augmentation_type) ===
Augmentation_type 'entity_replace': Friedman's chi2=12.300, p=0.006423
Augmentation_type 'expand': Friedman's chi2=4.443, p=0.2174
Augmentation_type 'explain_simple': Friedman's chi2=2.544, p=0.4673
Augmentation_type 'identity': Friedman's chi2=14.550, p=0.002245
```

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Augmentation_type 'negation': Friedman's chi2=8.250, p=0.04112
Augmentation_type 'noise': Not all models have data for all prompts.
Augmentation_type 'paraphrase': Friedman's chi2=14.700, p=0.002092
Augmentation_type 'question_gen': Not all models have data for all prompts.
Augmentation_type 'shuffle': Friedman's chi2=12.450, p=0.00599
Augmentation_type 'summarize': Friedman's chi2=4.950, p=0.1755
Augmentation_type 'synonym': Friedman's chi2=4.650, p=0.1993

=== Friedman's Test for length_ratio (by augmentation_type) ===
Augmentation_type 'entity_replace': Friedman's chi2=8.089, p=0.04422
Augmentation_type 'expand': Friedman's chi2=20.550, p=0.0001305
Augmentation_type 'explain_simple': Friedman's chi2=14.850, p=0.001949
Augmentation_type 'identity': Friedman's chi2=4.950, p=0.1755
Augmentation_type 'negation': Friedman's chi2=5.430, p=0.1429
Augmentation_type 'noise': Not all models have data for all prompts.
Augmentation_type 'paraphrase': Friedman's chi2=15.450, p=0.00147
Augmentation_type 'question_gen': Not all models have data for all prompts.
Augmentation_type 'shuffle': Friedman's chi2=3.300, p=0.3476
Augmentation_type 'summarize': Friedman's chi2=17.100, p=0.000674
Augmentation_type 'synonym': Friedman's chi2=5.550, p=0.1357

=== Friedman's Test for bleu (by augmentation_type) ===
Augmentation_type 'entity_replace': Friedman's chi2=14.700, p=0.002092
Augmentation_type 'expand': Friedman's chi2=1.650, p=0.6481
Augmentation_type 'explain_simple': Friedman's chi2=4.050, p=0.2561
Augmentation_type 'identity': Friedman's chi2=15.000, p=0.001817
Augmentation_type 'negation': Friedman's chi2=12.450, p=0.00599
Augmentation_type 'noise': Not all models have data for all prompts.
Augmentation_type 'paraphrase': Friedman's chi2=11.400, p=0.009748
Augmentation_type 'question_gen': Not all models have data for all prompts.
Augmentation_type 'shuffle': Friedman's chi2=9.150, p=0.02736
Augmentation_type 'summarize': Friedman's chi2=4.200, p=0.2407
Augmentation_type 'synonym': Friedman's chi2=4.050, p=0.2561

=== Friedman's Test for cosine_similarity (by augmentation_type) ===
Augmentation_type 'entity_replace': Friedman's chi2=11.250, p=0.01045
Augmentation_type 'expand': Friedman's chi2=0.750, p=0.8614
Augmentation_type 'explain_simple': Friedman's chi2=1.800, p=0.6149
Augmentation_type 'identity': Friedman's chi2=15.000, p=0.001817
Augmentation_type 'negation': Friedman's chi2=6.600, p=0.0858
Augmentation_type 'noise': Not all models have data for all prompts.
Augmentation_type 'paraphrase': Friedman's chi2=13.950, p=0.002974
Augmentation_type 'question_gen': Not all models have data for all prompts.
Augmentation_type 'shuffle': Friedman's chi2=13.050, p=0.00453
Augmentation_type 'summarize': Friedman's chi2=5.850, p=0.1191
Augmentation_type 'synonym': Friedman's chi2=3.450, p=0.3273

=== Friedman's Test for wer (by augmentation_type) ===
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Augmentation_type 'entity_replace': Friedman's chi2=8.423, p=0.03803
Augmentation_type 'expand': Friedman's chi2=17.416, p=0.0005804
Augmentation_type 'explain_simple': Friedman's chi2=13.720, p=0.003312
Augmentation_type 'identity': Friedman's chi2=21.911, p=6.806e-05
Augmentation_type 'negation': Friedman's chi2=12.797, p=0.005096
Augmentation_type 'noise': Not all models have data for all prompts.
Augmentation_type 'paraphrase': Friedman's chi2=12.038, p=0.007253
Augmentation_type 'question_gen': Not all models have data for all prompts.
Augmentation_type 'shuffle': Friedman's chi2=12.600, p=0.005587
Augmentation_type 'summarize': Friedman's chi2=11.962, p=0.007514
Augmentation_type 'synonym': Friedman's chi2=16.269, p=0.0009986

=== Friedman's Test for char_diversity (by augmentation_type) ===
Augmentation_type 'entity_replace': Friedman's chi2=13.709, p=0.003329
Augmentation_type 'expand': Friedman's chi2=14.316, p=0.002505
Augmentation_type 'explain_simple': Friedman's chi2=8.250, p=0.04112
Augmentation_type 'identity': Friedman's chi2=15.450, p=0.00147
Augmentation_type 'negation': Friedman's chi2=14.760, p=0.002034
Augmentation_type 'noise': Not all models have data for all prompts.
Augmentation_type 'paraphrase': Friedman's chi2=14.400, p=0.002408
Augmentation_type 'question_gen': Not all models have data for all prompts.
Augmentation_type 'shuffle': Friedman's chi2=0.797, p=0.8501
Augmentation_type 'summarize': Friedman's chi2=0.375, p=0.9454
Augmentation_type 'synonym': Friedman's chi2=14.392, p=0.002417

=== Friedman's Test for type_token_ratio (by augmentation_type) ===
Augmentation_type 'entity_replace': Friedman's chi2=19.105, p=0.00026
Augmentation_type 'expand': Friedman's chi2=16.350, p=0.0009612
Augmentation_type 'explain_simple': Friedman's chi2=14.924, p=0.001883
Augmentation_type 'identity': Friedman's chi2=15.759, p=0.00127
Augmentation_type 'negation': Friedman's chi2=16.063, p=0.001101
Augmentation_type 'noise': Not all models have data for all prompts.
Augmentation_type 'paraphrase': Friedman's chi2=12.342, p=0.0063
Augmentation_type 'question_gen': Not all models have data for all prompts.
Augmentation_type 'shuffle': Friedman's chi2=18.974, p=0.0002768
Augmentation_type 'summarize': Friedman's chi2=4.378, p=0.2234
Augmentation_type 'synonym': Friedman's chi2=13.937, p=0.002993

=== Friedman's Test for bigram_overlap (by augmentation_type) ===
Augmentation_type 'entity_replace': Friedman's chi2=15.000, p=0.001817
Augmentation_type 'expand': Friedman's chi2=1.481, p=0.6867
Augmentation_type 'explain_simple': Friedman's chi2=5.388, p=0.1455
Augmentation_type 'identity': Friedman's chi2=14.550, p=0.002245
Augmentation_type 'negation': Friedman's chi2=8.924, p=0.03032
Augmentation_type 'noise': Not all models have data for all prompts.
Augmentation_type 'paraphrase': Friedman's chi2=9.115, p=0.0278
Augmentation_type 'question_gen': Not all models have data for all prompts.
Augmentation_type 'shuffle': Friedman's chi2=11.960, p=0.007521
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Augmentation_type 'summarize': Friedman's chi2=1.481, p=0.6867
Augmentation_type 'synonym': Friedman's chi2=4.038, p=0.2573

=== Friedman's Test for total_duration_ns (by augmentation_type) ===
Augmentation_type 'entity_replace': Friedman's chi2=22.950, p=4.136e-05
Augmentation_type 'expand': Friedman's chi2=24.000, p=2.498e-05
Augmentation_type 'explain_simple': Friedman's chi2=24.000, p=2.498e-05
Augmentation_type 'identity': Friedman's chi2=18.300, p=0.0003814
Augmentation_type 'negation': Friedman's chi2=22.950, p=4.136e-05
Augmentation_type 'noise': Not all models have data for all prompts.
Augmentation_type 'paraphrase': Friedman's chi2=16.950, p=0.0007237
Augmentation_type 'question_gen': Not all models have data for all prompts.
Augmentation_type 'shuffle': Friedman's chi2=16.650, p=0.0008341
Augmentation_type 'summarize': Friedman's chi2=11.100, p=0.0112
Augmentation_type 'synonym': Friedman's chi2=19.950, p=0.0001738

=== Friedman's Test for load_duration_ns (by augmentation_type) ===
Augmentation_type 'entity_replace': Friedman's chi2=22.950, p=4.136e-05
Augmentation_type 'expand': Friedman's chi2=21.600, p=7.9e-05
Augmentation_type 'explain_simple': Friedman's chi2=22.200, p=5.927e-05
Augmentation_type 'identity': Friedman's chi2=22.950, p=4.136e-05
Augmentation_type 'negation': Friedman's chi2=21.600, p=7.9e-05
Augmentation_type 'noise': Not all models have data for all prompts.
Augmentation_type 'paraphrase': Friedman's chi2=15.750, p=0.001276
Augmentation_type 'question_gen': Not all models have data for all prompts.
Augmentation_type 'shuffle': Friedman's chi2=22.200, p=5.927e-05
Augmentation_type 'summarize': Friedman's chi2=22.200, p=5.927e-05
Augmentation_type 'synonym': Friedman's chi2=21.750, p=7.353e-05

=== Friedman's Test for prompt_eval_count (by augmentation_type) ===
Augmentation_type 'entity_replace': Friedman's chi2=22.792, p=4.462e-05
Augmentation_type 'expand': Friedman's chi2=22.792, p=4.462e-05
Augmentation_type 'explain_simple': Friedman's chi2=23.734, p=2.838e-05
Augmentation_type 'identity': Friedman's chi2=22.792, p=4.462e-05
Augmentation_type 'negation': Friedman's chi2=22.792, p=4.462e-05
Augmentation_type 'noise': Not all models have data for all prompts.
Augmentation_type 'paraphrase': Friedman's chi2=22.792, p=4.462e-05
Augmentation_type 'question_gen': Not all models have data for all prompts.
Augmentation_type 'shuffle': Friedman's chi2=22.792, p=4.462e-05
Augmentation_type 'summarize': Friedman's chi2=22.792, p=4.462e-05
Augmentation_type 'synonym': Friedman's chi2=22.792, p=4.462e-05

=== Friedman's Test for prompt_eval_duration_ns (by augmentation_type) ===
Augmentation_type 'entity_replace': Friedman's chi2=21.750, p=7.353e-05
Augmentation_type 'expand': Friedman's chi2=21.600, p=7.9e-05
Augmentation_type 'explain_simple': Friedman's chi2=21.750, p=7.353e-05
Augmentation_type 'identity': Friedman's chi2=21.600, p=7.9e-05
Augmentation_type 'negation': Friedman's chi2=21.600, p=7.9e-05
```

```
Augmentation_type 'noise': Not all models have data for all prompts.
Augmentation_type 'paraphrase': Friedman's chi2=22.200, p=5.927e-05
Augmentation_type 'question_gen': Not all models have data for all prompts.
Augmentation_type 'shuffle': Friedman's chi2=21.750, p=7.353e-05
Augmentation_type 'summarize': Friedman's chi2=21.750, p=7.353e-05
Augmentation_type 'synonym': Friedman's chi2=21.750, p=7.353e-05

=== Friedman's Test for eval_count (by augmentation_type) ===
Augmentation_type 'entity_replace': Friedman's chi2=9.279, p=0.0258
Augmentation_type 'expand': Friedman's chi2=nan, p=nan
Augmentation_type 'explain_simple': Friedman's chi2=nan, p=nan
Augmentation_type 'identity': Friedman's chi2=5.211, p=0.157
Augmentation_type 'negation': Friedman's chi2=13.476, p=0.003712
Augmentation_type 'noise': Not all models have data for all prompts.
Augmentation_type 'paraphrase': Friedman's chi2=19.800, p=0.0001867
Augmentation_type 'question_gen': Not all models have data for all prompts.
Augmentation_type 'shuffle': Friedman's chi2=7.163, p=0.06687
Augmentation_type 'summarize': Friedman's chi2=18.917, p=0.0002845
Augmentation_type 'synonym': Friedman's chi2=14.053, p=0.002834

=== Friedman's Test for eval_duration_ns (by augmentation_type) ===
Augmentation_type 'entity_replace': Friedman's chi2=24.000, p=2.498e-05
Augmentation_type 'expand': Friedman's chi2=24.000, p=2.498e-05
Augmentation_type 'explain_simple': Friedman's chi2=24.000, p=2.498e-05
Augmentation_type 'identity': Friedman's chi2=15.600, p=0.001369
Augmentation_type 'negation': Friedman's chi2=22.950, p=4.136e-05
Augmentation_type 'noise': Not all models have data for all prompts.
Augmentation_type 'paraphrase': Friedman's chi2=18.450, p=0.0003552
Augmentation_type 'question_gen': Not all models have data for all prompts.
Augmentation_type 'shuffle': Friedman's chi2=13.650, p=0.003422
Augmentation_type 'summarize': Friedman's chi2=8.700, p=0.03356
Augmentation_type 'synonym': Friedman's chi2=19.950, p=0.0001738

=== Friedman's Test for tokens_per_second (by augmentation_type) ===
Augmentation_type 'entity_replace': Friedman's chi2=24.000, p=2.498e-05
Augmentation_type 'expand': Friedman's chi2=24.000, p=2.498e-05
Augmentation_type 'explain_simple': Friedman's chi2=24.000, p=2.498e-05
Augmentation_type 'identity': Friedman's chi2=24.000, p=2.498e-05
Augmentation_type 'negation': Friedman's chi2=24.000, p=2.498e-05
Augmentation_type 'noise': Not all models have data for all prompts.
Augmentation_type 'paraphrase': Friedman's chi2=24.000, p=2.498e-05
Augmentation_type 'question_gen': Not all models have data for all prompts.
Augmentation_type 'shuffle': Friedman's chi2=24.000, p=2.498e-05
Augmentation_type 'summarize': Friedman's chi2=24.000, p=2.498e-05
Augmentation_type 'synonym': Friedman's chi2=24.000, p=2.498e-05

C:\Users\parth\anaconda3\Lib\site-packages\scipy\stats\_stats_py.py:9427:
RuntimeWarning: invalid value encountered in scalar divide
```

```
  chisq = (12.0 / (k*n*(k+1)) * ssbn - 3*n*(k+1)) / c
C:\Users\parth\anaconda3\Lib\site-packages\scipy\stats\_stats_py.py:9427:
RuntimeWarning: invalid value encountered in scalar divide
  chisq = (12.0 / (k*n*(k+1)) * ssbn - 3*n*(k+1)) / c
```

[53]:
```
# Kolmogorov-Smirnov (K-S) Test (for all pairs of models)

from scipy.stats import ks_2samp
from itertools import combinations

for metric_col in metrics:
    print(f"\n=== Kolmogorov-Smirnov Test for {metric_col} (All Model Pairs)␣
 ↪===")
    models = df['model'].unique()
    for m1, m2 in combinations(models, 2):
        data1 = df[df['model'] == m1][metric_col].dropna()
        data2 = df[df['model'] == m2][metric_col].dropna()
        if len(data1) > 0 and len(data2) > 0:
            ks_stat, ks_p = ks_2samp(data1, data2)
            print(f"  {m1} vs {m2}: KS stat={ks_stat:.3f}, p={ks_p:.4g}")
        else:
            print(f"  {m1} vs {m2}: Not enough data.")
```

```
=== Kolmogorov-Smirnov Test for levenshtein_similarity (All Model Pairs) ===
  qwen2:0.5b vs smollm2:360m: KS stat=0.239, p=0.01307
  qwen2:0.5b vs gemma3:1b: KS stat=0.364, p=1.447e-05
  qwen2:0.5b vs granite3.1-moe:1b: KS stat=0.175, p=0.1334
  smollm2:360m vs gemma3:1b: KS stat=0.273, p=0.002746
  smollm2:360m vs granite3.1-moe:1b: KS stat=0.328, p=0.0001615
  gemma3:1b vs granite3.1-moe:1b: KS stat=0.430, p=1.772e-07

=== Kolmogorov-Smirnov Test for jaccard_similarity (All Model Pairs) ===
  qwen2:0.5b vs smollm2:360m: KS stat=0.125, p=0.4999
  qwen2:0.5b vs gemma3:1b: KS stat=0.386, p=3.018e-06
  qwen2:0.5b vs granite3.1-moe:1b: KS stat=0.261, p=0.005102
  smollm2:360m vs gemma3:1b: KS stat=0.432, p=9.537e-08
  smollm2:360m vs granite3.1-moe:1b: KS stat=0.350, p=4.452e-05
  gemma3:1b vs granite3.1-moe:1b: KS stat=0.309, p=0.0004753

=== Kolmogorov-Smirnov Test for length_ratio (All Model Pairs) ===
  qwen2:0.5b vs smollm2:360m: KS stat=0.352, p=3.046e-05
  qwen2:0.5b vs gemma3:1b: KS stat=0.352, p=3.046e-05
  qwen2:0.5b vs granite3.1-moe:1b: KS stat=0.552, p=2.635e-12
  smollm2:360m vs gemma3:1b: KS stat=0.250, p=0.007959
  smollm2:360m vs granite3.1-moe:1b: KS stat=0.234, p=0.01648
  gemma3:1b vs granite3.1-moe:1b: KS stat=0.263, p=0.004845
```

```
=== Kolmogorov-Smirnov Test for bleu (All Model Pairs) ===
  qwen2:0.5b vs smollm2:360m: KS stat=0.114, p=0.6237
  qwen2:0.5b vs gemma3:1b: KS stat=0.409, p=5.664e-07
  qwen2:0.5b vs granite3.1-moe:1b: KS stat=0.266, p=0.004142
  smollm2:360m vs gemma3:1b: KS stat=0.432, p=9.537e-08
  smollm2:360m vs granite3.1-moe:1b: KS stat=0.250, p=0.00845
  gemma3:1b vs granite3.1-moe:1b: KS stat=0.419, p=3.87e-07

=== Kolmogorov-Smirnov Test for cosine_similarity (All Model Pairs) ===
  qwen2:0.5b vs smollm2:360m: KS stat=0.159, p=0.2161
  qwen2:0.5b vs gemma3:1b: KS stat=0.398, p=1.325e-06
  qwen2:0.5b vs granite3.1-moe:1b: KS stat=0.188, p=0.09014
  smollm2:360m vs gemma3:1b: KS stat=0.489, p=6.79e-10
  smollm2:360m vs granite3.1-moe:1b: KS stat=0.289, p=0.001383
  gemma3:1b vs granite3.1-moe:1b: KS stat=0.307, p=0.0005372

=== Kolmogorov-Smirnov Test for wer (All Model Pairs) ===
  qwen2:0.5b vs smollm2:360m: KS stat=0.284, p=0.001555
  qwen2:0.5b vs gemma3:1b: KS stat=0.330, p=0.000125
  qwen2:0.5b vs granite3.1-moe:1b: KS stat=0.474, p=4.679e-09
  smollm2:360m vs gemma3:1b: KS stat=0.261, p=0.004732
  smollm2:360m vs granite3.1-moe:1b: KS stat=0.266, p=0.004142
  gemma3:1b vs granite3.1-moe:1b: KS stat=0.317, p=0.0003074

=== Kolmogorov-Smirnov Test for char_diversity (All Model Pairs) ===
  qwen2:0.5b vs smollm2:360m: KS stat=0.216, p=0.03278
  qwen2:0.5b vs gemma3:1b: KS stat=0.523, p=2.457e-11
  qwen2:0.5b vs granite3.1-moe:1b: KS stat=0.165, p=0.1801
  smollm2:360m vs gemma3:1b: KS stat=0.602, p=3.508e-15
  smollm2:360m vs granite3.1-moe:1b: KS stat=0.170, p=0.1528
  gemma3:1b vs granite3.1-moe:1b: KS stat=0.562, p=8.939e-13

=== Kolmogorov-Smirnov Test for type_token_ratio (All Model Pairs) ===
  qwen2:0.5b vs smollm2:360m: KS stat=0.125, p=0.4999
  qwen2:0.5b vs gemma3:1b: KS stat=0.716, p=5.142e-22
  qwen2:0.5b vs granite3.1-moe:1b: KS stat=0.309, p=0.0004753
  smollm2:360m vs gemma3:1b: KS stat=0.773, p=3.786e-26
  smollm2:360m vs granite3.1-moe:1b: KS stat=0.230, p=0.01977
  gemma3:1b vs granite3.1-moe:1b: KS stat=0.815, p=1.146e-28

=== Kolmogorov-Smirnov Test for bigram_overlap (All Model Pairs) ===
  qwen2:0.5b vs smollm2:360m: KS stat=0.068, p=0.9876
  qwen2:0.5b vs gemma3:1b: KS stat=0.409, p=5.664e-07
  qwen2:0.5b vs granite3.1-moe:1b: KS stat=0.211, p=0.03958
  smollm2:360m vs gemma3:1b: KS stat=0.432, p=9.537e-08
  smollm2:360m vs granite3.1-moe:1b: KS stat=0.223, p=0.02583
  gemma3:1b vs granite3.1-moe:1b: KS stat=0.339, p=8.87e-05
```

```
=== Kolmogorov-Smirnov Test for total_duration_ns (All Model Pairs) ===
  qwen2:0.5b vs smollm2:360m: KS stat=0.693, p=1.659e-20
  qwen2:0.5b vs gemma3:1b: KS stat=0.977, p=5.354e-48
  qwen2:0.5b vs granite3.1-moe:1b: KS stat=0.926, p=2.924e-39
  smollm2:360m vs gemma3:1b: KS stat=0.795, p=5.799e-28
  smollm2:360m vs granite3.1-moe:1b: KS stat=0.533, p=1.9e-11
  gemma3:1b vs granite3.1-moe:1b: KS stat=0.897, p=1.893e-36

=== Kolmogorov-Smirnov Test for load_duration_ns (All Model Pairs) ===
  qwen2:0.5b vs smollm2:360m: KS stat=0.989, p=6.119e-50
  qwen2:0.5b vs gemma3:1b: KS stat=0.989, p=6.119e-50
  qwen2:0.5b vs granite3.1-moe:1b: KS stat=0.988, p=1.766e-47
  smollm2:360m vs gemma3:1b: KS stat=0.989, p=6.119e-50
  smollm2:360m vs granite3.1-moe:1b: KS stat=0.116, p=0.578
  gemma3:1b vs granite3.1-moe:1b: KS stat=0.988, p=1.766e-47

=== Kolmogorov-Smirnov Test for prompt_eval_count (All Model Pairs) ===
  qwen2:0.5b vs smollm2:360m: KS stat=1.000, p=3.477e-52
  qwen2:0.5b vs gemma3:1b: KS stat=0.068, p=0.9876
  qwen2:0.5b vs granite3.1-moe:1b: KS stat=1.000, p=1.051e-49
  smollm2:360m vs gemma3:1b: KS stat=0.989, p=6.119e-50
  smollm2:360m vs granite3.1-moe:1b: KS stat=1.000, p=1.051e-49
  gemma3:1b vs granite3.1-moe:1b: KS stat=1.000, p=1.051e-49

=== Kolmogorov-Smirnov Test for prompt_eval_duration_ns (All Model Pairs) ===
  qwen2:0.5b vs smollm2:360m: KS stat=0.250, p=0.007959
  qwen2:0.5b vs gemma3:1b: KS stat=1.000, p=3.477e-52
  qwen2:0.5b vs granite3.1-moe:1b: KS stat=0.875, p=3.991e-34
  smollm2:360m vs gemma3:1b: KS stat=1.000, p=3.477e-52
  smollm2:360m vs granite3.1-moe:1b: KS stat=0.898, p=1.456e-36
  gemma3:1b vs granite3.1-moe:1b: KS stat=0.988, p=1.766e-47

=== Kolmogorov-Smirnov Test for eval_count (All Model Pairs) ===
  qwen2:0.5b vs smollm2:360m: KS stat=0.341, p=6.249e-05
  qwen2:0.5b vs gemma3:1b: KS stat=0.136, p=0.3883
  qwen2:0.5b vs granite3.1-moe:1b: KS stat=0.244, p=0.01078
  smollm2:360m vs gemma3:1b: KS stat=0.455, p=1.437e-08
  smollm2:360m vs granite3.1-moe:1b: KS stat=0.535, p=1.513e-11
  gemma3:1b vs granite3.1-moe:1b: KS stat=0.155, p=0.2388

=== Kolmogorov-Smirnov Test for eval_duration_ns (All Model Pairs) ===
  qwen2:0.5b vs smollm2:360m: KS stat=0.739, p=1.327e-23
  qwen2:0.5b vs gemma3:1b: KS stat=0.920, p=3.198e-40
  qwen2:0.5b vs granite3.1-moe:1b: KS stat=0.938, p=1.104e-40
  smollm2:360m vs gemma3:1b: KS stat=0.784, p=4.822e-27
  smollm2:360m vs granite3.1-moe:1b: KS stat=0.625, p=6.761e-16
  gemma3:1b vs granite3.1-moe:1b: KS stat=0.898, p=1.456e-36
```

```
=== Kolmogorov-Smirnov Test for tokens_per_second (All Model Pairs) ===
  qwen2:0.5b vs smollm2:360m: KS stat=1.000, p=3.477e-52
  qwen2:0.5b vs gemma3:1b: KS stat=1.000, p=3.477e-52
  qwen2:0.5b vs granite3.1-moe:1b: KS stat=1.000, p=1.051e-49
  smollm2:360m vs gemma3:1b: KS stat=1.000, p=3.477e-52
  smollm2:360m vs granite3.1-moe:1b: KS stat=1.000, p=1.051e-49
  gemma3:1b vs granite3.1-moe:1b: KS stat=1.000, p=1.051e-49
```

[54]:
```python
# Spearman's Rank Correlation for All Metrics

# How to interpret:

# Correlation coefficient in [-1, 1]:
# Closer to 1 = strong positive monotonic relation
# Closer to -1 = strong negative
# Close to 0 = no monotonic association

# P-value:
# p < 0.05 means significant correlation
# p > 0.05 means correlation is not statistically significant

import pandas as pd
from scipy.stats import spearmanr

# List of your numeric metrics
metrics = [
    'levenshtein_similarity', 'jaccard_similarity', 'length_ratio', 'bleu',
    'cosine_similarity', 'wer', 'char_diversity', 'type_token_ratio',␣
 ↪'bigram_overlap',
    'total_duration_ns', 'load_duration_ns', 'prompt_eval_count',␣
 ↪'prompt_eval_duration_ns',
    'eval_count', 'eval_duration_ns', 'tokens_per_second'
]

# Subset dataframe to just these columns, drop rows with any NaN
metrics_df = df[metrics].dropna()

# Compute Spearman correlation (returns correlation matrix and p-value matrix)
corr, pval = spearmanr(metrics_df, axis=0)

# To DataFrame for easier reading:
corr_df = pd.DataFrame(corr, index=metrics, columns=metrics)
pval_df = pd.DataFrame(pval, index=metrics, columns=metrics)

# Print or display the correlation matrix
print("Spearman Correlation Matrix:")
print(corr_df.round(3))
```

```
print("\nSpearman Correlation P-value Matrix:")
print(pval_df.round(3))
```

```
Spearman Correlation Matrix:
                        levenshtein_similarity  jaccard_similarity  \
levenshtein_similarity                   1.000               0.329
jaccard_similarity                       0.329               1.000
length_ratio                            -0.240               0.301
bleu                                     0.403               0.921
cosine_similarity                        0.325               0.843
wer                                     -0.538              -0.454
char_diversity                          -0.226              -0.262
type_token_ratio                        -0.060              -0.352
bigram_overlap                           0.447               0.911
total_duration_ns                       -0.079               0.340
load_duration_ns                         0.131               0.198
prompt_eval_count                       -0.183              -0.053
prompt_eval_duration_ns                  0.041               0.369
eval_count                              -0.440               0.245
eval_duration_ns                        -0.115               0.305
tokens_per_second                       -0.172              -0.227

                        length_ratio   bleu  cosine_similarity    wer  \
levenshtein_similarity        -0.240  0.403              0.325 -0.538
jaccard_similarity             0.301  0.921              0.843 -0.454
length_ratio                   1.000  0.253              0.279  0.494
bleu                           0.253  1.000              0.816 -0.448
cosine_similarity              0.279  0.816              1.000 -0.399
wer                            0.494 -0.448             -0.399  1.000
char_diversity                 0.168 -0.173             -0.087  0.456
type_token_ratio              -0.270 -0.379             -0.507 -0.069
bigram_overlap                 0.218  0.972              0.794 -0.462
total_duration_ns              0.218  0.365              0.368  0.031
load_duration_ns               0.127  0.253              0.273  0.082
prompt_eval_count             -0.277 -0.101             -0.137 -0.257
prompt_eval_duration_ns       -0.142  0.373              0.360 -0.238
eval_count                     0.625  0.203              0.238  0.413
eval_duration_ns               0.298  0.335              0.341  0.112
tokens_per_second              0.023 -0.273             -0.263  0.072

                        char_diversity  type_token_ratio  bigram_overlap  \
levenshtein_similarity          -0.226            -0.060           0.447
jaccard_similarity              -0.262            -0.352           0.911
length_ratio                     0.168            -0.270           0.218
bleu                            -0.173            -0.379           0.972
cosine_similarity               -0.087            -0.507           0.794
wer                              0.456            -0.069          -0.462
```

| | | | |
|---|---|---|---|
| char_diversity | 1.000 | -0.238 | -0.213 |
| type_token_ratio | -0.238 | 1.000 | -0.342 |
| bigram_overlap | -0.213 | -0.342 | 1.000 |
| total_duration_ns | 0.324 | -0.561 | 0.302 |
| load_duration_ns | 0.307 | -0.512 | 0.247 |
| prompt_eval_count | -0.242 | 0.394 | -0.111 |
| prompt_eval_duration_ns | 0.223 | -0.445 | 0.345 |
| eval_count | 0.397 | -0.294 | 0.145 |
| eval_duration_ns | 0.372 | -0.568 | 0.268 |
| tokens_per_second | -0.266 | 0.511 | -0.246 |

| | total_duration_ns | load_duration_ns | \ |
|---|---|---|---|
| levenshtein_similarity | -0.079 | 0.131 | |
| jaccard_similarity | 0.340 | 0.198 | |
| length_ratio | 0.218 | 0.127 | |
| bleu | 0.365 | 0.253 | |
| cosine_similarity | 0.368 | 0.273 | |
| wer | 0.031 | 0.082 | |
| char_diversity | 0.324 | 0.307 | |
| type_token_ratio | -0.561 | -0.512 | |
| bigram_overlap | 0.302 | 0.247 | |
| total_duration_ns | 1.000 | 0.285 | |
| load_duration_ns | 0.285 | 1.000 | |
| prompt_eval_count | 0.078 | -0.728 | |
| prompt_eval_duration_ns | 0.703 | 0.457 | |
| eval_count | 0.505 | 0.117 | |
| eval_duration_ns | 0.971 | 0.239 | |
| tokens_per_second | -0.841 | -0.292 | |

| | prompt_eval_count | prompt_eval_duration_ns | \ |
|---|---|---|---|
| levenshtein_similarity | -0.183 | 0.041 | |
| jaccard_similarity | -0.053 | 0.369 | |
| length_ratio | -0.277 | -0.142 | |
| bleu | -0.101 | 0.373 | |
| cosine_similarity | -0.137 | 0.360 | |
| wer | -0.257 | -0.238 | |
| char_diversity | -0.242 | 0.223 | |
| type_token_ratio | 0.394 | -0.445 | |
| bigram_overlap | -0.111 | 0.345 | |
| total_duration_ns | 0.078 | 0.703 | |
| load_duration_ns | -0.728 | 0.457 | |
| prompt_eval_count | 1.000 | 0.061 | |
| prompt_eval_duration_ns | 0.061 | 1.000 | |
| eval_count | 0.091 | 0.314 | |
| eval_duration_ns | 0.052 | 0.604 | |
| tokens_per_second | 0.043 | -0.608 | |

| | eval_count | eval_duration_ns | tokens_per_second |
|---|---|---|---|

| | | | |
|---|---|---|---|
| levenshtein_similarity | -0.440 | -0.115 | -0.172 |
| jaccard_similarity | 0.245 | 0.305 | -0.227 |
| length_ratio | 0.625 | 0.298 | 0.023 |
| bleu | 0.203 | 0.335 | -0.273 |
| cosine_similarity | 0.238 | 0.341 | -0.263 |
| wer | 0.413 | 0.112 | 0.072 |
| char_diversity | 0.397 | 0.372 | -0.266 |
| type_token_ratio | -0.294 | -0.568 | 0.511 |
| bigram_overlap | 0.145 | 0.268 | -0.246 |
| total_duration_ns | 0.505 | 0.971 | -0.841 |
| load_duration_ns | 0.117 | 0.239 | -0.292 |
| prompt_eval_count | 0.091 | 0.052 | 0.043 |
| prompt_eval_duration_ns | 0.314 | 0.604 | -0.608 |
| eval_count | 1.000 | 0.548 | -0.119 |
| eval_duration_ns | 0.548 | 1.000 | -0.830 |
| tokens_per_second | -0.119 | -0.830 | 1.000 |

Spearman Correlation P-value Matrix:

| | levenshtein_similarity | jaccard_similarity | \ |
|---|---|---|---|
| levenshtein_similarity | 0.000 | 0.000 | |
| jaccard_similarity | 0.000 | 0.000 | |
| length_ratio | 0.000 | 0.000 | |
| bleu | 0.000 | 0.000 | |
| cosine_similarity | 0.000 | 0.000 | |
| wer | 0.000 | 0.000 | |
| char_diversity | 0.000 | 0.000 | |
| type_token_ratio | 0.263 | 0.000 | |
| bigram_overlap | 0.000 | 0.000 | |
| total_duration_ns | 0.145 | 0.000 | |
| load_duration_ns | 0.015 | 0.000 | |
| prompt_eval_count | 0.001 | 0.326 | |
| prompt_eval_duration_ns | 0.450 | 0.000 | |
| eval_count | 0.000 | 0.000 | |
| eval_duration_ns | 0.033 | 0.000 | |
| tokens_per_second | 0.001 | 0.000 | |

| | length_ratio | bleu | cosine_similarity | wer | \ |
|---|---|---|---|---|---|
| levenshtein_similarity | 0.000 | 0.000 | 0.000 | 0.000 | |
| jaccard_similarity | 0.000 | 0.000 | 0.000 | 0.000 | |
| length_ratio | 0.000 | 0.000 | 0.000 | 0.000 | |
| bleu | 0.000 | 0.000 | 0.000 | 0.000 | |
| cosine_similarity | 0.000 | 0.000 | 0.000 | 0.000 | |
| wer | 0.000 | 0.000 | 0.000 | 0.000 | |
| char_diversity | 0.002 | 0.001 | 0.106 | 0.000 | |
| type_token_ratio | 0.000 | 0.000 | 0.000 | 0.201 | |
| bigram_overlap | 0.000 | 0.000 | 0.000 | 0.000 | |
| total_duration_ns | 0.000 | 0.000 | 0.000 | 0.565 | |
| load_duration_ns | 0.018 | 0.000 | 0.000 | 0.128 | |

| | | | | |
|---|---|---|---|---|
| prompt_eval_count | 0.000 | 0.061 | 0.011 | 0.000 |
| prompt_eval_duration_ns | 0.008 | 0.000 | 0.000 | 0.000 |
| eval_count | 0.000 | 0.000 | 0.000 | 0.000 |
| eval_duration_ns | 0.000 | 0.000 | 0.000 | 0.039 |
| tokens_per_second | 0.674 | 0.000 | 0.000 | 0.184 |

| | char_diversity | type_token_ratio | bigram_overlap | \ |
|---|---|---|---|---|
| levenshtein_similarity | 0.000 | 0.263 | 0.000 | |
| jaccard_similarity | 0.000 | 0.000 | 0.000 | |
| length_ratio | 0.002 | 0.000 | 0.000 | |
| bleu | 0.001 | 0.000 | 0.000 | |
| cosine_similarity | 0.106 | 0.000 | 0.000 | |
| wer | 0.000 | 0.201 | 0.000 | |
| char_diversity | 0.000 | 0.000 | 0.000 | |
| type_token_ratio | 0.000 | 0.000 | 0.000 | |
| bigram_overlap | 0.000 | 0.000 | 0.000 | |
| total_duration_ns | 0.000 | 0.000 | 0.000 | |
| load_duration_ns | 0.000 | 0.000 | 0.000 | |
| prompt_eval_count | 0.000 | 0.000 | 0.040 | |
| prompt_eval_duration_ns | 0.000 | 0.000 | 0.000 | |
| eval_count | 0.000 | 0.000 | 0.007 | |
| eval_duration_ns | 0.000 | 0.000 | 0.000 | |
| tokens_per_second | 0.000 | 0.000 | 0.000 | |

| | total_duration_ns | load_duration_ns | \ |
|---|---|---|---|
| levenshtein_similarity | 0.145 | 0.015 | |
| jaccard_similarity | 0.000 | 0.000 | |
| length_ratio | 0.000 | 0.018 | |
| bleu | 0.000 | 0.000 | |
| cosine_similarity | 0.000 | 0.000 | |
| wer | 0.565 | 0.128 | |
| char_diversity | 0.000 | 0.000 | |
| type_token_ratio | 0.000 | 0.000 | |
| bigram_overlap | 0.000 | 0.000 | |
| total_duration_ns | 0.000 | 0.000 | |
| load_duration_ns | 0.000 | 0.000 | |
| prompt_eval_count | 0.150 | 0.000 | |
| prompt_eval_duration_ns | 0.000 | 0.000 | |
| eval_count | 0.000 | 0.031 | |
| eval_duration_ns | 0.000 | 0.000 | |
| tokens_per_second | 0.000 | 0.000 | |

| | prompt_eval_count | prompt_eval_duration_ns | \ |
|---|---|---|---|
| levenshtein_similarity | 0.001 | 0.450 | |
| jaccard_similarity | 0.326 | 0.000 | |
| length_ratio | 0.000 | 0.008 | |
| bleu | 0.061 | 0.000 | |
| cosine_similarity | 0.011 | 0.000 | |

```
wer                              0.000                  0.000
char_diversity                   0.000                  0.000
type_token_ratio                 0.000                  0.000
bigram_overlap                   0.040                  0.000
total_duration_ns                0.150                  0.000
load_duration_ns                 0.000                  0.000
prompt_eval_count                0.000                  0.261
prompt_eval_duration_ns          0.261                  0.000
eval_count                       0.091                  0.000
eval_duration_ns                 0.338                  0.000
tokens_per_second                0.425                  0.000

                         eval_count  eval_duration_ns  tokens_per_second
levenshtein_similarity        0.000             0.033              0.001
jaccard_similarity            0.000             0.000              0.000
length_ratio                  0.000             0.000              0.674
bleu                          0.000             0.000              0.000
cosine_similarity             0.000             0.000              0.000
wer                           0.000             0.039              0.184
char_diversity                0.000             0.000              0.000
type_token_ratio              0.000             0.000              0.000
bigram_overlap                0.007             0.000              0.000
total_duration_ns             0.000             0.000              0.000
load_duration_ns              0.031             0.000              0.000
prompt_eval_count             0.091             0.338              0.425
prompt_eval_duration_ns       0.000             0.000              0.000
eval_count                    0.000             0.000              0.027
eval_duration_ns              0.000             0.000              0.000
tokens_per_second             0.027             0.000              0.000
```

[55]:
```python
import pandas as pd

metrics = [
    'levenshtein_similarity', 'jaccard_similarity', 'length_ratio', 'bleu',
    'cosine_similarity', 'wer', 'char_diversity', 'type_token_ratio',␣
 ↪'bigram_overlap',
    'total_duration_ns', 'load_duration_ns', 'prompt_eval_count',␣
 ↪'prompt_eval_duration_ns',
    'eval_count', 'eval_duration_ns', 'tokens_per_second'
]

# Filter to relevant columns, drop rows with any missing value in those columns
df_manova = df[['model', 'augmentation_type'] + metrics].dropna()
print(df_manova.shape)


## MANOVA by Model Only
```

```
# Interpretation
# Look at the line for model in Wilks' lambda, Pillai's trace, Hotelling-Lawley␣
 ↪trace, and Roy's largest root.
# The Pr > F column gives the p-value for the global null: "All LLMs have the␣
 ↪same mean vector of metrics."
# p < 0.05 means at least one LLM is different on at least one combination of␣
 ↪metrics.

from statsmodels.multivariate.manova import MANOVA

formula = ' + '.join(metrics) + ' ~ model'
maov = MANOVA.from_formula(formula, data=df_manova)
print(maov.mv_test())
```

```
(344, 18)
                     Multivariate linear model
============================================================================

----------------------------------------------------------------------------
         Intercept          Value   Num DF  Den DF    F Value   Pr > F
----------------------------------------------------------------------------
            Wilks' lambda    0.0002 4.0000 337.0000 397563.2643 0.0000
           Pillai's trace    0.9998 4.0000 337.0000 397563.2643 0.0000
   Hotelling-Lawley trace 4718.8518 4.0000 337.0000 397563.2643 0.0000
      Roy's greatest root 4718.8518 4.0000 337.0000 397563.2643 0.0000
----------------------------------------------------------------------------

----------------------------------------------------------------------------
         model             Value    Num DF   Den DF    F Value   Pr > F
----------------------------------------------------------------------------
          Wilks' lambda    0.0000 12.0000  891.9097  14012.8080 0.0000
         Pillai's trace    2.7918 12.0000 1017.0000   1136.1569 0.0000
 Hotelling-Lawley trace 4114.5409 12.0000  585.4521 115258.0057 0.0000
    Roy's greatest root 4062.1679  4.0000  339.0000 344268.7303 0.0000
============================================================================
```

[56]:
```
#MANOVA by Model and Augmentation Type (Factorial MANOVA)

formula = ' + '.join(metrics) + ' ~ model + augmentation_type'
maov2 = MANOVA.from_formula(formula, data=df_manova)
print(maov2.mv_test())
```

```
                     Multivariate linear model
============================================================================

----------------------------------------------------------------------------
```

```
        Intercept          Value   Num DF  Den DF    F Value   Pr > F
-----------------------------------------------------------------------
          Wilks' lambda    0.0007 4.0000 327.0000 118474.4430 0.0000
         Pillai's trace    0.9993 4.0000 327.0000 118474.4430 0.0000
  Hotelling-Lawley trace 1449.2287 4.0000 327.0000 118474.4430 0.0000
     Roy's greatest root 1449.2287 4.0000 327.0000 118474.4430 0.0000
-----------------------------------------------------------------------

-----------------------------------------------------------------------
         model            Value    Num DF  Den DF    F Value   Pr > F
-----------------------------------------------------------------------
          Wilks' lambda    0.0000 12.0000 865.4522  18710.9331 0.0000
         Pillai's trace    2.8161 12.0000 987.0000   1259.7218 0.0000
 Hotelling-Lawley trace 4490.8920 12.0000 567.9526 122058.1435 0.0000
    Roy's greatest root 4389.9069  4.0000 329.0000 361069.8449 0.0000
-----------------------------------------------------------------------

-------------------------------------------------------------------------
     augmentation_type     Value    Num DF    Den DF   F Value  Pr > F
-------------------------------------------------------------------------
           Wilks' lambda  0.0624  40.0000  1241.8012  33.4889  0.0000
          Pillai's trace  2.1361  40.0000  1320.0000  37.8209  0.0000
   Hotelling-Lawley trace  3.8299  40.0000   903.4338  31.1871  0.0000
      Roy's greatest root  1.6174  10.0000   330.0000  53.3743  0.0000
=========================================================================
```

[57]:
```
# MANOVA With Interaction Term

formula = ' + '.join(metrics) + ' ~ model * augmentation_type'
maov3 = MANOVA.from_formula(formula, data=df_manova)
print(maov3.mv_test())
```

```
                     Multivariate linear model
=======================================================================

-----------------------------------------------------------------------
        Intercept          Value   Num DF  Den DF   F Value   Pr > F
-----------------------------------------------------------------------
          Wilks' lambda    0.0017 4.0000 297.0000 42399.6330 0.0000
         Pillai's trace    0.9983 4.0000 297.0000 42399.6330 0.0000
  Hotelling-Lawley trace 571.0388 4.0000 297.0000 42399.6330 0.0000
     Roy's greatest root 571.0388 4.0000 297.0000 42399.6330 0.0000
-----------------------------------------------------------------------

-----------------------------------------------------------------------
         model            Value    Num DF  Den DF   F Value   Pr > F
-----------------------------------------------------------------------
```

```
          Wilks' lambda   0.0001 12.0000 786.0796  1896.7015 0.0000
         Pillai's trace   2.2510 12.0000 897.0000   224.6345 0.0000
 Hotelling-Lawley trace 475.3388 12.0000 515.4541 11730.9296 0.0000
    Roy's greatest root 464.4037  4.0000 299.0000 34714.1754 0.0000
-------------------------------------------------------------------

-------------------------------------------------------------------
      augmentation_type    Value   Num DF   Den DF  F Value  Pr > F
-------------------------------------------------------------------
             Wilks' lambda 0.0221 40.0000 1128.0447  48.8563 0.0000
            Pillai's trace 2.4002 40.0000 1200.0000  45.0087 0.0000
    Hotelling-Lawley trace 7.5527 40.0000  819.4412  55.8374 0.0000
       Roy's greatest root 4.4835 10.0000  300.0000 134.5065 0.0000
-------------------------------------------------------------------

-------------------------------------------------------------------
   model:augmentation_type Value   Num DF    Den DF  F Value Pr > F
-------------------------------------------------------------------
             Wilks' lambda 0.0071 120.0000 1183.2579 24.4081 0.0000
            Pillai's trace 3.4426 120.0000 1200.0000 61.7649 0.0000
    Hotelling-Lawley trace 8.2202 120.0000 1016.8090 20.2479 0.0000
       Roy's greatest root 3.2635  30.0000  300.0000 32.6351 0.0000
===================================================================
```

[58]:
```
#Post-hoc Exploration: Which Metrics Are Responsible?

#If MANOVA is significant, run univariate ANOVA for each metric:


import statsmodels.api as sm
from statsmodels.formula.api import ols

for metric in metrics:
    model = ols(f'{metric} ~ model', data=df_manova).fit()
    anova_table = sm.stats.anova_lm(model, typ=2)
    print(f"ANOVA for {metric}:\n", anova_table, "\n")
```

```
ANOVA for levenshtein_similarity:
              sum_sq     df          F        PR(>F)
model      3.910936    3.0  26.939258  1.177641e-15
Residual  16.453291  340.0        NaN           NaN

ANOVA for jaccard_similarity:
              sum_sq     df          F        PR(>F)
model      2.841742    3.0  27.623562  5.199714e-16
Residual  11.659036  340.0        NaN           NaN
```

```
ANOVA for length_ratio:
              sum_sq     df         F    PR(>F)
model      1.649220    3.0  7.683006  0.000056
Residual  24.327932  340.0       NaN       NaN

ANOVA for bleu:
              sum_sq     df          F        PR(>F)
model      3.799517    3.0  32.509959  1.687200e-18
Residual  13.245540  340.0        NaN           NaN

ANOVA for cosine_similarity:
              sum_sq     df          F        PR(>F)
model      3.033348    3.0  26.071969  3.336757e-15
Residual  13.185790  340.0        NaN           NaN

ANOVA for wer:
              sum_sq     df          F        PR(>F)
model      4.486417    3.0  15.780305  1.249106e-09
Residual  32.221214  340.0        NaN           NaN

ANOVA for char_diversity:
             sum_sq     df        F        PR(>F)
model     0.773419    3.0  25.9462  3.882705e-15
Residual  3.378306  340.0      NaN           NaN

ANOVA for type_token_ratio:
             sum_sq     df           F        PR(>F)
model     0.456416    3.0  107.263221  6.956917e-49
Residual  0.482245  340.0         NaN           NaN

ANOVA for bigram_overlap:
              sum_sq     df          F        PR(>F)
model      3.389387    3.0  30.396176  1.967544e-17
Residual  12.637461  340.0        NaN           NaN

ANOVA for total_duration_ns:
                sum_sq     df           F        PR(>F)
model     3.833423e+21    3.0  114.62624  2.664217e-51
Residual  3.790185e+21  340.0        NaN           NaN

ANOVA for load_duration_ns:
                sum_sq     df         F    PR(>F)
model     6.248601e+18    3.0  0.399618  0.753364
Residual  1.772131e+21  340.0       NaN       NaN

ANOVA for prompt_eval_count:
                 sum_sq     df            F         PR(>F)
model     145085.989429    3.0  1572.665804  6.740158e-199
```

```
Residual   10455.545455  340.0          NaN            NaN

ANOVA for prompt_eval_duration_ns:
                 sum_sq     df           F         PR(>F)
model     1.914607e+20    3.0  825.644883  1.072120e-155
Residual  2.628112e+19  340.0         NaN            NaN

ANOVA for eval_count:
                 sum_sq     df          F        PR(>F)
model      7829.789852    3.0  19.476164  1.124172e-11
Residual  45562.163636  340.0        NaN           NaN

ANOVA for eval_duration_ns:
                 sum_sq     df           F        PR(>F)
model     2.324837e+21    3.0  158.921245  2.226701e-64
Residual  1.657938e+21  340.0         NaN           NaN

ANOVA for tokens_per_second:
                sum_sq     df            F  PR(>F)
model     1418.920496    3.0  97736.33144     0.0
Residual     1.645355  340.0          NaN     NaN
```

[59]:
```
# Interpreting Model-wise Effect Sizes
#1. Mean Difference (mean_diff):
#This is simply the difference in mean value of a metric between two models.

#Positive value: model_1 has a higher mean than model_2 for that metric.

#Negative value: model_2 has a higher mean.

#2. Cohen's d (cohens_d):
#Cohen's d quantifies the magnitude of the difference (in standard deviations)␣
 ↪between two distributions.

#Conventional thresholds for interpretation:

#    |d| < 0.2: Negligible difference (practically equivalent)

#    0.2  |d| < 0.5: Small effect (slight difference)

#     0.5  |d| < 0.8: Medium effect (moderate difference)

#   |d|  0.8: Large effect (substantial/practically meaningful difference)

# Sign of d:
```

```
# Positive: model_1 performs better (higher mean for positively oriented␣
 ↪metrics, e.g., similarity, tokens/sec).

# Negative: model_2 performs better.

# 3. Metric Orientation
# For similarity/accuracy metrics (bleu, levenshtein_similarity,␣
 ↪cosine_similarity, etc.): Higher is better.

# For error or duration metrics (wer, total_duration_ns, etc.): Lower is better.

# If Cohen's d is positive and the metric is an error/duration, model_1 has␣
 ↪worse performance.
```

[60]:
```
import pandas as pd
import numpy as np
from itertools import combinations
from scipy.stats import ttest_ind

def cohens_d(x, y):
    nx, ny = len(x), len(y)
    pooled_std = np.sqrt(((nx-1)*np.var(x, ddof=1) + (ny-1)*np.var(y, ddof=1)) /
 ↪ (nx + ny - 2))
    return (np.mean(x) - np.mean(y)) / pooled_std

metrics = [
    "total_duration_ns", "load_duration_ns", "prompt_eval_count",␣
 ↪"prompt_eval_duration_ns", "eval_count",
    "eval_duration_ns", "tokens_per_second", "levenshtein_similarity",␣
 ↪"jaccard_similarity", "length_ratio",
    "bleu", "cosine_similarity", "wer", "char_diversity", "type_token_ratio",␣
 ↪"bigram_overlap"
]

model_list = sorted(df['model'].unique())
results = []


for metric in metrics:
    for m1, m2 in combinations(model_list, 2):
        x = df[df['model'] == m1][metric].dropna()
        y = df[df['model'] == m2][metric].dropna()
        if len(x) > 1 and len(y) > 1:
            d = cohens_d(x, y)
            mean_diff = x.mean() - y.mean()
            t_stat, p_val = ttest_ind(x, y, equal_var=False)
            results.append({
```

```
                'metric': metric,
                'model_1': m1,
                'model_2': m2,
                'mean_model_1': x.mean(),
                'mean_model_2': y.mean(),
                'std_model_1': x.std(ddof=1),
                'std_model_2': y.std(ddof=1),
                'n_model_1': len(x),
                'n_model_2': len(y),
                'mean_diff': mean_diff,
                'cohens_d': d,
                'ttest_pval': p_val
            })

effectsize_df = pd.DataFrame(results)

pd.set_option('display.max_rows', None)
print("Table: Pairwise Model Comparisons Across All Metrics-Means, Standard␣
  ↪Deviations, Sample Sizes, Mean Differences, Cohen's d, and t-Test p-values")
display(effectsize_df)
```

Table: Pairwise Model Comparisons Across All Metrics-Means, Standard Deviations, Sample Sizes, Mean Differences, Cohen's d, and t-Test p-values

| | metric | model_1 | model_2 | \ |
|---|---|---|---|---|
| 0 | total_duration_ns | gemma3:1b | granite3.1-moe:1b | |
| 1 | total_duration_ns | gemma3:1b | qwen2:0.5b | |
| 2 | total_duration_ns | gemma3:1b | smollm2:360m | |
| 3 | total_duration_ns | granite3.1-moe:1b | qwen2:0.5b | |
| 4 | total_duration_ns | granite3.1-moe:1b | smollm2:360m | |
| 5 | total_duration_ns | qwen2:0.5b | smollm2:360m | |
| 6 | load_duration_ns | gemma3:1b | granite3.1-moe:1b | |
| 7 | load_duration_ns | gemma3:1b | qwen2:0.5b | |
| 8 | load_duration_ns | gemma3:1b | smollm2:360m | |
| 9 | load_duration_ns | granite3.1-moe:1b | qwen2:0.5b | |
| 10 | load_duration_ns | granite3.1-moe:1b | smollm2:360m | |
| 11 | load_duration_ns | qwen2:0.5b | smollm2:360m | |
| 12 | prompt_eval_count | gemma3:1b | granite3.1-moe:1b | |
| 13 | prompt_eval_count | gemma3:1b | qwen2:0.5b | |
| 14 | prompt_eval_count | gemma3:1b | smollm2:360m | |
| 15 | prompt_eval_count | granite3.1-moe:1b | qwen2:0.5b | |
| 16 | prompt_eval_count | granite3.1-moe:1b | smollm2:360m | |
| 17 | prompt_eval_count | qwen2:0.5b | smollm2:360m | |
| 18 | prompt_eval_duration_ns | gemma3:1b | granite3.1-moe:1b | |
| 19 | prompt_eval_duration_ns | gemma3:1b | qwen2:0.5b | |
| 20 | prompt_eval_duration_ns | gemma3:1b | smollm2:360m | |
| 21 | prompt_eval_duration_ns | granite3.1-moe:1b | qwen2:0.5b | |
| 22 | prompt_eval_duration_ns | granite3.1-moe:1b | smollm2:360m | |

```
prompt_eval_duration_ns         qwen2:0.5b       smollm2:360m
              eval_count          gemma3:1b  granite3.1-moe:1b
              eval_count          gemma3:1b         qwen2:0.5b
              eval_count          gemma3:1b       smollm2:360m
              eval_count  granite3.1-moe:1b         qwen2:0.5b
              eval_count  granite3.1-moe:1b       smollm2:360m
              eval_count         qwen2:0.5b       smollm2:360m
       eval_duration_ns          gemma3:1b  granite3.1-moe:1b
       eval_duration_ns          gemma3:1b         qwen2:0.5b
       eval_duration_ns          gemma3:1b       smollm2:360m
       eval_duration_ns  granite3.1-moe:1b         qwen2:0.5b
       eval_duration_ns  granite3.1-moe:1b       smollm2:360m
       eval_duration_ns         qwen2:0.5b       smollm2:360m
      tokens_per_second          gemma3:1b  granite3.1-moe:1b
      tokens_per_second          gemma3:1b         qwen2:0.5b
      tokens_per_second          gemma3:1b       smollm2:360m
      tokens_per_second  granite3.1-moe:1b         qwen2:0.5b
      tokens_per_second  granite3.1-moe:1b       smollm2:360m
      tokens_per_second         qwen2:0.5b       smollm2:360m
 levenshtein_similarity          gemma3:1b  granite3.1-moe:1b
 levenshtein_similarity          gemma3:1b         qwen2:0.5b
 levenshtein_similarity          gemma3:1b       smollm2:360m
 levenshtein_similarity  granite3.1-moe:1b         qwen2:0.5b
 levenshtein_similarity  granite3.1-moe:1b       smollm2:360m
 levenshtein_similarity         qwen2:0.5b       smollm2:360m
     jaccard_similarity          gemma3:1b  granite3.1-moe:1b
     jaccard_similarity          gemma3:1b         qwen2:0.5b
     jaccard_similarity          gemma3:1b       smollm2:360m
     jaccard_similarity  granite3.1-moe:1b         qwen2:0.5b
     jaccard_similarity  granite3.1-moe:1b       smollm2:360m
     jaccard_similarity         qwen2:0.5b       smollm2:360m
           length_ratio          gemma3:1b  granite3.1-moe:1b
           length_ratio          gemma3:1b         qwen2:0.5b
           length_ratio          gemma3:1b       smollm2:360m
           length_ratio  granite3.1-moe:1b         qwen2:0.5b
           length_ratio  granite3.1-moe:1b       smollm2:360m
           length_ratio         qwen2:0.5b       smollm2:360m
                  bleu          gemma3:1b  granite3.1-moe:1b
                  bleu          gemma3:1b         qwen2:0.5b
                  bleu          gemma3:1b       smollm2:360m
                  bleu  granite3.1-moe:1b         qwen2:0.5b
                  bleu  granite3.1-moe:1b       smollm2:360m
                  bleu         qwen2:0.5b       smollm2:360m
       cosine_similarity          gemma3:1b  granite3.1-moe:1b
       cosine_similarity          gemma3:1b         qwen2:0.5b
       cosine_similarity          gemma3:1b       smollm2:360m
       cosine_similarity  granite3.1-moe:1b         qwen2:0.5b
       cosine_similarity  granite3.1-moe:1b       smollm2:360m
```

| | | | |
|---|---|---|---|
| 71 | cosine_similarity | qwen2:0.5b | smollm2:360m |
| 72 | wer | gemma3:1b | granite3.1-moe:1b |
| 73 | wer | gemma3:1b | qwen2:0.5b |
| 74 | wer | gemma3:1b | smollm2:360m |
| 75 | wer | granite3.1-moe:1b | qwen2:0.5b |
| 76 | wer | granite3.1-moe:1b | smollm2:360m |
| 77 | wer | qwen2:0.5b | smollm2:360m |
| 78 | char_diversity | gemma3:1b | granite3.1-moe:1b |
| 79 | char_diversity | gemma3:1b | qwen2:0.5b |
| 80 | char_diversity | gemma3:1b | smollm2:360m |
| 81 | char_diversity | granite3.1-moe:1b | qwen2:0.5b |
| 82 | char_diversity | granite3.1-moe:1b | smollm2:360m |
| 83 | char_diversity | qwen2:0.5b | smollm2:360m |
| 84 | type_token_ratio | gemma3:1b | granite3.1-moe:1b |
| 85 | type_token_ratio | gemma3:1b | qwen2:0.5b |
| 86 | type_token_ratio | gemma3:1b | smollm2:360m |
| 87 | type_token_ratio | granite3.1-moe:1b | qwen2:0.5b |
| 88 | type_token_ratio | granite3.1-moe:1b | smollm2:360m |
| 89 | type_token_ratio | qwen2:0.5b | smollm2:360m |
| 90 | bigram_overlap | gemma3:1b | granite3.1-moe:1b |
| 91 | bigram_overlap | gemma3:1b | qwen2:0.5b |
| 92 | bigram_overlap | gemma3:1b | smollm2:360m |
| 93 | bigram_overlap | granite3.1-moe:1b | qwen2:0.5b |
| 94 | bigram_overlap | granite3.1-moe:1b | smollm2:360m |
| 95 | bigram_overlap | qwen2:0.5b | smollm2:360m |

| | mean_model_1 | mean_model_2 | std_model_1 | std_model_2 | n_model_1 | \ |
|---|---|---|---|---|---|---|
| 0 | 1.637960e+10 | 1.051084e+10 | 3.626593e+09 | 3.985375e+09 | 88 | |
| 1 | 1.637960e+10 | 7.160551e+09 | 3.626593e+09 | 1.498693e+09 | 88 | |
| 2 | 1.637960e+10 | 1.105877e+10 | 3.626593e+09 | 3.707344e+09 | 88 | |
| 3 | 1.051084e+10 | 7.160551e+09 | 3.985375e+09 | 1.498693e+09 | 80 | |
| 4 | 1.051084e+10 | 1.105877e+10 | 3.985375e+09 | 3.707344e+09 | 80 | |
| 5 | 7.160551e+09 | 1.105877e+10 | 1.498693e+09 | 3.707344e+09 | 88 | |
| 6 | 3.576625e+08 | 4.339573e+08 | 2.094949e+09 | 3.690988e+09 | 88 | |
| 7 | 3.576625e+08 | 7.873249e+07 | 2.094949e+09 | 2.526810e+08 | 88 | |
| 8 | 3.576625e+08 | 2.220176e+08 | 2.094949e+09 | 1.883081e+09 | 88 | |
| 9 | 4.339573e+08 | 7.873249e+07 | 3.690988e+09 | 2.526810e+08 | 80 | |
| 10 | 4.339573e+08 | 2.220176e+08 | 3.690988e+09 | 1.883081e+09 | 80 | |
| 11 | 7.873249e+07 | 2.220176e+08 | 2.526810e+08 | 1.883081e+09 | 88 | |
| 12 | 6.051136e+01 | 1.113750e+02 | 4.651027e+00 | 7.350837e+00 | 88 | |
| 13 | 6.051136e+01 | 6.019318e+01 | 4.651027e+00 | 4.977785e+00 | 88 | |
| 14 | 6.051136e+01 | 8.168182e+01 | 4.651027e+00 | 4.970130e+00 | 88 | |
| 15 | 1.113750e+02 | 6.019318e+01 | 7.350837e+00 | 4.977785e+00 | 80 | |
| 16 | 1.113750e+02 | 8.168182e+01 | 7.350837e+00 | 4.970130e+00 | 80 | |
| 17 | 6.019318e+01 | 8.168182e+01 | 4.977785e+00 | 4.970130e+00 | 88 | |
| 18 | 3.247818e+09 | 1.986857e+09 | 3.790890e+08 | 3.220527e+08 | 88 | |
| 19 | 3.247818e+09 | 1.438463e+09 | 3.790890e+08 | 1.865101e+08 | 88 | |
| 20 | 3.247818e+09 | 1.446045e+09 | 3.790890e+08 | 1.714840e+08 | 88 | |

| | | | | | |
|---|---|---|---|---|---|
| 21 | 1.986857e+09 | 1.438463e+09 | 3.220527e+08 | 1.865101e+08 | 80 |
| 22 | 1.986857e+09 | 1.446045e+09 | 3.220527e+08 | 1.714840e+08 | 80 |
| 23 | 1.438463e+09 | 1.446045e+09 | 1.865101e+08 | 1.714840e+08 | 88 |
| 24 | 5.450000e+01 | 5.842500e+01 | 1.127911e+01 | 5.204124e+00 | 88 |
| 25 | 5.450000e+01 | 5.244318e+01 | 1.127911e+01 | 1.289735e+01 | 88 |
| 26 | 5.450000e+01 | 4.521591e+01 | 1.127911e+01 | 1.433703e+01 | 88 |
| 27 | 5.842500e+01 | 5.244318e+01 | 5.204124e+00 | 1.289735e+01 | 80 |
| 28 | 5.842500e+01 | 4.521591e+01 | 5.204124e+00 | 1.433703e+01 | 80 |
| 29 | 5.244318e+01 | 4.521591e+01 | 1.289735e+01 | 1.433703e+01 | 88 |
| 30 | 1.277295e+10 | 8.088102e+09 | 2.695345e+09 | 7.398549e+08 | 88 |
| 31 | 1.277295e+10 | 5.641965e+09 | 2.695345e+09 | 1.415511e+09 | 88 |
| 32 | 1.277295e+10 | 9.388878e+09 | 2.695345e+09 | 3.048139e+09 | 88 |
| 33 | 8.088102e+09 | 5.641965e+09 | 7.398549e+08 | 1.415511e+09 | 80 |
| 34 | 8.088102e+09 | 9.388878e+09 | 7.398549e+08 | 3.048139e+09 | 80 |
| 35 | 5.641965e+09 | 9.388878e+09 | 1.415511e+09 | 3.048139e+09 | 88 |
| 36 | 4.273367e+00 | 7.226096e+00 | 3.561606e-02 | 4.246905e-02 | 88 |
| 37 | 4.273367e+00 | 9.319944e+00 | 3.561606e-02 | 1.127637e-01 | 88 |
| 38 | 4.273367e+00 | 4.833095e+00 | 3.561606e-02 | 5.736037e-02 | 88 |
| 39 | 7.226096e+00 | 9.319944e+00 | 4.246905e-02 | 1.127637e-01 | 80 |
| 40 | 7.226096e+00 | 4.833095e+00 | 4.246905e-02 | 5.736037e-02 | 80 |
| 41 | 9.319944e+00 | 4.833095e+00 | 1.127637e-01 | 5.736037e-02 | 88 |
| 42 | 3.679158e-01 | 9.707118e-02 | 3.630858e-01 | 1.097015e-01 | 88 |
| 43 | 3.679158e-01 | 1.198860e-01 | 3.630858e-01 | 1.267990e-01 | 88 |
| 44 | 3.679158e-01 | 1.812696e-01 | 3.630858e-01 | 1.740149e-01 | 88 |
| 45 | 9.707118e-02 | 1.198860e-01 | 1.097015e-01 | 1.267990e-01 | 80 |
| 46 | 9.707118e-02 | 1.812696e-01 | 1.097015e-01 | 1.740149e-01 | 80 |
| 47 | 1.198860e-01 | 1.812696e-01 | 1.267990e-01 | 1.740149e-01 | 88 |
| 48 | 3.546248e-01 | 1.860765e-01 | 3.107514e-01 | 1.117166e-01 | 88 |
| 49 | 3.546248e-01 | 1.367926e-01 | 3.107514e-01 | 8.982222e-02 | 88 |
| 50 | 3.546248e-01 | 1.347398e-01 | 3.107514e-01 | 1.343297e-01 | 88 |
| 51 | 1.860765e-01 | 1.367926e-01 | 1.117166e-01 | 8.982222e-02 | 80 |
| 52 | 1.860765e-01 | 1.347398e-01 | 1.117166e-01 | 1.343297e-01 | 80 |
| 53 | 1.367926e-01 | 1.347398e-01 | 8.982222e-02 | 1.343297e-01 | 88 |
| 54 | 1.095413e+00 | 1.026081e+00 | 2.090215e-01 | 1.511895e-01 | 88 |
| 55 | 1.095413e+00 | 1.198417e+00 | 2.090215e-01 | 3.287786e-01 | 88 |
| 56 | 1.095413e+00 | 1.033046e+00 | 2.090215e-01 | 3.272457e-01 | 88 |
| 57 | 1.026081e+00 | 1.198417e+00 | 1.511895e-01 | 3.287786e-01 | 80 |
| 58 | 1.026081e+00 | 1.033046e+00 | 1.511895e-01 | 3.272457e-01 | 80 |
| 59 | 1.198417e+00 | 1.033046e+00 | 3.287786e-01 | 3.272457e-01 | 88 |
| 60 | 3.051032e-01 | 7.407307e-02 | 3.461640e-01 | 9.534950e-02 | 88 |
| 61 | 3.051032e-01 | 5.953811e-02 | 3.461640e-01 | 8.590007e-02 | 88 |
| 62 | 3.051032e-01 | 6.096194e-02 | 3.461640e-01 | 1.295520e-01 | 88 |
| 63 | 7.407307e-02 | 5.953811e-02 | 9.534950e-02 | 8.590007e-02 | 80 |
| 64 | 7.407307e-02 | 6.096194e-02 | 9.534950e-02 | 1.295520e-01 | 80 |
| 65 | 5.953811e-02 | 6.096194e-02 | 8.590007e-02 | 1.295520e-01 | 88 |
| 66 | 5.728996e-01 | 3.985247e-01 | 2.586751e-01 | 1.549181e-01 | 88 |
| 67 | 5.728996e-01 | 3.658572e-01 | 2.586751e-01 | 1.736955e-01 | 88 |
| 68 | 5.728996e-01 | 3.319037e-01 | 2.586751e-01 | 1.807902e-01 | 88 |

| | | | | | |
|---|---|---|---|---|---|
| 69 | 3.985247e-01 | 3.658572e-01 | 1.549181e-01 | 1.736955e-01 | 80 |
| 70 | 3.985247e-01 | 3.319037e-01 | 1.549181e-01 | 1.807902e-01 | 80 |
| 71 | 3.658572e-01 | 3.319037e-01 | 1.736955e-01 | 1.807902e-01 | 88 |
| 72 | 8.781957e-01 | 9.144531e-01 | 4.386598e-01 | 2.030163e-01 | 88 |
| 73 | 8.781957e-01 | 1.165567e+00 | 4.386598e-01 | 2.397264e-01 | 88 |
| 74 | 8.781957e-01 | 1.046012e+00 | 4.386598e-01 | 2.881699e-01 | 88 |
| 75 | 9.144531e-01 | 1.165567e+00 | 2.030163e-01 | 2.397264e-01 | 80 |
| 76 | 9.144531e-01 | 1.046012e+00 | 2.030163e-01 | 2.881699e-01 | 80 |
| 77 | 1.165567e+00 | 1.046012e+00 | 2.397264e-01 | 2.881699e-01 | 88 |
| 78 | 2.378138e-01 | 1.262054e-01 | 1.270707e-01 | 8.308520e-02 | 88 |
| 79 | 2.378138e-01 | 1.519883e-01 | 1.270707e-01 | 9.223367e-02 | 88 |
| 80 | 2.378138e-01 | 1.196035e-01 | 1.270707e-01 | 8.893092e-02 | 88 |
| 81 | 1.262054e-01 | 1.519883e-01 | 8.308520e-02 | 9.223367e-02 | 80 |
| 82 | 1.262054e-01 | 1.196035e-01 | 8.308520e-02 | 8.893092e-02 | 80 |
| 83 | 1.519883e-01 | 1.196035e-01 | 9.223367e-02 | 8.893092e-02 | 88 |
| 84 | 8.790624e-01 | 9.697264e-01 | 5.318228e-02 | 2.606863e-02 | 88 |
| 85 | 8.790624e-01 | 9.547864e-01 | 5.318228e-02 | 3.235066e-02 | 88 |
| 86 | 8.790624e-01 | 9.612689e-01 | 5.318228e-02 | 3.241982e-02 | 88 |
| 87 | 9.697264e-01 | 9.547864e-01 | 2.606863e-02 | 3.235066e-02 | 80 |
| 88 | 9.697264e-01 | 9.612689e-01 | 2.606863e-02 | 3.241982e-02 | 80 |
| 89 | 9.547864e-01 | 9.612689e-01 | 3.235066e-02 | 3.241982e-02 | 88 |
| 90 | 2.815986e-01 | 6.352941e-02 | 3.447873e-01 | 8.072161e-02 | 88 |
| 91 | 2.815986e-01 | 4.801648e-02 | 3.447873e-01 | 6.967926e-02 | 88 |
| 92 | 2.815986e-01 | 5.264021e-02 | 3.447873e-01 | 1.249316e-01 | 88 |
| 93 | 6.352941e-02 | 4.801648e-02 | 8.072161e-02 | 6.967926e-02 | 80 |
| 94 | 6.352941e-02 | 5.264021e-02 | 8.072161e-02 | 1.249316e-01 | 80 |
| 95 | 4.801648e-02 | 5.264021e-02 | 6.967926e-02 | 1.249316e-01 | 88 |

| | n_model_2 | mean_diff | cohens_d | ttest_pval |
|---|---|---|---|---|
| 0 | 80 | 5.868757e+09 | 1.543774 | 1.851807e-18 |
| 1 | 88 | 9.219049e+09 | 3.322503 | 2.961777e-43 |
| 2 | 88 | 5.320826e+09 | 1.450927 | 7.612526e-18 |
| 3 | 88 | 3.350292e+09 | 1.133510 | 2.123763e-10 |
| 4 | 88 | -5.479307e+08 | -0.142610 | 3.589552e-01 |
| 5 | 88 | -3.898222e+09 | -1.378640 | 2.599088e-15 |
| 6 | 80 | -7.629477e+07 | -0.025743 | 8.711013e-01 |
| 7 | 88 | 2.789301e+08 | 0.186939 | 2.182100e-01 |
| 8 | 88 | 1.356449e+08 | 0.068101 | 6.520340e-01 |
| 9 | 88 | 3.552248e+08 | 0.139150 | 3.929292e-01 |
| 10 | 88 | 2.119397e+08 | 0.073381 | 6.450656e-01 |
| 11 | 88 | -1.432851e+08 | -0.106653 | 4.811095e-01 |
| 12 | 80 | -5.086364e+01 | -8.355988 | 1.886816e-90 |
| 13 | 88 | 3.181818e-01 | 0.066052 | 6.618335e-01 |
| 14 | 88 | -2.117045e+01 | -4.398394 | 9.215735e-69 |
| 15 | 88 | 5.118182e+01 | 8.227185 | 2.036954e-92 |
| 16 | 88 | 2.969318e+01 | 4.775472 | 1.143564e-62 |
| 17 | 88 | -2.148864e+01 | -4.320228 | 8.524955e-68 |
| 18 | 80 | 1.260961e+09 | 3.571155 | 4.332445e-54 |

| | | | | |
|---|---|---|---|---|
| 19 | 88 | 1.809355e+09 | 6.056565 | 5.553946e-74 |
| 20 | 88 | 1.801773e+09 | 6.124173 | 1.976495e-72 |
| 21 | 88 | 5.483942e+08 | 2.109349 | 1.038791e-25 |
| 22 | 88 | 5.408122e+08 | 2.124975 | 1.991987e-25 |
| 23 | 88 | -7.582020e+06 | -0.042321 | 7.792561e-01 |
| 24 | 80 | -3.925000e+00 | -0.440030 | 3.929526e-03 |
| 25 | 88 | 2.056818e+00 | 0.169771 | 2.616884e-01 |
| 26 | 88 | 9.284091e+00 | 0.719752 | 3.956313e-06 |
| 27 | 88 | 5.981818e+00 | 0.597979 | 1.086813e-04 |
| 28 | 88 | 1.320909e+01 | 1.202731 | 8.598580e-13 |
| 29 | 88 | 7.227273e+00 | 0.530006 | 5.608076e-04 |
| 30 | 80 | 4.684852e+09 | 2.322765 | 8.042706e-29 |
| 31 | 88 | 7.130989e+09 | 3.312522 | 7.247881e-46 |
| 32 | 88 | 3.384076e+09 | 1.176189 | 5.715397e-13 |
| 33 | 88 | 2.446137e+09 | 2.136695 | 1.594519e-28 |
| 34 | 88 | -1.300776e+09 | -0.574309 | 1.895015e-04 |
| 35 | 88 | -3.746913e+09 | -1.576699 | 1.068502e-18 |
| 36 | 80 | -2.952729e+00 | -75.657094 | 2.351837e-248 |
| 37 | 88 | -5.046577e+00 | -60.352309 | 7.143902e-168 |
| 38 | 88 | -5.597273e-01 | -11.723850 | 2.343432e-120 |
| 39 | 88 | -2.093848e+00 | -24.141375 | 7.374049e-136 |
| 40 | 88 | 2.393001e+00 | 47.087189 | 1.213217e-223 |
| 41 | 88 | 4.486849e+00 | 50.155324 | 2.266405e-191 |
| 42 | 80 | 2.708446e-01 | 0.990177 | 1.254957e-09 |
| 43 | 88 | 2.480298e-01 | 0.912055 | 2.128155e-08 |
| 44 | 88 | 1.866462e-01 | 0.655580 | 2.813334e-05 |
| 45 | 88 | -2.281481e-02 | -0.191771 | 2.130662e-01 |
| 46 | 88 | -8.419844e-02 | -0.572931 | 2.215139e-04 |
| 47 | 88 | -6.138362e-02 | -0.403181 | 8.268438e-03 |
| 48 | 80 | 1.685483e-01 | 0.708777 | 5.856547e-06 |
| 49 | 88 | 2.178322e-01 | 0.952357 | 7.156117e-09 |
| 50 | 88 | 2.198850e-01 | 0.918539 | 1.420898e-08 |
| 51 | 88 | 4.928390e-02 | 0.488751 | 2.087465e-03 |
| 52 | 88 | 5.133669e-02 | 0.413728 | 7.620339e-03 |
| 53 | 88 | 2.052788e-03 | 0.017965 | 9.052994e-01 |
| 54 | 80 | 6.933222e-02 | 0.377251 | 1.422422e-02 |
| 55 | 88 | -1.030037e-01 | -0.373898 | 1.425727e-02 |
| 56 | 88 | 6.236747e-02 | 0.227144 | 1.340200e-01 |
| 57 | 88 | -1.723359e-01 | -0.663170 | 2.045843e-05 |
| 58 | 88 | -6.964743e-03 | -0.026906 | 8.577026e-01 |
| 59 | 88 | 1.653711e-01 | 0.504160 | 1.010546e-03 |
| 60 | 80 | 2.310301e-01 | 0.891690 | 2.877416e-08 |
| 61 | 88 | 2.455651e-01 | 0.973697 | 4.123433e-09 |
| 62 | 88 | 2.441413e-01 | 0.934135 | 1.008213e-08 |
| 63 | 88 | 1.453496e-02 | 0.160571 | 3.025711e-01 |
| 64 | 88 | 1.311113e-02 | 0.114452 | 4.534489e-01 |
| 65 | 88 | -1.423834e-03 | -0.012954 | 9.316383e-01 |
| 66 | 80 | 1.743749e-01 | 0.808729 | 3.295645e-07 |

```
67          88  2.070424e-01   0.939730   4.274439e-09
68          88  2.409959e-01   1.079940   2.947308e-11
69          88  3.266754e-02   0.197954   1.993847e-01
70          88  6.662102e-02   0.394272   1.101874e-02
71          88  3.395348e-02   0.191526   2.056260e-01
72          80 -3.625732e-02  -0.104465   4.867596e-01
73          88 -2.873716e-01  -0.812987   3.021430e-07
74          88 -1.678158e-01  -0.452184   3.166699e-03
75          88 -2.511143e-01  -1.126018   8.851207e-12
76          88 -1.315585e-01  -0.523574   7.346167e-04
77          88  1.195558e-01   0.451057   3.188480e-03
78          80  1.116084e-01   1.029719   2.334366e-10
79          88  8.582543e-02   0.773014   8.445384e-07
80          88  1.182103e-01   1.077858   3.172278e-11
81          88 -2.578296e-02  -0.292993   5.836048e-02
82          88  6.601907e-03   0.076590   6.195587e-01
83          88  3.238487e-02   0.357459   1.882925e-02
84          80 -9.066401e-02  -2.133571   3.319137e-28
85          88 -7.572400e-02  -1.720351   7.128313e-22
86          88 -8.220651e-02  -1.866547   2.014489e-24
87          88  1.494001e-02   0.505958   1.152241e-03
88          88  8.457504e-03   0.286037   6.316345e-02
89          88 -6.482510e-03  -0.200169   1.859940e-01
90          80  2.180692e-01   0.852688   9.730077e-08
91          88  2.335821e-01   0.939098   1.306125e-08
92          88  2.289584e-01   0.882943   5.028529e-08
93          88  1.551292e-02   0.206462   1.863744e-01
94          88  1.088920e-02   0.102523   4.995303e-01
95          88 -4.623723e-03  -0.045711   7.621899e-01
```

[61]:
```
#What does Cliff's Delta calculate?
#It measures the probability that a randomly selected value from group A will␣
 ↪be larger than a randomly selected value from group B, minus the reverse␣
 ↪probability.

#Values range from -1 (all values in A are less than all in B) to +1 (all␣
 ↪values in A are greater than all in B).

#0 means the two groups completely overlap (no effect).

#Positive values: A tends to be greater than B.

#Negative values: A tends to be less than B.


# Interpretation
```

```
# Large effect (|delta|  0.474): Models are substantially different for this␣
 ↪metric

# Medium/small effect: Moderate/small practical differences

# Negligible: Distributions largely overlap
```

[62]:
```
import pandas as pd
import numpy as np
from itertools import combinations

# --- Pure Python Cliff's Delta implementation ---
def cliffs_delta(x, y):
    """
    Computes Cliff's delta and magnitude for two arrays/lists x, y.
    Returns delta (float) and magnitude (str: 'negligible', 'small', 'medium',␣
 ↪'large')
    """
    x, y = np.asarray(x), np.asarray(y)
    n_x, n_y = len(x), len(y)
    more = sum(xi > yj for xi in x for yj in y)
    less = sum(xi < yj for xi in x for yj in y)
    delta = (more - less) / (n_x * n_y)
    # Magnitude thresholds (Romano et al., 2006)
    adelta = abs(delta)
    if adelta < 0.147:
        magnitude = "negligible"
    elif adelta < 0.33:
        magnitude = "small"
    elif adelta < 0.474:
        magnitude = "medium"
    else:
        magnitude = "large"
    return delta, magnitude



# --- Specify metrics for which to compute Cliff's delta ---
metrics = [
    "total_duration_ns", "load_duration_ns", "prompt_eval_count",␣
 ↪"prompt_eval_duration_ns", "eval_count",
    "eval_duration_ns", "tokens_per_second", "levenshtein_similarity",␣
 ↪"jaccard_similarity", "length_ratio",
    "bleu", "cosine_similarity", "wer", "char_diversity", "type_token_ratio",␣
 ↪"bigram_overlap"
]
```

```python
# --- Get unique models ---
model_list = sorted(df['model'].unique())

# --- Compute Cliff's delta for all pairs and all metrics ---
results = []
for metric in metrics:
    for m1, m2 in combinations(model_list, 2):
        x = df[df['model'] == m1][metric].dropna()
        y = df[df['model'] == m2][metric].dropna()
        if len(x) > 1 and len(y) > 1:
            delta, magnitude = cliffs_delta(x, y)
            results.append({
                'metric': metric,
                'model_1': m1,
                'model_2': m2,
                "cliffs_delta": delta,
                "magnitude": magnitude,
                "mean_model_1": x.mean(),
                "mean_model_2": y.mean(),
                "n_model_1": len(x),
                "n_model_2": len(y)
            })

cliffs_df = pd.DataFrame(results)

# Show all rows (optional, for Jupyter)
pd.set_option("display.max_rows", None)
print("Table: Cliff's Delta Effect Sizes for Pairwise Model Comparisons Across␣
  ↪All Metrics")
display(cliffs_df)
```

Table: Cliff's Delta Effect Sizes for Pairwise Model Comparisons Across All Metrics

| | metric | model_1 | model_2 | \ |
|---|---|---|---|---|
| 0 | total_duration_ns | gemma3:1b | granite3.1-moe:1b | |
| 1 | total_duration_ns | gemma3:1b | qwen2:0.5b | |
| 2 | total_duration_ns | gemma3:1b | smollm2:360m | |
| 3 | total_duration_ns | granite3.1-moe:1b | qwen2:0.5b | |
| 4 | total_duration_ns | granite3.1-moe:1b | smollm2:360m | |
| 5 | total_duration_ns | qwen2:0.5b | smollm2:360m | |
| 6 | load_duration_ns | gemma3:1b | granite3.1-moe:1b | |
| 7 | load_duration_ns | gemma3:1b | qwen2:0.5b | |
| 8 | load_duration_ns | gemma3:1b | smollm2:360m | |
| 9 | load_duration_ns | granite3.1-moe:1b | qwen2:0.5b | |
| 10 | load_duration_ns | granite3.1-moe:1b | smollm2:360m | |
| 11 | load_duration_ns | qwen2:0.5b | smollm2:360m | |
| 12 | prompt_eval_count | gemma3:1b | granite3.1-moe:1b | |

| | | | |
|---|---|---|---|
| 13 | prompt_eval_count | gemma3:1b | qwen2:0.5b |
| 14 | prompt_eval_count | gemma3:1b | smollm2:360m |
| 15 | prompt_eval_count | granite3.1-moe:1b | qwen2:0.5b |
| 16 | prompt_eval_count | granite3.1-moe:1b | smollm2:360m |
| 17 | prompt_eval_count | qwen2:0.5b | smollm2:360m |
| 18 | prompt_eval_duration_ns | gemma3:1b | granite3.1-moe:1b |
| 19 | prompt_eval_duration_ns | gemma3:1b | qwen2:0.5b |
| 20 | prompt_eval_duration_ns | gemma3:1b | smollm2:360m |
| 21 | prompt_eval_duration_ns | granite3.1-moe:1b | qwen2:0.5b |
| 22 | prompt_eval_duration_ns | granite3.1-moe:1b | smollm2:360m |
| 23 | prompt_eval_duration_ns | qwen2:0.5b | smollm2:360m |
| 24 | eval_count | gemma3:1b | granite3.1-moe:1b |
| 25 | eval_count | gemma3:1b | qwen2:0.5b |
| 26 | eval_count | gemma3:1b | smollm2:360m |
| 27 | eval_count | granite3.1-moe:1b | qwen2:0.5b |
| 28 | eval_count | granite3.1-moe:1b | smollm2:360m |
| 29 | eval_count | qwen2:0.5b | smollm2:360m |
| 30 | eval_duration_ns | gemma3:1b | granite3.1-moe:1b |
| 31 | eval_duration_ns | gemma3:1b | qwen2:0.5b |
| 32 | eval_duration_ns | gemma3:1b | smollm2:360m |
| 33 | eval_duration_ns | granite3.1-moe:1b | qwen2:0.5b |
| 34 | eval_duration_ns | granite3.1-moe:1b | smollm2:360m |
| 35 | eval_duration_ns | qwen2:0.5b | smollm2:360m |
| 36 | tokens_per_second | gemma3:1b | granite3.1-moe:1b |
| 37 | tokens_per_second | gemma3:1b | qwen2:0.5b |
| 38 | tokens_per_second | gemma3:1b | smollm2:360m |
| 39 | tokens_per_second | granite3.1-moe:1b | qwen2:0.5b |
| 40 | tokens_per_second | granite3.1-moe:1b | smollm2:360m |
| 41 | tokens_per_second | qwen2:0.5b | smollm2:360m |
| 42 | levenshtein_similarity | gemma3:1b | granite3.1-moe:1b |
| 43 | levenshtein_similarity | gemma3:1b | qwen2:0.5b |
| 44 | levenshtein_similarity | gemma3:1b | smollm2:360m |
| 45 | levenshtein_similarity | granite3.1-moe:1b | qwen2:0.5b |
| 46 | levenshtein_similarity | granite3.1-moe:1b | smollm2:360m |
| 47 | levenshtein_similarity | qwen2:0.5b | smollm2:360m |
| 48 | jaccard_similarity | gemma3:1b | granite3.1-moe:1b |
| 49 | jaccard_similarity | gemma3:1b | qwen2:0.5b |
| 50 | jaccard_similarity | gemma3:1b | smollm2:360m |
| 51 | jaccard_similarity | granite3.1-moe:1b | qwen2:0.5b |
| 52 | jaccard_similarity | granite3.1-moe:1b | smollm2:360m |
| 53 | jaccard_similarity | qwen2:0.5b | smollm2:360m |
| 54 | length_ratio | gemma3:1b | granite3.1-moe:1b |
| 55 | length_ratio | gemma3:1b | qwen2:0.5b |
| 56 | length_ratio | gemma3:1b | smollm2:360m |
| 57 | length_ratio | granite3.1-moe:1b | qwen2:0.5b |
| 58 | length_ratio | granite3.1-moe:1b | smollm2:360m |
| 59 | length_ratio | qwen2:0.5b | smollm2:360m |
| 60 | bleu | gemma3:1b | granite3.1-moe:1b |

| | | | | |
|---|---|---|---|---|
| 61 | bleu | gemma3:1b | qwen2:0.5b | |
| 62 | bleu | gemma3:1b | smollm2:360m | |
| 63 | bleu | granite3.1-moe:1b | qwen2:0.5b | |
| 64 | bleu | granite3.1-moe:1b | smollm2:360m | |
| 65 | bleu | qwen2:0.5b | smollm2:360m | |
| 66 | cosine_similarity | gemma3:1b | granite3.1-moe:1b | |
| 67 | cosine_similarity | gemma3:1b | qwen2:0.5b | |
| 68 | cosine_similarity | gemma3:1b | smollm2:360m | |
| 69 | cosine_similarity | granite3.1-moe:1b | qwen2:0.5b | |
| 70 | cosine_similarity | granite3.1-moe:1b | smollm2:360m | |
| 71 | cosine_similarity | qwen2:0.5b | smollm2:360m | |
| 72 | wer | gemma3:1b | granite3.1-moe:1b | |
| 73 | wer | gemma3:1b | qwen2:0.5b | |
| 74 | wer | gemma3:1b | smollm2:360m | |
| 75 | wer | granite3.1-moe:1b | qwen2:0.5b | |
| 76 | wer | granite3.1-moe:1b | smollm2:360m | |
| 77 | wer | qwen2:0.5b | smollm2:360m | |
| 78 | char_diversity | gemma3:1b | granite3.1-moe:1b | |
| 79 | char_diversity | gemma3:1b | qwen2:0.5b | |
| 80 | char_diversity | gemma3:1b | smollm2:360m | |
| 81 | char_diversity | granite3.1-moe:1b | qwen2:0.5b | |
| 82 | char_diversity | granite3.1-moe:1b | smollm2:360m | |
| 83 | char_diversity | qwen2:0.5b | smollm2:360m | |
| 84 | type_token_ratio | gemma3:1b | granite3.1-moe:1b | |
| 85 | type_token_ratio | gemma3:1b | qwen2:0.5b | |
| 86 | type_token_ratio | gemma3:1b | smollm2:360m | |
| 87 | type_token_ratio | granite3.1-moe:1b | qwen2:0.5b | |
| 88 | type_token_ratio | granite3.1-moe:1b | smollm2:360m | |
| 89 | type_token_ratio | qwen2:0.5b | smollm2:360m | |
| 90 | bigram_overlap | gemma3:1b | granite3.1-moe:1b | |
| 91 | bigram_overlap | gemma3:1b | qwen2:0.5b | |
| 92 | bigram_overlap | gemma3:1b | smollm2:360m | |
| 93 | bigram_overlap | granite3.1-moe:1b | qwen2:0.5b | |
| 94 | bigram_overlap | granite3.1-moe:1b | smollm2:360m | |
| 95 | bigram_overlap | qwen2:0.5b | smollm2:360m | |

| | cliffs_delta | magnitude | mean_model_1 | mean_model_2 | n_model_1 | n_model_2 |
|---|---|---|---|---|---|---|
| 0 | 0.809943 | large | 1.637960e+10 | 1.051084e+10 | 88 | 80 |
| 1 | 0.993027 | large | 1.637960e+10 | 7.160551e+09 | 88 | 88 |
| 2 | 0.778151 | large | 1.637960e+10 | 1.105877e+10 | 88 | 88 |
| 3 | 0.934943 | large | 1.051084e+10 | 7.160551e+09 | 80 | 88 |
| 4 | -0.219034 | small | 1.051084e+10 | 1.105877e+10 | 80 | 88 |
| 5 | -0.594525 | large | 7.160551e+09 | 1.105877e+10 | 88 | 88 |
| 6 | 0.975000 | large | 3.576625e+08 | 4.339573e+08 | 88 | 80 |
| 7 | 0.977531 | large | 3.576625e+08 | 7.873249e+07 | 88 | 88 |
| 8 | 0.977531 | large | 3.576625e+08 | 2.220176e+08 | 88 | 88 |
| 9 | -0.975000 | large | 4.339573e+08 | 7.873249e+07 | 80 | 88 |
| 10 | -0.015909 | negligible | 4.339573e+08 | 2.220176e+08 | 80 | 88 |

| | | | | | | |
|---|---|---|---|---|---|---|
| 11 | 0.977273 | large | 7.873249e+07 | 2.220176e+08 | 88 | 88 |
| 12 | -1.000000 | large | 6.051136e+01 | 1.113750e+02 | 88 | 80 |
| 13 | 0.030088 | negligible | 6.051136e+01 | 6.019318e+01 | 88 | 88 |
| 14 | -0.999742 | large | 6.051136e+01 | 8.168182e+01 | 88 | 88 |
| 15 | 1.000000 | large | 1.113750e+02 | 6.019318e+01 | 80 | 88 |
| 16 | 1.000000 | large | 1.113750e+02 | 8.168182e+01 | 80 | 88 |
| 17 | -1.000000 | large | 6.019318e+01 | 8.168182e+01 | 88 | 88 |
| 18 | 0.978125 | large | 3.247818e+09 | 1.986857e+09 | 88 | 80 |
| 19 | 1.000000 | large | 3.247818e+09 | 1.438463e+09 | 88 | 88 |
| 20 | 1.000000 | large | 3.247818e+09 | 1.446045e+09 | 88 | 88 |
| 21 | 0.906250 | large | 1.986857e+09 | 1.438463e+09 | 80 | 88 |
| 22 | 0.935227 | large | 1.986857e+09 | 1.446045e+09 | 80 | 88 |
| 23 | -0.059143 | negligible | 1.438463e+09 | 1.446045e+09 | 88 | 88 |
| 24 | -0.100000 | negligible | 5.450000e+01 | 5.842500e+01 | 88 | 80 |
| 25 | 0.121772 | negligible | 5.450000e+01 | 5.244318e+01 | 88 | 88 |
| 26 | 0.428848 | medium | 5.450000e+01 | 4.521591e+01 | 88 | 88 |
| 27 | 0.237358 | small | 5.842500e+01 | 5.244318e+01 | 80 | 88 |
| 28 | 0.580966 | large | 5.842500e+01 | 4.521591e+01 | 80 | 88 |
| 29 | 0.324768 | small | 5.244318e+01 | 4.521591e+01 | 88 | 88 |
| 30 | 0.803693 | large | 1.277295e+10 | 8.088102e+09 | 88 | 80 |
| 31 | 0.887138 | large | 1.277295e+10 | 5.641965e+09 | 88 | 88 |
| 32 | 0.715909 | large | 1.277295e+10 | 9.388878e+09 | 88 | 88 |
| 33 | 0.909943 | large | 8.088102e+09 | 5.641965e+09 | 80 | 88 |
| 34 | -0.284091 | small | 8.088102e+09 | 9.388878e+09 | 80 | 88 |
| 35 | -0.608471 | large | 5.641965e+09 | 9.388878e+09 | 88 | 88 |
| 36 | -1.000000 | large | 4.273367e+00 | 7.226096e+00 | 88 | 80 |
| 37 | -1.000000 | large | 4.273367e+00 | 9.319944e+00 | 88 | 88 |
| 38 | -1.000000 | large | 4.273367e+00 | 4.833095e+00 | 88 | 88 |
| 39 | -1.000000 | large | 7.226096e+00 | 9.319944e+00 | 80 | 88 |
| 40 | 1.000000 | large | 7.226096e+00 | 4.833095e+00 | 80 | 88 |
| 41 | 1.000000 | large | 9.319944e+00 | 4.833095e+00 | 88 | 88 |
| 42 | 0.417898 | medium | 3.679158e-01 | 9.707118e-02 | 88 | 80 |
| 43 | 0.362087 | medium | 3.679158e-01 | 1.198860e-01 | 88 | 88 |
| 44 | 0.196152 | small | 3.679158e-01 | 1.812696e-01 | 88 | 88 |
| 45 | -0.105256 | negligible | 9.707118e-02 | 1.198860e-01 | 80 | 88 |
| 46 | -0.385085 | medium | 9.707118e-02 | 1.812696e-01 | 80 | 88 |
| 47 | -0.251033 | small | 1.198860e-01 | 1.812696e-01 | 88 | 88 |
| 48 | 0.249148 | small | 3.546248e-01 | 1.860765e-01 | 88 | 80 |
| 49 | 0.462939 | medium | 3.546248e-01 | 1.367926e-01 | 88 | 88 |
| 50 | 0.522082 | large | 3.546248e-01 | 1.347398e-01 | 88 | 88 |
| 51 | 0.292330 | small | 1.860765e-01 | 1.367926e-01 | 80 | 88 |
| 52 | 0.377273 | medium | 1.860765e-01 | 1.347398e-01 | 80 | 88 |
| 53 | 0.098011 | negligible | 1.367926e-01 | 1.347398e-01 | 88 | 88 |
| 54 | 0.263494 | small | 1.095413e+00 | 1.026081e+00 | 88 | 80 |
| 55 | -0.307076 | small | 1.095413e+00 | 1.198417e+00 | 88 | 88 |
| 56 | 0.133135 | negligible | 1.095413e+00 | 1.033046e+00 | 88 | 88 |
| 57 | -0.453409 | medium | 1.026081e+00 | 1.198417e+00 | 80 | 88 |
| 58 | -0.008381 | negligible | 1.026081e+00 | 1.033046e+00 | 80 | 88 |

| | | | | | | |
|---|---|---|---|---|---|---|
| 59 | 0.319473 | small | 1.198417e+00 | 1.033046e+00 | 88 | 88 |
| 60 | 0.381960 | medium | 3.051032e-01 | 7.407307e-02 | 88 | 80 |
| 61 | 0.506586 | large | 3.051032e-01 | 5.953811e-02 | 88 | 88 |
| 62 | 0.537319 | large | 3.051032e-01 | 6.096194e-02 | 88 | 88 |
| 63 | 0.231534 | small | 7.407307e-02 | 5.953811e-02 | 80 | 88 |
| 64 | 0.279403 | small | 7.407307e-02 | 6.096194e-02 | 80 | 88 |
| 65 | 0.048425 | negligible | 5.953811e-02 | 6.096194e-02 | 88 | 88 |
| 66 | 0.388068 | medium | 5.728996e-01 | 3.985247e-01 | 88 | 80 |
| 67 | 0.510072 | large | 5.728996e-01 | 3.658572e-01 | 88 | 88 |
| 68 | 0.584065 | large | 5.728996e-01 | 3.319037e-01 | 88 | 88 |
| 69 | 0.177273 | small | 3.985247e-01 | 3.658572e-01 | 80 | 88 |
| 70 | 0.297301 | small | 3.985247e-01 | 3.319037e-01 | 80 | 88 |
| 71 | 0.131973 | negligible | 3.658572e-01 | 3.319037e-01 | 88 | 88 |
| 72 | 0.149290 | small | 8.781957e-01 | 9.144531e-01 | 88 | 80 |
| 73 | -0.323735 | small | 8.781957e-01 | 1.165567e+00 | 88 | 88 |
| 74 | -0.070377 | negligible | 8.781957e-01 | 1.046012e+00 | 88 | 88 |
| 75 | -0.598580 | large | 9.144531e-01 | 1.165567e+00 | 80 | 88 |
| 76 | -0.296733 | small | 9.144531e-01 | 1.046012e+00 | 80 | 88 |
| 77 | 0.299458 | small | 1.165567e+00 | 1.046012e+00 | 88 | 88 |
| 78 | 0.514773 | large | 2.378138e-01 | 1.262054e-01 | 88 | 80 |
| 79 | 0.427686 | medium | 2.378138e-01 | 1.519883e-01 | 88 | 88 |
| 80 | 0.525956 | large | 2.378138e-01 | 1.196035e-01 | 88 | 88 |
| 81 | -0.140341 | negligible | 1.262054e-01 | 1.519883e-01 | 80 | 88 |
| 82 | 0.097869 | negligible | 1.262054e-01 | 1.196035e-01 | 80 | 88 |
| 83 | 0.228564 | small | 1.519883e-01 | 1.196035e-01 | 88 | 88 |
| 84 | -0.903835 | large | 8.790624e-01 | 9.697264e-01 | 88 | 80 |
| 85 | -0.819473 | large | 8.790624e-01 | 9.547864e-01 | 88 | 88 |
| 86 | -0.845558 | large | 8.790624e-01 | 9.612689e-01 | 88 | 88 |
| 87 | 0.266051 | small | 9.697264e-01 | 9.547864e-01 | 80 | 88 |
| 88 | 0.148438 | small | 9.697264e-01 | 9.612689e-01 | 80 | 88 |
| 89 | -0.113895 | negligible | 9.547864e-01 | 9.612689e-01 | 88 | 88 |
| 90 | 0.371165 | medium | 2.815986e-01 | 6.352941e-02 | 88 | 80 |
| 91 | 0.477402 | large | 2.815986e-01 | 4.801648e-02 | 88 | 88 |
| 92 | 0.509168 | large | 2.815986e-01 | 5.264021e-02 | 88 | 88 |
| 93 | 0.175142 | small | 6.352941e-02 | 4.801648e-02 | 80 | 88 |
| 94 | 0.225426 | small | 6.352941e-02 | 5.264021e-02 | 80 | 88 |
| 95 | 0.051911 | negligible | 4.801648e-02 | 5.264021e-02 | 88 | 88 |

[63]:
```
#Association between Text Augmentation with Prompts

import pandas as pd
import numpy as np
from scipy.stats import chi2_contingency
from sklearn.metrics import mutual_info_score

# Optional: If you don't have `theils_u` function, define it:
def theils_u(x, y):
```

```
    """Theil's U (uncertainty coefficient U(x|y)), asymmetric"""
    s_xy = mutual_info_score(x, y)
    s_x = entropy_from_series(x)
    return 0 if s_x == 0 else s_xy / s_x

def entropy_from_series(s):
    p = s.value_counts(normalize=True)
    return -np.sum(p * np.log2(p + 1e-12))

# Build contingency table
contingency = pd.crosstab(df['augmentation_type'], df['prompt_id'])

# 1. Chi-square Test
chi2_stat, p_chi2, dof, expected = chi2_contingency(contingency)

# 2. Cramér's V
n = contingency.values.sum()
cramers_v = np.sqrt(chi2_stat / (n * (min(contingency.shape) - 1)))

# 3. Mutual Information Score
mi = mutual_info_score(df['augmentation_type'], df['prompt_id'])

# 4. Theil's U (both directions, asymmetric)
u_aug_given_prompt = theils_u(df['augmentation_type'], df['prompt_id'])
u_prompt_given_aug = theils_u(df['prompt_id'], df['augmentation_type'])

# 5. Summary Table
summary = pd.DataFrame({
    'Test': [
        "Chi-square Test of Independence",
        "Cramér's V (Effect Size)",
        "Mutual Information Score",
        "Theil's U (Augmentation|Prompt)",
        "Theil's U (Prompt|Augmentation)"
    ],
    'Statistic/Value': [
        f"Chi2={chi2_stat:.5g}, dof={dof}, p={p_chi2:.5g}",
        f"{cramers_v:.5f}",
        f"{mi:.5f}",
        f"{u_aug_given_prompt:.5f}",
        f"{u_prompt_given_aug:.5f}"
    ]
})

print("Association between Text Augmentation with Prompts")
display(summary)
```

```
Association between Text Augmentation with Prompts
```

```
                                Test            Statistic/Value
0  Chi-square Test of Independence  Chi2=0.86154, dof=70, p=1
1         Cramér's V (Effect Size)                    0.01892
2         Mutual Information Score                    0.00125
3  Theil's U (Augmentation|Prompt)                    0.00036
4  Theil's U (Prompt|Augmentation)                    0.00042
```

[64]:
```
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt
import seaborn as sns

# 2. Select Numeric Columns for PCA
numeric_cols = [
    'total_duration_ns', 'load_duration_ns', 'prompt_eval_count',
    'prompt_eval_duration_ns', 'eval_count', 'eval_duration_ns',
    'tokens_per_second', 'levenshtein_similarity', 'jaccard_similarity',
    'length_ratio', 'bleu', 'cosine_similarity', 'wer', 'char_diversity',
    'type_token_ratio', 'bigram_overlap'
]
df_numeric = df[numeric_cols].dropna()

# 3. Standardize
scaler = StandardScaler()
X_scaled = scaler.fit_transform(df_numeric)

# 4. PCA
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)
df_pca = pd.DataFrame(X_pca, columns=['PC1', 'PC2'])

# 5. KMeans
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
clusters = kmeans.fit_predict(X_pca)
df_pca['Cluster'] = clusters

# 6. Plotting
plt.figure(figsize=(8, 6))
sns.scatterplot(data=df_pca, x='PC1', y='PC2', hue='Cluster', palette='Set2',␣
 ↪s=80)
plt.title('KMeans Clustering on PCA-Reduced Augmentation Metrics')
plt.xlabel('Principal Component 1')
plt.ylabel('Principal Component 2')
plt.grid(True)
plt.tight_layout()
```

```
plt.show()



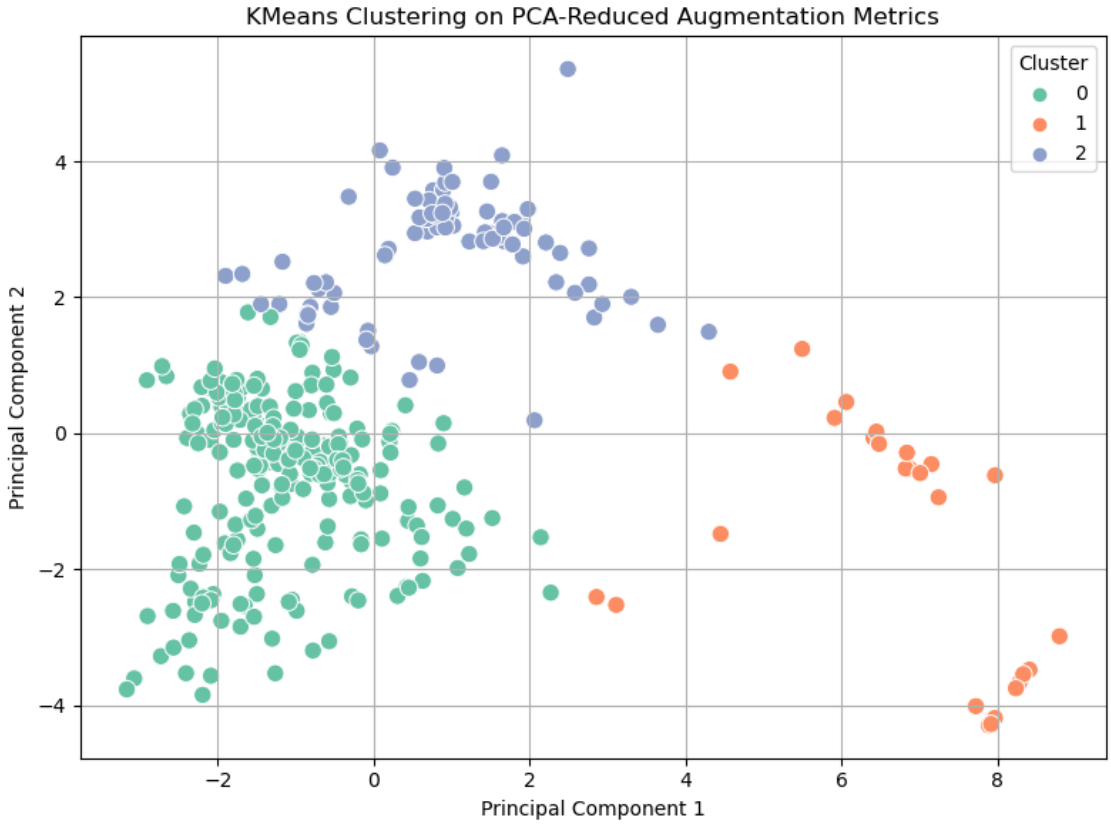
# Print the cluster assignments for the first few rows and the centroids
print("Cluster assignments (first 10):")
print(df_pca['Cluster'].head(10))

print("\nKMeans cluster centers in PCA space:")
print(kmeans.cluster_centers_)

# Count the number of points in each cluster
print("\nNumber of points in each cluster:")
print(df_pca['Cluster'].value_counts())

loadings = pd.DataFrame(
    pca.components_.T,  # shape: (n_features, n_components)
    index=numeric_cols,
    columns=['PC1', 'PC2']
)
print(loadings)
```

```
C:\Users\parth\anaconda3\Lib\site-
packages\joblib\externals\loky\backend\context.py:110: UserWarning: Could not
find the number of physical cores for the following reason:
[WinError 2] The system cannot find the file specified
Returning the number of logical cores instead. You can silence this warning by
setting LOKY_MAX_CPU_COUNT to the number of cores you want to use.
  warnings.warn(
  File "C:\Users\parth\anaconda3\Lib\site-
packages\joblib\externals\loky\backend\context.py", line 199, in
_count_physical_cores
    cpu_info = subprocess.run(
               ^^^^^^^^^^^^^^
  File "C:\Users\parth\anaconda3\Lib\subprocess.py", line 548, in run
    with Popen(*popenargs, **kwargs) as process:
         ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\parth\anaconda3\Lib\subprocess.py", line 1026, in __init__
    self._execute_child(args, executable, preexec_fn, close_fds,
  File "C:\Users\parth\anaconda3\Lib\subprocess.py", line 1538, in
_execute_child
    hp, ht, pid, tid = _winapi.CreateProcess(executable, args,
                       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
C:\Users\parth\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1436:
UserWarning: KMeans is known to have a memory leak on Windows with MKL, when
there are less chunks than available threads. You can avoid it by setting the
environment variable OMP_NUM_THREADS=2.
  warnings.warn(
```




```
Cluster assignments (first 10):
0    0
1    0
2    0
3    0
4    0
5    0
6    0
7    0
8    0
9    0
Name: Cluster, dtype: int32

KMeans cluster centers in PCA space:
[[-1.10796476 -0.66622526]
 [ 6.82928688 -1.79858054]
 [ 0.9291741   2.67848547]]

Number of points in each cluster:
Cluster
0    238
```

```
2     78
1     28
Name: count, dtype: int64
                             PC1       PC2
total_duration_ns        0.226049  0.368875
load_duration_ns         0.028109  0.105715
prompt_eval_count       -0.091955 -0.058772
prompt_eval_duration_ns  0.271370  0.210033
eval_count               0.062268  0.354597
eval_duration_ns         0.230526  0.390617
tokens_per_second       -0.216960 -0.162517
levenshtein_similarity   0.313847 -0.260757
jaccard_similarity       0.370734 -0.155544
length_ratio             0.048719  0.257775
bleu                     0.374915 -0.149463
cosine_similarity        0.342834 -0.069804
wer                     -0.272953  0.314114
char_diversity           0.010643  0.364538
type_token_ratio        -0.228997 -0.236165
bigram_overlap           0.372093 -0.165159
```

[ ]: